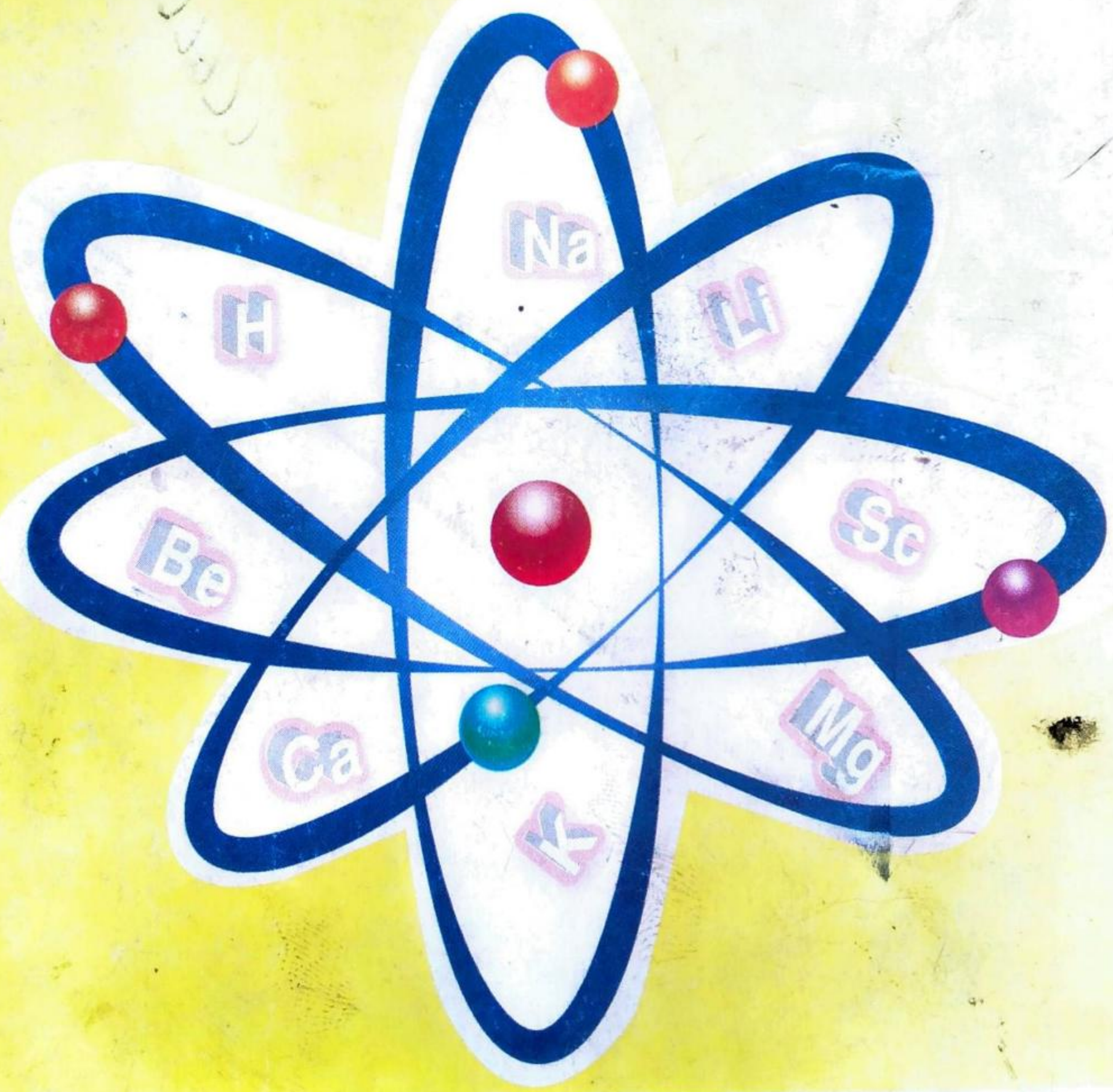
Extreme Series

CHEMISTRY

GRADE 9-10



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UNIT 1

Structure of the Atom

Unit Outline:

1.1 Dalton's Atomic Theory

1.2 Modern Atomic Thoery

1.3 Discovery of Fundamental Subatomic Particles

1.4 Atomic Models

1.5 Electron Configuration

Introduction

An atom is the smallest characteristics particle of an element that takes part in chemical reactions

Historical development of the atomic theory

Democritus

- ➦ proposed the first atomic theory
- ➦ believed that matter was discontinuous (division has limit)

Arstotle

- ➦ was opponent of Democritus
- ➦ believed that matter was discontinuous
- ➦ division has no limit

1.1 Dalton's Atomic theory

Dalton was the founder of modern atomic theory. He carried out a series of experiments on the nature of gases and the way how elements combine together to form compounds.

The main postulates of Dalton's atomic theory are:

1. All substances are made up of very tiny particles called atoms.
2. Atoms are indivisible and indestructible
3. All atoms of a given element are identical in mass and other properties.
4. Atoms of one element differ from atoms of any other element in their properties.
5. Two or more different elements combine in a fixed whole number ratio to form compounds.
6. A chemical reaction involves a change not in the atoms themselves but in the way atoms are combined to form compounds,

Note:

postulate 2 and 3 were known to be incorrect by later work

1.2 The modern atomic theory

This theory can be summarized as follows:

1. Elements are made up of small particles called atoms.
2. Atoms cannot be created or destroyed during ordinary chemical reactions.
3. Atoms are divisible. They can be subdivided into electrons, protons and neutrons
4. All atoms of the same element have different masses
5. Atoms of different element are different
6. Atoms combine in small whole number ratio to form compounds.

Illustrative Example

1. The Greek philosopher that proposed the first atomic theory was:

   A. Arstotle
   B. Democritus
   C. Socritos
   D. Empedoeles

   Answer: B

2. How Dalton's atomic theory explained three laws of chemical reactions.

Explanation:

i Law of conservation of mass since a chemical reaction involves only the separation, union and rearrangement of atom, mass is conserved in a chemical reaction.

ii Law of definite proportions – according to Dalton's atomic theory, since atoms combine in small whole number ratios, a compound should contain the same elements combined in a definite ratio by mass.

iii Law of multiple proportions – since atoms combine in whole number ratios, the masses of one element combining with a fixed mass of the other element must be in whole number ratio.

3. Give some examples that illustrates the law of multiple proportion

Explanation:

➦ SnO and SnO_2

- ➦ CO and CO_2
- ➦ N_2O, NO, N_2O_3, NO_2 and N_2O_5

4. List two postulates of Dalton that were rejected by the modern atomic theory.

Explanation:

- ➦ Atoms are indivisible
- ➦ Atoms of the same element have identical mass.

1.3 Discovery of fundamental subatomic particles

1. Electron

- ➦ was discovered by J.J. Thomson
- ➦ Was the first to be identified
- ➦ is negatively charged particle

The charge to mass ratio of an electron was determined by J.J. Thomson and found to be -1.76×10^8c/g

The charge of an electron was found by R. Millikan using oil drop experiment. It is -1.6×10^{-19}C

2. Nucleus

Nuclei of some elements are not stable. They spontaneously emit radiations. This phenomenon is called radioactivity.

Radioactive substances emit three types of radiations

Table 1.1. Different radiations, their charge, symbol and relative mass.

Radiation	Symbol	Charge	Relative mass	Composition
Alpha particles	α	+2	4	Similar to the helium nucleus
Beta particles	β	-1	0	Identical to the electron
Gamma rays	γ	0	0	high energy radiation

Ernest Rutherford bombarded a thin gold foil with alpha radiation from radioactive element. From the experiment, he got the following conclusions:

- Most of the alpha particles passed through the foil undeflected. This indicated that most space in atom is empty.
- Some particles were scattered through moderate angles. This is due to positive charge.
- A few alpha particles were deflected back. This suggested that there is a heavy dense particle at the center.

Neutrons are chargless particles discovered by James Chad Wick

Makeup of the nucleus

Nucleus contains two heavy subatomic particles **protons** and **neutrons**. Since protons and neutrons make up the nuclei of atoms, they are collectively known as **nucleons**

Table 1.2. Properties of subatomic particles

Subatomic particle	Symbol	mass in gram	Approximate relative mass (amu)	Charge
Electron	e^-	9.11×10^{-28}	0	Negative
Proton	p^+	1.672×10^{-24}	1	Positive
Neutron	n^0	1.675×10^{-24}	1	Neutral

Illustrative Example

5. Which particle among the following has the highest charge to mass ratio? (EGSECE)

   A. neutron
   B. nucleus
   C. electron
   D. proton

Explanation: Since electron has the smallest mass, the highest charge to mass ratio corresponds to it.

Answer: C

6. What was determined from R-Millikan's oil drop experiment?

   A. Charge of electron
   B. Mass of an electron
   C. charge of proton
   D. mass of proton

Answer: A

7. In Ruthertord's experiment, most of the alpha particles passed through the gold foil. What does this indicate?

   A. The presence of positive charge

   B. most of the space in an atom is empty

   C. the space occupied by heavy positive center is small

   D. the center of an atom contains electron.

   Answer: B

8. An electron has (EGSECE...2007 E.C)

   A. a charge -1 and a smaller mass than a proton

   B. a charge of +1 and a smaller mass than a proton

   C. a charge of -1 and a the same massosa proton

   D. a charge of +1 and the same mass as a proton

   Answer: A

9. Calculate the mass of an electron from its charge and charge to mass ratio.

   Solution:

   $$\frac{e}{m} = -1.76 \times 10^{8} c / g$$

   $$\ell = -1.60 \times 10^{-19} c$$

   $$m_e = \frac{e}{e / m_\ell}$$

   $$m_e = \frac{-1.60 \times 10^{-19} c}{-1.76 \times 10^{8} c / g} = 9.09 \times 10^{-28} g$$

10. Define cathode rays and list their properties.

Explanation: cathode rays are also called an electron beam or streams of electrons observed in vacuum tube in the discharge tube experiments.

Cathode rays:

- travel in straight line
- are negatively charged
- have mass
- are found in all elements

11. What are the two regions of an atom?

Explanation:

- Atomic shell – has electron
- Nucleus – has proton and neutron

Atomic number and mass number

Atomic number (Z) is the number of protons in an atom.

For a neutral atom:

Atomic number = proton number = electron number

Mass number (A) is the sum of atomic number and neutron number.

$$A = Z + n \text{ or } A = p + n$$

Example

Aluminum has 13 protons, 13 electrons and 14 neutrons. Find its mass number.

Solution:

$A = p + n$

$A = 13 + 14$

$A = 27$

Mass number is equal to number of nucleons

Isotopes and atomic mass

Isotopes are atoms of the same element with the same number of protons but different number of neutrons.

Representation of isotopes.

i **Nuclear symbolic representation:** ${}^{A}_{Z}X$

$A \rightarrow$ mass number $\qquad Z \rightarrow$ atomic number

$X \rightarrow$ symbol

Example

Carbon has three isotopes represented as ${}^{12}_{6}C$, ${}^{13}_{6}C$ *and* ${}^{14}_{6}C$

ii **Hyphen notation**

X – A

Example

C – 12, C – 13 and C – 14

Atomic mass

By international agreement, an atom of the carbon isotope called carbon – 12 has exactly 12 atomic mass units (amu)

mass of one carbon – 12 atom = 12amu

$$1\text{amu} = \frac{mass\ of\ one\ carbon - 12\ atom}{12}$$

$$1\ \text{amu} = 1.66 \times 10^{-24}\text{g}$$

$$\text{Atomi mass of an element} = \frac{A_1 f_1 + A_2 f_2 + ---- A_n f_n}{100}$$

- $A_1, A_2 \ldots A_n$ are atomic mass of first, second, …n^{th} isotope respectively .
- $f_1, f_2, \ldots f_n$ are percentage abundance of first, second, …..n^{th} isotope respectively.

$$\text{Atomic mass of an element} = A_1 x_1 + A_2 x_2 + \cdots A_n x_n$$

- $x_1, x_2, - - - - x_n$ are fraction abundance of first , second - - - - n^{th} isotope.

Example

The atomic masses of the two stable isotopes of boron, ${}^{10}_{5}B$ (19.78%) and ${}^{11}_{5}B$ (80.22%) are 10.0129 amu and 11.0093 amu respectively. Calculate the average atomic mass of boron?

Solution:

$A_1 = 10.0129$ amu $\qquad f_1 = 19.78\%$

$A_2 = 11.0093$ amu $\qquad f_2 = 80.22\%$

$$Av.atomic\ mass\ B = \frac{A_1 f_1 + A_2 f_2}{100}$$

$$Av.atomic\ mass\ B = \frac{10.0129 \times 19.78 + 11.0093 \times 80.22}{100}$$

$$= \frac{198.055 + 883.166}{100}$$

$= 10.8$amu

Illustrative Example

12. Given the chemical symbol of a certain element as ${}^{50}_{23}Z$, how many neutrons does it contain?

A. 73 B. 27 C. 23 D. 50

Answer: B

13. The number of neutrons in the nucleus of an iron atom $\left({}^{56}_{26}Fe\right)$ is (EGSECE)

A. 82 B. 56 C. 30 D. 26

Solution:

$A = p + n$

$n = A - P$

$n = 56 - 26 = 30$

Answer: C

14. Isotopes are atoms of the some element that differ in their.

A. Number of protons
B. Number of electrons
C. mass number
D. atomic number

Explanation:

Isotopes are atoms of the same element that differ in neutron number. mass number and physical property but similar in atomic number. proton number. electron number and chemical property.

Answer: C

15. Which of the following is true about an atom?

A. It has equal number of protons. electrons and neutrons
B. All parts of an atoms have uniform density
C. An atom is the smallest particle of a substance that takes part in a chemical reaction
D. Most of the parts of an atom are occupied by sub-atomic particles.

Answer: C

16. Gallium has two naturally occurring isotopes. Information about the two isotopes is given in the table below

Isotope	Atomic mass (amu)	Percentage abundance (%)
Ga – 69	68.93	60.11
Ga – 71	70.92	39.89

What is the average atomic mass of gallium (EGSECE)

A. 69.54 B. 70.29 C. 70.00 D. 69.72

Solution:

$$Av.\,atomic\,mass = \frac{A_1 f_1 + A_2 f_2}{100}$$

$$= \frac{68.93 \times 160.11 + 70.92 \times 39.89}{100}$$

$$= 69.72amu$$

Answer: D

17. The number of nucleons in $^{238}_{92}U$ is

A. 92 B. 238 C. 146 D. 330

Explanation: number of nucleons is the same as mass number

Answer: B

18. Determine the number of electrons and neutrons in

a. $^{71}_{31}Ga^{+3}$

b. $^{128}_{52}Te^{2-}$

Solution:

a. $e = p - charge$

$e = 31 - (+3)$

$e = 28$

b. $e = p - charge$

$e = 52 - (-2)$

$e = 54$

$n = A - P$

$n = 71 - 31$

$n = 40$

$n = A - P$

$n = 128 - 52$

$n = 76$

19. Chlorine has two isotopes, Cl – 35 and Cl – 37 with average atomic mass 35.5. Find the percentage abundance of each isotope.

Solution:

$A_1 = 35$

$A_2 = 37$

$f_1 + f_2 = 100$

$f_2 = 100 - f_1$

$$Av.\,atomic\,mass = \frac{A_1 f_1 + A_2 f_2}{100}$$

$$35.5 = \frac{35 f_1 + 37 f}{100} \quad \text{but } f_2 = 100 - f_1$$

$$35.5 = \frac{35 f_1 + 37(100 - f_1)}{100}$$

$3550 = 35f_1 + 3700 - 37f_1$

$3550 - 3500 = 35f_1 - 37f_1$

$-150 = -2f_1$

$f_1 = 75\%$ $f_2 = 100 - 75 = 25\%$

20. The atomic masses of ${}^{6}_{3}Li$ *and* ${}^{7}_{3}Li$ are 6.0151 amu and 7.0160 amu with fraction abundance of 0.07493 and 0.92507 respectively. Find the average atomic mass of lithium.

Solution:

$A_1 = 6.0151\text{amu}, \quad x_1 = 0.07493$

$A_2 = 7.0160\text{amu}, \quad x_2 = 0.092507$

Av. Atomic mass $= A_1 x_1 + A_2 x_2$

Li

$= 6.0151 \times 0.07493 + 7.0160 \times 0.92507$

$= 0.45 + 6.49 \quad = 6.94\text{amu}$

21. A charged atom has 15 protons, 16 neutrons and 18 electrons. Write nuclear symbolic representation for this atom using a symbol P.

Solution: First find the mass number and charge of the atom.

$A = P + n$	charge $= P - e$
$A = 15 + 16$	charge $= 16 - 18$
$A = 31$	charge $= -3$

Thus, symbolic representation $= {}^{31}_{15}P^{3-}$

22. Is there an atom with more than one protons but no neutron?

Explanation: No, because neutrons reduce the electrostatic force of repulsion between protons.

1.4 Atomic models

The ideas about atomic model have been changed many times and we summarize these into five models.

1. The Dalton model

➦ He thought of atoms as solid indivisible particles. This model is also called the **billiard ball model**

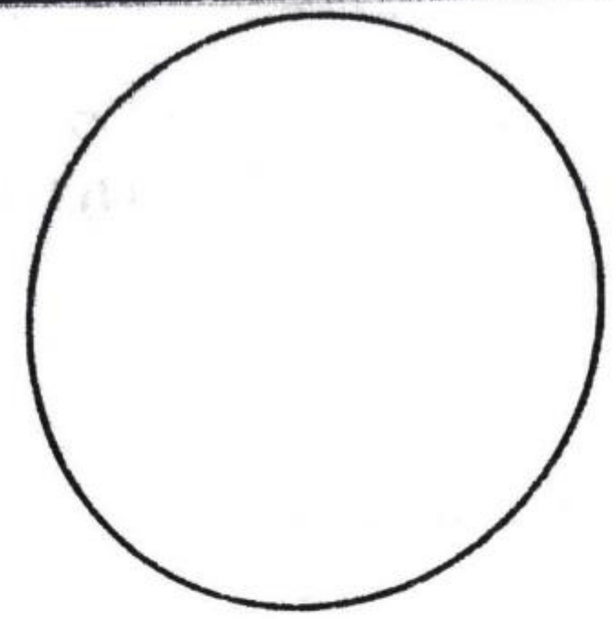


Figure 1.1 Dalton's atom

2. The Thomson model

He suggested a **plum – pudding model** of an atom in which electrons are scattered throughout a positively charged sphere.

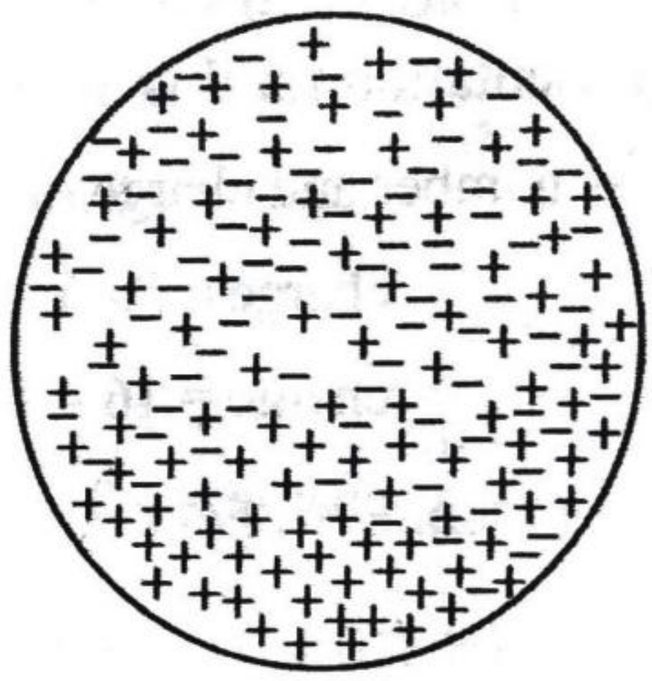


Figure 1.2 The Thomson atom

3. The Rutherford model

This model of an atom is sometimes called the **planetary** model. This **model** of an atom has a tiny, dense, positively charged nucleus which electrons whirled at high speeds and at relatively high distance from it.

4. The Bohr model

Bohr amended Rutherford's model in two ways

i The electrons in an atom can exist only in a restricted number of stable **orbits** in which they neither absorb or emit energy

ii An electron can jump from one orbit to another by absorbing energy and fall by emitting energy.

Bohr model gives the energy levels of a hydrogen atom. These energy levels can be designated by the numbers 1,2,3,----- or letters K,L,M,N----etc.

K – shell – 1st energy level

L – shell – 2nd energy level

M – shell – 3rd energy level

N – shell – 4th energy level

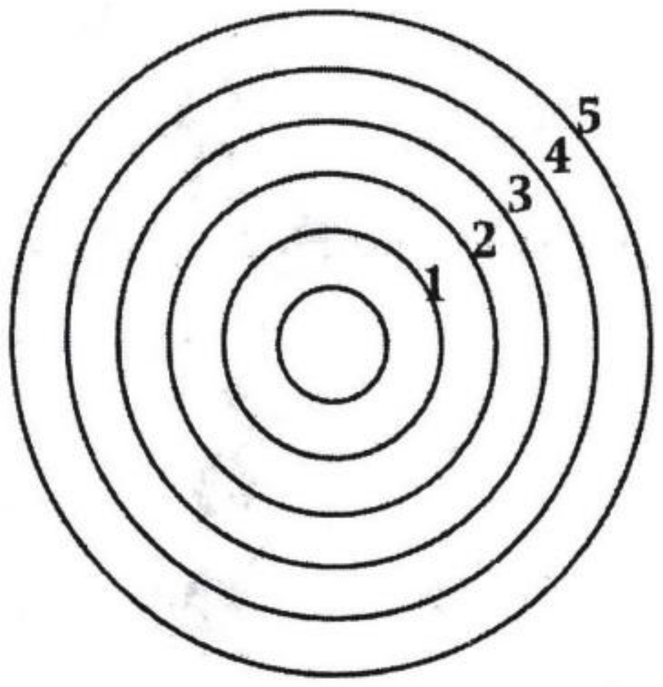


Figure 1.3 The first five shells in the Bohr model of the hydrogen atom.

Bohr also suggested that different orbit /energy levels/ of an atom can only accommodate a certain number of electrons

The maximum number of electrons in any orbit can be given by the formula $2n^2$ where, n = 1, 2, 3, 4- - -

For example, the number of electrons in the first, second and third energy levels are:

- n = 1, number of electrons = $2n^2 = 2(1)^2 = 2$ electrons
- n = 2, number of electrons = $2n^2 = 2(2)^2 = 8$ electrons
- n = 3, number of electrons = $2(n)^2 = 2(3^2) = 18$ electrons

But, regardless of its maximum capacity, *any outer most shell of an atom cannot hold more than eight electrons.*

Bohr's Diagrammatic representation using shells in some atoms.

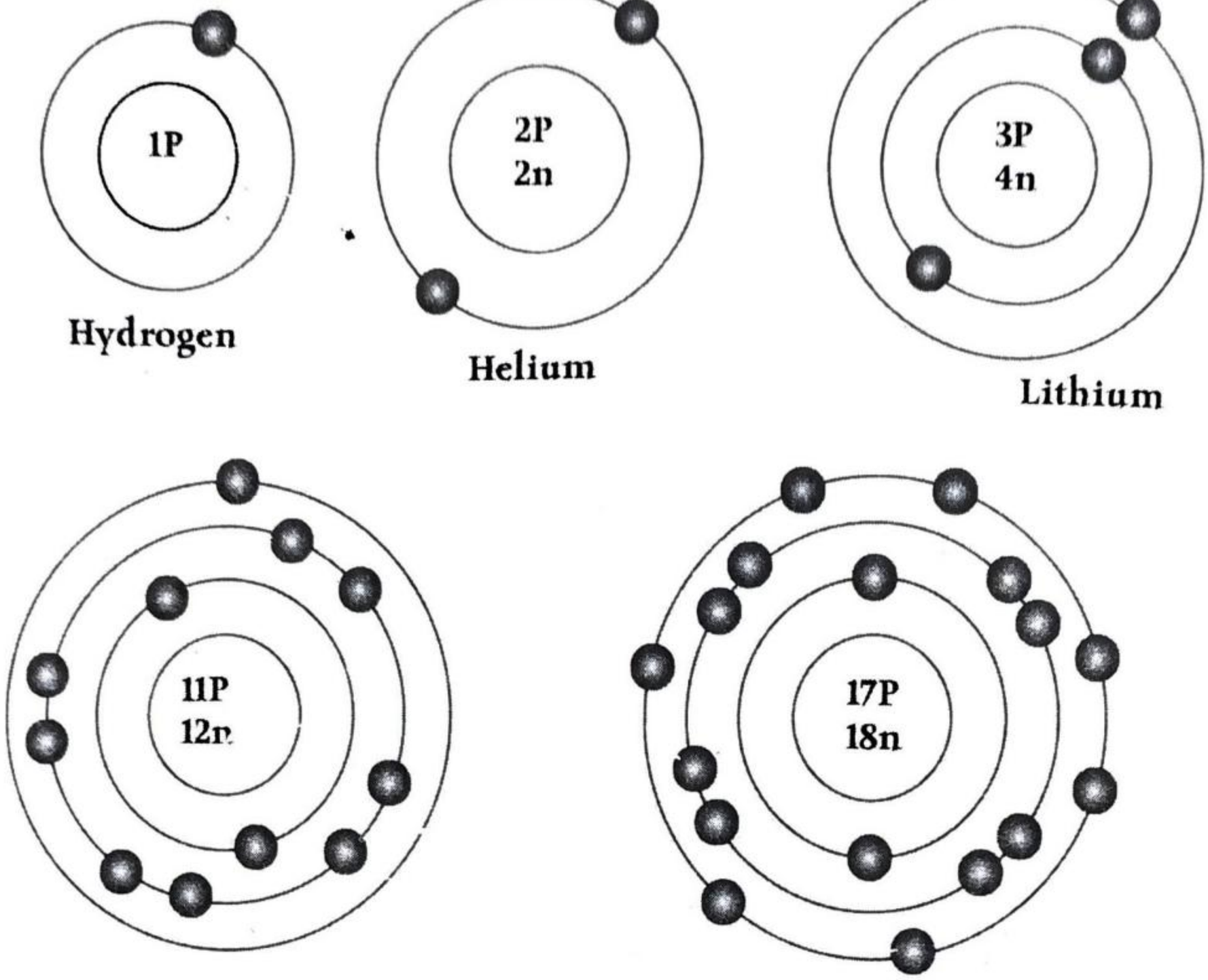


Figure 1.4 Diagramatic representation of sub atomic particles

Illustrative Examples

23. Whose model was discarded as a result of Rutherford's model?

A. Dalton's model
B. Thomson's model
C. Bohr's morel
D. Quantum model

Explanation: Electrons and protons have opposite charges and cannot be found together. This idea was explained by Rutherford.

Answer: B

24. Which of the following statements best describes Bohr's postulate about the electronic structures of an atom? (EGSECE)

A. Electrons occupy restricted orbits and their transition between the orbits is accomparied by their absorption or emission of energy
B. Electrons occupy random space around the nucleus and transition is not allowed for the electrons under any circumstance

C. When the electrons fall from higher (excited) state to the lower (ground) state, they absorb energy

D. When the electrons jump from a lower to a higher energy state, they emit a fixed amount of energy.

Explanation: Bohr's postulates include the following:

- ➦ The hydrogen atom has only certain allowable energy levels called orbits.
- ➦ The atom does not radiate energy while in one of its orbit.
- ➦ The electron moves from lower orbit to higher orbit by absorbing energy.
- ➦ The electron falls from higher orbit to lower orbit by emitting energy

Answer: A

25. Which model of the atom can be pictured as "planetary atom"? **(EGSECE)**

A. Bohr's model
C. Thomson's model
B. Rutherford's model
D. Dalton's model

Answer: B

26. What is the maximum number of electrons in the N – shell?

A. 32 B. 18 C. 16 D. 8

Solution: N – shell is the fourth shell (n = 4)

number of electrons = $2n^2 = 2(4)^2 = 32$

Answer: A

27. The atomic models of Thomson, Dalton, Rutherford and Bohr, all support which of the following statements? (EGSECE)

A. All matter is composed of atoms

B. Atoms are mostly composed of empty space.

C. Atoms are composed of electrons and protons

D. Electrons orbits the nucleus of an atom at distinct energy levels.

Answer: A

28. Why did scientists consider Rutherford's nuclear model of the atom incomplete?

Explanation: Rutherford's model is said to be incomplete because his model lacked details on how electrons of an atom are arranged around the nucleus

29. What were the main limitations of Bohr's model of the atom?

Explanation: The main defects of Bohr's theory were:

- Bohr thought that electrons move in orbits around the nucleus. Today, they are described in terms of the probability of being found in certain regions of space around the nucleus.
- His model was not successful for many electron atoms.

5. The Quantum mechanical model of the atom.

- It is also called **Shrödinger** or **wave mechanical model**
- It is the latest explanation which describes electrons to be in constant motion somewhere outside the nucleus.
- The exact position and speed of an electron around the nucleus cannot be obtained at the same time. (uncertainity principle)
- The region in space **around the nucleus** where high probability of getting an electron is called **orbital**
- This model introduced sub – shells designated by small letters, s,p, d, f, g,- - - -

Number of orbitals = n^2

Number of sub–shell = n

Number of electrons in an orbital = 2

Number of electrons in main shell = $2n^2$

Table 1.3. Energy sub–levels for the first four main energy level.

Main energy level		Number of sub– levels	Types of sub – level	Maximum number of electrons in main shell
Principal quantum number (n)	Symbol			
1	K	1	s	2
2	L	2	s,p	8
3	M	3	s,p,d	18
4	N	4	s, p,d, f	32

Table 1.4. number of orbitals and maximum electrons in sub-shell

Sub-shell	Number of orbital	number of electrons
s	1	2
p	3	6
d	5	10
f	7	14

Illustrative Examples

30. The total number of orbitals in the fourth main energy level (n = 4) is (EGSECE)

A. 2 B. 4 C. 8 D. 16

Solution: number of orbitals = $n^2 = 4^2 = 16$

Answer: D

31. Which one of the following atomic models was developed with a proposition that electrons can be described only interms of the probability of their location?

A. Rutherford's model
B. Dalton's model
C. Quantum mechanical model
D. Bohr's model

Answer: C

32. According to the quantum mechanical model of the atom **(EGSECE...2007)**

A. electrons are embedded in an atom and remain stationary
B. the atom is a positively charged solid sphere
C. orbitals are represented as clouds of charge
D. electrons can revolve in all orbits with all possible radii

Answer: C

33. Which sequence represents a correct order of the historical development that led to the modern model of the atom? (EGSECE)

A. Most of the atom is empty → the atom is a hard sphere → electrons exist in orbitals outside the nucleus
B. The atom is a hard sphere → most of the atom is empty → electrons exist in orbitals outside the nucleus

C. Most of the atom is empty → electrons exist outside the nucleus → the atom is a hard sphere

D. The atom is a hard sphere → electrons exist in orbitals outside the nucleus → most of the atom is empty

Answer: B

34. A particular shaped volume of space in an atom where the probability of finding an electron is maximum is known as

A. an orbital
B. an orbit
C. a sub energy level
D. a shell

Answer: A

35. Determine the number of sub shells and orbitals in the 5^{th} main shell?

Solution:

➦ Number of shells = n = 5 (s, p, d, f, g)

➦ Number of orbitals = $n^2 = 5^2 = 25$

36. In quantum mechanical model concept, each energy level is described by a number called _____

Answer: principal quantum number

37. What is the relationship between principal quantum number (n), size and energy of orbital?

Explanation: As the value of n increases the size and energy of orbital also increase.

1.5 Electron configuration

➦ It is the arrangement of electrons about the nucleus of an atom according to quantum mechanical model description

➦ There are different types of electron configurations:

i **Main shell electron configuration**

Example

For Na (z = 11) : 2, 8, 1

ii Sub – shell electron configuration

➦ This is followed by **afbua principle** (building up principle). This principle states that sub – shells with lowest energy filled with electrons first.

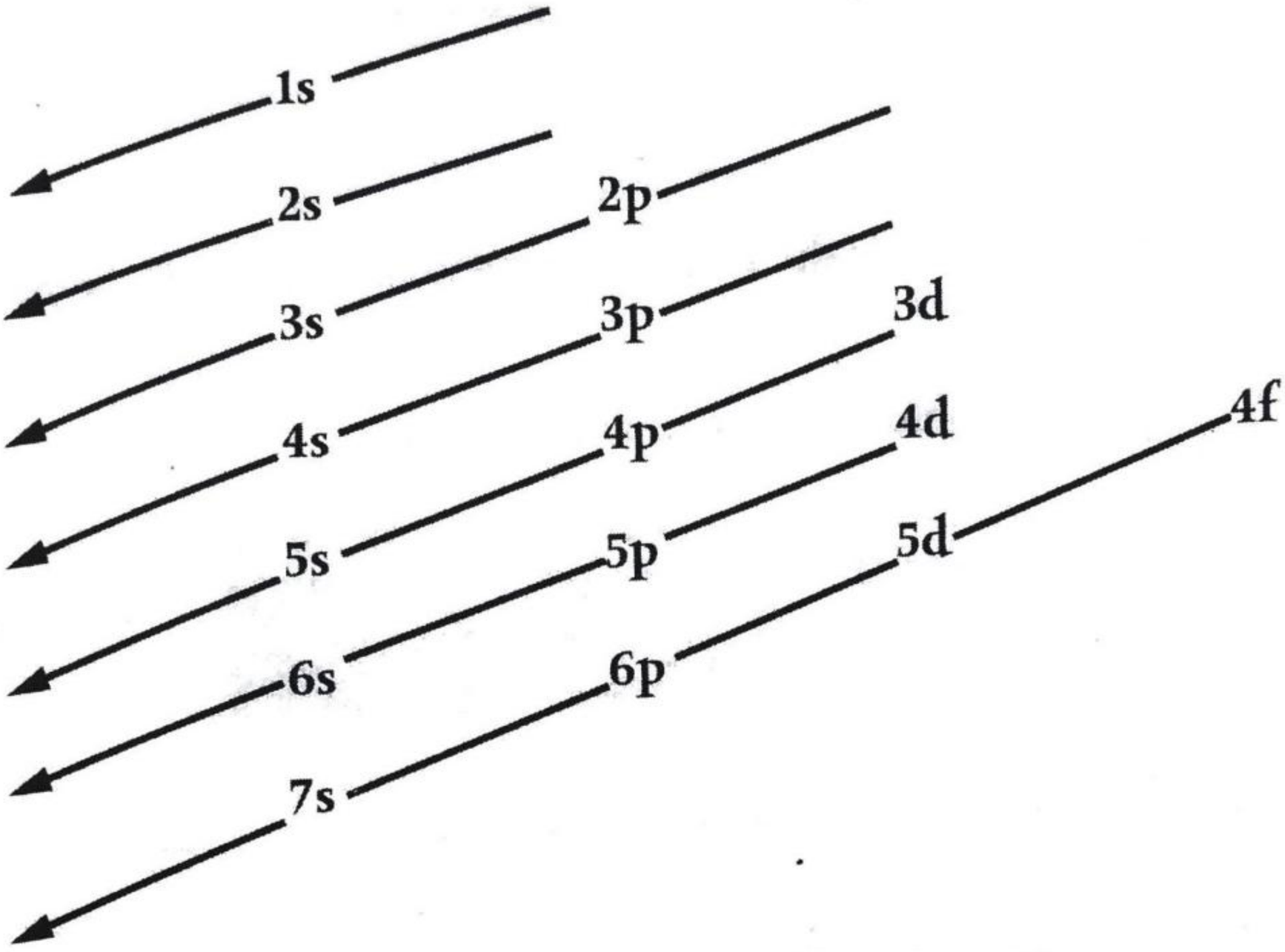


Figure 1.5 diagonal rule for filling electrons

➦ If you list the sub shells following the arrows, you can get electron configuration of most atoms.

Example

Write electron configuration for:

a. Si (Z = 14)

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

b. Fe (Z = 26)

$1s^2\ 2s^2\ 2p^6\ 3p^6\ 4s^2\ 3d^6$

iii Noble gas Core configuration

➦ This is using noble gases interms of inner electrons (all electrons out of valence electrons)

➦ The outer most occupied shell of an atom is called **valence shell**

➦ The electrons that enter in the valence shell are called **valence electrons.**

Example

Consider Mg (z = 12)

➦ $1s^2\ 2s^2\ 2p^6\ 3s^2$

- The third shell is the valence shell and there are 2 electrons in this shell
- The core noble gas configuration becomes:

[Ne] $3s^2$ because a noble gas with 10 electrons is neon.

Species that have the same electron configuration are called isoelectronic species.

Example

➦ Cl^-, Ar, K^+ and Ca^{+2} are isoelectronic because they have the same number of electrons

Illustrative Examples

38. Which of the following is the correct ground state electronic configuration for $_{28}Ni$? (EGSECE)

A. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 4p^6\ 3d^2$

B. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^9$

C. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$

D. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^8$

Explanation:→ $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^8$

Answer: D

39. How many electrons enter into 3d sub – shell of an atom whose atomic number is 21?

A. 9 B. 3 C. 0 D. 1

Explanation: → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^1$

Thus, there is one electron is 3d

Answer: D

40. Which of the following is the correct electron configuration for an atom having atomic number 24?

A. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^4$

B. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$

C. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$

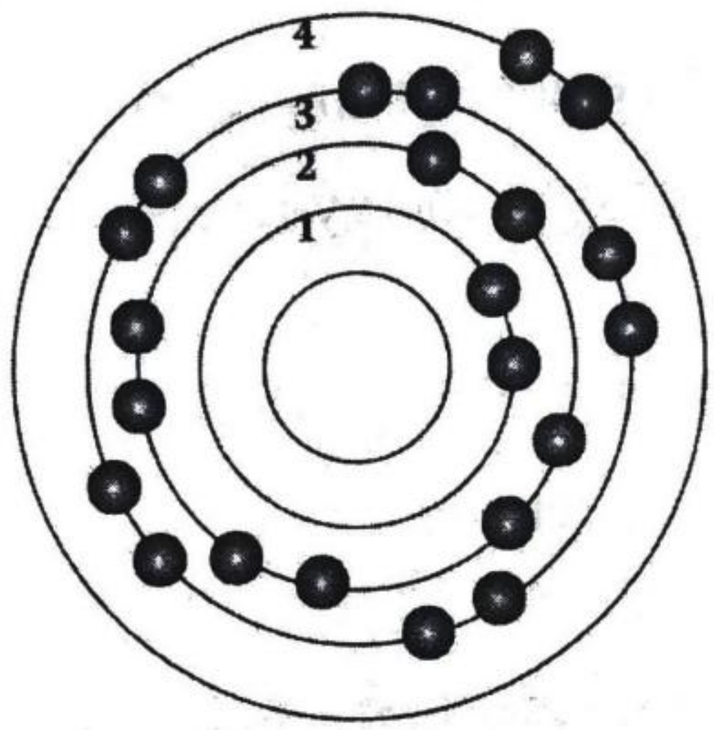
D. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$

Explanation: Chromium (z = 24) and copper (z = 29) violate the building up principle to have half – filled and fulfilled stable orbitals respectively.

- Cr (z = 24): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$
- Cu (z = 29) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$

Answer: D

41. The following diagram shows Bohr model for a certain atom.



In which energy level lsl do the valence electrons located?

A. 2 and 3

B. 1

C. 3 and 4

D. 4

Explanation: The valence electrons are found in the outer most shell i.e. in the 4^{th} shell.

Answer: D

42. In the notation $3d^1$ for the electron configuration, 3, d and 1 respectively stand for:

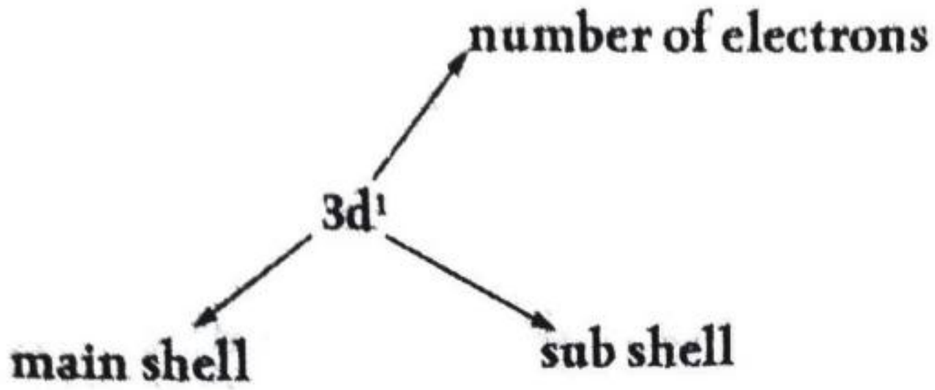
A. Main energy level, sub shell and number of electron

B. Number of electron, sub shell and main energy

C. Main energy level, number of electrons and sub shell

D. Sub – shell, main energy level and number of electrons.

Explanation:



Answer: A

43. Which of the following is true about an element with electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$?

    A. It has 5 valence electrons

    B. It has atomic number 17

    C. Its last electrons enter 3s sub – level

    D. Its electrons occupy 4 energy levels.

Explanation:

- ➦ The third shell is the valence shell.
  - Valence electrons = 2 + 5 = 7
- ➦ Z = 2 + 2 + 6 + 2 + 5 = 17
- ➦ The last electrons enter in the 3p sub shell

Answer: B

44. In which sub shell does the last electron enter for an element with atomic number 31?

    A. 4s B. 4P C. 4d D. 4f

Explanation: → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^1$

Answer: B

45. Which group contains chemical species with the same electronic configuration? (Atomic number, Mg =12, Na =11, A1=13, S =16, cl = 17 Ca = 20, K = 19) (EGSECE)

    A. Mg, Ca, Na C. Mg^{+2}, Ca^{+2}, Na^{+}

    B. K^{+}, Al^{3+}, S^{2-} D. S^{2-}, Cl^{-}, K^{+}

Explanation: chemical species with same electron number are termed as isoelectronic.

Answer: D

46. The atomic number of an element with valence electron configuration of $4s^2\ 4p^6$ is

Explanation:

➦ Write electronic configuration:

- $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6$

$Z = 2 + 2 + 6 + 2 + 6 + 2 + 10 + 6 = 36$

47. Write electron configuration by sing subshells for:

a. $_7N^{3-}$ b. $_{31}Ga^{+3}$ c. $_{37}Rb$

Explanation:

a. $e = z - \text{charge} = 7 - (-3) = 10$

➦ $1s^2\ 2s^2\ 2p^6$

b. $e = z - \text{charge} = 31 - (+3) = 28$

➦ $1s^2 + 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$, 4s and 4p electrons are removed first.

c. $e = z = 37$

➦ $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^1$

48. A certain ion of an atom has a charge of –2 and electronic configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. What is the atomic number of the atom?

Explanation:

$e = z - \text{charge}$

$z = e + \text{charge}$

$z = 18 + (-2) = 16$

49. Write electronic configuration by using noble gases as a core for:

a. $_{15}P$ b. $_{27}Co$ c. $_{38}Sr$

Explanation:

a. $\underbrace{1s^2 2s^2 2p^6}_{[Ne]}\ 3s^2\ 3p^3 = [Ne]3s^2\ 3p^3$

b. $\underbrace{1s^2 2s^2 2p^6\ 3s^2\ 3p^6}_{[Ar]}\ 4s^2\ 3d^7 = [Ar]4s^2\ 3d^7$

c. $\underbrace{1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6}_{[Kr]} 5s^2 = [Kr]5s^2$

50. Write electron configuration by using main shells for

A. $_{20}Ca$ B. $_{9}F$

Explanation:

A. 2, 8, 8, 2 B. 2, 7

Solved Problems

51. An albitray element "X" has two naturally occurring isotopes. Information about the two isotopes are shown in the table below.

Isotope	Atomic mass in amu	Percentage abundance (%)
X – 40	40	80
X – 44	44	20

What is the average atomic mass of element x?

Answer: 40.80amu

52. Natural copper is a mixture of only two isotopes one of which has 29 protons, 34 neutrons and a relative abundance of 69.17%. the relative atomic mass of natural copper is 63.6. How many neutrons must be there is the second isotope?

Solution:

$A_1 = P + n_1 = 29 + 34 = 63$

$f_1 = 69.17\%$

$A_2 = ?$

$f_2 = 100 - f_1 = 100 - 69.17 = 30.83\%$

$$\text{Av. Atomic mass} = \frac{A_1 f_1 + A_2 f_2}{100}$$

$$63.6 = \frac{63 \times 69.17 + A_2(30.83)}{100}$$

$6360 = 4357.71 + A_2(30.83)$

$$A_2 = \frac{6360 - 4357.71}{30.83} = 65$$

$$A_2 = P + n_2$$

$$n_2 = A_2 - p = 65 - 29 = 36$$

Thus, the number of neutrons in the second isotope is 36.

53. Why do isotopes of an element have similar chemical properties?

Explanation: Because isotopes of an element have the same numbers valence electrons.

54. Carbon has two isotopes, C – 12 and C – 13, that exist in 9:1 abundance ratio. Determine the average atomic mass of carbon.

Solution: convert the ratio into fraction abundance

$$X_1 = \frac{9}{9+1} = 0.9$$

$$X_2 = \frac{1}{9+1} = 0.1$$

Average atomic mass $= A_1 x_1 + A_2 x_2$

$= 12 \times 0.9 + 13 \times 0.1 = 12.1$ amu

55. The last electron of an atom forms a $3d^2$– configuration.

a. How many protons does an atom of the element have?

b. Write the type of sub shells involved for accommodating the electrons of the atom.

c. Total number of s – electron = _____; p – electrons = _____

Explanation:

a. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^2$

$P = 2 + 2 + 6 + 2 + 6 + 2 + 2 = 22$

b. s, p, and d

c. s – electrons = 8

p – electrons = 12

56. Write electron configuration for

a. Mn^{+4} (Z = 25)

b. Cr^{+2} (Z = 24)

Explanation: for elements whose last electrons enter in the d – sub shell first the s- electrons then d – electrons are removed.

a. $e = 25 - 4 = 21$

➦ $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3$

b. $e = 24 - 2 = 22$

➦ $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^4$

57. An atom has electron configuration of $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^4$. If its mass number is 128, what will be the number electrons and neutrons?

Explanation: for a neutral atom:

$$p = e = 2 + 2 + 6 + 2 + 6 + 2 + 10 + 4 = 34$$

$$n = A - P = 128 - 34 = 94$$

58. Explain the term model.

Explanation: model is the mind's picture of what a real thing would look like. From ideas and experiences mind construct 'picture' about things which cannot be easily seen directly.

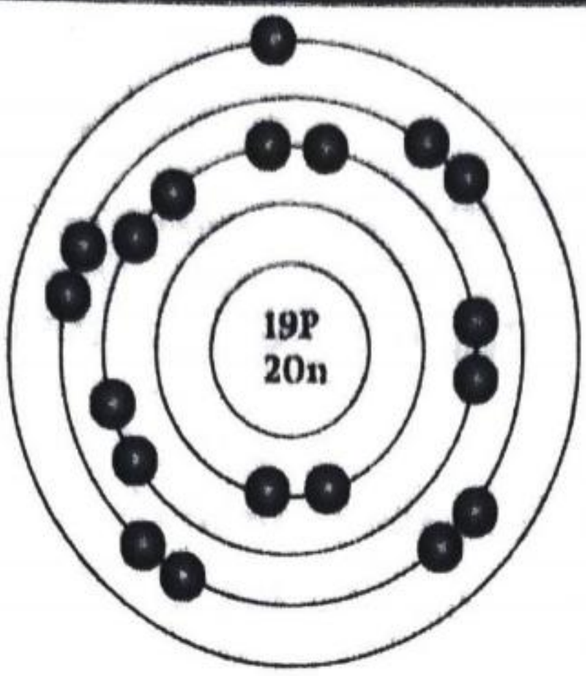
59. Draw Bohr diagram for

a. $^{39}_{19}K$ b. $^{73}_{32}Ge$

Explanation: determine the number of neutrons first and write main shell configuration

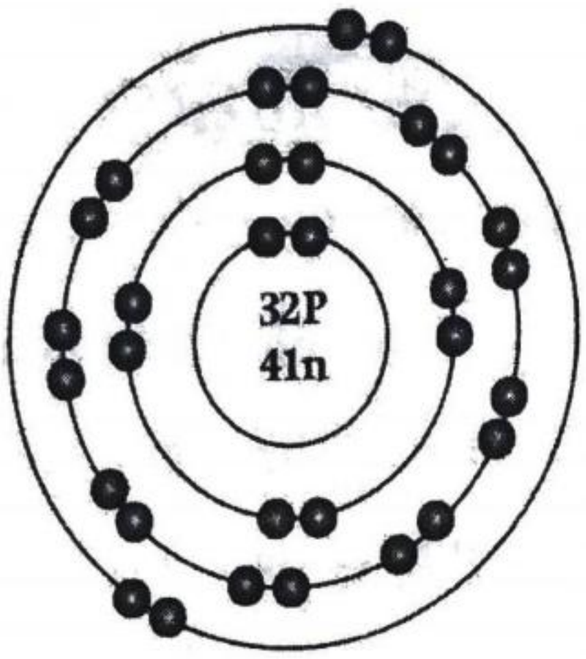
a. $n = A - P = 39 - 19 = 20$

main shell configuration: 2, 8, 8, 1



b. $n = A - P = 37 - 32 = 41$

main shell configuration = 2, 8, 18, 4



60. Two particles A and B have the following composition.

A: 15 protons, 16 neutrons, 15 electrons

B: 15 protons, 16 neutrons or 18 electrons

a. What is the relationship between these particles

b. Will these particles have similar chemical properties?

Explanation:

a. B is anion of A

b. No, because they have different electrons.

61. Identify the following subatomic particles

a. The number of these in the nucleus is equal to the atomic number.

b. The particle that is gained or lost when ions are formed

c. The particle that is not found in the nucleus

d. The particle that has a lower mass than the others.

e. The particle that has no charge.

Answer:

a) proton
b) electron
c. electron
d. electron
e. neutron

62. An atom has an atomic number 34. How many electrons will it have in its valence shell?

Explanation: write electron configuration

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^4$

Valence shell configuration: $4s^2\ 4p^4$

Valence electrons = 2 + 4 = 6

Supplementary exercise

I. True/false type questions

1. Different atoms of the same element have same number of protons
2. Different elements can have the same number of electrons
3. Different elements can have the same number of neutrons
4. Atomic mass always increases with atomic number.
5. Beta particles have mass and charge
6. The mass of an atom is mainly concentrated in the nucleus.
7. The 4p – sublevel comes before 3d – sublevel.
8. The f – sublevel contains seven orbitals.
9. According to Dalton, an atom is a solid, indestructible sphere.
10. The fundamental particle not present in ordinary hydrogen atom is neutron.

II. Matching

A	B
11. Dalton's model	A. orbit
12. Thomson's model	B. orbital
13. Rutherford's model	C. planetary model
14. Bohr's model	D. billiard ball model
15. Quantum mechanical model	E. plum – pudding model

III. Multiple choice type questions

16. Which one of the following is the smallest particle of a substance that takes part in a chemical reaction?
    A. An atom
    B. Anion
    C. A formula unit
    D. A molecule
17. Which of the following experimental findings in the past confirmed that most of the parts of an atom is an empty space?
    A. The discovery of the magnetic and electrical behavior of cathode rays.
    B. The discovery of the electro neutrality of the atom
    C. The deflection of the cathode rays in the discharge tube towards the positive terminal of the electric field
    D. The passage of most of the alpha particles through a thin gold foil.
18. Which of the following statements is accepted both in Dalton's and modern atomic theory?
    A. Atoms of the same element may have different masses
    B. Atoms cannot be subdivided, created or destroyed
    C. Atoms of different elements combine in simple whole number ratios.
    D. Atoms of a given element are identical in mass and other properties
19. An ion that consists of 7 protons, 7 neutrons and 10 electrons has a net charge of
    A. 3+ B. 4+ C. 4 – D. 3 –
20. Which statement compares the masses of two subatomic particles?
    A. The mass of a proton is greater than the mass of an electron
    B. The mass of a proton is greater than the mass of neutron
    C. The mass of an electron is greater than the mass of a neutron
    D. The mass of an electron is greater than the mass of a proton.
21. Which of the following ideas is introduced by quantum mechanical model of an atom?
    A. The structure of an atom resembles the solar system.
    B. For atoms of the known elements, there are four types of sublevels
    C. Electrons revolve around the nucleus in a circular orbit called shell
    D. Electrons are embedded in a positively charged sphere.

22. The number of protons, electrons and neutrons respectively in $^{65}_{30}Zn^{+2}$ are

A. 30, 30, 35
C. 30, 28, 65
B. 30, 28, 35
D. 30, 30 65

23. According to the modern atomic theory,

A. Atoms of the same element can never combine
B. Atoms are divisible, but not indestructible
C. Atoms of different elements can have the same atomic number
D. Atoms of the same element are exactly identical in properties.

24. The electron configuration for copper ($_{29}$Cu) is:

A. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$
C. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$
B. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$
D. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^9$

25. The charge to mass ratio of an electron was determined by

A. J.J. Thomson
C. R. Millikan
B. E. Rutherford
D. Dalton

26. Which of the following postulates of Dalton's atomic theory was disproved by the modern atomic theory?

A. Two or more atoms combine to form a compound
B. All atoms of a given element are identical
C. Atoms of one element differ from atoms of another element
D. All substances are made up of atoms

27. Which of the following is true about an element with electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$?

A. It has 5 valence electrons
B. It has atomic number 18
C. Its last electrons enters in the 3s sub – shell
D. Its electrons occupy 4 energy levels.

28. The aufbua principle is one that governs

A. vapour pressure
C. electronic configuration
B. proton distribution
D. atomic models

29. An atom has 83 protons, 83 electrons and 126 neutrons. What is its mass number?

A. 83
B. 126
C. 166
D. 209

30. How many different types of sub-shells are found in the fifth main shell?

A. 5 B. 10 C. 15 D. 20

31. Which sub-shell is found in all main shell?

A. s B. p C. d D. f

32. The maximum number of electrons in the second shell is

A. 4 B. 8 C. 2 D. 16

33. A mono-atomic ion that has 13 protons and a +3 charge

A. Has 10 neutrons
B. Has 16 electrons
C. has a symbol Ne
D. is isoelectronic with Ne

34. A neutral atom A at its ground state has half – filled 4d sub – shell. What is the atomic number of A?

A. 48 B. 43 C. 38 D. 33

35. Which element corresponds to the electron configuration of [Ne] $3s^2$ $3p^3$?

A. $_{14}Si$ B. $_{15}P$ C. $_{16}S$ D. $_{13}Al$

36. A subshell of chlorine which is not completely filled is

A. 3s B. 2s C. 3p D. 2p

37. The elements $^{14}_{6}C$ and $^{14}_{7}N$ have the same

A. Chemical properties
B. Mass number
C. proton number
D. neutron number

38. An atom has the electron configuration of $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$ $3d^{10}$ $4p^2$. If its mass number is 75, what will be its neutron number?

A. 32 B. 75 C. 45 D. 43

39. What is the total number subatomic particles in $^{108}_{47}Ag$?

A. 47
B. 108
C. 155
D. 61

40. The atomic number of an atom is always is equal to

A. Proton number
B. Mass number
C. neutron number
D. number of nucleus

41. Which electron configuration is not correct?

A. 2, 8, 3
B. 2, 8, 9, 1
C. 2, 8, 18, 1
D. 2, 8, 8, 2

42. Which of the following has the highest energy?

A. Gamma ray
B. X – ray
C. alpha particle
D. beta particle

43. Which of the following atomic particles have almost the same mass as a neutron?

A. Proton
B. Electron
C. Hydrogen ion, H^+
D. A and C

44. Alpha particles which has a mass of two protons and two neutrons.

A. H
B. He^{2+}
C. Li
D. Be^{2+}

45. An atom is electrically neutral because it contains:

A. An equal number of electrons and protons
B. An equal number of electrons and neutrons
C. More neutrons than electrons
D. More electrons than protons

46. The atoms ${}^{40}_{20}A$ *and* ${}^{41}_{x}B$ are isotopes of the same element. the value of x in the representation of B is:

A. 2 B. 10 C. 15 D. 20

47. A model in chemistry is used in the following

A. Giving a mind's picture of an idea
B. Predicting tow possible ways a reaction can take place
C. Showing the mass of an atom
D. Experimenting for the first time

48. The relative atomic mass of neon is 20.2. It consists of the isotopes $^{22}_{10}Ne$ and $^{20}_{10}Ne$. $^{20}_{10}Ne$ has a percentage abundance in naturally occurring neon of:

A. 90 B. 80 C. 0.9 D. 50

IV. Give short answer

49. Determine the number of electrons, protons and neutrons

a. $^{197}_{78}Pt$ b. $^{97}_{42}Mo^{+2}$ c. $^{75}_{33}As^{3}$

50. Write electron configuration by using noble gases as a core for:

a. Br (z = 35)

b. Cd (z = 48)

51. Write electron configuration by using main shells for

a. P(z = 15) b. b. S^{2-} (z = 16)

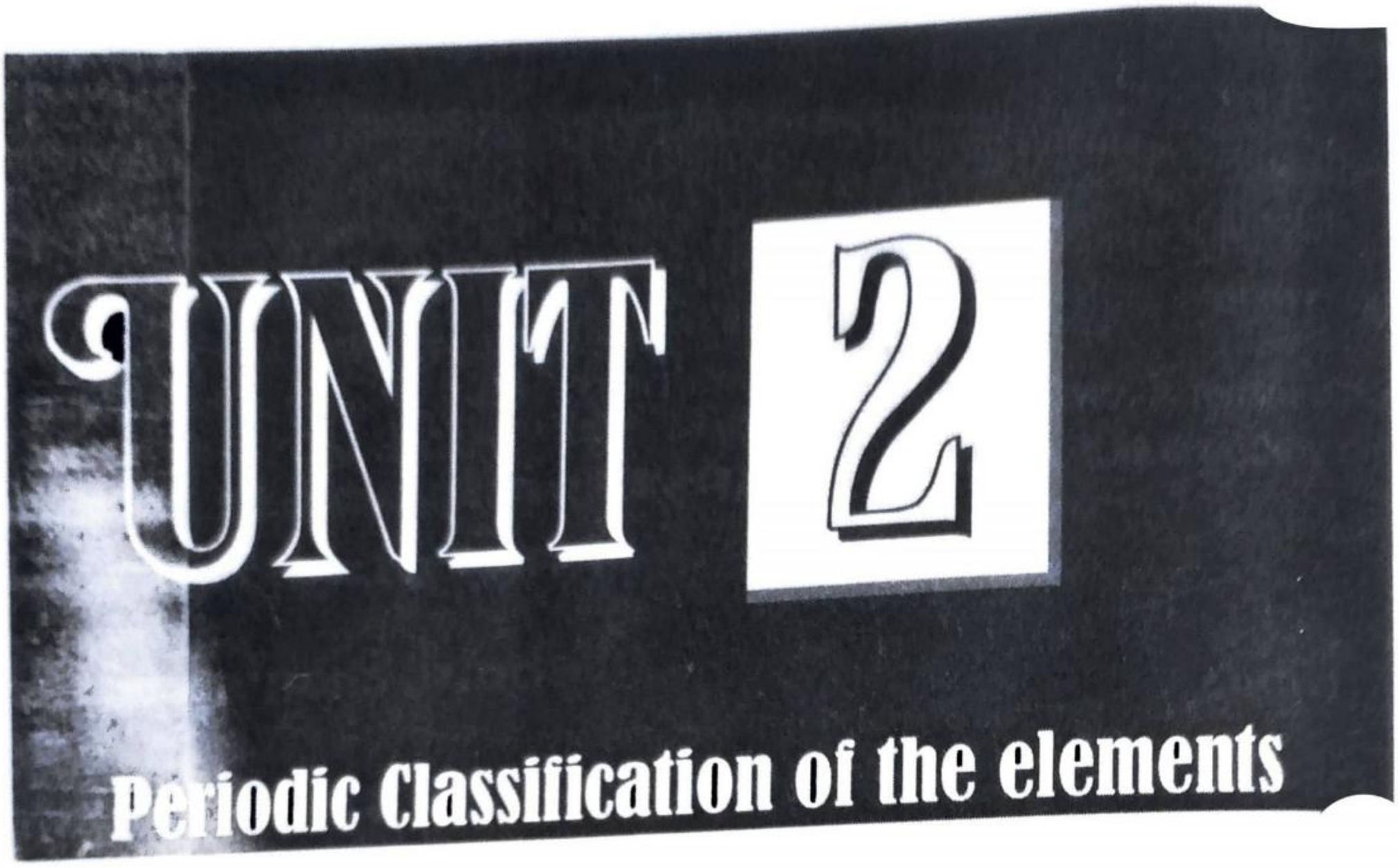
52. The two bromine isotopes ^{80}Br and ^{81}Br exist in 1:1 ratio. Calculate the average atomic mass of bromine.

53. The two naturally occurring isotopes of antimony, ^{121}Sb (121.149) (57.21%) and ^{122}sb 122.904amu, respectively. What is the average atomic mass of Sb?

54. Copper has two naturally occuring isotopes.

Isotop	% abundance	Atomic mass in amu	Average atomic mass
1	69.150	x	63.546
2	30.850	x + 2	

What is the value of x?



Unit Outline:

2.1 Historical development of the periodic table

2.2 The Modern Periodic law

2.3 General Characterstics with in groups and Periods

2.4 Advantages of Periodic Table

Introduction

A systematic classification of elements allows us to grasp the maximum possible information with short time and least effort. The importance of periodic classification of elements lies in the fact that an element which is a member of a certain group will show the major characteristics of the family without losing its unique properties.

2.1 Historical development of the periodic law

1. Dobereiner's Triads

- The law states that when elements in a triad are arranged in order of increasing atomic masses, the middle element have the average atomic mass of the other two.
- Dobereiner's work was the first systematic grouping of elements that revated atomic masses to other properties.

Consider the following Dobereiner's triads

a. ^{7}Li , ^{23}Na , ^{39}K

$$\text{Atomic mass of Na} = \frac{\text{Atomic mass of Li} + \text{Atomic mass K}}{2}$$

$$= 23$$

b. ^{32}S, ^{79}Se, ^{128}Te

$$\textit{Atomic Mass of Se} = \frac{\text{Atomic mass of S} + \text{Atomic mass Te}}{2}$$

$$= \frac{32+128}{2} = 80$$

80 is almost equal to 79

Limitation

- The law of triad was not applied even for all the elements discovered at that time.

2. Newland's law of octaves

- It states that when elements are arranged in increasing order of their atomic masses, every eighth element, staring from a given

one had similar properties to the first element.

Table 2.1. Arrangement of elements in Newlands law of octaves.

Li	Be	B	C	N	O	F
Na	Mg	Al	Si	P	S	Cl
K	Ca					

According to the law of octaves, for example, Li and Na have similar properties.

Limitations

- ➦ This law does not work after calcium.
- ➦ Failed to explain when noble gases were discovered because they became the ninth element.

Illustrative Example

1. The earliest of the attempts to classify elements was that of (EGSECE)
   A. Mendeleev's periodic classification
   B. Newlands law of octaves
   C. Dobreiner's triads
   D. Meyer's classification

   Answer: C

2. What is the similarity between law of triads and octaves?

Explanation: Both arranged elements in order of increasing atomic masses.

3. Show the Dobereiner's triad for chlorine, bromine and iodine.

Explanation:

$^{35.5}Cl$, ^{80}Br, ^{127}I

$$Atomic\ Mass\ of\ Br = \frac{atomic\ mass\ of\ Cl + atomic\ mass\ I}{2}$$

$$= 81.25$$

Thus, 81.25 is almost equal to 80. Therefore, Cl, Br and I form one triad.

4. Explain the contribution of Newland's law of octaves to the development of the modern periodic table.

Explanation: Newland proposed for the first time periodicity in the properties of elements.

3. Periodic law

- The periodic law was proposed by Mendeleev, a Russian and L. Meyer, a German scientist.
- Periodic law states that, the properties of elements is a period functions of their atomic masses.
- Mendeleev organized his table in columns.

Advantages of Mendeleev's periodic table

- Leaving gap for undiscovered elements initiated other scientists to discover them.

Drawbacks of Mendeleev's periodic table

- No positions for isotopes
- Wrong order of the atomic masses of some elements.

2.2 The modern periodic law

Henry Mosely discovered a new properties of elements known as **atomic number**. Every element has unique atomic number.

- The modern period law states that, when elements are arranged in increasing order of atomic number, there occurs a periodic repetition of physical and chemical properties.

Groups and periods

i Groups

- They are also called **families** because they contain elements that have similar chemical properties.
- They are vertical columns in the periodic table.
- There are 18 vertical columns but 16 groups.
- The 16 groups classified into two. These are 8 main groups (IA-VIIIA) and 8 sub groups (IB-VIIIB)

Table 2.2. Common names of main group elements

Group	Common name
IA	Alkali metals except hydrogen
IIA	Alkaline earth metals
IIIA	Boron family
IVA	Carbon family
VA	Nitrogen family
VIA	Oxygen family (chalcogens)
VIIA	Halogens (salt formers)
VIIIA	Noble gases or 'O' group

ii Periods

- They are also called series because elements in the same period have consecutive atomic numbers.
- There seven periods

Table 2.3. Periods and the number of elements they contain

Period	Number of elements
1	2, H and He
2	8, from Li to Ne
3	8, from Na to Ar
4	18, from K to Kr
5	18, from Rb to Xe
6	32, from Cs to Rn
7	Incomplete from Fr - ?

For 'A' groups
Group number = valence electrons
For all elements
Period number = valence shell or number of shells

Example

Find the group and period of

a. $_{6}C$

b. $_{17}Cl$

***Explanation*:** First write electron configuration and identify the number of valence electrons and valence shell.

a. $1s^2\ 2s^2\ 2p^2$

Number of valence electrons = 2 + 2 = 4

$\Rightarrow$ group = IVA

The second shell is the valence shell

$\Rightarrow$ period = 2

b. : $1s^2\ 2s^2\ 2p^6\ \ 3s^2\ 3p^5$

The third shell is the valence shell and there are 7 electrons in this shell.

$\Rightarrow$ group = VIIA

$\Rightarrow$ Period = 3

For 'B' group elements, use the following simple formula

Valence shell configuration	Group
$ns^2\ (n\text{-}1)d^1$	2 + 1 = IIIB
$ns^2\ (n\text{-}1)d^2$	2 + 2 = IVB
$ns^2\ (n\text{-}1)d^3$	2 + 3 = VB
$ns^1\ (n\text{-}1)d^5$	1 + 5 = VIB
$ns^2\ (n\text{-}1)d^5$	2 + 5 = VIIB
$ns^2\ (n\text{-}1)d^6$	2 + 6 = VIIIB
$ns^2\ (n\text{-}1)d^7$	(2 + 7)-1 = VIIIB
$ns^2\ (n\text{-}1)d^8$	(2 + 8)-2 = VIIIB
$ns^1\ (n\text{-}1)d^{10}$	(1 + 10)-10 = IB
$ns^2\ (n\text{-}1)d^{10}$	(2 + 10)-10 = IIB

Example

Find the group and period of

a. $_{25}Mn$

b. $_{29}Cu$

***Explanation*:**

a. $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^2\,3d^5$

Valence shell = 4

$\Rightarrow$ period = 4

Group = 2 + 5 = VIIB

b. $1s^7\,2s^2\,2p^6\,3s^2\,3p^6\,4s^1\,3d^{10}$

Valence shell = 4

$\Rightarrow$ period = 4

Group = (1 + 10) - 10 = IB

Illustrative Example

5. According to the modern periodic law: (EGS ECE)

   A. The properties of the elements are periodic function of their atomic masses.

   B. No two elements can have the same atomic mass

   C. No two elements can have the same mass number

   D. The properties of the elements are periodic functions of their atomic number

***Explanation*:** According to modern periodic law:

- Elements are arranged based on atomic number
- No two elements can have the same atomic number

Answer: D

6. How many elements are there in the fifth period of the periodic table?

   A. 2 B. 18 C. 8 D. 32

Answer: B

7. How many periods are there in the modern periodic table? **(EGSECE...2007)**

   A. 16 B. 18 C. 7 D. 8

Answer: C

8. To which group and period of the periodic table does an element with atomic number 32 belong?

   A. Group IIB and period 2

   B. Group IIA and period 2

   C. Group IVA and period 4

   D. Group IIIA and period 4

Explanation:

$Z=32$, $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^2$

Valence shell configuration: $4s^2\ 4p^2$

$\Rightarrow$ period = 4

group = 2 + 2 = IVA

Answer: C

9. Which of the following pairs does not show the relationship between electron configuration and modern periodic table? (**EGSECE**)

A. Mass number group name

B. Position of last electron block type

C. Valence electrons – group number

D. Number of energy levels of an atom – period number

Answer: A

10. An element has the valence electron configuration of $2s^2 2p^3$. To which group and period does it belong?

A. IIA and 5

B. VA and 2

C. IVA and 4

D. III A and 2

Explanation:

Valence shell = 2= period

Valence electrons = 2 + 3 = 5

$\Rightarrow$ group = VA

Answer B

11. The electron configuration of an element that is found in group IIA and period 4 is _____

***Explanation*:**

- ➦ Group = 11A implies the valence electrons = 2
- ➦ Period = 4 means the valence shell is the fourth shell.

$\Rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$ is the electron configuration of the element.

12. Determine the group and period of the element with atomic number

a. 37 b. 44 c. 56

Explanation:

a. Write electron configuration : [Kr] $5s^1$

Group = IA, period = 5

b. Group = 2 + 6 = VIIIB , Period = 4

c. Group = IIA, Period = 6

13. Luthar Meyer and Dimitri Mendeleev proposed similar periodic tables. However, Mendeleev is generally given more credit for his periodic table. Why?

***Explanation*:**

- ➦ This is because Mendeleev left gaps in his table for undiscovered elements and was able to predict the properties of these elements

14. Which group in the modern periodic table contains elements in the three states of matter at room temperature?

***Explanation*:**

Group VIIA or the halogens

Chlorine and flourine = gases

Bromine = liquid

Iodine and astatine = solid

15. The shortest and longest periods respectively are _____ and _____

Answer period 1 and 6

16. Why group IA elements except hydrogen are commonly known as alkali metals?

***Explanation*:** Because they react with water and form soluble bases called alkalis.

Classification of elements

Based on their properties, elements can be classified in to three

1. Metals

- They are electropositive elements
- All are solids at room temperature except mercury which is liquid.
- They are good conductors of heat and electricity
- They have high melting and boiling points
- They are malleable, ductile and shiny
- They are found to the left, middle and bottom of the periodic table

2. Non-metals

- They are electronegative elements
- They exist in the three states of matter
- They have low boiling and melting points
- They are poor conductors of electricity except graphite
- They are found to the right side of the periodic table

3. Metalloids

- They are all solids
- They are semi-conductors
- They are found between metals and non-metals along the zigzag line

Based on electronic arrangement, elements are categorized in to three:

1. Representative Elements

- They are 'A' group elements, that is found from IA – VIIIA
- They are s-block and p-block elements (their last electrons enter in s or p – sub levels)
- They are metals, non-metals and metalloids.

2. Transition elements

- They are 'B' group elements
- They are d – block elements
- They are all metals.
- They are found between IIA and IIIA
- They form colored and complex compounds.
- They have variable valence numbers

Examples

- Iron (II) and Iron (III)
- Copper (I) and copper (II)

3. Inner transition elements

- They are also called rare earth metals.
- They are f-block elements
- They are all metals
- They are found below the periodic table into two separate rows
- They are composed of the lanthanide and actinide series
- They are found in the 6^{th} and 7^{th} periods.

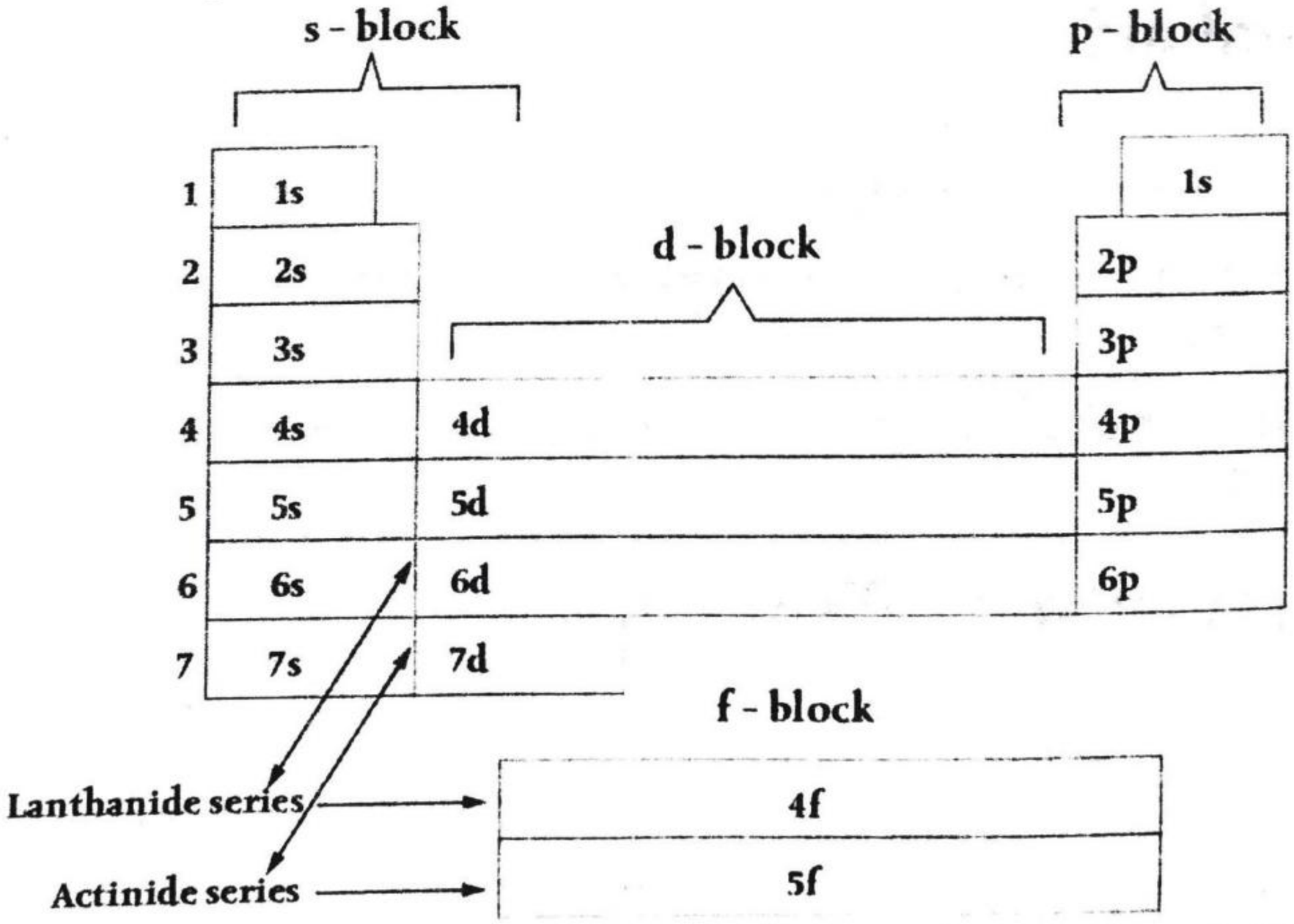


Figure 2.1 Classification of elements in modern periodic table based on sub-shell

Illustrative Example

17. The modern periodic table is constructed by arranging the elements according to their increasing order of protons. How is the actual structure of periods, groups and blocks designed in relation to the electronic structure of the main group elements? (EGSECE)

A. Groud state electronic configurations of the elements in their main energy levels can be related to group, period and block of the elements

B. The number of main energy levels corresponds to the group number of the elements in the periodic table.

C. Ground state electronic configurations of the elements in their sub-energy levels can be related to group, period and block of the elements.

D. The number of electrons in the valence shells corresponds to the block of the elements in the periodic table.

Explanation:

Valence electrons = group

Number of shells = period

The electron in the last sub shell = block

Answer: C

18. If the electronic configuration of a certain element is given as $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^5$, what are the block, the period and group number of the element in the periodic table respectively?

A. s, 3 and IVA
B. d,4 and IIB
C. p, 4 and VIIA
D. f, 3 and VIIIB

Explanation: Valence shell electron configuration = $4s^2\ 4p^5$

Group = 2 + 5 = VIIA

Period = 4

block = p

Answer: C

19. Elements whose outer most shell electrons fall in the f-orbitals are called

A. Transition elements
B. Rate earth metals
C. Metallic elements
D. Alkaline earth metals.

Explanation: Rare earth metals also called inner transition metals are f-block elements

Answer: B

20. Given element 'X' that has an electron configuration of $5s^2\ 4d^{10}\ 5p^6$. Which of the following is true regarding this element?

A. X is metallic element
B. X is d-block element
C. X is electropositive element
D. X is a p-block element

Explanation: Metals usually have valence electrons from 1 to 3. Since the last electrons enter in the p-sub shell, X is p-block element with 7 valence electrons. Thus, it is a halogen

Answer: D

21. In which of the following blocks of the periodic table does an element with atomic number 14 located? (EGSECE)

A. f B. d C. p D. s

Explanation: Write electron configuration using sub levels.

$\rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

Thus, it is p-block element

Answer: C

22. Suppose an element Y has an electron configuration of $1s^2\ 2s^2\ 2p^6\ 3d^5\ 4s^1$ and has a mass number of 52. Which of the following is true about the element Y?

A. It is an s-block element
B. It has 26 neutrons
C. It is an alkali metal
D. It has 24 protons.

Explanation:

The last electron enters in the d-subshell according to building up principle. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$

$\Rightarrow P = 24$

block = d

Answer: D

23. Which of the following elements is an s-block element?

A. $_{29}Cu$ B. $_{17}Cl$ C. $_{11}Na$ D. $_{30}Zn$

Explanation:

Cu – $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$, d – block

Cl – $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$, p – block

Na – $1s^2\ 2s^2\ 2p^6\ 3s^1$, s – block

Zn – $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}$, d – block

Answer: C

24. An element is located in the third period and group VA of the periodic table. Which of the following is true about this element?

A. It is a metallic element

B. It forms a basic oxide

C. It has electron losing tendency

D. It forms a trivalent ion

Explanation: The element in period 3 and group VA has an electron configuration of $1s2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

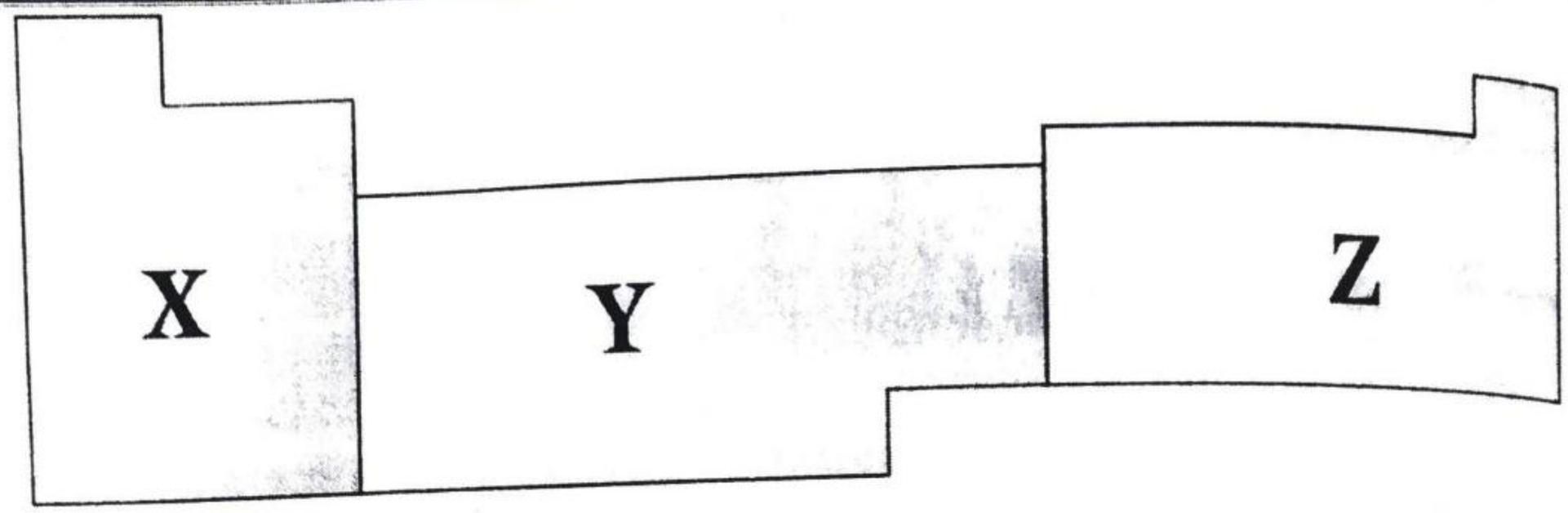
➦ Number of valence electrons = 2 + 3 = 5

Thus, it is non-metallic element and forms acidic oxide.

Its valence number = 8 – 5 = 3

Answer: D

25. Which of the following is true about positions indicated by X, Y, Z and K in the periodic table below? (**EGSECE**)



K

A. All elements in which the last electron occupies the p-orbitals are placed in position Z.

B. 'B' group elements are placed in position Y and Z.

C. Elements in which the last electron goes to d-orbitals occupy position K.

D. All 'A' group elements are placed in position D.

Explanation:

- ➦ Z = p – block elements
- ➦ X = s – block elements
- ➦ Y = d – block elements
- ➦ K = f – block elements
- ➦ X and Y represent 'A' group elements.
- ➦ Y is 'B' group elements

Answer: A

26. Classify the following elements are representative, transition or inner transition.

a. 38^{Sr} b. 42^{Mo} c. 34^{Se} d. 58^{Ce}

Explanation: Write electron configuration using sub shells to classify therm.

a. $[Kr]5s^2$ – representative element

b. [Kr] $5s^1$ $4d^5$- transition element

c. [Ar] $4s^2$ $3d^{10}$ $4p^4$ - representative element

d. [Xe] $6s^2$ $4f^2$ - inner transition element

27. Write general valence shell electron configuration for 'A' group elements.

Explanation:

Group	General valence shell electron configuration
IA	ns^1
IIA	ns^2
IIIA	ns^2 np^1
IVA	ns^2 np^2
VA	ns^2 np^3
VIA	ns^2 np^4
VIIA	ns^2 np^5
VIIIA	ns^2 np^6

28. List the main characteristics of transition elements

Answer:

- ➦ All are metals
- ➦ They have variable valencies
- ➦ They form coloured compounds
- ➦ They have ability to form complex compounds

29. Most of the chemical elements are _____

Answer: Metals

2.3 General characteristics within groups and periods

1. Number of valence electrons

- ➦ Number of valence electrons remain same down a group and increases across a period

2. Nuclear charge

- ➦ It is the total positive electric charge in the nucleus of an atom.
- ➦ It increases down a group and across a period.

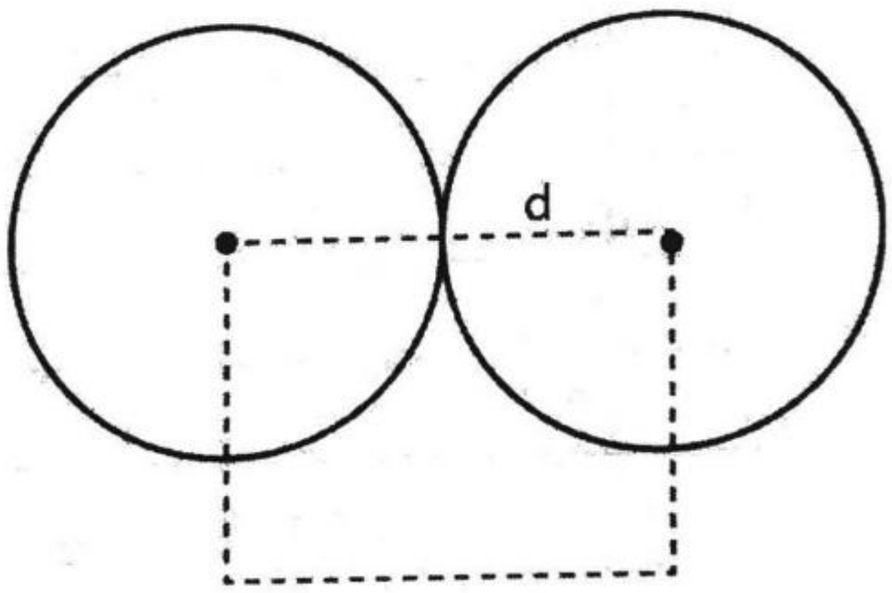
For example:

P = 11

Nuclear charge = + 11

3. Atomic size (Atomic radius)

- ➦ It may be defined as half the distance between two chemically bonded atoms in a molecule



Atomic radius (r) = $\frac{d}{2}$

For example, the distance between the nuclei in a Br_2 is $2.28A^0$.

Thus, Br = $\frac{2.28A^0}{2} = 1.14A^0$

- ➦ The atomic radius of an atom is very short. As a result it is measured with angstrom [A^0]

  $1A^0 = 10^{-10}m = 10^{-8}cm$

- ➦ Atomic size increases down a group because number of shells increases but decreases across a period because effective nuclear charge increases.

  Zeff = z – s

Z – atomic number, s – number of inner electrons.

Table 2.4. Atomic radii of group IA metals

Element	Electron configuration	Number of shells	Atomic radius (A^0)
Li	2,1	2	1.52
Na	2,8,1	3	1.86
K	2,8,8,1	4	2.27
Rb	2,8,18,8,1	5	2.46
Cs	2,8,18,18,8,1	6	2.65

Note:

1. The atomic size of positive ion is less than that of its neutral form.

   Example: $k^+ < k$

2. The atomic size of negative ion is greater than its neutral form.

   Example: $F^- > F$

4. Ionization energy (IE)

- It is the amount of energy required to remove an electron from isolated gaseous atom.
- $X(g) + IE \rightarrow X^+(g) + e^-$

There are different types of ionization energies.

- First ionization energy (IE_1) removes the first valence electrons.
- Second ionization energy (IE_2) removes the second valence electron.
- Third ionization energy (IE_3) removes the third valence electron
- Fourth ionization energy (IE_4) removes the fourth valence electron

$$A(g) + IE_1 \longrightarrow A^+(g) + e^-$$

$$A^+(g) + IE_2 \longrightarrow A^{+2}(g) + e^-$$

$$A^{+2}(g) + IE_3 \longrightarrow A^{+3}(g) + e^-$$

$$A^{+3}(g) + IE_4 \longrightarrow A^{+4}(g) + e^-$$

Ionization energy decreases down a group but increases across a period from left to right

Factors effecting ionization energy

i **Atomic size**

➦ As size increases ionization energy decreases and vice versal.

ii **Effective nuclear charge**

➦ Smaller effective nuclear charge makes removal of electron easier.

iii **Types of electrons**

➦ In the same atomic shell, the order of difficulty to remove an electron is:

s-electron > p-electron > d-electron > f-electron

iv **Shielding/screening/of inner electrons.**

➦ As the number of inner electrons increases, the ionization energy decreases.

Example:

Determine the number of inner electrons and compare the ionization energy of Li and Na

Explanation:

Li: $1s^2\ 2s^1$ inner shell = $1s^2$

number of inner electrons = 2

Na: $1s^2\ 2s^2\ 2p^6\ 3s^1$ inner shells = $1s^2\ 2s^2\ 2p^6$

Number of inner electrons = 2 + 2 + 6 = 10

Thus, Na has lower ionization energy than Li

v **Stability of orbitals**

➦ Fulfilled and half-filled orbitals are stable and require higher amount of energy to remove electron from them.

metals are characterized by low ionization energy but non-metals with high ionization energy

$_{1}H$ 1312							$_{2}He$ 2372
$_{3}Li$ 520	$_{4}Be$ 900	$_{5}B$ 801	$_{6}C$ 1086	$_{7}N$ 1402	$_{8}O$ 1314	$_{9}F$ 1618	$_{10}Ne$ 2018
$_{11}Na$ 496	$_{12}Mg$ 738	$_{13}Al$ 578	$_{14}Si$ 789	$_{15}P$ 1012	$_{16}S$ 1000	$_{17}Cl$ 1251	18Ar 1521

Figure 2.6. First ionization energies of the first eighteen elements in kilojoule per mole.

Note:

- Ionization energy of beryllium is greater than boron due to full filled orbital in beryllium
- Ionization energy of nitrogen is greater than oxygen due to half - filled orbital in nitrogen
- In period 3, ionization energy of Mg is greater than Al and P greater than S with similar reason as the above.

Illustrative Example

30. Which of the following has the largest atomic radii?

A. $_{11}Na^{+}$ B. $_{12}Mg^{2+}$ C. $_{13}Al^{+3}$ D. $_{14}Si^{+4}$

Explanation: In the same period, the more positive the ion, the smaller is the atomic radii.

Answer: A

31. Which of the following has the smallest atomic radii?

A. $_{6}C^{4-}$ B. $_{7}N^{3-}$ C. $_{8}O^{2-}$ D. $_{9}F^{-}$

Explanation: In the same period, the more negative the anion, the larger is the atomic radii

Answer: D

32. Which of the following is the correct increasing order of the atomic size of the elements Mg, B, K, Cl and C? (Z, Mg = 12, B = 5, L = 6, K = 19, Cl = 17)

A. B < C < Cl < K < Mg

B. C < B < Cl < Mg < K

C. Cl < C < B < K < Mg

D. Mg < K < Cl < C < B

Explanation:

- B and C are found in period 2 and size of C < B
- Mg and Cl are found in period 3 and size of Cl < Mg
- K is found in period 4 and has the largest atomic radius.

$\Rightarrow$ C < B < Cl < Mg < K

Answer: B

33. The reason for the decrease in atomic radius as you go from left to right in a row of the periodic table is due to the. **(EGSECE...2007 E.C)**

A. decrease in the effective nuclear charge

B. increase in the number of neutrons

C. increase in the nuclear charge

D. addition of extra electron shells

Answer: C

34. Which of the following elements has the lowest ionization energy? (Z, K = 19, Na = 11, Rb = 37, C_s = 55)

A. K

B. Na

C. Cs

D. Rb

Explanation: Down a group, ionization energy decreases.

Answer: C

35. Increasing order of ionization energy for the elements Be, B, N, and O is

A. B < Be < O < N

B. Be < B < N < O

C. O < N < B < Be

D. N < O < Be < B

Explanation: Across a period from left to highest, ionization energy

increases, however, due to stability of fulfilled and half-filled orbitals the ionization energy of:

$\Rightarrow$ Be > B and N > O

Answer: A

36. Consider the ionization energies of aluminum.

$Al(g) \xrightarrow{478kJ} Al^{+}(g) \xrightarrow{1817kJ} A1^{2+}(g) \xrightarrow{2745kJ} Al^{3+}(g) \xrightarrow{11578kJ} Al^{4+}(g)$ why fourth ionization energy is the highest?

Explanation: This is because Al^{4+} has the smallest atomic radii.

37. List the factors that affect ionization energy.

Answer:

- Atomic size
- Stability of orbitals
- Type of electrons effect

38. Identify the shielding shells, shielding electrons, nuclear charge and effective nuclear charge for:

    a. Al (z = 13)

    b. N (z = 7)

    c. Rb (z = 37)

Explanation: write electron configuration

a. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

- Shielding shells = K and L shells
- Number of shielding electrons = 2 + 2 + 6 = 10
- Nuclear charge = 2 + 2 + 6 + 2 + 1 = +13
- Zeff = z – s = 13 – 10 = + 3

b. $1s^2\ 2s^2\ 2p^3$

- Shielding shell = K - shell
- Number of shielding electrons = 2
- Nuclear charge = 2 + 2 + 3 = + 7
- Zeff = z – s = 7 – 2 = + 5

c. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^1$

- ➦ Shielding shells = K – shell, L – shell, M-shell and N-shell
- ➦ Number of shielding electrons = 2+2+6+2+6+2+10+6 = 36
- ➦ Nuclear charge = + 37
- ➦ Zeff = z – s = 37 – 36 = + 1

39. Compare the ionization energy of the orbitals below.

    4s, 4p, 4d, 4f

    Answer:

    4s > 4p > 4d > 4f

40. Determine the number of shells in

    a. O (z = 8)

    b. Ar (z = 18)

    c. Sr (z = 38)

Explanation: Write electron configuration by using main shells.

a. 2,6

- ➦ Number of shells = 2

b. 2,8,8

- ➦ Number of shells = 3

c. 2, 8, 18, 8, 2

- ➦ Number of shells = 5

5. **Electro negativity** of an atom is the ability of an atom to attract or pull electrons to itself when it is in a molecule.

- ➦ It has no unit.
- ➦ The most electronegative atom is fluorine
- ➦ Pauling devised a scale to compare the electronegativities of elements. This scale assigned an electronegativity of 4 to the most electronegative element, **fluorine**.
- ➦ The element with the least electronegativity value is **cesium** that is **0.7**.

$_{1}H$ 2.1							$_{2}He$ 0
$_{3}Li$ 1.0	$_{4}Be$ 1.5	$_{5}B$ 2.0	$_{6}C$ 2.5	$_{7}N$ 3.0	$_{8}O$ 3.5	$_{9}F$ 4.0	$_{10}Ne$ 0
$_{11}Na$ 0.9	$_{12}Mg$ 1.2	$_{13}Al$ 1.5	$_{14}Si$ 1.8	$_{15}P$ 2.1	$_{16}S$ 2.5	$_{17}Cl$ 3.0	18Ar 0

Figure 2.2 Electronegativities of the first eighteen elements in the periodic table

Electronegativity decreases down a group and increases across a period from left to right

6. Electron affinity (EA) of an atom is defined as the energy lost or gained when one mole of the gaseous atoms add on one mole of electrons to form one mole of singly charged anions.

Example:

$Cl(g) + e^- \longrightarrow Cl^-(g)$, EA = – 348kJ/mol

Electron affinity increases across a period and decreases down a group.

Generally the first electron affinity is negative for electronegative sets. The second electron affinity has a positive energy value.

For example, the first and second electron affinities for oxygen are given as follows:

$$O(g) + e \rightarrow O(g), EA_1 = -141\ KJ/mol$$
$$\underline{O(g) + e \rightarrow O^{2-}(g), EA_2 = +798\ KJ/mol}$$
$$O(g) + 2e \rightarrow O^{2-}(g), EA = +657\ KJ/mol$$

H -73							He +54
Li -57	Be 66	B -15	C -123	N -31	O -141	F -333	Ne +99

Na	Mg	Al	Si	P	S	Cl	Ar
-21	+67	-26	-135	-60	-200	-348	
						Br	
						-325	
						I-297	

Figure 2.3

- First electron affinities of some elements of the periodic table

7. Metallic Character/electropositivity/

- Metals lose electrons to form cations. The more easily a metal can lose electrons, the more metallic it is.

Metallic character increases down a group but decreases across a period

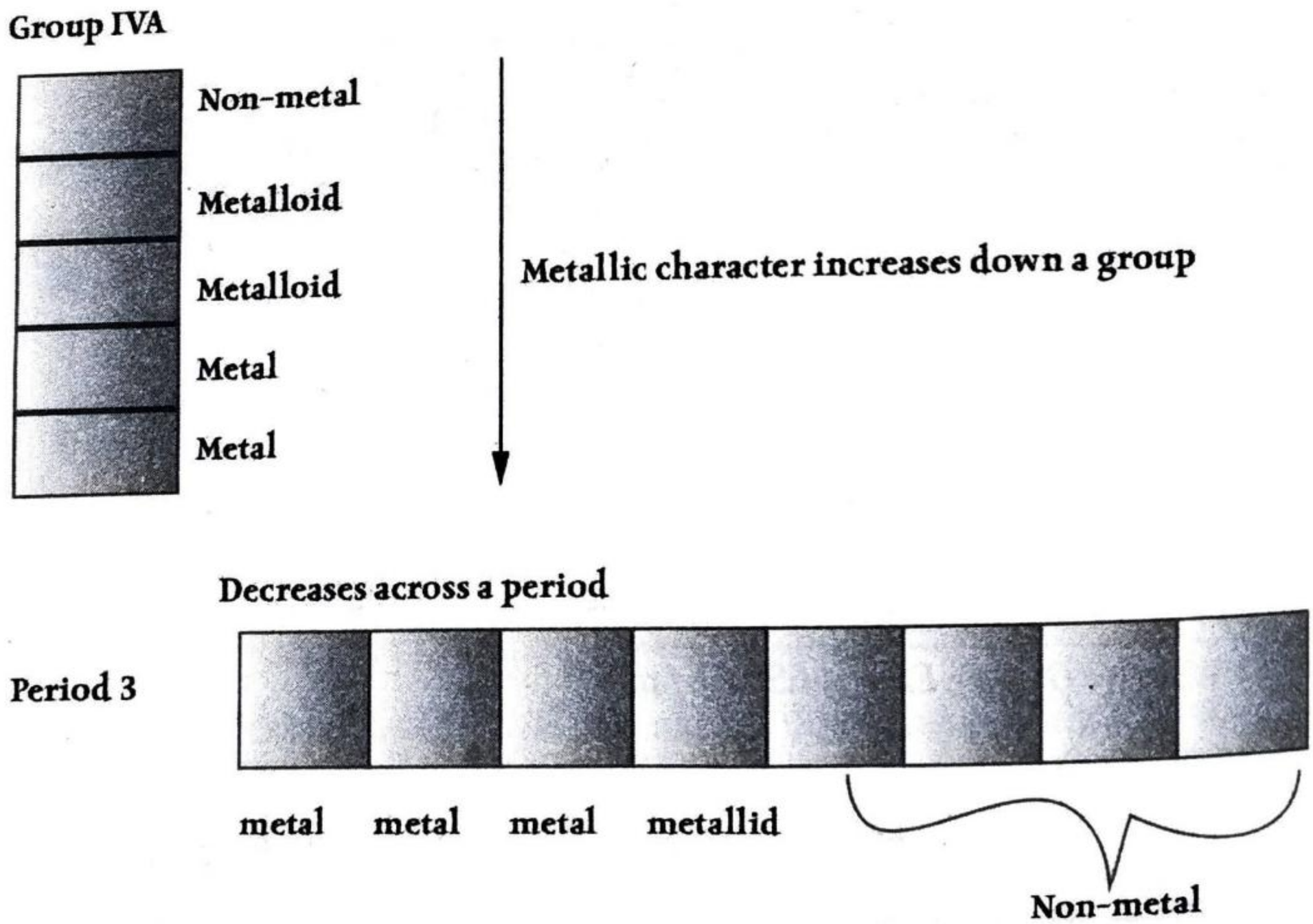
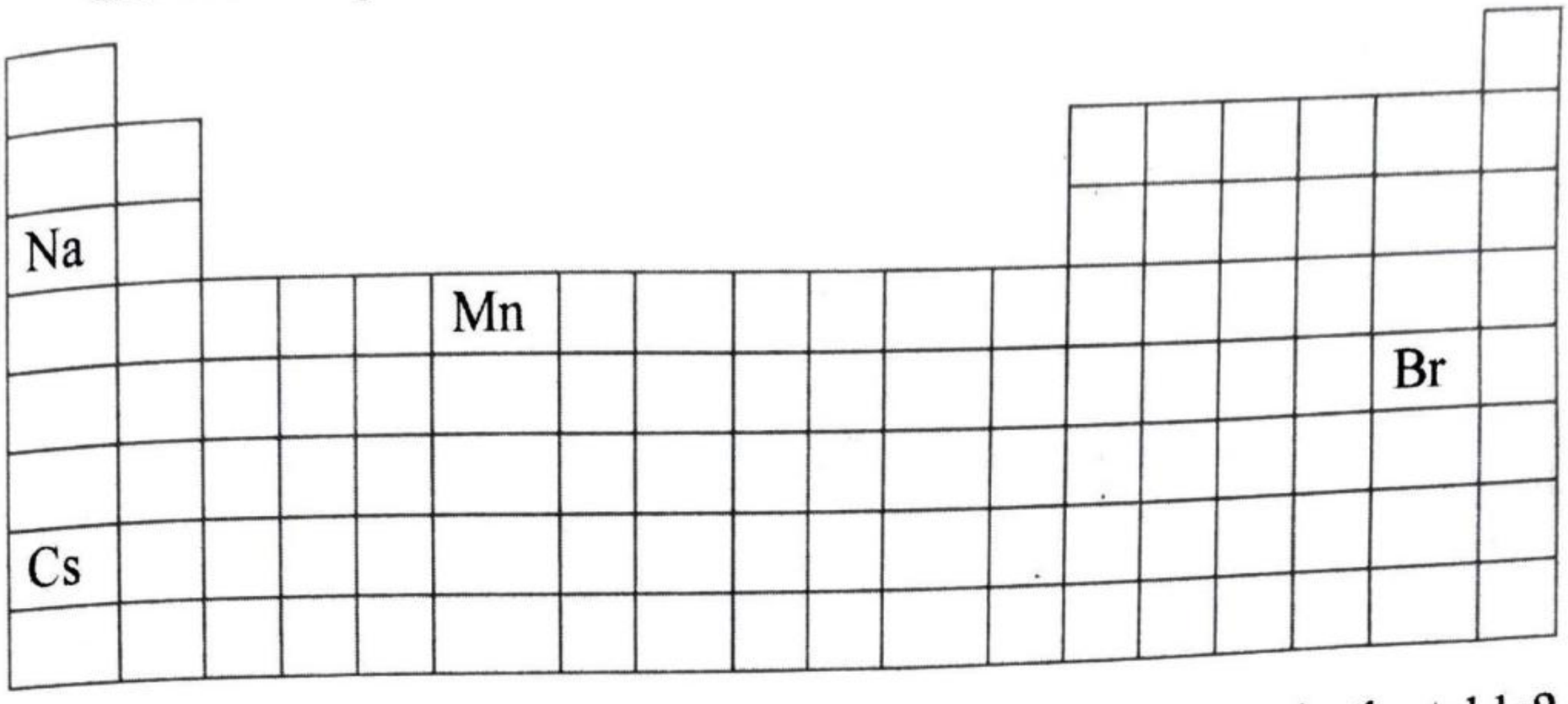


Figure 2.4 Change in metallic character in group IVA and period 3.

Illustrative Example

41. Refer to the figure below that shows part of the modern periodic table and answer question that follows.



42. Which of the following statements is true about elements in the table? (EGSECE)

A. Na has lower ionization energy than Cs

B. Br is larger than Mn

C. Br is the most electronegatives

D. Cs has more electron affinity than Br

Explanation: Non –metals are more electronegative than metals.

Answer: C

43. Which of the following is the correct order of electronegativities of the elements C, N, O and F?

A. C > N > O > F

B. C > F > O > N

C. F > O > N > C

D. O > N > F > C

Explanation: Electronegativity increases from left to right in a given period.

Answer: C

44. Which of the following properties of elements increases down a group?

A. Metallic character

B. Electronegativity

C. Electron affinity

D. Valence electrons

Answer: A

45. Which of the following is the most metallic element?

A. Na B. K C. Mg D. Ca

Explanation: Group IA metals are the most metallic element and down a group metallic character increases.

Answer: B

46. One of the following element has the highest electron affinity. Which one?

A. F B. Cl C. O D. N

Explanation: Chlorine exceptionally has the highest negative electron affinity.

Answer: B

47. Which of the following is generally true about elements in the same period?

A. Elements at the left side of the period have high ionization energy

B. Metallic property decreases from left to right

C. Number of shells and number of valence electrons of the element remain the same

D. Element at the right side of the period have larger size.

Answer: B

48. The metal with the lowest electronegativity value is

A. sodium C. cesium

B. potassium D. Lithium

Explanation: Cesium has the smallest electronegativity value that is 0.7.

Answer: C

49. Which of the following elements are characterized by low electronegativity and electron affinity?

A. metals C. metalloids

B. non-metals D. non

Answer: A

2.4 Diagonal Relationship

Shown by the second and third period elements.

Li and Mg	
Be and Al	have diagonal relationship
B and Si	

2.5 Advantages of Periodic table

- It puts elements with similar properties in groups.
- It is useful in predicting properties of elements and their compounds.
- It is used to predict formula of compounds
- It is used to determine position of elements.
- It is used to refer name, symbol, atomic number and atomic mass of elements.

Illustrative Example

50. A chemist can predict all of the following from the modern periodic table EXCEPT (**EGSECE**)

A. The formula of compound that the element can form

B. Chemical properties of elements

C. Colour of compounds

D. Position of elements.

Answer: C

51. The proton number of element X is 6. The proton number of element Y is 9. What is the formula of a compound of these elements?

A. X_2Y_3 B. X_3Y_2 C. XY_3 D. XY_4

Explanation: Write electron configuration and determine the valence number

X: 2, 4

Valence number = 8 – number of valence

for non-metals electrons

$= 8 - 4 = 4$

$Y = 2, 7$

Valence number $= 8 - 7 = 1$

Formula: X^4 $Y^1 = XY_4$

Answer: D

52. Which of the following pairs of elements have diagonal relationship?

A. Aluminum and magnesium
B. Boron and silicon
C. Sodium and potassium
D. Sulphur and phosphorus

Answer: B

53. Classify oxides of period 3 as acidic, basic, or amphoteric oxides.

Na_2O, MgO, Al_2O_3, SiO_2, P_2O_5, SO_2

Explanation: Na_2O and MgO = basic oxides

Al_2O_3 = amphoteric oxide

SiO_2, P_2O_5 and SO_2 = acidic oxides.

Across a period metallic character decreases.

Solved Problems

54. What was Mosely's contribution to the modern form of periodic table?

Explanation: Henry Mosely discovered atomic number using x-ray which is unique characteristic of each element. He arranged elements based on atomic number. As a result, it becomes very easy to study the properties of elements.

55. Explain why there are no p-block elements in the first period of the periodic table?

Explanation: The p-sub shell does not exist in the first energy shell.

56. Write the electron configuration and determine the atomic number of an element that corresponds to the following descriptions.

a. The noble gas in the fourth period

b. The group IA element in the fourth period

c. The group VIIA element in the fifth period

Explanation:

a. Noble gases are found in group VIIIA

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6$

$Z = 2 + 2 + 6 + 2 + 6 + 2 + 10 + 6 = 36$

b. $1s^2\ 2s^2\ 2p6\ 3s^2\ 3p^6\ 4s^1$

$Z = 2 + 2 + 6 + 2 + 6 + 1 = 19$

c. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^2\ 4d^{10}\ 5p^5$

$Z = 2 + 2 + 6 + 2 + 6 + 2 + 10 + 6 + 2 + 10 + 5 = 53$

57. Classify the following as s-block, p-block d-block or f-block elements.

a. Ba (Z = 56)

b. Zr (Z = 40)

c. As (Z = 33)

d. U(Z = 92)

Explanation:

a. [Xe] $6s^2$ – s-block

b. [Kr] $5s^2\ 4d^2$ – d-block

c. [Ar] $4s^2\ 3d^{10}\ 4p^3$ – p-block

d. [Rn] $7s^2\ 5f^4$ – f-block

58. Identify the element with the following descriptions.

a. Smallest atomic radius in group IIIA

b. Most metallic in group VA

c. Highest first ionization energy in group IIA

d. Largest atomic size in period 3

Explanation:

a. Boron

b. Bismith

c. Berylium

d. Sodium

59. The distance between the two nuclei in iodine molecule (I_2) is $2.66A^0$. What is the atomic radius of the iodine atom.

Explanation: Atomic radius is half the distance between the two nuclei of identical atoms that are covalently bonded.

➦ Atomic radius of I = $\frac{1}{2} \times 2.66A^0 = 1 = 1.33A^0$

60. Arrange the following species in decreasing order of atomic size

a. Be, C, Li, N

b. O^{2-}, Na^{+}, N^{3-}, F^{-}

c. B, Al, N, P

d. Be^{+2}, Mg^{+2}, Ca^{+2}, Sr^{+2}

Answer:

a. Li > Be > C > N

b. $N^{3-} > O^{2-} > F^{-} > Na^{+}$

c. Al > P > B > N

d. $Sr^{+2} > Ca^{+2} > Mg^{+2} > Be^{+2}$

61. Arrange the following in increasing order of ionization energy.

a. Mg, Mg^{+}, Mg^{+2}, Mg^{3+}

b. Al, Mg, S, P

c. Li, Na, K, Rb

Answer:

a. $Mg < Mg^{+} < Mg^{2+} < Mg^{3+}$

b. Al < Mg < S < P

c. Rb < K < Na < Li

62. The atomic radii of lithium, sodium, magnesium and aluminum are 0.134, 0.154, 0.145 and 0.130nm respectively. Arrange these elements in increasing order of atomic size.

Explanation: As magnitude increases, atomic radii also increases.

➦ Al < Li < Mg < Na

63. Why noble gases have extremely low (almost zero) electron affinity?

Explanation: Due to their stable electron configuration and low tendency to accept an additional electron.

64. Halogens have the highest electron affinities (highest negative value) why?

Explanation: Halogens need only one electron to be stable. As a result,

they have a high tendency to gain an electron and form stable negative ions.

65. Select smaller ion in each of the following pairs.

a. K^+ or Li^+ b. Au^+ or Au^{+3} c. p^{3-} or N^{3-}

Answer:

a. Li^+ b. Au^{3+} c. N^{3-}

66. Classify the group IVA elements as metal, non-metal and metalloids.

Answer:

carbon – non-metal

silicon and germanium – metalloids

tin and lead – metals

Supplementary Exercise

I. True/False

1. Newland arranged elements based on atomic mass.
2. Periodic table is useful in predicting the properties of compounds.
3. Elements in the same period are characterized by the same number of valence electrons.
4. Periods 2 and 3 have the same number of elements.
5. The element with electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$ is a metal.
6. The period 6 inner transition series fills the 4f-orbitals and are called actinide series.
7. An element with the valence electron configuration $2s^2\ 2p^3$ belongs to period 5.
8. Elements in the same period have consecutive atomic number.
9. Mendeleev's law states that properties of elements is a periodic function of their atomic number.
10. Elements in the same group have similar chemical properties.

II. Matching

A	B
11. Alkali metals	A. $ns^2 np^6$
12. Alkaline earth metals	B. $ns^2 np^5$
13. Boron family	C. ns^2
14. Carbon family	D. ns^1
15. Nitrogen family	E. $ns^2 np^2$
16. Chalcogens	F. $ns^2 np^4$
17. Halogens	G. ns^2np^3
18. Noble gases	H. $ns^2 np^1$
	I. $ns^2 (n-1) d^1$
	J. $ns^2(n-1)d^2$

III. Multiple Choice

19. The element with atomic number 25 belongs to

A. s-block C. d-block

B. p-block D. f-block

20. The property of the element with atomic number 12 resembles that of the element with atomic number

A. 11 B. 13 C. 1 D. 20

21. Which of the following elements would you expect to have chemical properties most similar to that of S?

A. P C. Se

B. Cl D. Na

22. Which of the following are arranged correctly in order of increasing radius?

A. $F^- < Cl^- < Br^- < I^-$

B. $O^{2-} < F^- < Ne < Na^+$

C. $Ca^{+2} < Cl^- < K^+ < Ar$

D. $Rb^+ < K^+ < Na^+ < Li^+$

23. Neglecting hydrogen, non-metals are located in

A. s-block
C. d-block
B. p-block
D. s and p-block

24. Carbon and lead belong to the same group in the periodic table. But carbon is a non-metallic element while lead is a metallic element. The reason is that:

A. carbon and lead have different number of valence electrons
B. carbon is more dense than lead.
C. lead is more electronegative than carbon.
D. lead is bigger than carbon atom

25. An element with a $4s^2\ 3d^2$ – valence configuration is:

A. s – block metal
B. found in period 2 of the periodic table
C. found in group IIIA of the periodic table
D. definitely a transition metal

26. In any group of the periodic table, elements,

A. become more reactive in moving down a group
B. tend to accept electron more readily as atomic number increases.
C. Tend to lose electrons more readily as atomic size increases
D. become less reactive as atomic number increases.

27. Which of the following elements has the greatest tendency to lose an electron if energy is supplied?

A. potassium
C. oxygen
B. calcium
D. fluorine

28. Which of the following elements has the smallest electron affinity?

A. carbon
C. neon
B. oxygen
D. flourine

29. Which of the following set of elements have the same outer most electron configuration

I. H, He, Be

II. H, Li, Be

III. H, Li, Na

IV. He, Ne, Ar

A. I and IV
B. III
C. I and III
D. IV

30. Which of the following is a noble gas?

A. oxygen
B. ammonia
C. neon
D. hydrogen

31. A periodic property

A. changes according to periods
B. repeats itself in the same manner with increase in atomic number
C. changes according to ionization energy
D. changes with temperature and mass number

32. Which of the following statements is not true?

I. valence electrons play the same role as core electrons

II. valence electrons are responsible for the chemical properties of elements.

III. The maximum number of elements in the second period are 8

IV. All periods have eighteen elements.

A. I and IV
B. I, II, III
C. I, III, IV
D. II and IV

33. Transition elements are

I. found between group IA and IIIA

II. all metals or semi-metals

III. all metals

IV. found between group IIA and IIIA

A. I,III
B. III
C. IV
D. III,IV

34. Hydrogen can be placed in group IA because

I. can lose one electron in its reactions

II. has one electron in its outer shell

III. can gain one electron to complete its outershell

IV. has an oxidation number of +2.

A. IV
B. IV and I
C. I, II, IV
D. II, III

35. Neon has a higher ionization energy than oxygem because neon has

A. An inert gas structure
B. a bigger atomic radius
C. greater screening effect
D. a higher electronegativity

36. All of the following arranged elements based on their atomic masses EXCEPT

A. Doberiener
B. Henry Mosely
C. Mendeleev
D. Newland

37. Given elements with their atomic radii (in A^0): X = 1.52, Y = 1.86, Z = 2.27, W = 1.04 which of the following is true about these elements?

A. Z has the highest ionization energy
B. W is the least electro positive element
C. X has the greatest electron affinity
D. Y is the most electron affinity

38. Elements not found in nature, synthesized in nuclear reactions and involving completion of the 5f electronic orbitals are

A. the lanthanides
B. the actinides
C. the transition metals
D. the halogens

39. Non-metals when compared to metals

A. are less electronegative and have smaller size
B. have greater ionization energy and smaller size
C. have lower ionization energy and smaller size
D. are less electronegative and have smaller size

40. Which of the following groups consists of entirely metals?

A. IVA

B. IA

C. IIA

D. VIA

41. Which one is the following is the most non-metallic elements?

A. Mg

B. Si

C. Al

D. p

42. The most metallic element among the following is

A. Na

B. Li

C. K

D. Rb

43. Which of the following sets of atomic numbers represent three different groups in the modern periodic table?

A. 10, 12, 14

B. 4, 10, 12

C. 3, 11, 19

D. 7, 24, 15

44. All of the following decreases down a group EXCEPT

A. electropositivity

B. electronnegativity

C. Electron affinity

D. Ionization energy

45. Which sequence of atomic numbers represent elements which have similar chemical properties?

A. 4, 12, 20

B. 3, 4, 5

C. 11,19,20

D. 16,17,18

IV. Give Short Answer

46. List the factors that affect electron affinity

47. Predict which member of each pair has larger size.

a) F or Ne

b) Ca or Ca^{+}

c) Br or Br^{-}

d) Fe^{+3} or Fe^{+2}

e) Be or B

f) O^{-} or O^{2-}

48. Determine which member has higher ionization energy.

a. Ca or Sr

b. N or F

c. Mg^{+} or Ca^{+}

d. Cr^{+2} or Cr^{+3}

49. Arranger the following in ascending order of electronegativity

a. As, Sn, Pb, Ge

b. F, Cl, Br, I

c. O, N, F Cl

d. Li, Be, C, B

50. Find the period and group of

a. Ar

b. V

c. Sb

UNIT 3

CHEMICAL BONDING AND INTERMOLECULAR FORCES

Unit Outline:

3.1 Ionic Bonding

3.2 Covalent Bonding

3.3 Metallic Bonding

3.4 Intermolecular forces

Introduction

When atoms interact to form a chemical bond only their outer regions come into contact.

For this reason, when we study chemical bonding, we are usually concerned primarily with valence electrons of the atoms.

Valence electrons are found in the outer most shell of atoms.

There are three types of chemical bonds. These are

1. Ionic bonding
2. Covalent bonding
3. Metallic bonding

3.1 Ionic Bonding

- It is also called **electrovalent bond** because it is attraction force between positive ions and negative ions.
- It is usually formed between non-metals and metals
- It is formed between atoms that have large size and small size.
- It is formed by complete transfer of valence electrons. As a result positive and negative ions are formed

1. Positive ions (cations)

- They are formed when atoms lose electrons

Example:

$Fe \longrightarrow Fe^{2+} + 2e^-$

- They have lower electrons than protons.

They have smaller size than their neutral form.

2. Negative ions (anions)

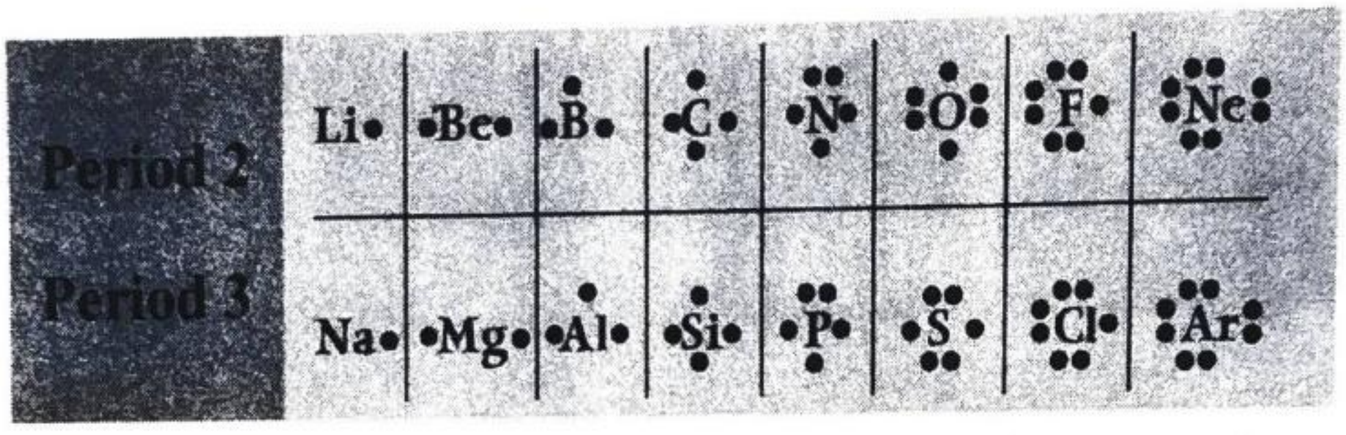
- They are formed by gaining of valence electrons.

Example:

$F + e^- \longrightarrow F^-$

- They have higher number of electrons than protons
- They have larger size than their neutral form.

The elements symbol represents the nucleus and inner electrons but the surrounding dots represent the valence electrons as shown below for period 2 and 3 periods.



Formation of ionic bonds

- Ionic bond is formed between atoms that have low ionization energy and high electron negativity.

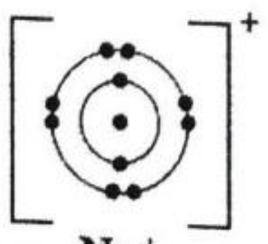
Example

a. Between calcium and oxygen

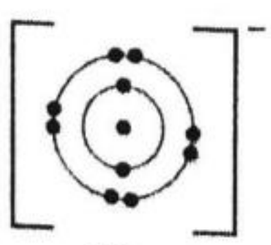
b. Between magnesium and nitrogen

Formation of sodium fluoride using Bohr model

➦ $^{23}_{11}N$ 2, 8, 1



➦ $^{19}_{9}F$ 2, 7



Naming ionic compounds

- Compounds formed by ionic bond are termed as **ionic compounds**.
- For naming binary ionic compounds, the name of the metal remains same while that of the none metal ends with the suffix – i d e.

Example

- NaI – sodium iodide
- CaS – calcium sulphide
- Ba_3N_2 – Barium nitride

For naming ionic compound of poly atomic ions, the name of the metal followed by the name of polyatomic ion.

Example

- Na_2CO_3 – sodium carbonate
- K_2SO_4 – potassium sulphate
- $Mg(NO_3)_2$ – magnesium nitrate

For ionic compounds containing transition metals state their valence number by using Roman number

Example

- CuO – Copper (II) oxide
- Cu_2O – copper (I) oxide
- $FeSO_3$ – Iron (11) sulphite

Illustrative Examples

1. Why do atoms combine with chemical bonds to form compounds? Because
   A. They will attain the lowest energy
   B. They will attain the highest energy
   C. Most atoms are more stable when they exist free
   D. They want to be together

Explanation: Most atoms are less stable when they are free and combine together to lower and become more stable

Answer: A

2. Which type of electrons participate in bond formation.

   A. For metals all valence electrons

   B. For non – metals only unpaired electrons

   C. For non-metals all valence electrons

   D. A and B are correct

Explanation: For metals the total number of valence of electrons lost to form cations while the non-metals only unpaired electrons became paired in ionic bond formation.

Answer: D

3. The formula of ionic compound between group IIA metals "M" and group VIA non-metals "X" will be

   A. MX B. M_2X_2 C. MgX_2 D. M_2X_3

Explanation: The number of valence electrons of metals is same as their valence number but valence number but valence number of non – metals of equal to number of unpaired electrons of the non – metal or eight minus number of valence electrons.

➡ $M^{2}\times X^{2} = M_2X_2 =$ MX The formula of ionic compound is always empirical formula.

Answer: A

4. In which of the following combination ionic bond is not formed?

   A. Lithium and fluorine

   B. Aluminum and oxygen

   C. Sulphur and chlorine

   D. Calcium and phosphorus

Answer: C

5. What type of bond is formed between atoms of being greatly differing electro negativities?

   A. Covalent bond

   B. Ionic bond

   C. Dative bond

   D. Metallic bond

Explanation: Atoms with greatly different electro negativities are metals and non-metals. The type of bond formed between these atoms is ionic bond

Answer: B

6. What type of bond is formed between elements having electron configuration of $1s^2\ 2s^1$ and $1s^2\ 2s^2\ 2p^5$ respectively?

   A. Ionic bond
   B. Covalent bond
   C. coordinate covalent bond
   D. metallic bond

Explanation: These two elements are found in group IA (alkali metals) and group VIIA (halogens). Thus the type of ionic bond formed is ionic.

Answer: A

7. Which one of the following statements is not true about ionic compounds (EGSECE)

   A. One of the bond forming atom is a metal and the other a non-metal
   B. Electrons must be transferred from valence shell of one atom in to valence shell of another atom.
   C. Cations and anions are formed as a result of the bond formation.
   D. The two bond forming atoms should have equal attraction force.

Explanation: For transfer of electrons the bonded atoms must have different attraction forces or electro negativities.

Answer: D

8. In which of the following compounds are both ions are iso electronic with the argon atom? (EGSECE, 2001E.C)

   A. $CaCl_2$
   B. NaF
   C. KBr
   D. Al_2O_3

Explanation: Isoelectronic species have the same number of electrons. Argon has 18 electrons

Answer: A

9. Write the name of the following ionic compounds.

a. RbBr
b. SrO
c. $Zn Cl_2$
d. Fe_2O_3
e. FeO
f. $Al_2 (SO_4)_3$

Explanation:

a. Rubidium bromide
b. Strontiumoxide
c. Zinc chloride
d. Iron (III) oxide
e. Iron (II) oxide
f. Aluminum sulphate

10. Write the formula of an ionic compound formed between

a. Mg and P
b. Li and N
c. Mg and S
d. Al and F

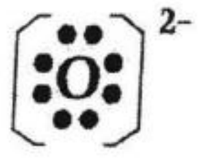
Answer:

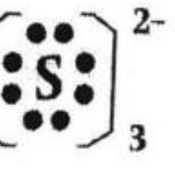
a. Mg_3P_2
b. Li_3N
c. MgS
d. AlF_3

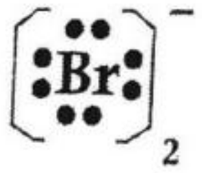
11. Write Lewis structure for the following ionic compounds.

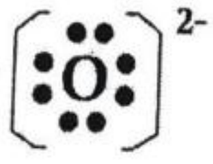
a. BaO
b. $CaBr_2$
c. Al_2S_3
d. Na_2O

Answer:

a. Ba^{2+}  $^{2-}$

c. Al_2^{3+}  $^{2-}_{3}$

b. Ca^{2+}  $^{-}_{2}$

d. Na_2^{+}  $^{2-}$

12. What is the difference between a Lewis symbol and a Lewis structure?

Answer:

➥ A Lewis symbol consists of a chemical symbol together with dots which are placed around the symbol. A Lewis structure is a combination of Lewis symbols that represents other transfer of electrons on ionic bonding or sharing of electrons in covalent bonding.

13. Draw the Lewis symbols for the following elements.

a. Potassium

b. Calcium

c. Bromine

d. Krypton

Answer

a. K•

b. Ca:

c. :Br:

d. :Kr:

14. Write the changes that take place in the formation of NaCl.

Answer:

$Na(s) \longrightarrow Na(g)$

$Na(g) \longrightarrow Na^{+}(g) + e$

$Cl_2(g) \longrightarrow 2Cl(g)$

$Cl(g) + e^{-} \longrightarrow Cl^{-}(g)$

$Na^{+}(g) + Cl^{-}(g) \longrightarrow NaCl(s)$

15. Draw Bohr's diagrammatic representation for

a. Magnesium chloride

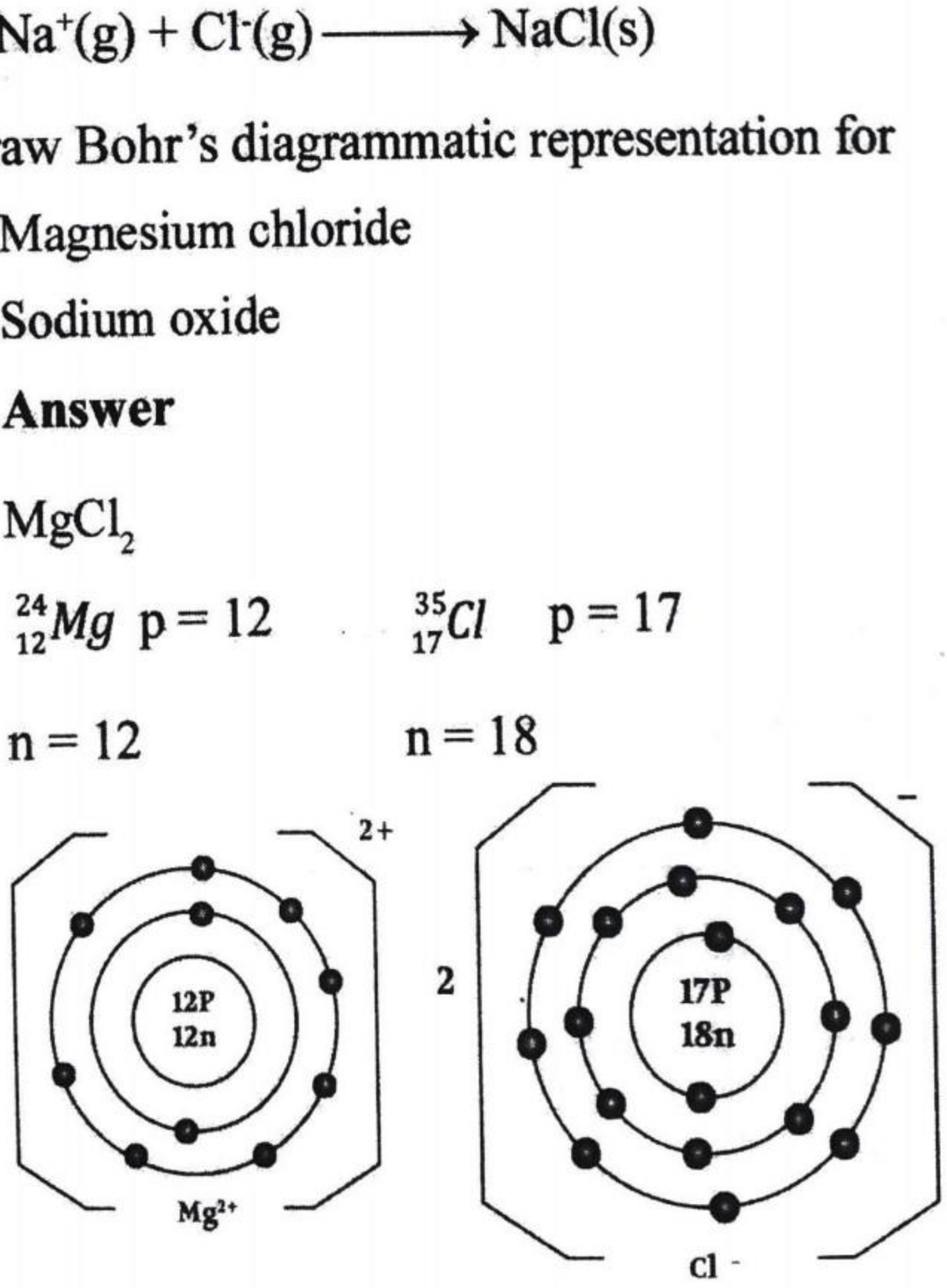
b. Sodium oxide

Answer

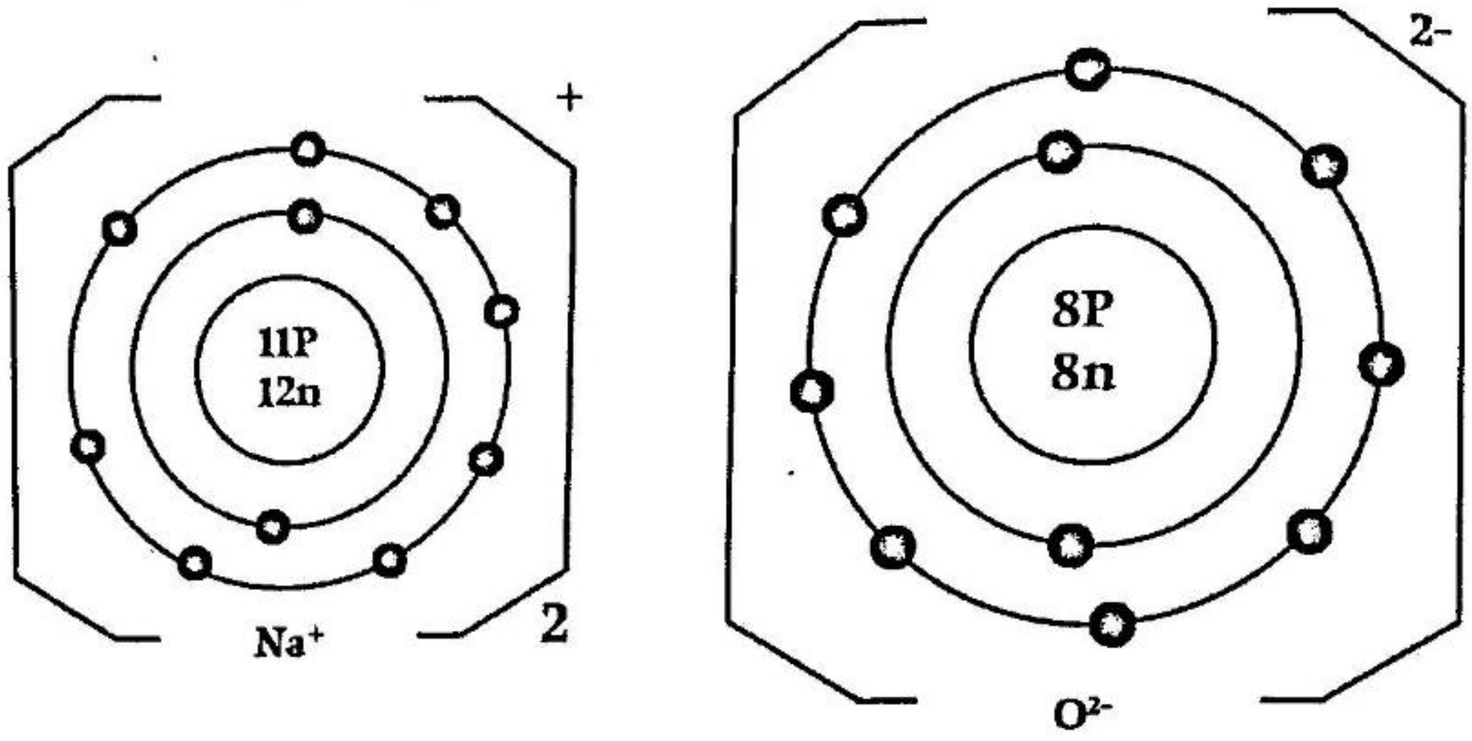
a. $MgCl_2$

$^{24}_{12}Mg$ p = 12 $^{35}_{17}Cl$ p = 17

n = 12 n = 18



b. $^{23}_{11}Na$ p = 11, n = 12 $^{16}_{8}O$ p = 8, n = 8



Properties of ionic compounds

- They are aggregates of positive and negative ions

Example:

consider NaCl

Na^+ Cl^- Na^+ Cl^- Na^+ Cl^- Na^+ Cl^-
Cl^- Na^+ Cl^- Na^+ Cl^- Na^+ Cl^- Na^+
Na^+ Cl^- Na^+ Cl^- Na^+ Cl^- Na^+ Cl^-
Cl^- Na^+ Cl^- Na^+ Cl^- Na^+ Cl^- Na^+
Na^+ Cl^- Na^+ Cl^- Na^+ Cl^- Na^+ Cl^-

- They are usually soluble in water but not soluble in organic solvents such as ethanol, benzene, carbontetrachloride etc.

Example:

NaCl is soluble in water but insoluble in benzenze or ethanol.

- They have high melting and boiling points.
- They are hard and brittle
- They have high density

- They donot contain molecules
- They are non – volatile compounds
- They transmit electricity in molten or aqueous state.
- They are all crystalline solids meaning have regular arrangement of particles.

Illustrative Examples

16. Ionic compounds are generally . . . (EGSECE. 2003E.C)

A. Soft molecular solids
B. Rigid molecular solids
C. Volatile molecular liquids
D. Closely packed dense solids.

Answer: D

17. Which of the following does not transmits electricity?

A. KCl (s)
B. KCl (aq)
C. KCl (l)
D. Cu(s)

Solution: Ionic compounds do not transmit electricity in solid state because ions are not free to carry charge. All metals transmit electricity in three states of matter.

s – solid, ℓ = molten and (aq) means aqueous solution

Answer: A

18. Which of the following is the only solvent that dissolves ionic compounds?

A. C_2H_5OH
B. C_6H_6
C. H_2O
D. C_7H_8

Answer: C

19. Arrange the following in increasing order of melting point. LiF, NaF, RbF, KF

Solution: As size increases strength of metallic bond decreases. This leads to decrease in melting point.

$\Rightarrow$ RF < KF < NaF

20 What are the charge carriers in ionic compounds?

Solution:

Cations and anions.

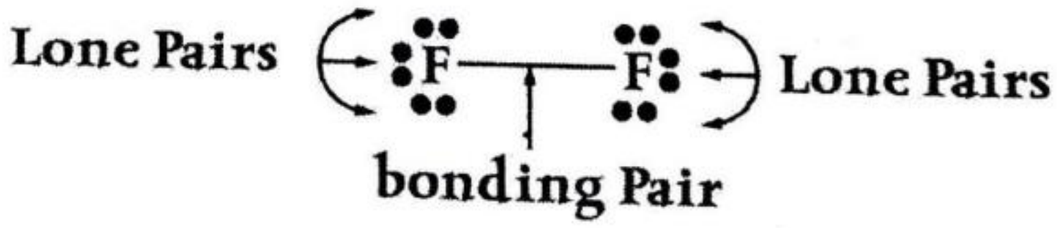
21. Ionic compounds are rigid crystalline solids. What is the difference between crystalline solid and amorphous solid?

Solution

Crystalline solid	Amorphous solid
➦ regular pattern of particles	▪ irregular pattern of particles
➦ has fixed melting and boiling points	▪ has variable melting and boiling points.
Example	**Example**
Ionic compounds	Glass, rubber, plastics

3.2 Covalent Bonding /electron pair bond/

- ➦ It is formed by sharing of electrons between elements that have similar electro negativities,
- ➦ It is attraction force between shared electrons and the nuclei of bond atoms.
- ➦ The shared electrons localized between the bonded atoms are called **bonding pairs (bp)**
- ➦ The unshared electrons are termed as **lone pairs or non – bonding pairs (LP)**

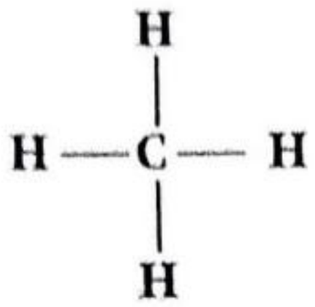


A Lewis structure is a representation of covalent bonding using Lewis dot symbol in which shared electron pairs are shown either as lines or as pairs of dots between two atoms and lone pairs as pairs of dots are shown on individual atoms.

- ➦ Two atoms held together by one electron pair are said to be joined by a **single bond.**

Example:

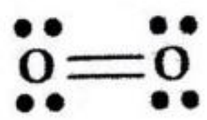
H_2, H — H

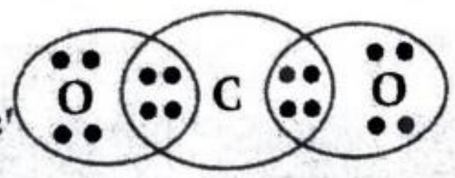
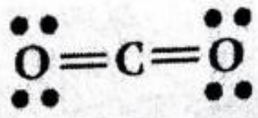
CH_4 

➦ If two atoms share two pairs of electrons, the covalent bond is called a **double bond**

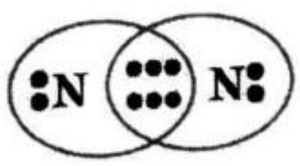
Example:

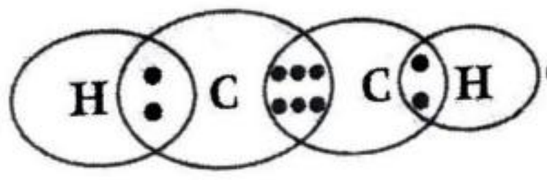
O_2:



$\Rightarrow CO_2$,  or 

➦ **A triple bond** arises when two atoms share three pairs of electrons.

N_2,  or

C_2H_2,  or H—C≡C—H

Bond strength: Triple bond > double bond > single bond

Bond Length: Single bond > double bond > triple bond

Lewis Structure for Covalent Compounds

To draw Lewis structure of covalent compounds follow the following rules.

1. Determine the total number of valence electrons and pair them.

Example

SCl_4 $_{16}S$, 2, 8, 6

Valence electrons = 6

$_{17}Cl$ 2, 8, 7

Valence electrons = 7

$\Rightarrow S: 1 \times 6 = 6$

$Cl: 4 \times 7 = \underline{28}$

Total number of valence electrons = 34

pairs = 34/2 = 17

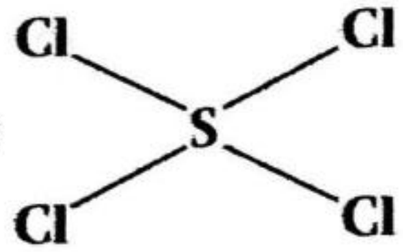
2. Identify the central atom and draw skeletal structure.

Centeral atom is more positive or larger in size

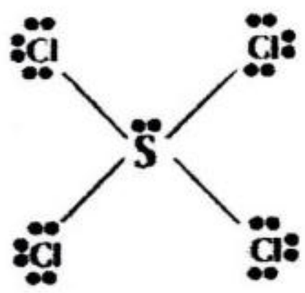
S is centeral atom for the above example

Hydrogen and fluorine cannot be centeral atom because hydrogen has very small size and fluorine is the most electronegative element

Skeletal structure



3. Fulfill the octet of the terminal atoms except hydrogen and put any remaining pairs of electrons on the central atom.



Lp = 13

bp = 4

Thus, the above structure is the Lewis structure of SCl_4

4. Make multiple bond to fulfill the octet of some molecules if necessary.

Example:

SO_3

$S: 6 \times 1 = 6$

$O: 3 \times 6 = 18$

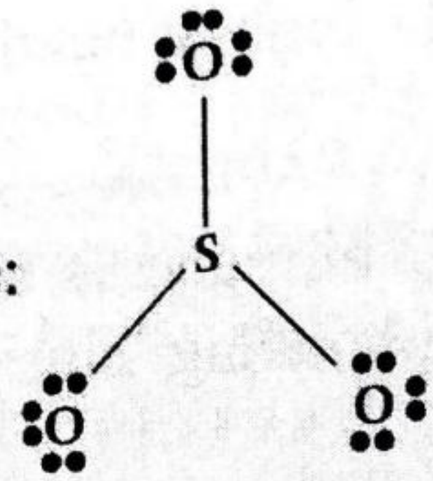
Total valence = 24

electrons

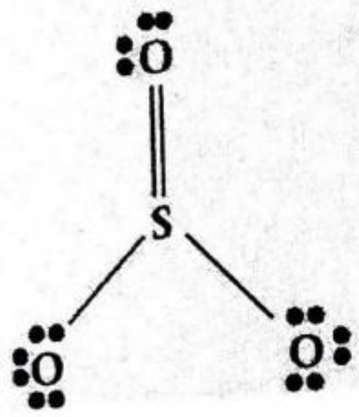
pairs = 24/2 = 12

Centeral atom: S

Skeletal structure:



No electron pair left. Therefore, use any one lone pair from one of the three oxygen atoms and make double bond.



bp = 4

Lp = 8

Illustrative Examples

22. How many lone pairs are there in the Lewis structure of CCl_4? (EGSECE)

A. O B. 9 C. 10 D. 12

Explanation:

$_6$C: 2, 4

C: 1 × 4 = 4

Cl: 4 × 7 = 28

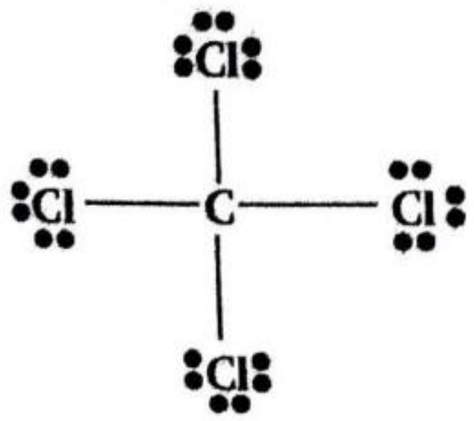
$_{12}$Cl: 2, 8, 7

Pairs = $3\frac{2}{2}$

= 16

Total valence electrons = 32

Centeral atom = C



bp = 4

Lp = 12

Answer:D

23. Which of the following has the highest bond energy?

A. A triple bond
B. A double bond
C. A single bond
D. none

Explanation: Bond energy and bond strength are directly related. Triple bond is the strongest bond and has the highest bond energy.

Answer: A

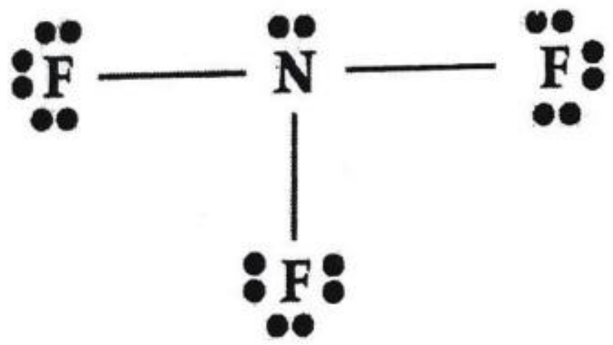
24. In the Lewis formula for NF_3, nitrogen is surrounded by ____ bonding pairs and ____ non-bonding electron pair respectively. (EGSECE 2003 E.C)

A. 3.1
B. 1,3
C. 2, 2
D. 4,0

Explanation:

$_7$N: 2, 5 N: $1 \times 5 = 5$

$_9$F: 2, 7F: $3 \times 7 = 21$



Total valence electrons = 26

Pairs = 26/2 = 13

Centeral atom = N

Lp = 1 on N

bp = 3

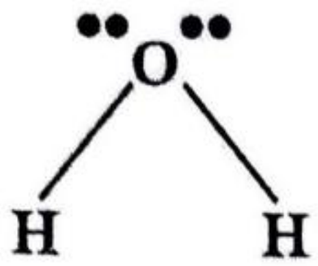
Answer: A

25. Draw Lewis structure and determine the number of bonding pairs and Lone pairs for the following molecules.

a. H_2O
b. NH_4^+
c. HCN
d. SF_6
e. BH_3
f. $Cl\bar{O}_3$

Solution:

a. H : $2 \times 1 = 2$



$O : 1 \times 6 = \underline{6}$

8

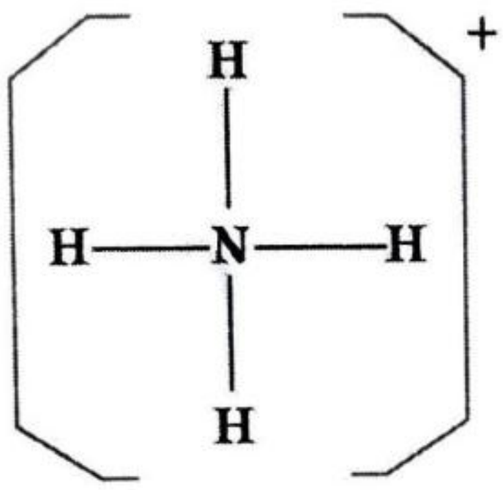
Pairs of electrons = 8/2 = 4

Lp = 2

bp = 2

b. $N : 1 \times 5 = 5$

$H : 4 \times 1 = 4$



Charge: $\underline{-1}$

8

Pairs of electrons = $\frac{8}{2} = 4$

Lp = O

bp = 4

Note:

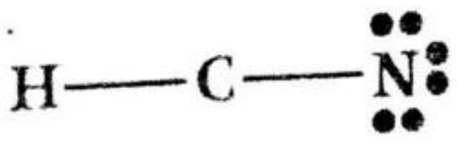
Use closed interval bracket for polyatomic ions.

c. $H : 1 \times 1 = 1$

$C : 1 \times 4 = 4$

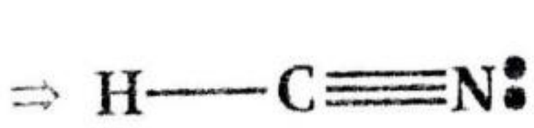
$N : 1 \times 5 = \frac{5}{10}$

Pairs = $\frac{10}{2} = 5$



Central atom = C

Make multiple bond between carbon and nitrogen to fulfill octet of carbon



Lp = 1

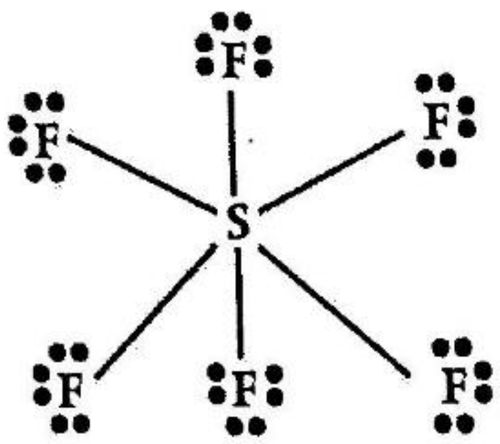
bp = 4

d. $S : 1 \times 6 = 6$

$$F : 6 \times 7 = \frac{42}{48}$$

$$\text{Pairs} = \frac{48}{2} = 24$$

Central atom: S

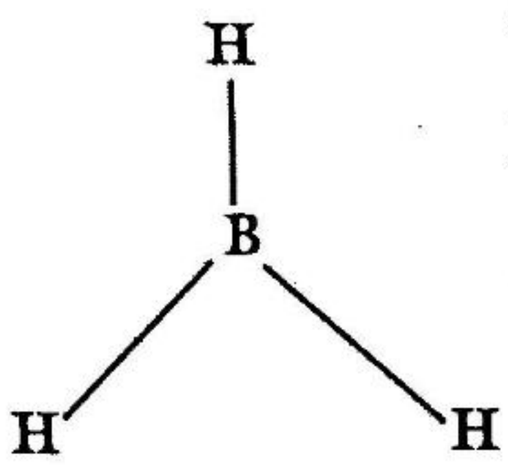


Lp = 18

bp = 6

e. $_5B$ 2, 3

$B : 1 \times 3 = 3$



$$H : 3 \times 1 = \frac{3}{6}$$

Pairs = 6/2 = 3

Centeral atom : B

Lp = 0

bp = 3

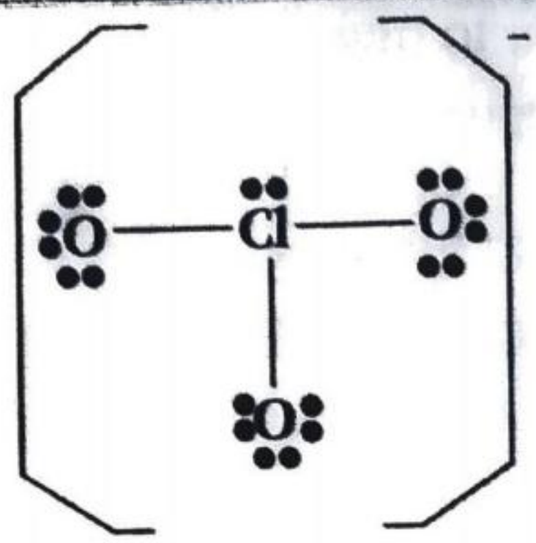
f. $Cl : 1 \times 7 = 7$

$O : 3 \times 6 = 18$

Change $\frac{+1}{26}$

Pairs = 26/2 = 13

Ceral atom : Cl



Lp = 10

bp = 3

26. Hydrogen and fluorine cannot serve as centeral atom. Why?

Explanation: Hydrogen has only one valence electron and fluorine is the most electronegative element.

27. Determine the number of lone pairs on the centeral atom for the following species.

a. H_3O^+

b. ICl_2^-

c. BrF_5

d. $SiCl_4$

Solution:

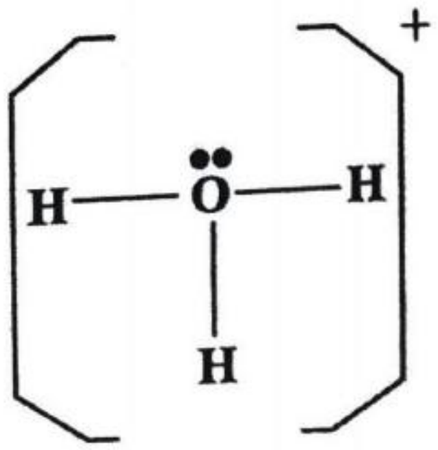
a. H : $3 \times 1 = 3$

O : $1 \times 6 = 6$

Charge: $^{-1}/_{8}$

Pairs = $^{8}/_{2} = 4$

Centeral atom = 0



Lp = 1

bp = 3

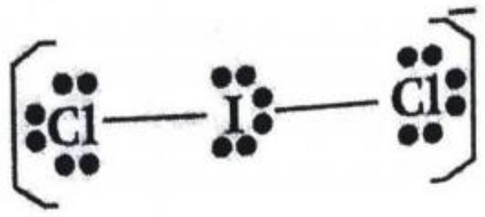
b. I : $1 \times 7 = 7$

Cl : $2 \times 7 = 14$

$$\text{Charge} = \frac{+1}{22}$$

$$\text{Pairs} = \frac{22}{2} = 11$$

Centeral atom = I



There are 3 lone pairs on I.

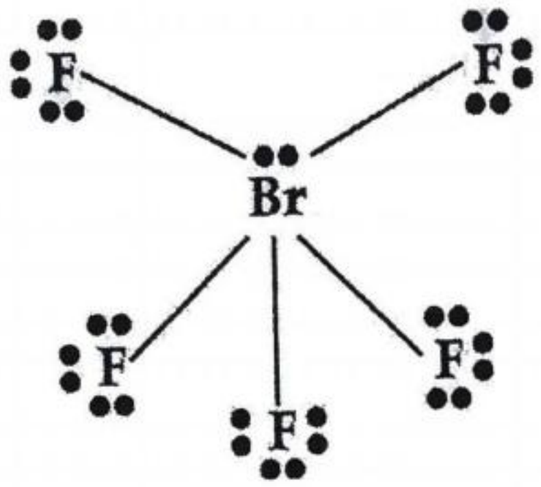
c. Br : $1 \times 7 = 7$

F : $5 \times 7 = \underline{35}$

42

Pairs = $\frac{42}{2} = 21$

Centeral atom = Br



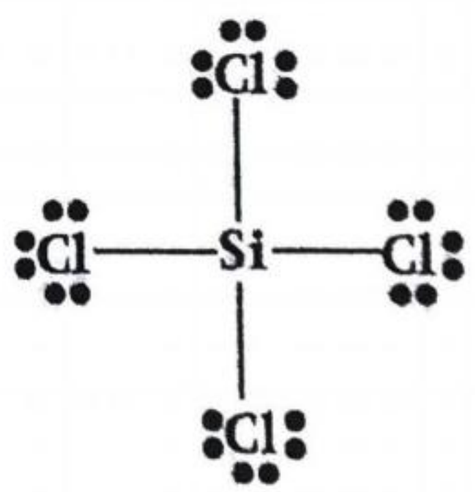
One Lone pair on Br

d. $_{14}$Si: 2, 8, 4

Si : $1 \times 4 = 4$

Cl : $4 \times 7 = \underline{28}$

32



Pairs = 32/2 = 16

Centeral atom = Si

No lone pair on Si

28. State the roles of centeral atom in covalent compounds.

Answer:

A central atom is bonded to two or more terminal atoms. The arrangement of lone pairs and bonding pairs on the centeral atom is used to determine the structure of the molecule.

29. Why there is gap or space between bonding pairs and lone pairs, when we draw Lewis structure for a given covalent compound?

Explanation: Bonding pairs and lone pairs are electrons with the same charge. There is repulsion between likes charges. That is the reason for gaps between these pairs of electrons. The order of repulsion is:

Lone pair – Lone pair > Lone – pair-bonding pair > bonding – pair – bonding – pair.

Coordinate covalent bond

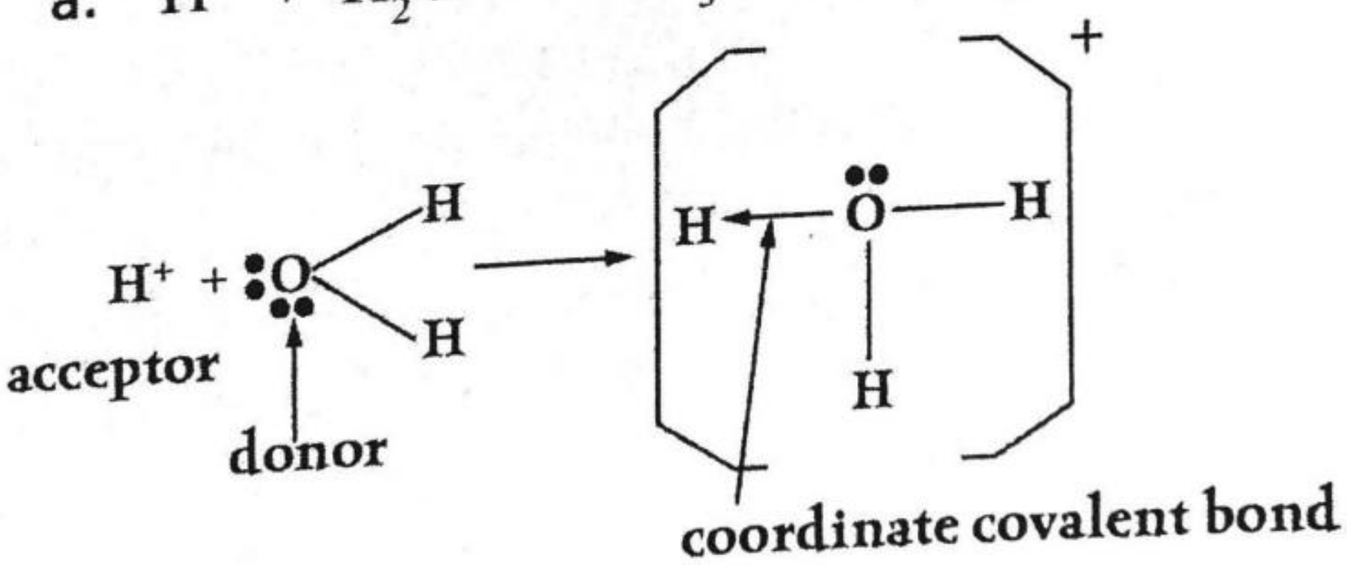
- It is also called **dative** or donor acceptor bond
- It is formed when the shared electrons are obtained only from one of the bonded atom.

$$X + :Y \longrightarrow X \leftarrow Y$$

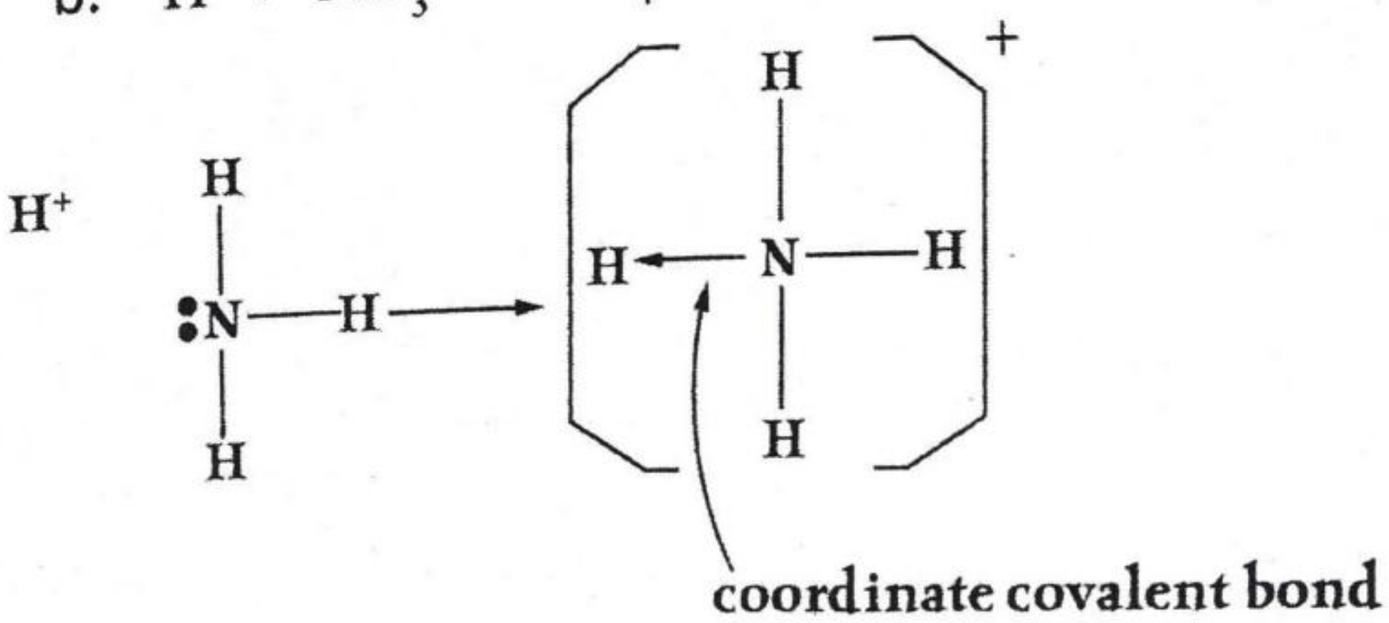
Acceptor donar Coordiante covalent bond

Example

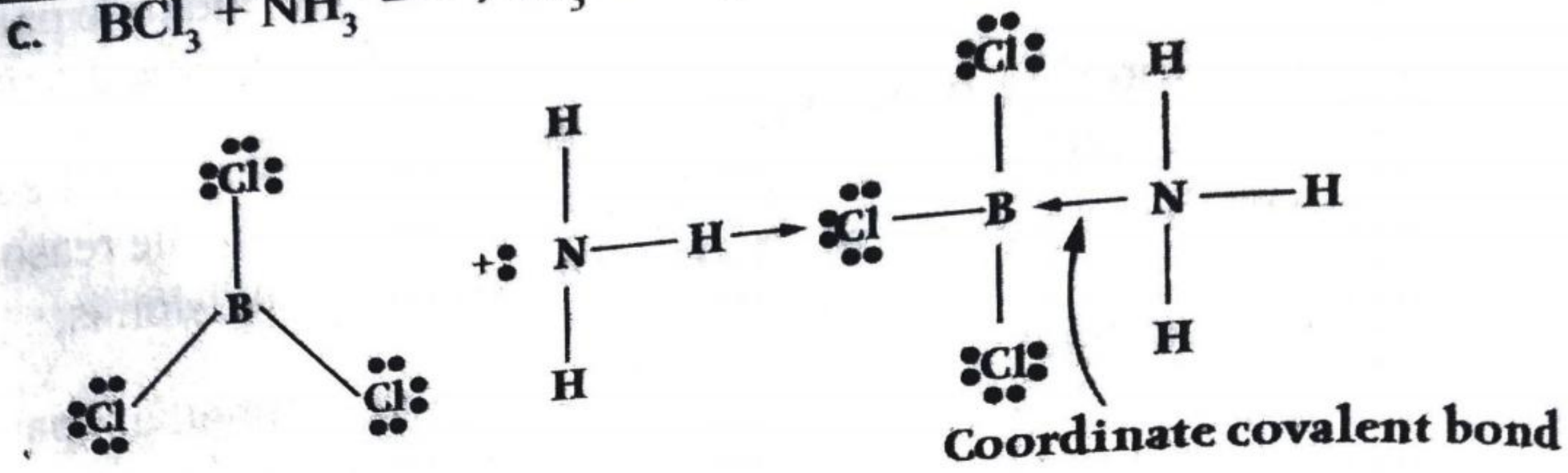
a. $H^+ + H_2O \longrightarrow H_3O^+$

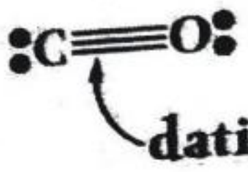


b. $H^+ + NH_3 \rightarrow NH_4$



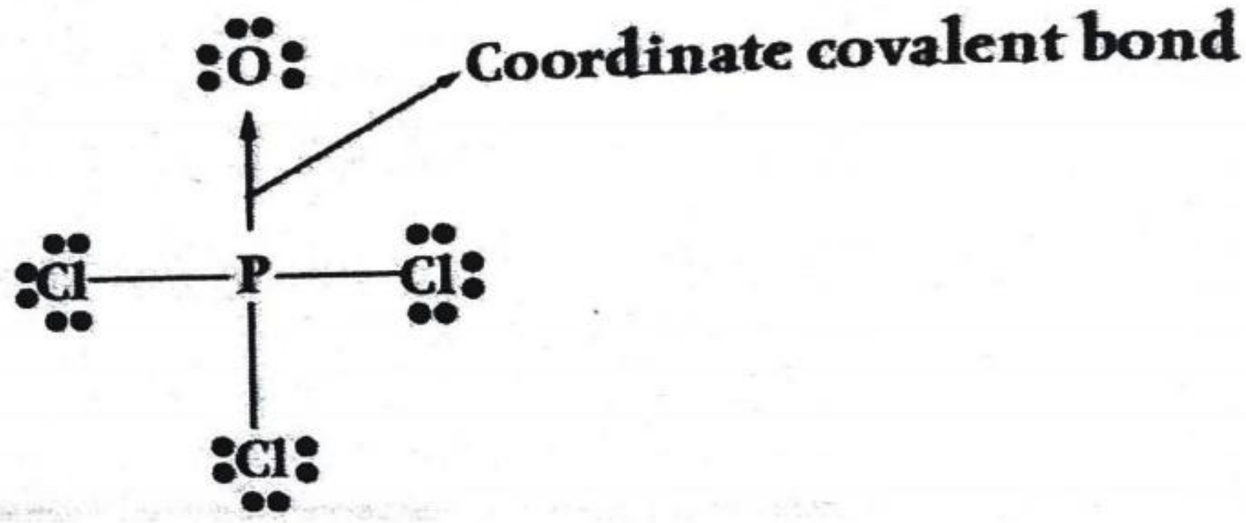
c. $BCl_3 + NH_3 \longrightarrow Cl_3BNH_3$



d. CO 

e. $POCl_3$ Coordinate covalent bond

$POCl_3$



Polar and non – polar covalent bonds

1. **Polar covalent bond** is formed between two atoms that have different electro negativities.
   - The more electronegative element has partial negative charge (σ-) and the more positive atom has partial positive charge (σ+)
   - This bond is partially ionic bond

Example

a. HF : $\overset{\sigma+}{H}——\overset{\sigma-}{F}$

b. HCl $\overset{\sigma+}{H}——\overset{\sigma-}{Cl}$

Note:

The larger the electro negativity difference between the bonded atoms the higher is the polarity.

Example:

HF > HCl > HBr > HI

$HF > H_2O > NH_3$

- Molecules formed by polar covalent bond are called **polar covalent molecules.**

Example:

H_2O, NCl_3, PCl_3, SO_2, HF, HCl, HI, HBr, CH_3Cl etc

2. **Non – polar covalent bond** is formed between atoms that have similar electro negativities.

Molecules formed by this bond are called non-polar molecules

Example

CH_4, BCl_3, H_2, O_2, Cl_2, Br_2, I_2, N_2 etc

Properties of covalent compounds

- They are often gases, liquids or solids.

Example:

H_2, O_2, Cl_2, F_2, N_2, CH_4, C_2H_6 – gases

H_2O, Br_2, C_2H_5OH – liquids

I_2, $C_{12}H_{22}O_{11}$ - solids

- They are molecules
- They have relatively lower melting point and boiling point than ionic compounds.
- They are less dense than ionic compounds.
- They are generally volatile
- Polar covalent compounds are soluble in polar solvents and non – polar covalent compounds are soluble in non – polar solvents.
- Polar covalent compounds have higher boiling point than non – polar covalent compounds.

Example:

Boiling point of ICl is higher than boiling point of I_2.

- Non – polar covalent compounds are non – electrolytes.
- Some polar covalent compounds are electrolytes.

Illustrative Examples

30. Unequal distribution of the shared electrons between two atom results in the formation of (EGSECE)

A. Ionic bond
B. Polar covalent bond
C. non – polar covalent bond
D. coordinate covalent bond

Explanation: If electrons are shared unequally between the bonded atoms due to electro negativity difference the bond is polar covalent bond.

Answer: B

31. Which of the following hypothetical reactions forms coordinate covalent bond? (EGSECE)

A. $G: + H^+ \longrightarrow [G : H^+]^+$
B. $E. + F. \longrightarrow [E^+ F^-]$
C. $C^+ D^- \longrightarrow C^+ D^-$
D. $A. + .B \longrightarrow A : B$

Explanation: In coordinate covalent bond formation, only one atom contributes the shared electrons.

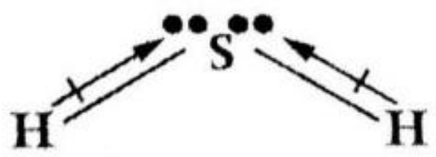
Answer: A

32. Which of the following molecules is polar covalent molecule? (EGSECE)

A. H_2S
B. CO_2
C. CH_4
D. NaCl

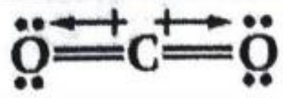
Explanation: Draw Lewis structure and use vectors to identify wether a given molecule is polar or not. If the vector sum is different from zero it is polar. If the vector sum is zero it is non – polar.

H_2S:



H_2S is polar

$\mu \# O$
μ = dipole moment

CO_2  $\mu=O$

$\therefore$ CH_4 is non – polar

NaCl is ionic compound

Answer: A

33. The most polar bond among the following is

A. Si - - - - - O

B. Si - - - - - F

C. Si - - - - - Cl

D. Si - - - - - S

Explanation: The greater the electro negativity difference between the bonded atoms the higher the polarity of the bond. Flourine is the most electronegative element.

Answer: B

34. Which of the following contains both ionic and covalent bond?

A. SF_4

B. NaCl

C. KCN

D. CF_4

Explanation: Ionic and covalent bonds are found together when metals or ammoniemioz bonded with polyatomic ions such as CN^-, OH^-, CO^{2-}_3, SO^{2-}_4, SO_3, PO^{3-}_4 etc

Answer: C

35. Which one of the following is true about covalent compounds?

A. They are molecular substances

B. They are soluble in polar solvents

C. They are shatter when hammered with external force

D. They are solids at ordinary conditions.

Answer: A

36. Which of the following molecule is non – polar with polar bond?

A. H_2O

B. NF_3

C. CH_4

D. CCl_4

Explanation:

- Non – polar covalent compounds with polar bonds are compounds whose dipole moment is zero but there is electro negativity difference between centeral atom and terminal atoms.
- H_2O and HF are polar covalent molecules.
- CH_4 is non – polar covalent molecule with non – polar bond.

$\Rightarrow CCl_4$ is non-polar with polar bond.

Answer: D

37. What is the solubility of Naphthalene, graphite and iodine in water, ethanol, hexane and benzene.

Explanation:

Substances	Solubility in			
	Water	Hexane	Ethanol	Benzene
Naphthalene	insoluble	soluble	insoluble	soluble
Graphite	insoluble	insoluble	insoluble	insoluble
Iodine	insoluble	soluble	soluble	soluble

38. List some polar covalent compounds that transmit electricity.

Answer:

Ammonia, acids such as HCl, HI, H_2SO_4, HI etc

39. Classify the following covalent compounds as polar or non – polar covalent compounds.

a) CS_2 d) C_6H_6 g) SiF_4

b) H_3O^+ e) NH_3 h) PCl_5

c) $CHCl_3$ f. C_7H_8 i) SF_2

Answer:

a, d, f, g and h are non – polar covalent compounds

b, c, e and i are polar covalent compounds

Covalent compounds that violet the octet rule

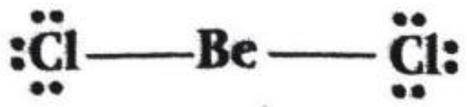
Moleculates that do not contain eight electrons around the centeral atom violets the octet rule. These include:

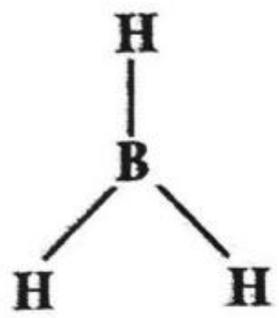
1. Molecules below octet

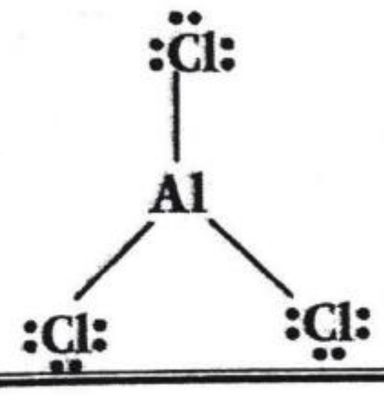
These type of molecules contain less than eight electrons on the centeral atom.

Example:

$BeCl_2$, BH_3, $AlCl_3$ etc

$BeCl_2$  4 electrons on Be

BH_3  6 electrons on B

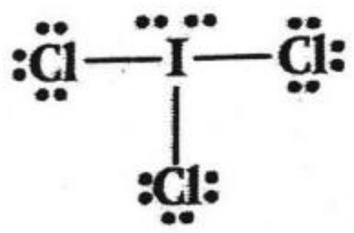
$AlCl_3$  6 electrons on Al

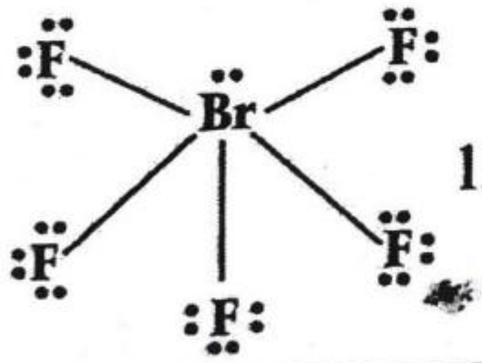
2. Molecules above octet

These molecules have greater than eight electrons on the centeral atom.

Example:

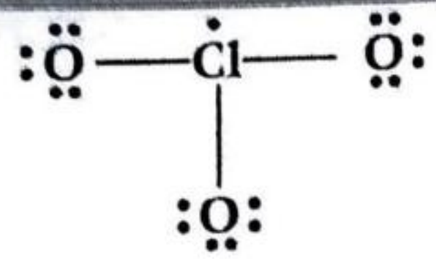
PCl_5, SF_6, BrF_5, ICl_3 etc

ICl_3  10 electrons I

BrF_5  12 electrons on Br

3. Molecules with odd number of electrons

All molecules with odd number of electrons violet the octet rule because eight is even number.

ClO_3 

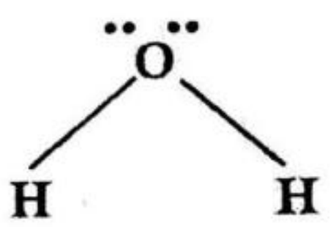
7 electrons on Cl

Illustrative Examples

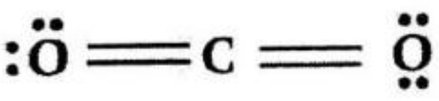
40. Which of the following molecules violates the octet rule? (EGSECE)

A. H_2O
B. CO_2
C. SO_2
D. BH_3

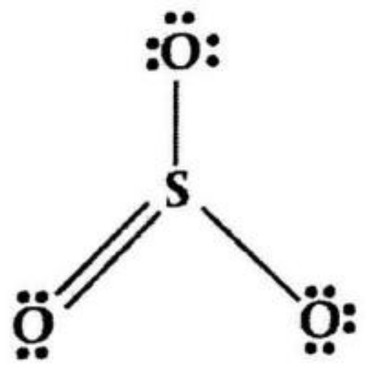
Explanation: Draw Lewis structure and count the number of electrons around the centeral atom. If the number of electrons is not equal to eight such molecule violate the octet rule.

H_2O 

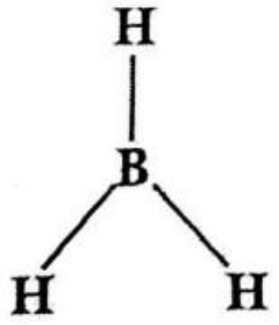
8 electrons on O

CO_2 

8 electrons on C

SO_2 

8 electrons on S

BH_3 

6 electrons on B

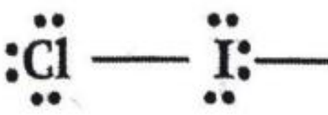
Answer: D

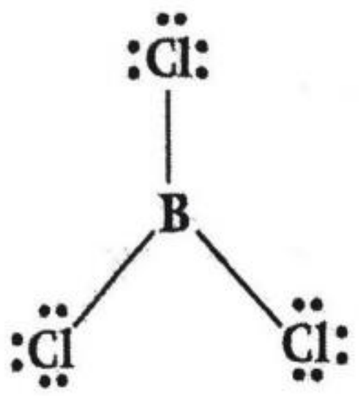
41. Classify the following molecules as below octet, above octet or odd number of electrons.

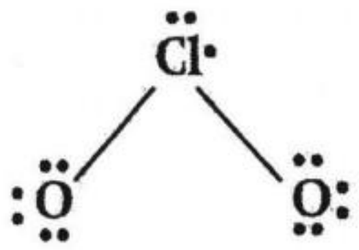
a) BeH_2
b) ICl_2^-
c) BCl_3
d) ClO_2
e) ICl_4^+
f) NO

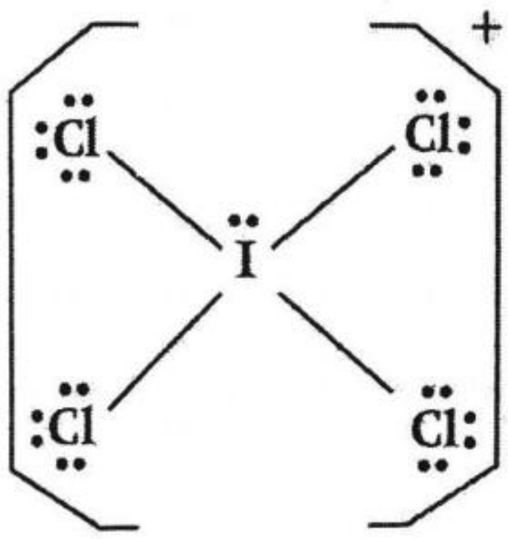
Explanation:

a. H——Be——H below octet

b.  above octet

c.  below octet

d.  odd number of electrons

e.  above octet

f. odd number of electrons

3.3 Metallic Bonding

- It is bonding in metals.
- It is electrostatic force of attraction between metal cations (kernels) and the delocalized valence electrons.

Figure 3.1 structure of metallic bond

Factors affecting strength of metallic bond

There are two factors that affect the strength of metallic bonding. These are:

1. Atomic radii of cations
2. Number of valence electrons of metals

1. Atomic radii of cations

- As size of the cations increases strength of metallic bond decreases.

Example:

Metallic bond strength of: Li > Na > K > Rb

: Al > Mg > Na

2. Number of valence electrons

- As the number of valence electrons increases, metallic bond strength also increases.

Example:

Al > Mg > Na

Metallic bond accounts for the following physical properties of metals.

- ➦ They are malleable – hammered into sheet
- ➦ They are ductile – drawn into wire
- ➦ They transmit heat and electricity because they have free electrons that carry charge.
- ➦ They have high melting and boiling points.
- ➦ The melting points of metals cover a broad range from – 38.9^{0}c (mercury) to 3410^{0}c (tungsten)

Illustrative Examples

42. Which of the following metals has the highest melting point?

A. Mg B. Ca C. Sr D. Ba

Explanation:

- As size increases strength of metallic bond strength decreases. Magnesium has the smallest size.

Answer: A

43. Which one of the following is not true of a metallic bond? (EGSECE)

A. Electrons are free to move throughout the metal structure.

B. Strength of a metallic bond is the same for all metals.

C. The valence electrons in a metal structure are shared among two or more atoms.

D. Metallic bond gives rise to excellent electrical conductivity

Explanation:

Due to size and number of valence electrons different metals have different metallic bond strength.

Answer B

44. Which of the following is the most metallic element?

A. Li B. K C. Na D. Rb

Explanation: As size increases metallic character also increases but metallic bond strength decreases.

Answer: D

45. The metal with the highest melting point is _____

Answer: Tungsten (3410^0c)

46. Arrange the following metals is increasing order of melting point.

a. Mg, Na, Al, K b. K, Rb, Na, Cs

Answer:

a. K < Na < Mg < Al b. Cs < Rb < K < Na

47. Which member is more metallic?

a. Al or B

b. Li or Be

c. Mg or Sr

Answer:

a. Aℓ

b. Li

c. Sr

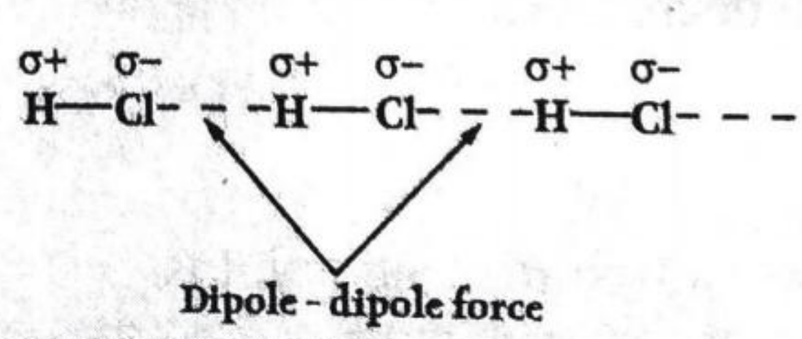
3.4 Intermolecular Forces

- These are forces that hold molecules together.
- These are attractions between one molecule and a neighboring molecule
- They are weaker than chemical bonds.
- There are three types of intermolecular forces. These are
  1. Dipole – dipole force
  2. Hydrogen bond
  3. London force

1. Dipole – dipole force

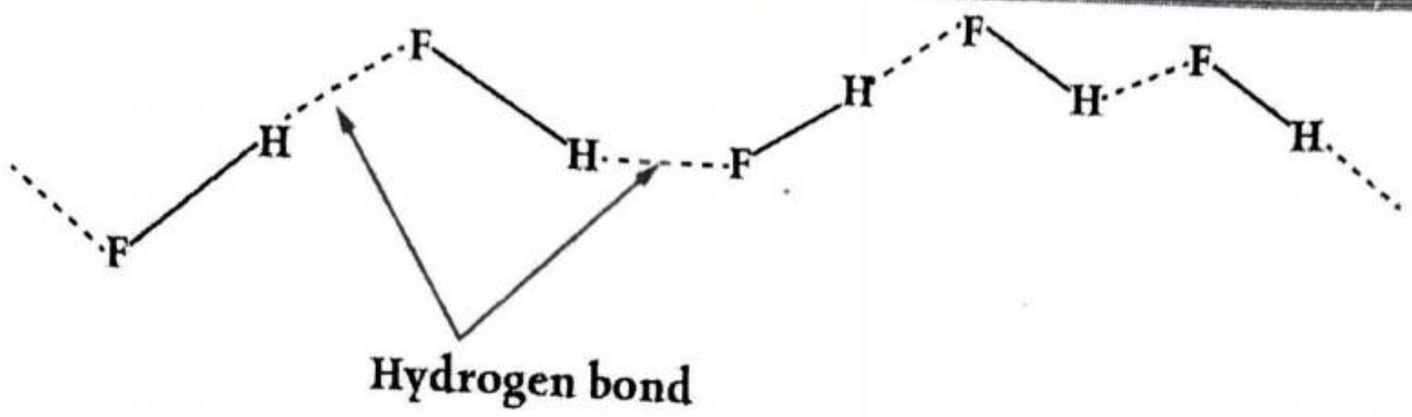
- Dipole – dipole forces are forces that act between polar molecules, that is, between molecules that posess dipole moments
- The larger the dipole moments the greater the dipole – dipole forces.

Example



2. Hydrogen bond

- Hydrogen bond is formed when hydrogen in one molecules bonded with oxygen, fluorine and nitrogen in another molecule.
- It is a special type of dipole – dipole interaction between the hydrogen atom in a polar bond such as N — H, O — H or F—H and an electronegative O, N or F atom.



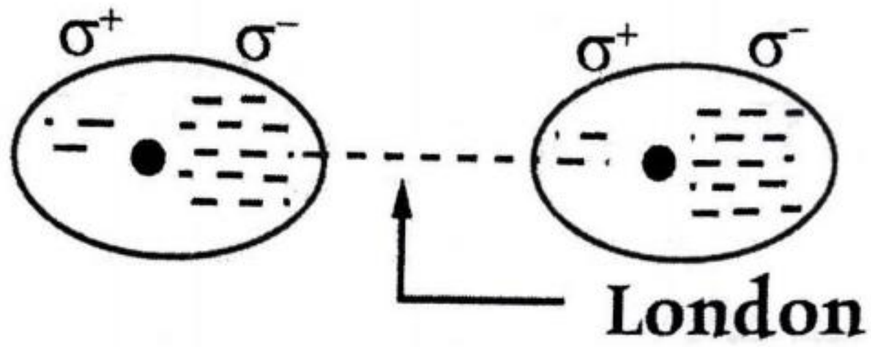
Hydrogen bond

➦ Hydrogen bond is the **strongest** intermolecular force.

3. London force/ Dispersion forces/

➦ It is the intermolecular force found between non – polar molecules.

It is attraction force that results when the electrons in two adjacent atoms occupy positions that make the atom form induced dipole.



London force

➦ It is the **weakest** intermolecular force.

➦ It is very important for industrial production of **nitrogen** and **oxygen** gases by fractional distillation.

The strength of London force is affected by size. As size increases, strength of London force also increases.

In contrast to intermolecular forces, **intramolecular forces** hold atoms together in a molecule. Intramolecular forces stabilize individual molecules whereas intermolecular forces are primarily responsible for the bulk properties of matter (for example melting point and boiling point)

➦ Generally, intermolecular forces are much weaker than intramolecular forces.

Illustrative Examples

48. Which of the following is not intermolecular force?

A. dipole – dipole force
B. London force
C. hydrogen bond
D. covalent bond

Explanation: Covalent bond is intramolecular force.

Answer: D

49. Which of the following has the highest boiling point?

A. ICl
B. I_2
C. Cl_2
D. N_2

Explanation:

- ICl is polar molecule. In polar molecule, dipole – dipole force exists. This force is stronger than London force.

Answer: A

50. The following table shows substance of comparable molecular masses with their dominant intermolecular forces.

Substance	Dominant intermolecular force
A	London force
B	Hydrogen bond
C	Dipole – dipole force.

Which of the following is true about the substances? (EGSECE… 2004E.C)

A. "B" has the strongest intermolecular force of attraction.
B. "C" has weaker intermolecular force of attraction than "A".
C. "C" has the weakest intermolecular force of attraction.
D. "A" has higher boiling point than "B".

Explanation: Hydrogen bond is the strongest intermolecular force.

Answer: A

51. In which substances are attractive force between the particles to be weakest?

A. helium
B. water
C. Ice
D. Diamond

Explanation: Noble gases are non – polar gases and London force holds them together. Helium is a noble gas.

Answer: A

52. Which of the following molecules has the strongest forces of attraction?

A. NH_3 C. CH_4

B. O_2 D. SO_2

***Explanation*:**

- O_2 and CH_4 – London force
- SO_2 – dipole – dipole force
- NH_3 – hydrogen bond

Answer: A

53. Which of the following molecules can NOT form hydrogen bonding with its own molecule |s|? (EGSECE)

A. HF C. NH_3

B. H_2O D. PCl_5

Explanation: PCl_5 cannot form hydrogen bond with its own molecule.

Answer: D

54. Which one has higher boiling point.

a. H_2O or H_2S? c. ICl or I_2?

b. CH_4 or CH_3 CH_3? d. NaCl or CCl_4?

***Explanation*:**

a. H_2O because of hydrogen bond in H_2O but not in H_2S.

b. CH_3 CH_3 as size increases, the strength of London force increases.

c. ICl because it polar but I_2 is non – polar.

d. NaCl because of ionic bond.

55. What type of intermolecular force is dominant in the following molecules?

a) HBr f) H_2O

b) $CHCl_3$ g) CH_3NH_2

c) CH_3OH h) Ne

d) NH_3 i) CH_4

e) PCl_3 j) N_2

Explanation:

- a, b, e - - - dipole – dipole force.
- c, d, f, g - - - -hydrogen bond
- h, i, j, - - - - London force

Solved Problems

56. Arrange the following in increasing order of melting point.

NaCl, NaF, NaBr, NaI

Explanation: As size increases melting point of ionic compounds decreases.

$\Rightarrow$NaI < NaBr < NaCl < NaF

57. Draw Lewis structure of the following ionic compounds.

a. K_3N

b. $SrBr_2$

c. Al_2O_3

d. $FeCl_3$

Answer:

a. $K_3^+\left[\ddot{N}\right]^{3-}$

b. $Sr^{2+}\left[\ddot{Br}\right]^{-}_{2}$

c. $Al_2^{3+}\left[\ddot{O}\right]^{2-}_{3}$

d. $Fe^{3+}\left[\ddot{Cl}\right]^{-}_{3}$

58. Write the formula of ionic compounds formed between

a. Al and N

b. Ca and P

c. Rb and F

d. Ga and Cl

Answer:

a) AlN b)Ca_3P_2 c) RbF d)$GaCl_3$

59. Name the following ionic compounds.

a. BaI_2 b. CsCl c. CuF_2 d. $FeBr_3$

Answer:

a) Bariumiodide

b) Cesium chloride

c) Copper (II) fluoride

d) Iron (III) bromide

60. Identify the ionic compounds that violate the octet rule among the following.

a) NaCl c) $CuCl_2$ e) MgI_2

b) LiF d) ZnO f) FeO

Explanation: If there are less than or greater than eight valence of the cations such type of ionic compounds violate the octet rule.

a. NaCl $_{11}Na^+$, $1s^2\ 2s^2\ 2p^6$, 8 valence electrons
⇒ NaCl obeys octet rule

b. LiF $3Li+\ 1s^2$, 2 valence electrons in Li+
⇒ LiF violates the octet rule

c. $CuCl_2$, $_{29}Cu^{2+}$, $1s^2 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^9$, 17 valence electron in Cu^{2+}
⇒ $CuCl_2$ violates the octet rule.

d. ZnO, $30Zn^{2+}$: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$, 18 electrons in Zn^{2+}
⇒ ZnO violates the octet rule

e. MgI_2, $_{12}Mg^{2+}$ $1s^2\ 2s^2 2p^6$, 8 valence electrons in Mg^{2+}
⇒ MgI_2 obeys the octet rule

f. FeO, $_{26}Fe^{2+}$, $1s^2 2s^2\ 2p^6\ 35^6\ 3d^6$, 14 valence electrons in Fe^{2+}
⇒FeO violates the octet rule

61. What types of electrons are transfered and shared in ionic and covalent bonds formation.

Answer:

➦ Valence electrons of metals are transferred and unpaired electrons of non – metals are shared.

62. State the type of bonding – ionic, covalent or metallic you would expect in

a. O_3 c. BrO_2

b. $MgCl_2$ d. Fe

Answer:

➦ a and c covalent bond

➦ b ionic bond

➦ d metallic bond

63. Compare the bond length and bond energy of C – O, C == O and C≡O bonds

Answer:

Bond Length: C – O > C = O > C ≡ O

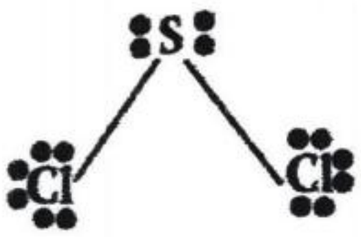
Bond energy: C ≡ O > C = O > C – O

64. Draw the Lewis structure and determine the number of bonding pairs and Lone pairs.

a) SCl_2 c) IF_3 e) PH^+_4

b) CH_3Cl d) NO^-_3 f) HNO_3

Answer:

1. 

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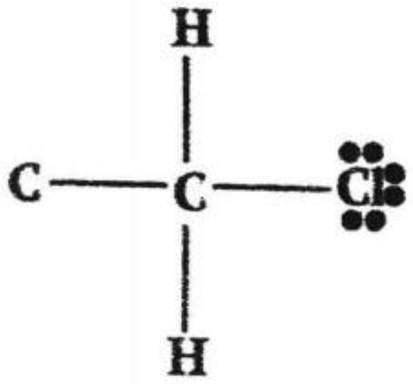
Lp=8

2. 

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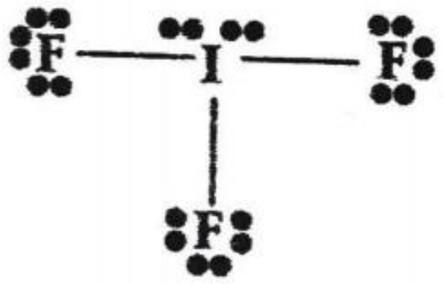
3. 

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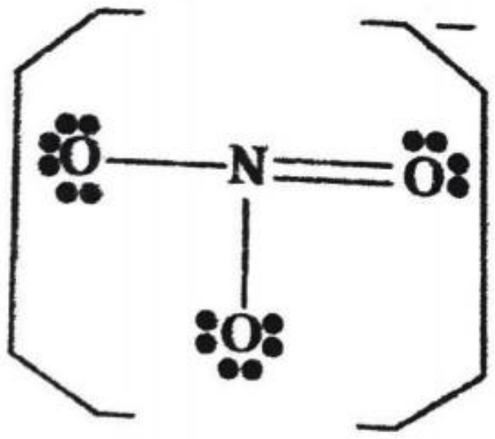
4. 

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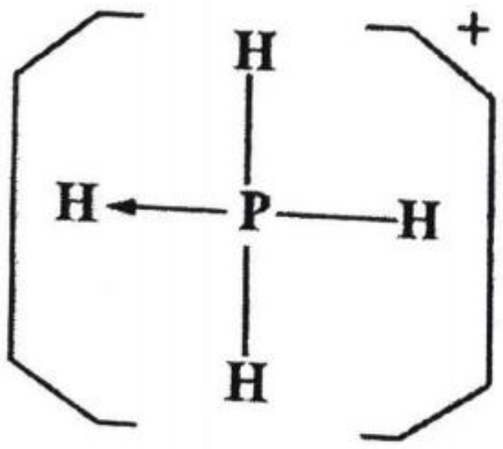

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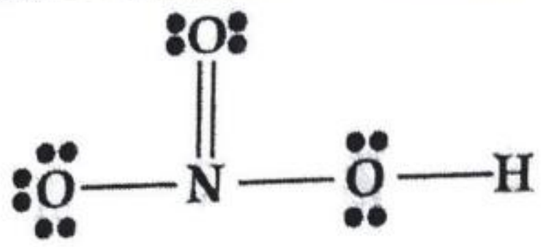
f.  bp=5 Lp=7

65. Classify the following molecules as polar or non- polar molecules.

a. SO_2 c. CH_3NH_2 e. SiH_4

b. H_2Se d. $CHCl_3$ f. CO_2

Answer:

- a, b, c and d are polar covalent molecules
- e and f are non – polar covalent compounds.

66. Arrange the following molecules in decreasing order of polarity. H_2O, HF, H_2S, HCl, PH_3, NH_3

Explanation: As electronegative difference between bonded atoms increases polarity also increases.

$\Rightarrow HF > H_2O > NH_3 > HCl > H_2S > PH_3$

67. List some groups of substances containing hydrogen bond.

Answer:

- Alcohols, carboxylic acids, amino acids, DNA, water, ammonia, hydrogen flouride

68. Explain the difference between intermolecular force and intramolecular force.

Explanation:

Intermolecular force	Intramolecular force
➦ found between molecules ➦ weak ➦ affect physical properties	➦ found within a molecule ➦ strong ➦ affect chemical properties

69. The bond energies of H – A, H – B, H – C and H – D bonds are 602, 830,411 and 213KJ/mol. Respectively Which bond is the most polar?

Explanation:

H – B is the most polar bond because as bond energy increases polarity also increases.

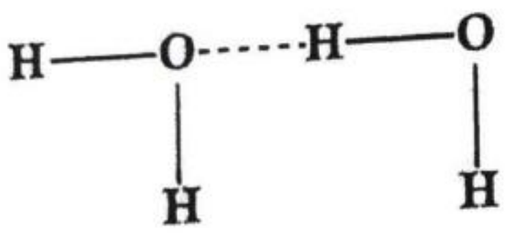
70. Arrange the boiling point of noble gases in decreasing order

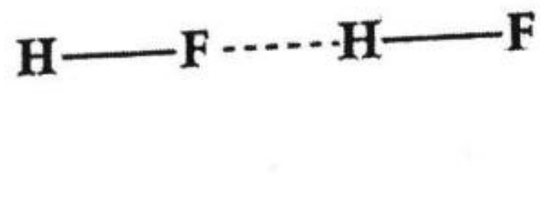
Explanation:

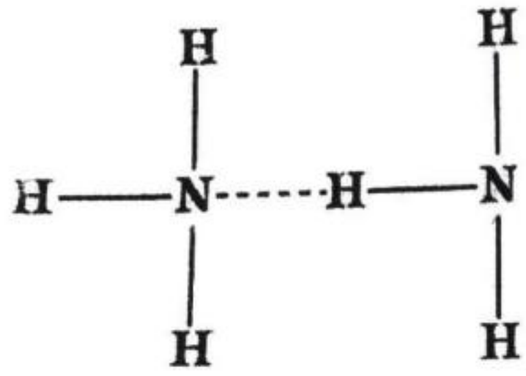
Rn > Xe > Kr > Ar > Ne > He

71. Explain the hydrogen bond formation between two molecules of water, ammonia and hydrogen fluoride.

Explanation:







Supplementary Exercise on Unit - 3

I. True/ False

1. In the formation of ionic bond, valence electrons are transferred
2. Common salt transmit electricity in solid state
3. Covalent bond is formed between atoms of non-metals.
4. Covalent bond is formed by sharing of valence electron
5. Ionic compounds are hard and malleable
6. Metallic bonding is formed between atoms of metals
7. A dative bond is formed when the shared electrons are supplied by one atom.

8. London force is the strongest intermolecular force
9. Intermolecular forces are weaker than intramolecular force
10. A Lewis dot symbol is same as a Lewis structure

II. Choose the best answer from the given alternatives

11. When three pairs of electrons are shared _____ is formed.
    A. double bond
    B. single bond
    C. triple bond
    D. ionic bond
12. Which of the following is electrovalent bond?
    A. Covalent bond
    B. Ionic bond
    C. hydrogen bond
    D. dative bond
13. Which of the following does not transmit electricity?
    A. Salt solution
    B. Gold
    C. graphite
    D. sugar solution
14. Which of the following molecule is the least polar?
    A. HF B. HCl C. HI D. HBr
15. Attraction force between induced dipoles is:
    A. London force
    B. Ion-dipole force
    C. dipole – dipole force
    D. chemical bond
16. Which of the following is a –polar compound?
    A. Cl_2 B. HF C. SiF_4 D. N_2
17. Which of the following molecules has no lone pairs on the centeral atom?
    A. H_2O B. SO_2 C. SO_3 D. OF_2
18. Which of the following is non – polar molecule with polar bonds?
    A. BCl_3 B. CH_4 C. H_2O D. $CHCl_3$
19. Which of the following has no hydrogen bond?
    A. ice
    B. dry ice
    C. amino acid
    D. alcohol

20. Elements A, B, and C have atomic numbers Z, Z + 1, and Z + 2 respectively. Which two elements combine to form ionic compound? (If the value of Z = 9)

A. A and B
B. C and B
C. C and A
D. B and A

21. Which of the following violates the octet rule?

A. MgF_2
B. KI
C. $FeCl_2$
D. CaO

22. Which of the following has the highest boiling point?

A. C_5H_{12}
B. C_6H_{14}
C. C_4H_8
D. C_7H_{16}

23. Which of the following metals has the lowest boiling point?

A. Rb
B. Li
C. Na
D. K

24. When viewed structurally, ionic compounds generally are

A. Single molecule
B. Gaint lattices
C. Ions surrounded by a sea of electrons
D. Giant molecules

25. Two atoms X and Y have electron configuration of $1s^22s^22p^63s^1$ and $1s^22s^22p^5$ respectively, The atoms from

A. An ionic compound of formula, X_3Y
B. A covalent compound of formula X_2Y_3
C. An ionic compound of formula XY.
D. A covalent compound of formula XY_3.

26. In which molecule is the London force likely to be the most important in determining the melting and boiling point?

A. N_2 B. NH_3 C. BH_3 D. PH_3

27. Which of the following compounds has the highest melting point?

A. KF
C. BH_3
B. NH_3
D. PH_3

28. How many lone pairs are the XeF_2?

A. 2
C. 6
B. 4
D. 9

29. Which of the following is a covalent compound?

A. AlF_3
C. Na_2O
B. SiO_2
D. $CaCO_3$

30. Which of the following has ionic, covalent and dative bonds?

A. KCl
C. NH_3
B. CO
D. NH_4Cl

31. An element with atomic number 19 would most likely combine with an element with atomic number _____ to form ionic compound

A. 10
C. 11
B. 9
D. 12

32. Which of the following is true about chemical bonding?

A. It involves valence electrons
B. It does not alter the electron configuration of the bonded atoms
C. It changes the nuclear composition of the bonded atoms.
D. It increases the potential energy of the bonded atoms.

33. Which of the following substances has the least ionic character?

A. LiCl
C. $BeCl_2$
B. NaCl
D. $MgCl_2$

34. Which of the following contains double bond?

A. N_2
C. Cl_2
B. NH_3
D. SO_3

35. The type of bond formed when BF_3 and NF_3 are combined is

A. covalent bond
C. Ionic bond
B. dative bond
D. Hydrogen bond

36. Which of the following compounds has only ionic bond?

A. NaBr
C. NH_4F
B. SO_3
D. NaCN

37. Which of the following covalent compounds obeys the octet rule?

A. ICl_3
C. ClF_3
B. BrF_3
D. PCl_3

38. Which of the following is the most metallic element?

A. Al
C. C
B. B
D. Si

39. Which bond is the least polar bond?

A. C - - - F
C. C - - - - N
B. C - - - - O
D. C - - - P

40. All of the following are intramolecular forces except one. Which one is the exception?

A. Ionic bond
C. Hydrogen bond
B. Covalent bond
D. metallic bond

III. Give short answer

41. The bond length of A – B, A – C, A – D and A – E is $0.5A^0$, $0.78A^0$, $1.6A^0$ and $0.42A^0$ respectively. Arrange these bonds in increasing order of bond strength

42. Which member in each pair expected to exhibit hydrogen bonding.

a. CH_3CH_2OH or CH_3OCH_3
b. CH_3CH_3 or CH_3COOH
c. HF or HCl

43. Arrange the following types of interaction in increasing order of bond energy.

    London force, hydrogen bond, ionic bond, dipole – dipole force.

44. Arrange the following in increasing order of boiling point

    N_2, Br_2, Mg, K

45. List the three types of intermolecular forces

UNIT 4

Chemical Reactions and Stochiometry

Unit Outline:

4.1 Laws of Chemical reactions

4.2 Chemical equations

4.3 Energy Changes in Chemical reactions

4.4 Types of Chemical reactions

4.5 Stoichiometry

4.6 Oxidation - reduction reaction

4.7 Rate of Chemical reaction

4.8 Chemical Equilibirum

Introduction

A chemical reaction involves a reorganization of the atoms in one or more substances. In an ordinary chemical reactions atoms are neither created nor destroyed. In other words, there must be the same number of each type of atom in the product and the reactant side.

A chemical reaction is the transformation of reactants (raw material) into products (new substances formed)

Reactants ⟶ products

(on the left side of the arrow) (on the right side of the arrow)

The arrow means "react to yield."

- All chemical reaction are chemical changes. Changes around us are classified into two. These are:

1. Chemical change
2. Physical change

1. Chemical Change

A change that produce new substances with totally different properties is called **chemical change**

Example:

- rusting of iron
- Fermentation
- Respiration
- Photosynthesis

Characteristics of chemical change

- It is the chemical property of the substance involved.
- New substances with new properties are formed.
- It is always accompanied by energy changes
- This change is generally difficult to reverse
- This change alters the composition of the substance.

2. Physical change

➦ It is a change in which no new substance is formed.

Example

➦ Melting of ice

➦ Freezing of water

➦ Tearing of paper

➦ Dissolving of salt in water etc

Characteristics of physical change

➦ It is a change in physical property

➦ No new substance is formed

➦ It is easily reversed by physical means

➦ Energy changes are not necessary

➦ The composition of substance is not altered

Illustrative Examples

1. Which of the following changes is an example of a chemical change?

A. Decomposition of copper carbonate
B. Evaporation of water
C. Melting of fat
D. dissolving of salt in water

Explanation: Decomposition of compounds is a chemical reaction. As a result, it is chemical change.

Answer: A

2. Which of the following is change of state?

A. Evaporation
B. Breaking of glass
C. Tearing of paper
D. dissolving sugar in water

Explanation: Change of state means conversion of one state of matter in to another state

Solid ⟶ liquid, melting

Solid ⟶ gas, sublimation

liquid ⟶ gas, evaporation

liquid ⟶ solid, freezing

gas ⟶ solid, deposition

gas ⟶ liquid, condensation

Answer: A

3. Which of the following changes is not the characteristics of physical change?

A. Chemical change
B. Change of state
C. mechanical change
D. Dissolution

Answer: A

4. Which of the following is not the characteristics of chemical change?

A. New substance is formed
B. Alters composition of substances
C. Energy is absorbed or released
D. Easily reversed

Explanation: A chemical change is difficult to reverse.

Answer: D

5. Define chemical reaction

Answer: A chemical reaction is the transformation of reactants into products

6. What is the difference between reactants and products?

Answer: Reactants are raw materials while products are new substances formed.

7. List the three types f physical changes.

Answer:

➦ Change of state
➦ Mechanical change
➦ Dissolution

8. Classify the following as physical or chemical changes.

a. Burning of wood
b. Sublimation of iodine
c. Condensation of water vapor
d. Heating of sugar
e. Fermenting tella
f. Souring of milk
g. Decaying of food
h. Formation of cloud
i. Melting of sulphur

Answer:

- a,d,e,f and g are examples of chemical change
- b,c,h and i are examples of physical change.

4.1 Laws of chemical reactions

There are three fundamental laws of chemical reactions. These are:-

1. Law of conservation of mass
2. Law of definite proportion
3. Law of multiple proportion

1. Law of conservation of mass

- It is also called law of indestructibility of matter.
- This law states that "during ordinary chemical reactions atoms neither created nor destroyed"
- According to this law in a balanced chemical reactions atoms, mass and charge remain same on both sides of the reaction.

Example

$C + O_2 \longrightarrow CO_2$

Mass of C = 12g mass of CO_2 = 44g

Mass of O_2 = 32g

Mass of reactants = mass of CO_2 = 44g

(12 + 32)g = 44g

44g = 44g

number of atoms on the reactant side = number of in the product atoms side

1 c-atom + 2 O – atoms = 1 C – atom + 2 O – atom

3 = 3

2. Law of definite proportion

It is also called law of constant composition.

- It states that 'a pure compound contains element in a fixed ratio by mass or atoms."

Example:

- In water (H_2O)

  The mass ratio of H to O is 1:8

  The atom ratio of H to O is 2:1

- In hydrogen peroxide (H_2O_2):

  The mass ratio of H to O is 1:16

  The atom ratio of H to O is 1:1

3. The Law of multiple proportion

- This law states that "When two elements form more than one compound, the masses of one element that combines with a fixed mass of the other element is a small whole number ratio."

- The compounds below illustrate the law of multiple proportion.

  - H_2O and H_2O_2
  - SnO and SnO_2
  - CO and CO_2
  - PbO and PbO_2
  - FeO and Fe_2O_3
  - SO_2 and SO_3
  - N_2O, NO NO_2, N_2O_3 and N_2O_5

Consider CO and CO_2, by fixing the mass of carbon it is possible to express the mass of oxygen in small number ratio

Formula	C : O	C : O	Ration of oxygen dividing by the smallest
CO	12 : 16	$\frac{12}{12}:\frac{16}{12}$	$1:\frac{\frac{16}{12}}{\frac{12}{16}}=1$
CO_2	12 : 32	$\frac{12}{12}:\frac{32}{12}$	$1:\frac{\frac{32}{12}}{\frac{12}{16}}=2$

Illustrative Examples

9. A 36g sample of pipe water and 36g sample of rain water were purified and subjected to electric current separately. Both samples of water released 4g of hydrogen and 32g of oxygen after the electrons was accomplished. This finding confirms the law of . . . (EGSECE)

A. Multiple proportion
C. mass action
B. Conservation of mass
D. constant composition

Explanation: The process that decomposes a sample of water into hydrogen and oxygen is electrolysis $2H_2O \rightarrow 2H_2 + O_2$

$2 \times 18g \quad 4g \quad 32g$

➦ Mass of reactant is equal to mass of product this law is the law of conservation of mass.

Answer: B

10. "A compound always contains the same elements combined together in the same proportion by mass." This is according to the law of . . . (EGSECE….)

A. Conservation of energy
C. Multiple proportion
B. Definite composition
D. Conservation of mass

Answer: B

11. It is possible that different masses of oxygen with ratio 1:2 combine with constant mass of tin to form SnO and SnO_2 respectively. Which law governs these two possible chemical combinations between atoms of tin and oxygen? (EGSECE…)

A. The law of constant composition
C. The law of mass action
B. The law of conservation of mass.
D. The law of multiple proportion

Answer: D

12. A sample of water taken from ice, rain or sea has the same ratio by mass of hydrogen and oxygen. This chemical fact explains the law of:

A. Conservation of mass
C. Definite proportion
B. Mass action
D. multiple proportion

Answer: C

13. How the five oxides of nitrogen illustrate the law of multiple proportions

Explanation: (Atomic mass of N = 14, O = 16)

Compound	Molecular formula	Mass ration of N:O	Ratio of oxygen
Dinitrogen monoxide	N_2O	$2:16 \Rightarrow \frac{28}{28}:\frac{16}{28} \Rightarrow 1:\frac{4}{7}$	1
Nitrogen monoxide	NO	$14:16 \Rightarrow \frac{28}{28}:\frac{48}{28} \Rightarrow 1:\frac{8}{7}$	2
Dinitrogen trioxide	N_2O_3	$28:48 \Rightarrow \frac{28}{28}:\frac{48}{28} \Rightarrow 1:\frac{16}{7}$	3
Nitrogen dioxide	N_2O	$14:32 \Rightarrow \frac{14}{14}:\frac{32}{14} \Rightarrow 1:\frac{16}{7}$	4
Dinitrogen pentoxide	N_2O_5	$28:80 \Rightarrow \frac{28}{28}:\frac{80}{28} \Rightarrow 1:\frac{20}{7}$	5

The mass of nitrogen is fixed, when you divide the mass of oxygen to the smaller you will get the following.

$\frac{4}{7} \div \frac{4}{7} = 1$ $\quad$ $\frac{8}{7} \div \frac{4}{7} = 2$ $\quad$ $\frac{16}{7} \div \frac{4}{7} = 4$

$\frac{12}{7} \div \frac{4}{7} = 3$ $\quad$ $\frac{20}{7} \div \frac{4}{7} = 5$

Thus, the ratio of oxygen in the five oxides is 1:2:3:4:5

14. A 20g of sample of A combines with a 35g of B to form a compound AB_2. What are the masses of reactants and products? What can you conclude from the result?

Explanation: According to the law of conservation of mass, mass of reactants is equal to mass of products in an ordinary chemical reaction. There force, the masses of reactants, A and B = 20 + 35 = 55g; the mass of product, AB_2 is also 55g.

15. When mercury (II) oxide is heated, it demposes into mercury and oxygen. If 25g of mercury oxide decomposes, 4g of oxygen is produced. What is the percentage of mercury in mercury (II) oxide?

Exp'anation: mass of mercury (II) oxide = 25g, mass of O_2

= 4g. $2HgO \rightarrow 2Hg + O_2$

mass of HgO = mass Hg + mass O_2

mass of Hg = mass of HgO – mass O_2

= 25g – 4g

= 21g

$$\text{Percentage of Hg} = \frac{mass\,of\,Hg}{mass\,of\,HgO} \times 100$$

$$= \frac{21g}{25g} \times 100$$

$$= 84\%$$

4.2 Chemical equation

- It is a short hand representation of a chemical reaction by means of symbols and chemical formula.
- It is written to show the chemical changes that occur during a chemical reactions.
- **A chemical reaction** involves a reorganization of the atoms in one or more substances.
- It is the transformation of **reactants** in to new substances called **products**

Example:

$2H_2(g) + O_2(g) \longrightarrow 2H_2O\,(l)$

- Some time it is important to write the physical states of reactants in bracket by using small letters, **s - for solid, ℓ – for liquid and g – for gas.**
- The arrow between reactants means "**react to yield.**"
- The quantitative meaning of chemical equation means the amount of products and reactants.

Example:

consider combustion of methane:

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$

- 1mole CH_4 + 2 moles $O_2 \longrightarrow$ 1mole $CO_2(g)$ + 2moles $H_2O(\ell)$
- 16g CH_4 + 64g of $O_2 \longrightarrow 44gCO_2$ + 36g H_2O

$6.02\times10^{23}\times$ molecules of $CH_4 \rightarrow 6.02\times10^{23}$molecules of CO_2

\+ +

$2\times6.02\times10^{23}$ molecules O_2 $2\times6.02\times10^{23}$molecules of H_2O

- The qualitative meaning of a chemical equation means the identify (name) of reactants and products.

Balancing chemical equation

- In a balanced chemical equations, atoms and masses are conserved but moles or molecules are not necessary conserved.

Some methods of balancing chemical equations are

1. The inspection /counting/ method (LCM)
2. The least common multiple method
3. Algebraic method.
4. The oxidation number method.
5. The ion – electron method.

- The last two will be studied in grade 12 for natural science students and they are used to balance oxidation reduction reaction

1. The inspection method

- Simple chemical reactions can be balanced by using the inspection or trial and error method.
- Place the smallest whole number coefficients for the unbalanced atoms.

Example

$H_2 + O_2 \longrightarrow H_2O$

Hydrogen is balanced but not oxygen. Place 2 in front of H_2O.

➦ $H_2 + O_2 \longrightarrow 2H_2O$, now oxygen is balanced but not hydrogen place 2 in front of H_2 to balance it.

➦ $2H_2 + O_2 \longrightarrow 2H_2O$

Lastly check wether the number of atoms and masses are conserved or not.

Reactants	Product
$H = 2 \times 2 = 4$	$H = 2 \times 2 = 4$
$O = 1 \times 2 = 2$	$O = 2 \times 1 = 2$
$m = 4 + 32 = 36g$	$mass = 2 \times 18g = 36g$

2. The Least common multiple method

This method is also used to balance simple chemical reactions.

- ➦ To balance by LCM method, place total valence number of each atom above it
- ➦ Find the LCM of each total valence number.
- ➦ Divide the LCM by each total valence number to obtain the coefficients of reactants and products.

Example

$Fe + O_2 \longrightarrow Fe_2O_3$, here the valence number of iron is 3 while that of oxygen is 2. Total valence number means subscript times valence number

➦ $\overset{3}{Fe} + \overset{4}{O_2} \rightarrow \overset{6}{Fe_2}\overset{6}{O_3}$

The LCM of 3,4 and 6 is 12

➦ $12 \div 3 = 4, \quad 12 \div 4 = 3, \quad 12 \div 6 = 2$

$$\overset{3}{Fe} + \overset{4}{O_2} \xrightarrow[LCM]{12} \overset{6}{Fe_2}\overset{6}{O_3}$$

➦ $4Fe + 3O_2 \longrightarrow 2Fe_2O_3$

Check: There are four atoms of iron and 6 atoms of oxygen on both sides of the reaction.

3. Algebraic method

- ➦ This method needs the knowledge of mathematics. We place small letters a,b,c,- - - as coefficients and then find their values by assuming the value of the first letter to be one.
- ➦ It is used to balance even complicated chemical reactions.

Example

$N_2 + bH_2 \rightarrow cNH_3$

N: $2a = c$ assume $a = 1$

H : $2b = 3C$

➦ $2a = c$ Thus, $a = 1$

$2 \times 1 = c$ $b = 3$

$C = 2$ $c = 2$

➦ $2b = 3c$

$2b = 3 \times 2$

$2b = 6$

$b = 3$

substitute these numbers interms of the letters

➦ $N_2 + 3H_2 \longrightarrow 2NH_3$

Atoms and masses are conserved.

Illustrative Examples

16. A balanced chemical equation gives the following informations, EXCEPT (EGSECE. . . 2004)

A. The nature of substances involved in the reaction

B. The amount of substances involved in the reaction

C. The physical states of substances involved in the reaction.

D. The rate of chemical reaction

Explanation: From a balanced chemical reaction we can understand the nature, physical state and amount of substances but not the speed at which reactants disappear and products appear.

Answer: D

17. In every balanced chemical equation, each side of the equation has the same numbers of (EGSECE)

A. Coefficients

B. Moles

C. molecules

D. atoms

Answer: D

18. When the reaction below is correctly balanced, what would be the coefficient of HNO_3? (EGSECE 1999 E.C)

$Cu + HNO_3 \rightarrow Cu(NO_3)_2 + NO + H_sO$

A. 6

B. 10

C. 8

D. 4

Explanation: Let us balance by algebraic method.

$aCu + bHNO_3 \longrightarrow c\,Cu(NO_3)_2 + dNO + eH_2O$

i. Cu: $a = c$

ii. H: $b = 2e$

Assume $a = 1$

$a = c = 1$

relate ii and iii

$2e = 2c + d$

iii. N: $b = 2c + d$

iv. O: $3b = 6c + d + e$

multiply the whole by 3

$a = 1$

$b = \frac{8}{3}$

$c = 1$

$2e = 2(1) + d$ $\quad d = \frac{2}{3}$

$d = 2e - 2 \ldots v$ $\quad e = \frac{4}{3}$

relate iv and v

$3b = 6c + d + e$ $\quad a = 3 \times 1 = 3$

$3b = 6(1) + 2e - 2 + e$ $\quad b = \frac{8}{3} \times 3 = 8$

but, $b = 2e$ $\quad c = 3 \times 1 = 3$

$3(2e) = 6 + 3e - 2$ $\quad d = \frac{2}{3} \times 3 = 2$

$6e - 3e = 4$ $\quad e = \frac{4}{3} \times 3 = 4$

$$e = \frac{4}{3}$$

$$b = 2e = 2 \times \frac{4}{3} = \frac{8}{3}$$

$$d = 2e - 2 = \frac{2}{3}$$

$3Cu + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$

Answer: C

19. Given $Al_2(SO_4)_3 + ZnCl_2 \longrightarrow AlCl_3 + ZnSO_4$ what is the sum of the coefficients of reactants and products?

A. 9 C. 4

B. 6 D. 8

Explanation: This is simple chemical equation, we use LCM method to balance it

$$\overset{6}{Al}_2(\overset{6}{SO_4})_3 + \overset{2}{Zn}\overset{2}{Cl_2} \xrightarrow{6} \overset{3}{Al}\overset{3}{Cl_2} + \overset{2}{Zn}SO_4$$

$$Al_2(SO_4)_3 + 3ZnCl_2 \rightarrow 2AlCl_3 + 3ZnSO_4$$

Thus, sum of coefficients: 1 + 3 + 2 + 3 = 9

Answer: A

20. Balance the following chemical equation by using any method you like

a. $C_{10}H_{22} + O_2 \longrightarrow CO_2 + H_2O$

b. $Fe + Cl_2 \longrightarrow FeCl_3$

c. $HNO_3 + SO_2 + H_2O \longrightarrow H_2SO_4 + NO$

d. $HCl + K_2Cr_2O_7 \longrightarrow KCl + CrCl_3 + H_2O + Cl_2$

e. $CuO + NH_3 \longrightarrow Cu + N_2 + H_2O$

Answer:

a. $2C_{10}H_{22} + 31O_2 \longrightarrow 20CO_2 + 22H_2O$

b. $2Fe + 3Cl_2 \longrightarrow 2FeCl_3$

c. $2HNO_3 + 3SO_2 + 2H_2O \longrightarrow 3H_2SO_4 + 2NO$

d. $14HCl + K_2Cr_2O_7 \rightarrow 2KCl + 2CrCl_3 + 3Cl_2 + 7H_2O$

e. $3CuO + 2NH_3 \longrightarrow N_2 + 3Cu + 3H_2O$

21. Write balanced chemical equation for the reaction between nitrogen pentoxide and water to produce nitric acid.

Answer

➦ The word equation is:

Nitrogen pentoxide + water ⟶ nitric acid

➦ The formula equation becomes.

$N_2O_5 + H_2O \longrightarrow HNO_3$, balancing by inspection method gives

$N_2O_5 + H_2O \longrightarrow 2HNO_3$

22. Change the following chemical equations into word equations

a) $4NO_2 + O_2 \longrightarrow 2N_2O_5$

b) $2NaHCO_3 \longrightarrow Na_2CO_3 + CO_2 + H_2O$

Answer:

a. Nitrogen dioxide + oxygen→ Dinitrogen pentoxide

b. Sodium hydrogen carbonate ⟶ sodium carbonate carbondioxide+ water

23. What is the difference between chemical reaction and chemical equation?

Answer

A chemical reaction is a process of converting reactants into products where as a chemical equation is a short hand expression of a chemical reaction.

24. We change only coefficients to balance a chemical equation but not subscripts. Why?

Explanation: Changing coefficients doesnot change the identify of substances but when we change subscripts the identify of substance change even we may produce unknown formula so far.

4.3 Energy changes in chemical reactions

- A chemical reaction takes place either by absorbing or releasing energy. This absorbed or released energy is called **heat of reaction or enthalpy of reaction.**
- **Heat change of reaction (ΔH)** is the difference between heat content of product (**Hp**) and heat content of reactant (**Hr**)
  $\Delta Hrxn = Hp - Hr$

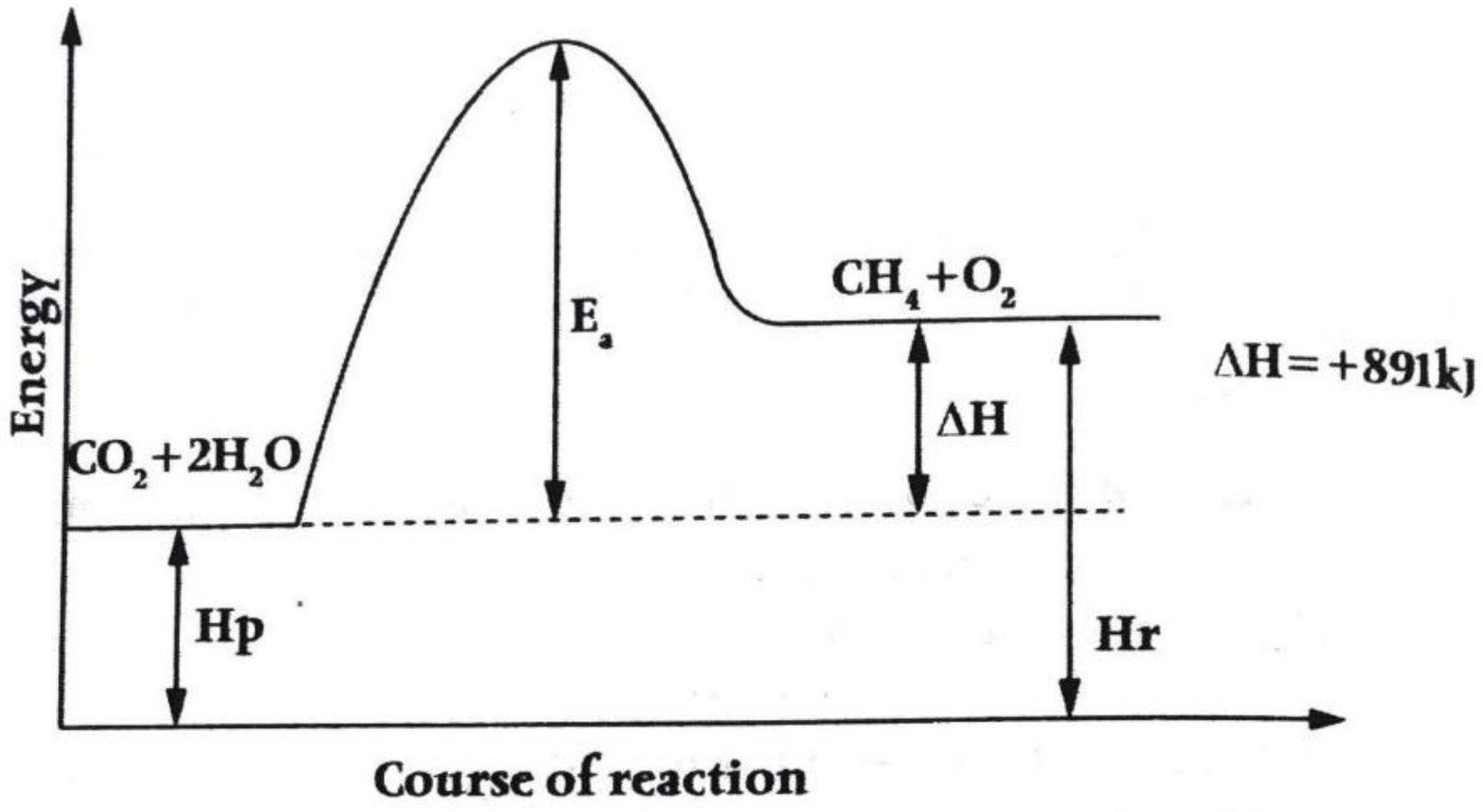
Endothermic reaction is a reaction that absorbs energy

- It has positive enthalpy of reaction
- Heat content of products is greater than that of reactants
- Heat is added to the reactant side.

Example

- $CO_2 + 2H_2O + 891kJ \longrightarrow CH_4 + 2O_2$
- Photosynthesis and many decomposition reactions are endothermic reactions.

The energy diagram for the above example



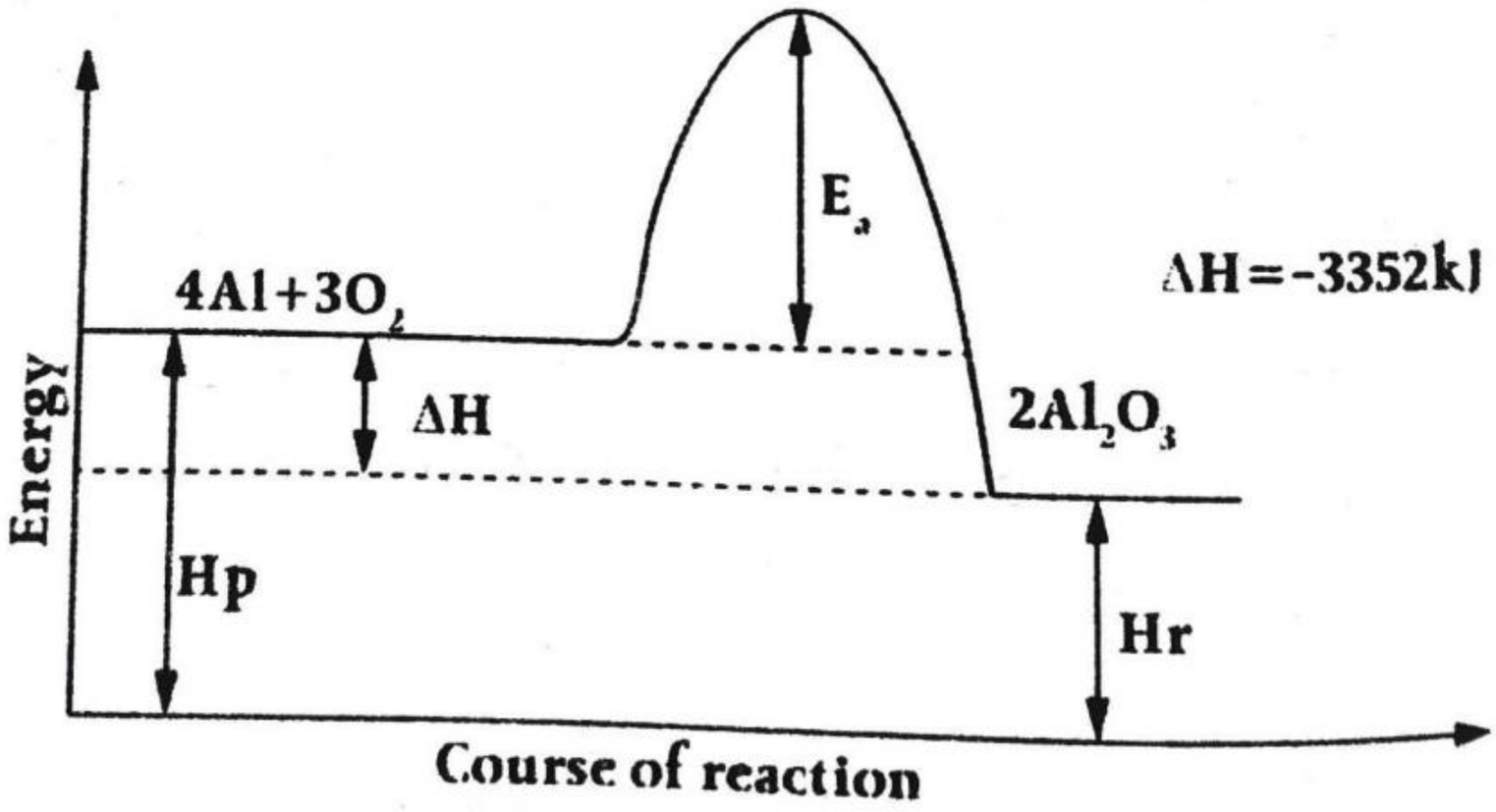
Exothermic reaction is a reaction that releases heat to the surrounding

- It has negative heat of reaction
- Heat content of reactants is greater than heat content of products
- Heat is added to the product side

Example

$4Al + 3O_2 \longrightarrow 2Al_2O_3 + 3352kJ$

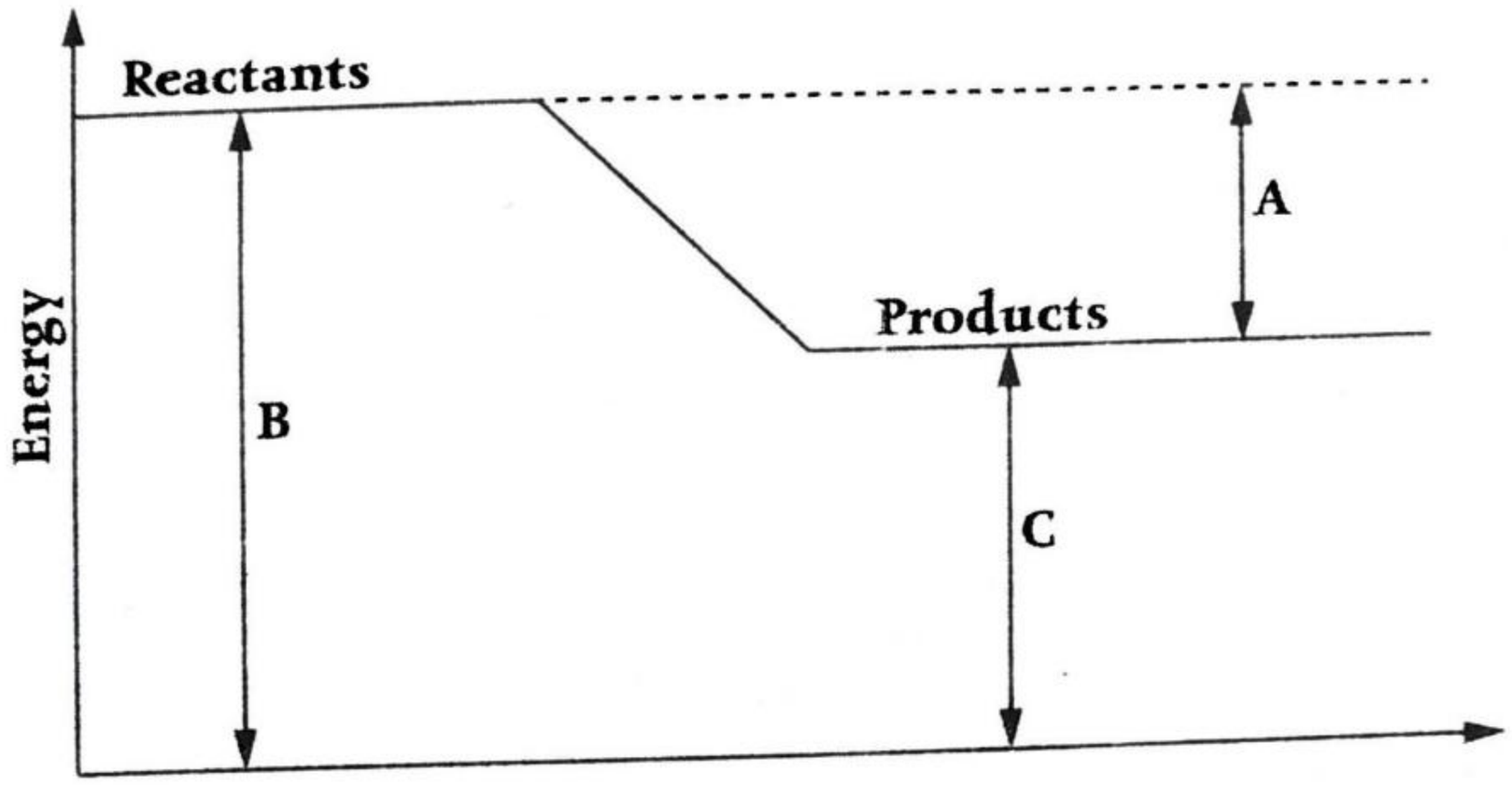
Respiration and combustion reactions are exothermic reactions



Activation energy (Ea) is the minimum amount of energy required for a reaction to occur.

Illustrative Examples

25. The following figure shows an energy diagram for a certain chemical reaction



Which of the following is true about the reaction shown in the diagram? (EGSECE 2004 E.C)

A. 'A' indicates enthalpy of reactants
B. 'B' indicates heat change of reaction
C. 'C' indicates heat content of product
D. The reaction is endothermic

Solution:

A – heat change of reaction
B – heat content of reactant
C – heat content of product
D – The reaction is exothermic

Answer: C

26. Assume that the energy of the products in a chemical reaction is 50kCal. This reaction could be exothermic if the energy of the reactants was

A. 50 kcal
B. 20 kcal
C. 60 kcal
D. 10 kcal

Explanation: For exothermic reaction $\Delta H = Hp - Hr < O$

➦ $Hp - Hr < O$

$50Kcal - Hr < O$

50KCal < Hr

Hr > 50 KCal

➦ The answer must be above 50Kcal.

Answer: C

27. Classify the following as endothermic or exothermic reaction.

a. $N_2(g) + 2O_2(g) \longrightarrow 2NO_2(g)$, $\Delta H = 66.4KJ$

b. $C(s) + O_2(g) \longrightarrow CO_2(g)$, $\Delta H = -394KJ$

c. $CaCO_3 \xrightarrow{heat} CaO + CO_2$

d. $Ca + H_2SO_4 \longrightarrow CaSO_4 + H_2 + heat$

Solution

➦ a and c are endothermic reactions

➦ b and d are exothermic reactions.

28. Write the difference between:

a. Endothermic reaction and endothermic process

b. Exothermic reaction and exothermic process

Solution

a. Endothermic reaction is a reaction that takes place by absorbing energy but endothermic process is a reaction or a physical change that gains energy.

➦ Melting, evaporation and sublimation are physical changes that absorb energy.

b. Exothermic reaction is a chemical change that absorbs energy but exothermic process is a chemical or physical change that absorb heat.

➦ Freezing, condensation and deposition are physical changes that release energy.

29. How would you identify weather a reaction is endothermic or exothermic in a laboratory?

Answer

By ùsing our skin, we can identify wether a reaction is endothermic or exothermic. If we feel hot it is exothermic but it we feel cold it is endothermic

30. Which reaction is used to produce energy?

Answer:

➦ It is exothermic reaction that generates energy

4.4 Types of chemical reaction

➦ There four types of reactions. These are

1. Combination reaction
2. Decomposition reaction
3. Single displacement reaction
4. Double displacement reaction

1. Combination reaction

➦ It is also called addition reaction or **synthesis**.

➦ Two or more reactants produce only one product

- $X + Y \longrightarrow XY$

Example

➦ $BaO + H_2O \longrightarrow Ba(OH)_2$

➦ $2H_2 + O_2 \longrightarrow 2H_2O$

2. Decomposition reaction

➦ It is also known as **analysis.**

➦ One reactant is converted in to two or more products

➦ $AB \longrightarrow A + B$

Example

➦ $2KClO_3 \longrightarrow 2KCl + 3O_2$

➦ $2NaHCO_3 \longrightarrow Na_2CO_3 + CO_2 + H_2O$

3. Single displacement reaction

- It is also known as substitution reaction
- A more reactive element displaces less reactive element
- $A + BC \longrightarrow AC + B$

A is more reactive than B and it can be metal or non – metal.

Example

- $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$
  - Zn is more reactive than H_2
- $Cu + ZnSO_4 \longrightarrow$ no reaction
  - This is because Cu is less reactive than zinc
- $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$

Note:

reactivity of halogens.

$F_2 > Cl_2 > Br_2 > I_2$

F_2– displaces all, Cl_2 only Br_2 and I_2, Br_2 displaces I_2 only.

4. Double displacement reaction

- It is also termed as **methatetical reaction**.
- It is a reaction between two compounds to produce another two compounds
- $AB + CD \longrightarrow AD + CB$

Example

- $AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$
- $NaOH + HBr \longrightarrow NaBr + H_2O$

All acid – base reactions are double displacement reactions but all double displacement reactions are not necessary acid – base reaction.

Illustrative Examples

31. Which of the following is an example of combination reaction (EGSECE 2004 E.C)

A. $BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2NaBr$

B. $Na_2O + H_2O \longrightarrow 2NaOH$

C. $Zn + 2HCl \longrightarrow ZnCl_2 + H_2$

D. $2NaHCO_3 \longrightarrow Na_2CO_3 + CO_2 + H_2O$

Answer: B

32. The types of reaction in which two or more products are formed form a single reactant is

A. Combination

B. Decomposition

C. Double displacement

D. single displacement

Answer: B

33. The reaction between chlorine gas and potassium iodide can be classified as:

A. Single displacement reaction

B. Double displacement reaction

C. Decomposition reaction

D. Combination reaction

Explanation: $Cl_2 + 2KI \rightarrow 2KCl + I_2$

Answer: A

34. Classify the following reactions as combination decomposition, single displacement or double displacement reaction

a. $4HNO_3 \longrightarrow 4NO_2 + 2H_2O + O_2$

b. $2Fe + 3Cl_2 \longrightarrow 2FeCl_3$

c. $Pb(NO_3)_2 + Na_2SO_4 \longrightarrow PbSO4 + 2NaNO_3$

d. $Br_2 + 2LiI \longrightarrow 2LiBr + I_2$

Answer

a. Decomposition reaction

b. Combination reaction

c. Double displacement reaction

d. Single displacement reaction

Importance of chemical change or reaction

Chemical reactions have the following importance.

1. Production of new substances
2. Production of energy. Exothermic reactions generate energy. For example, respiration creates energy for our bodies.

4.5 Stoichiometry

It is the study of the quantitative relationship between reactant and reactant, reactant and product or product and product in a balanced chemical equation. For example: The mass, relationships among reactants and products in a chemical reaction represent the stoichiometry of the reaction.

$$Number\,of\,mole(n) = \frac{given\,mass(m)}{molar\,mass(M)}$$

$$n = \frac{m}{M} \qquad m = n \cdot M$$

A mole is Avogadros number of particles

- 1 mole = 6.02×10^{23} particles
- Particles means atoms molecules, ions, formula units

Molar mass (M) is the mass of one mole of a substance.

Example

i Calculate the number of moles of 196g of H_2SO_4.(Atomic mass: S = 32, O = 16, H = 1)

- M = 2×1 + 1×32 + 4×16 = 98g/mol

m = 196g

$$n = \frac{m}{M} = \frac{196g}{98g/mol} = 2mol$$

ii Find the mass of 2.5moles of $CaCO_3$.
(atomic mass: Ca = 40)

➦ $M = 1\times40 + 1\times12 + 3\times18 = 100g/mol$
$m=n\times M = 2.5mol \times 100g/mol = 250g$

iii What is the mass of one atom of magnesium?
(Atomic mass Mg = 24)

➦ 1mole of Mg=Atomic mass of Mg $=6.02\times0^{23}$ atoms

$$m_{Mg} = \frac{M_{Mg} \times given number of atoms}{6.02\times10^{23} atoms}$$

$$= \frac{24g\times1atoms}{6.02\times10^{23} atom}$$

$$= 3.988 \times 10^{-23} g$$

iv Calculate the number of moles of 1.204×10^{23} molecules of water.

➦ $18g\ H_2O = 1mol\ H_2O = 6.02 \times 10^{23}$ molecules

$? = 1.204 \times 10^{23}$ molecules

➦ $$nH_2O = \frac{1mol\times1.204\times10^{23}}{6.02\times10^{23}} = 0.2mol$$

Types of stoichiometric relationships

1. mass – mass relationships

➦ Write balanced chemical equation and write the given mass above the formula and x above the formula of the required

➦ Write below the formula coefficient time molar mass.

➦ Set a proportion and solve for x.

Example

How many grams of carbondioxide is formed when 24gof oxygen reacts with excess carbon? (At. Mass. C = 12, O = 16)

➡ $\overset{24g}{C} + \underset{}{O_2} \rightarrow \overset{x}{CO_2}$

$1\times32g \quad 1\times44g$

$$\frac{24g}{32g} = \frac{x}{44g}$$

Solving for x, x = 33g

Thus, mass of CO_2 formed is 33g

2. Mole – mass relationship

- ➡ Either mole or mass is required. You follow the same method to find the answer.
- ➡ Write below the formula coefficient and given number of mole above the formula.

Example

Determine the number of moles of hydrogen that reacts with 7g of nitrogen to form ammonia.

➡ $\overset{7g}{N_2} + 3\overset{x}{H_2} \rightarrow 2NH_3$

$28 \quad 3mol$

➡ $$\frac{7g}{28g} = \frac{x}{3mol}$$

Thus the number of mole of H_2 is 0.75mol.

3. Mole – mole relationship

- ➡ Write balanced chemical equation and the given number of moles above the formula and x on the required. Write coefficients as number of mole below the formula.

Example

How many moles of hydrochloric acid reacts with excess aluminum to form 6 moles of hydrogen.

➡ $2Al + 6\underset{6}{\overset{x}{HCl}} \longrightarrow 2AlCl_3 + 3\underset{3}{\overset{6mol}{H_2}}$

- $\frac{x}{6} = \frac{6mol}{3}$

  $x = 12moles$

- Thus the number of moles of HCl required is 12 moles.

4. Mass – volume relationship

- **Molar volume** is the volume of one mole of any gas at standard temperature and pressure (STP)
- At STP, temperature = 0^0c, pressure = 1 atm, number of mole = 1 mole and volume is 22.4L.

  If the required is volume, write below the formula coefficient times 22.4L Below the mass given, write coefficient times molar mass,

Example

How many liters of carbondioxide gas is formed when 50g of calcium carbonate decomposed at STP?

$$\overset{50g}{CaCO_3}(s) \rightarrow CaO(s) + \overset{x}{CO_2}(g)$$

- $1 \times 100g$ $\qquad$ $1 \times 22.4L$
- $\frac{50g}{100g} = \frac{x}{22.4L}$

  $x = 11.2L$

Thus, the volume of CO_2 formed is 11.2L

5. Mole – volume relationship

To find number of mole, write coefficient below the formula and that of the given coefficient times 22.4L

Example

How many liters of oxygen gas reacts with 3 moles of aluminum to form aluminum oxide at STP?

$$\overset{3mol}{4Al(s)} + \overset{x}{3O_2(g)} \rightarrow 2Al_2(g) \rightarrow 2Al_2O_3(s)$$

$4mol \quad 3\times22.42$

$$\frac{3mol}{4mol} = \frac{x}{3\times22.4L}$$

$x = 50.4L$

Thus, the volume of oxygen is 50.4L

6. Volume – volume relationship

- Write coefficients below the formula of given and required and solve for X

Example

How many liters of carbondioxide gas will be formed when 44.8 liters of acetylene is burnt in excess oxygen?

$$\overset{44.8L}{2C_2H_2(g)} + SO_2(g) \rightarrow 4\overset{x}{CO_2} + 2H_2O(l)$$

$2 \qquad\qquad 4$

$$\frac{44.8L}{2} = \frac{x}{4}$$

$x = 89.6L$

Therefore, the volume of CO_2 formed is 89.6L.

Illustrative Examples

35. Given the reaction: $4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(l)$. How many grams of Fe_2O_3 are produced when 32g of iron reacts with sufficient amount of oxygen?

(At mass, Fe = 56, 0 = 16) (EGSECE. . . 2005E.C)

A. 72g

B. 45.7g

C. 91.4g

D. 106.7g

Solution:

$$\underset{4\times56}{\overset{32}{4Fe}} + 3O_2 \rightarrow \underset{2\times160}{\overset{x}{2Fe_2O_3}}$$

$$\frac{32g}{4\times56} = \frac{x}{2\times160}$$

$x = 45.7g$

Thus, mass of iron reacted was 45.7g.

Answer: B

36. Given the reaction:

$2KNO_3 \longrightarrow 2KNO_2 + O_2$, how many moles of potassium nitrate must be decomposed to produce 9 moles of oxygen gas? (EGSECE)

A. 18moles
C. 9 moles
B. 3moles
D. 2 moles

Solution

$$\frac{x}{2} = \frac{9\ \text{mol}}{1}$$

$x = 18$ moles

Answer: A

37. How many liters of carbonmonoxide at STP are required to react with 11.2L of hydrogen gas according to the equation

$CO(g) + 2H_2(g) \longrightarrow CH_3OH(1)$?

A. 44.8L
C. 11.2L
B. 22.4L
D. 5.6L

Solution

$$\underset{1}{\overset{x}{CO(g)}} \rightarrow \underset{2}{\overset{11.2L}{2H_2(g)}} \rightarrow CH_3OH(l)$$

$$\frac{x}{1} = \frac{11.2L}{2}, x = 5.6L$$

Answer: D

38. The equation for the reaction of sucrose with oxygen is given as follow:

$$C_{12}H_{22}O_{11} + 12O_2 \longrightarrow 12CO_2 + 11H_2O$$

How many liters of carbondioxide gas is produced at STP per gram of sucrose? (EGSECE. . . .1997 E.C) (At. mass C = 12, O = 16)

A. 0.78L B. 1.12L C. 0.48L D. 2.48L

Solution:

Per gram means per one gram

$$\overset{1g}{C_{12}H_{22}O_{11}} + 12O_2 \rightarrow 12\overset{x}{CO_2} + 11H_2O$$
$$\underset{1\times342g}{\quad}\qquad\qquad\qquad\underset{12\times22.4L}{\quad}$$

$$\frac{1g}{342g} = \frac{x}{12\times22.4L}$$

x = 0.786L

Therefore, the volume of carbondioxide is 0.786L.

Answer: A

39. Given the reaction

$CaC_2(s) + 2H_2O(s) \longrightarrow C_2H_2(g) + Ca(OH)_2$ (aq),what mass of calcium carbide is required to produce 5600mL of acetylene at STP? (At. Mass, Ca = 40 C = 12)

Solution

5600mL = 5.6L

$$\overset{x}{CaC_2}(s) + 2H_2O(1) \rightarrow \overset{5.6L}{C_2H_2}(g) + Ca(OH)_2(aq)$$
$$\underset{1\times642}{\quad}\qquad\qquad\qquad\underset{1\times22.4L}{\quad}$$

$$\frac{x}{64g} = \frac{5.6L}{22.4L}$$

x = 16g

Thus, 16g of calcium carbide is required to produce 5.6L of acetylene.

40. Aspirin (which many students take working on chemistry problems) is prepared by the reaction of salicylic acid ($C_7H_6O_3$) with acetic acid anhydride ($C_4H_6O_3$) according to the reaction $C_7H_6O_3 + C_4H_6O_3 \longrightarrow C_9H_8O_4 + C_2H_4O_2$

How many grams of salicylic acid must be used to prepare two aspirin tables, each containing 325mg of aspirin?

Solution

mass of one tablet = 325mg

mass of two tablets = 650mg

1mg = 0.01g

650mg = ?

➦ 650mg = 0.65g

Thus, mass of asprin produced = 0.65g

➦ $\overset{x}{C_7H_6O_3} + C_4H_6O_3 \rightarrow \overset{0.65g}{C_9H_8O_4} + C_2H_4O_2$

$\underset{1\times138}{\quad}\qquad\qquad\qquad\qquad\underset{1\times164}{\quad}$

$$\frac{x}{138} = \frac{0.65g}{164}$$

$x = 0.55g$

➦ Thus, the mass of salicylic acid required to produced two tablets of aspirin is 0.55g

41. Calculate the number of chloride ions in 2moles of $CaCl_2$ solution.

Solution

$CaCl_2 \longrightarrow Ca^{2+} + 2Cl^-$

1mol 2mol

2mol ?

➦ $n_{Cl^-} = 4$moles

1mole $Cl^- = 6.02 \times 10^{23}$

4moles $Cl^- = ?$

➦ number of $Cl^- = 4 \times 6.02 \times 10^{23}$

$= 2.408 \times 10^{24}$

42. Find the number of atoms in 22g of carbondioxide.

Solution

$$n_{CO_2} = \frac{m}{M} = \frac{22g}{44g/mol} = 0.5mol$$

number of atoms is $CO_2 = 0.5 \times (1+2) \times 6.02 \times 10^{23}$

$= 9.03 \times 10^{23}$ atoms.

43. The food we eat is degraded or broken down in our bodies to provide energy for growth and function. A general overall equation for this complex process represents the degradation of glucose to carbondioxide and water.

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$$

If 856g of $C_6H_{12}O_6$ are consumed by the body over a certain period, what is the mass of CO_2 produced?

Solution:

$$\overset{856g}{C_6H_{12}O_6} + 6O_2 \rightarrow 6\overset{x}{CO_2} + C_2H_4O_2$$

$$\underset{1\times180}{\quad} \qquad \underset{6\times44}{\quad}$$

$$\frac{856g}{180} = \frac{x}{6\times44}$$

x = 1255.5g = 1.25kg

Thus, amount of CO_2 produced was 1.25kg

44. On what law is stoichiometry based?

Answer

On the law of conservation of mass

45. The fertilizer ammoniumsulfate ($(NH_4)_2SO_4$) is prepared by the reaction between ammonia (NH_3) and sulphuric acid.

$$2NH_3(g) + H_2SO_4(aq) \longrightarrow (NH_4)_2\,SO_4(aq)$$

How many kilograms of NH_3 are needed to produce 1.0×10^5kg of $(NH_4)_2SO_4$?

Solution:

$$2\overset{x}{\underset{2\times17g}{NH_3}} + H_2SO_4(aq) \rightarrow \overset{1.0\times10^5\,kg}{\underset{1\times132g}{(NH_4)_2}}SO_4(aq)$$

$$\frac{x}{2\times17g} = \frac{1.0\times10^5\,kg}{132g}$$

$x = 2.6 \times 10^4$kg

Therefore, 2.6×10^4kg of ammonia is required to produce 1.0×10^5kg of ammonium sulphate

46. What mass of CH_4 formed when 8 moles of carbonmonoxide react with hydrogen

Solution

$$2\overset{8mol}{\underset{2}{CO}} + 2H_2 \rightarrow \overset{x}{\underset{1\times16g}{CH_4}} + CO_2$$

$$\frac{8mol}{2} = \frac{x}{16g}$$

x = 64g

Thus, mass of methane formed is 64g.

47. What is the importance of balanced chemical equation in solving stoichiometric calculations?

Solution:

Balanced chemical equation is used to find the mole ratio of reactants and products.

Limiting and Excess reactants

When a chemist carries out a reaction, the reactants are not usually present in exact **stoichiometric amounts**, that is, in the proportions indicated by the balanced equation

- The reactant used up first in a reaction is called the **limiting reactant**. The amount of product formed depends of this reactant.
- The other reactants present in quantities greater than those needed to react with the quantity of limiting reactant are called **excess reactants.**

Note

To identify the limiting reactant first determine the total number of moles of reactants. The one with smaller total number of mole is limiting reactant.

$$\text{Total number of mole } (n_T) = \frac{\text{given mass}}{\text{coefficent} \times \text{molar mass}}$$

$$n_T = \frac{m}{\text{coef} \times M}$$

Example

Given the reaction:

$2Al + Fe_2O_3 \longrightarrow Al_2O_3 + 2Fe$. If a mixture of 108g Al and 240g of Fe_2O_3 were ignited.

a. Identify the limiting reactant
b. Calculate the maximum amount of iron produced
c. How many grams of excess reactant left at the end of the reaction?

Solution

a. $n_T(Al) = \frac{m}{coef \times M} = \frac{108g}{2 \times 27g/mol} = 2mol$

$n_T(Fe_2O_3) = \frac{m}{coef \times M} = \frac{240g}{1 \times 160g/mol} = 1.5mol$

Therefore, Fe_2O_3 is the limiting reagent

b. The amount of product depends on the amount of limiting reactant.

$$2Al + \overset{240g}{\underset{1\times160}{Fe_2O_3}} \rightarrow Al_2O_3 + \overset{x}{\underset{2\times56}{2Fe}}$$

$$\frac{240g}{160} = \frac{x}{2 \times 56}$$

x = 168g of iron is produced

c. Aluminum is the excess reageant. Find the mass of Al required to react with 240g of Fe_2O_3. Then subtract this value to get the amount left.

➦ $\underset{2\times27g}{\overset{x}{2Al}} + \underset{1\times160g}{\overset{240g}{Fe_2O_3}} \rightarrow Al_2O_3 + 2Fe$

$$\frac{x}{2\times27g} = \frac{240g}{1\times160g}$$

x = mass of Al reacted = 81g.

➦ Mass of Al left = 108g – 81g = 27g

Percentage composition, Theoretical yield, actual yield and percentage yield

➦ $percentage\,of\,an\,element = \frac{mass\,of\,element}{mass\,of\,compound} \times 100$

➦ **Actual yield** is the amount of product obtained experimentally

➦ **Theoretical yield** is the amount of product obtained stoichiometrically from balanced chemical equation.

➦ $percentage\ yield = \frac{Actual\ yield}{Theoretical\ yield} \times 100$

Actual yield is always less than or equal to theoretical yield

Example

i Calculate the percentage of all elements in calcium carbonate. ($CaCO_3$)

Solution

Ca = 1 × 40 = 40g

C = 1 × 12 = 12g

O = 3 × 16 = <u>48g</u>

mass of compound = 100g

➦ $\%\,Ca = \frac{mass\,of\,Ca}{mass\,of\,CaCO_3} \times 100$

ii Kam

$$= \frac{40g}{100g} \times 100 = 40\%$$

➦ $\%C = \frac{mass\,of\,C}{mass\,of\,CaCO_3} \times 100$

$$= \frac{12g}{100g} \times 100 = 12\%$$

➦ $\%O = \frac{mass\,of\,O}{mass\,CaCO_3} \times 100$

$$= \frac{48g}{100g} \times 100 = 48\%$$

➦ The sum of percentage composition is 100.

ii A grade nine student heated 99.8g of copper in excess sulphur and obtained 80g of Cu_2S. Calculate the percentage yield.
(Atomic mass Cu = 64, S = 32)

$2Cu(s) + S(s) \rightarrow Cu_2S(s)$

Solution:

Actual yield = 80g

Find theoretical yield from balanced chemical equation.

$$\underset{2\times64g/mol}{\overset{99.8g}{2Cu(s)}} + S(s) \rightarrow \underset{1\times160g/mol}{\overset{x}{Cu_2S(s)}}$$

$$\frac{99.8g}{2\times64g} = \frac{x}{160g/mol}$$

x = mass of Cu_2S = 124.75g

➦ Theoretical yield = 124.75g

$$Percentage\ yield = \frac{Actual\ yield}{Theoretical\ yield} \times 100$$

$$= \frac{80g}{124.75g} \times 100$$

$$= 64.1\%$$

Illustrative Examples

48. Which of the following is true about the theoretical yield, actual and percentage yield? (EGSECE)

A. Percentage yield = theoretical yield × 100

B. When the amount of product is calculated from a balanced chemical equation the result is known as actual yield.

C. The actual yield is less than the theoretical yield.

D. The theoretical yield is the minimum amount of the product that can be produced from the quantities of the reactant used.

Explanation

- Percentage yield is actual yield divided by theoretical yield times 100.
- Actual yield is practically determined in the laboratory.
- Theoretical yield is a maximum amount of product obtained from balanced chemical equation.

Answer: C

49. When 32.5g of zinc reacted with excess hydrochloric acid, 0.4mol of hydrogen gas was formed, what was the percentage yield of hydrogen?

A. 80% B. 70% C. 60% D. 40%

Solution:

$$\underset{65g}{\overset{32.5g}{Zn}} + 2HCl \rightarrow ZnCl_2 + \underset{1\times 2g}{\overset{x}{H_2}}$$

$$\frac{32.5g}{65g} = \frac{x}{2g}$$

x = 1g mass of H_2 = Theoretical yield.

Actual yield = 0.4mol × 2g = 0.8g

- $Theoretical\ yield = \frac{Actual\ yield}{Theoretical\ yield} \times 100$

$$= \frac{0.8g}{1.0g} \times 100$$
$$= 80\%$$

Answer: A

50. Given the balanced equation that represent the reaction: $NaOH+HCl \rightarrow NaCl+H_2O$ Atomic mass: Na=23, Cl=35.5, O=16, H =1) What mass of sodium chloride is formed when 60g of NaOH is mixed with 73g of hydrochloric acid? (EGSECE 2006 E.C)

A. 39.0g
B. 36.5g
C. 117.0g
D. 87.8g

Solution:

First identify the limiting reactant and find the answer by mass – mass relationship.

$$n_T(NaOH) = \frac{m}{coeff\ M} = \frac{60g}{40g\,mol} = 1.5mol$$

$$n_T(HCl) = \frac{m}{coeff\ M} = \frac{73g}{1\times 36.5g/mol} = 2mol$$

NaOH is the limiting reactant

$$\overset{60g}{\underset{1\times40}{NaOH}} + HCl \rightarrow \overset{x}{\underset{1\times58.5}{NaCl}} + H_2O$$

$$\frac{60g}{40} = \frac{x}{58.5}$$

x = mass of NaCl = 87.75g

Answer: D

51. 2 moles of magnesium is heated in 3 moles of oxygen. How many moles of the excess reactant are left unreacted according to the following equation? $2Mg + O_2 \rightarrow 2MgO$

A. 1mol O_2
B. 1.5mol Mg
C. 2mol O_2
D. 1 mol Mg

Solution:

Divide the given number of moles to the coefficients and the reactant with smaller number of mole is limiting reactant.

➦ $2Mg + O_2 \rightarrow 2MgO$

$$n_T(Mg) = \frac{2mol}{2} = 1mol$$

$$n_T(O_2) = \frac{3mol}{1} = 3mol$$

➦ Mg is limiting reactant.

The number of moles of O_2 reacted is 1 mole. The unreacted number of moles is $= 3 - 1 = 2$ mol.

Answer: C

52. 1.6g of CaC_2 and 1.8g of H_2O were mixed and made to react according to the following reaction.

$$CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$$

The limiting reactant is

A. CaC_2
B. H_2O
C. C_2H_2
D. $Ca(OH)_2$

Solution:

$$n_r(CaC_2) = \frac{1.6g}{1 \times 64g/\text{mol}} = 0.025mol$$

$$n_r(H_2O) = \frac{1.8g}{2 \times 18g/mol} 0.05mol$$

CaC_2 is the limiting reactant

Answer: A

53. A 25g sample of aluminum sulphide contains 8.98g of aluminum. Calculate the percentage of sulphur in the sample.

Solution:

mass of sulphur = mass of sample – mass of aluminum

$$= 25g - 8.98g = 16.02g$$

$$\%sulphur = \frac{mass\,of\,sulphur}{mass\,of\,sample} \times 100$$

$$= \frac{16.02g}{25g} \times 100 = 64.1\%$$

54. If 5g of sulphurdioxide is actually formed from the reaction of 3g oxygen with sufficient sulphur, what is the percentage yield of SO_2?

Solution:

$$\underset{1\times32g}{S}+\overset{3g}{O_2}\rightarrow\underset{1\times64g}{\overset{x}{SO_2}}$$

$$\frac{3g}{32g}=\frac{x}{64g}$$

X = theoretical yield = 6g

Actual yield = 5g

$$\%yield=\frac{Actual\ yield}{Theoretical\ yield}\times100$$

$$=\frac{5g}{6g}\times100=83.3\%$$

55. When hematite is reduced to iron in a blast furnace, the percentage yield of iron is 78%. Calculate the actual mass of iron in kilogram that would be formed from the reduction of 1×10^9g of hematite. (At mass Fe = 56)

Solution

$$\underset{1\times160}{\overset{10^9}{Fe_2O_3}}+CO\rightarrow\underset{2\times56}{\overset{x}{2Fe}}+CO_2$$

$$\frac{10^9}{1\times160}\times\frac{x}{2\times56}$$

x = Theoretical yield of Fe = 7×10^8g

$$\%\text{yield}=\frac{Actual\ yield}{Theoretical\ yield}\times100$$

$$Actual\ yield=\frac{\%\ yield\times theoretical\ yield}{100}$$

$$=\frac{7\times10^8\ g\times78}{100}$$

$$=5.46\times10^8\text{grams}$$

Actual yield of iron in kilogram is 5.46×10^5kg of Fe

4.6 Oxidation – reduction Reaction

Oxidation number or oxidation state.

Oxidation number is the number of electrons gained or lost when an atom cambines with another atom.

Rules to determine oxidation number

1. The oxidation number of free atoms or molecules of the same atom is zero

Example

➦ Mg, Al, O_2, P_4, S_8, etc have zero oxidation state.

2. The oxidation number of ions of elements is the charge of the ions.

Example

➦ $Na^+ = +1$, $S^{2-} = -2$, $P^{3-} = -3$, $Al^{3+} = +3$ etc.

3. The oxidation number of alkali metals and alkaline earth metals in a compound are + 1 and + 2 respectively.

Example

NaCl, Na = +1

CaI_2, Ca = +2

4. The oxidation number of oxygen in most compounds is – 2. However. in

➦ Peroxides, it is – 1, H_2O_2, O = – 1

➦ Superoxides, it is -½ , KO_2, O = -½

➦ In compound with fluorine, it is + 2

example: OF_2, O = +2

5. The oxidation state of halogens in binary compounds is – 1

Example:

HCl, Cl = – 1

NaI, I = – 1

6. The oxidation state of hydrogen in most compounds is + 1 but in metal hydrides is – 1.

Example:

H_2O, H = + 1 NaH, H = – 1

7. The sum of oxidation state of elements in neutral compound is zero.
8. The sum of oxidation state of element in a polyatomic ion is equal to the charge on the ion.

Illustrative Examples

56. The oxidation number of Na, O_2, Cl^- and NH_4^+ respectively are (EGSECE.....2004E.C)

A. +1, – 4, - 1, + 1
B. 0, 0, -1, + 1
C. 0, -2, -1, + 5
D. + 1, -2, -1, +1

Solution

- The oxidation number of Na and O_2 is zero because they are free atom and molecules
- The oxidation state of Cl^- and NH_4^+ ions, is – 1 and + 1 respectively.

Answer: B

57. What is the oxidation number of sulphur in the molecule of $H_2S_2O_7$?

A. +2
B. +3
C. +5
D. +6

Solution

$\overset{+1}{H_2}\ \overset{x}{S_2}\ \overset{-2}{O_7}$

$2(1) + 2x + (7(-2)) = 0$

$2x + 2 - 14 = 0$

$2x = 12$

x = +6, Thus the oxidation number of sulphur in $H_2S_2O_7$ is + 6 **Answer: D**

58. The oxidation number of carbon in the compound $NaHCO_3$ is

A. -2 B. -4 C. $+6$ D. $+4$

Solution

$\overset{+1\ +1\ x\ -2}{NaHCO_3}$

$1+1+x+3(-2)=0$

$2+x-6=0$

$\underline{x=+4}$ Therefore the oxidation number of C in

$NaHCO_3$ is +4

Answer: D

59. Which one of the following statements is not true about oxidation number?

A. The oxidation number of hydrogen in CaH_2 is -1.

B. The oxidation number of oxygen in peroxide is -1

C. The sum of oxidation numbers of all atoms in a compound is positive or negative number

D. The oxidation number of a neutral atom and molecule is zero.

Answer: C

60. Determine the oxidation state of

a. Fe in Fe_3O_4

b. P in $K_4P_2O_7$

c. Fe in $Na_4(Fe(CN)6)$

d. S in $S_4O_6^{2-}$

e. N in NH_4^+

f. V in VO^+_2

g. Cr in $Cr_2O_7^{2-}$

h. F in NaF

Answer:

a. $+\frac{8}{3}$

b. $+5$

c. $+2$

d. $+5/2$

e) -3

f) + 5

g) + 6

h) -1

Oxidation and Reduction

Oxidation is loss of electrons. Use **oil** to remember oxidation easily

- O = oxidation, i = is, l = loss
- It is an increase in oxidation number.

Example

$Cu \longrightarrow Cu^{2+} + 2e^-$ $\qquad$ $2Cl^- \longrightarrow Cl_2 + 2e^-$

In the early time it was defined as the addition of oxygen to a substance or removal of hydrogen from substance

Reduction is gain of electrons. It is decrease in oxidation state. **Use Rig** to rember it easily that is R = reduction, i = is and g = gain,

Example

$Au^{3+} + 3e^- \longrightarrow Au$ $\qquad$ $O_2 + 2e^- \longrightarrow O^{2-}$

Oxidation – reduction reaction /redox/reaction

- It is a reaction that involves transfer of electrons.
- It is the combination of oxidation and reduction half – reactions.
- It is a reaction with elements that show decrease and increase in oxidation number.

Example of redox reaction

- $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$
- $Cu + 2Ag^+ \longrightarrow Cu^{2+} + 2Ag$
- $2H_2S + 2HNO_3 \longrightarrow 3S + 2NO + 4H_2O$

All reactions containing free elements or molecules are redox reaction

Reducing agent is a substance being oxidized but cause reduction by suppling electrons to the substance getting reduced.

- It loses electrons
- It shows increase in oxidation number

Oxidizing agent is a species being reduced but cause oxidation by gaining electrons from the substance oxidized

- It gains electrons
- It shows decrease in oxidation number.

Non – redox reactions are reactions that do not involve transfer of electrons. All double displacement reactions are non – redox reactions.

Example

$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

Disproportion reaction is a reaction in which the same substance is reduced and oxidized

All disproportion reactions are redox reactions but not the reverse.

Example

- $Cl_2 + H_2O \longrightarrow HCl + HClO$
- Here, Cl_2 is oxidized and reduced at the same time.

Illustrative Examples

61. What happens to a reducing agent in an oxidation – reduction reaction? (EGSECE. . .2007E.C)

A. It is reduced as it gains electron(s)

B. It is reduced as it loses electron(s)

C. It is oxidized as it gains electron(s)

D. It is oxidized as it loses electron(s)

Explanation:

It is oxidized as it loses electrons to the reduced substance.

Answer: D

62. Which of the following is an oxidation – reduction reaction?

A. $AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$

B. $2NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O$

C. $CuO + H_2 \longrightarrow Cu + H_2O$

D. $HCl + H_2O \longrightarrow H_3O^+ + Cl^-$

Answer: C

63. Which of the following statements about the reaction given below is true?

$14HCl + K_2Cr_2O_7 \longrightarrow 2KCl + 2CrCl_3 + 3Cl_2 + 7H_2O$

A. HCl is reducing agent

B. Hydrogen is oxidized

C. Chromium is oxidized

D. Potassium is reduced

Explanation:

$HCl^{-1} \longrightarrow Cl^0{}_2$

➦ Chlorine is oxidized

HCl is reducing agent

$K_2\overset{+6}{Cr}_2O_7 \rightarrow \overset{+3}{Cr}Cl$

➦ Chromimium is reduced

$K_2Cr_2O_7$ is oxidizing agent

Answer: A

64. What can be said about oxidizing agent in an oxidation reduction reaction? (EGSECE. . . 2006E.C)

A. It loses electron(s)

B. It is oxidized

C. Its charge increases

D. It is reduced

Explanation:

- Oxidizing agent:
- It is reduced
- It gains electron (s)
- Its charge decreases

Answer: D

65. Given the balanced equation representing an oxidation – reduction reaction?

$2Al + 3Cu^{2+} \longrightarrow 2Al^{3+} + 3Cu$

66. Which statement is true about this reaction?

A. Each Al^{3+} gains $2e^-$ and each Cu loses $3e^-$

B. Each Al^{3+} gains $3e^-$ and each Cu loses $2e^-$

C. Each Al loses $2e^-$ and each Cu^{2+} gains $3e^-$

D. Each Al loses $3e^-$ and each Cu^{2+} gains $2e^-$

Explanation: $Al \longrightarrow Al^{3+} + 3e^-$, oxidation

$Cu^{2+} + 2e^- \longrightarrow Cu$, reduction

➦ Each Al loses $3e^-$ and each Cu^{2+} gains $2e^-$.

Answer: D

67. In a redox reaction:

A. The oxidizing agent undergoes oxidation

B. Oxidation is decrease in oxidation number.

C. The oxidizing agent loses electrons

D. There is no oxidation without reduction

Answer: D

68. Which of the following half – reaction is reduction?

A. $Cr^{3+} + 3e^- \longrightarrow Cr$

B. $Cr \longrightarrow Cr^{3+} + 3e$

C. $Cr^{3+} \longrightarrow Cr + 3e$

D. $Cr(s) + 3e^- \longrightarrow Cr^{3-}$

Answer: A

69. Identify reduced, oxidized substance, reducing and oxidizing agents in

a. $3CuS + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 3S + 4H_2O + 2NO$

b. $3NO_2 + H_2O \longrightarrow 2HNO_3 + NO$

Answer:

a. Oxidized substance – S

reduced substance – N

reducing agent – CuS

oxidizing agent – HNO_3

b. Oxidized substance – N

reduced substance – N

reducing agent – NO_2

oxidizing agent – NO_2

70. Classify the following as oxidation half reaction or reduction half reaction

a. $2O^{2-} \longrightarrow O_2 + 4e^-$

b. $Pb^{2+} + 2e^- \longrightarrow Pb$

c. $Cr^3 + e^+ \longrightarrow Cr^{2+}$

d. $MnO^-_4 \longrightarrow MnO_2$

e. $SO^{2-}_3 \longrightarrow SO^{2-}_4$

f. $Cr_2O^{2-}_7 \longrightarrow CrO^{2-}_4$

Answer:

➦ a, and e are oxidation half reaction

➦ b, c, d are reduction half – reactions.

➦ f is neither reduction nor oxidation half – reaction.

4.7 Rate of Chemical reaction

The area of chemistry concerned with the speed, or rate at which a chemical reaction occurs is called **chemical kinetics.**

Rate of reaction is the change in concentration of a reactant or a product with time.

$$A \longrightarrow B$$

$$Rate = \frac{Change\ in\ concentration\ of\ A}{Change\ in\ time}$$

$$\boxed{r = \frac{-\Delta[A]}{\Delta t}}$$

$$Rate = \frac{change\ in\ concentration\ of\ B}{change\ in\ time}$$

$$\boxed{r = \frac{\Delta[B]}{\Delta t}}$$

Δ - Change

[ ] – concentration

➦ The negative sign (-) infront of the reactant concentration implies final concentration of a reactant is less than initial concentration

➦ The unit of concentration is molar (M) and time is second

- The unit of rate is molar per second. ($M. S^{-1}$)

The rate of a reaction is maximum at the beginning of a reaction

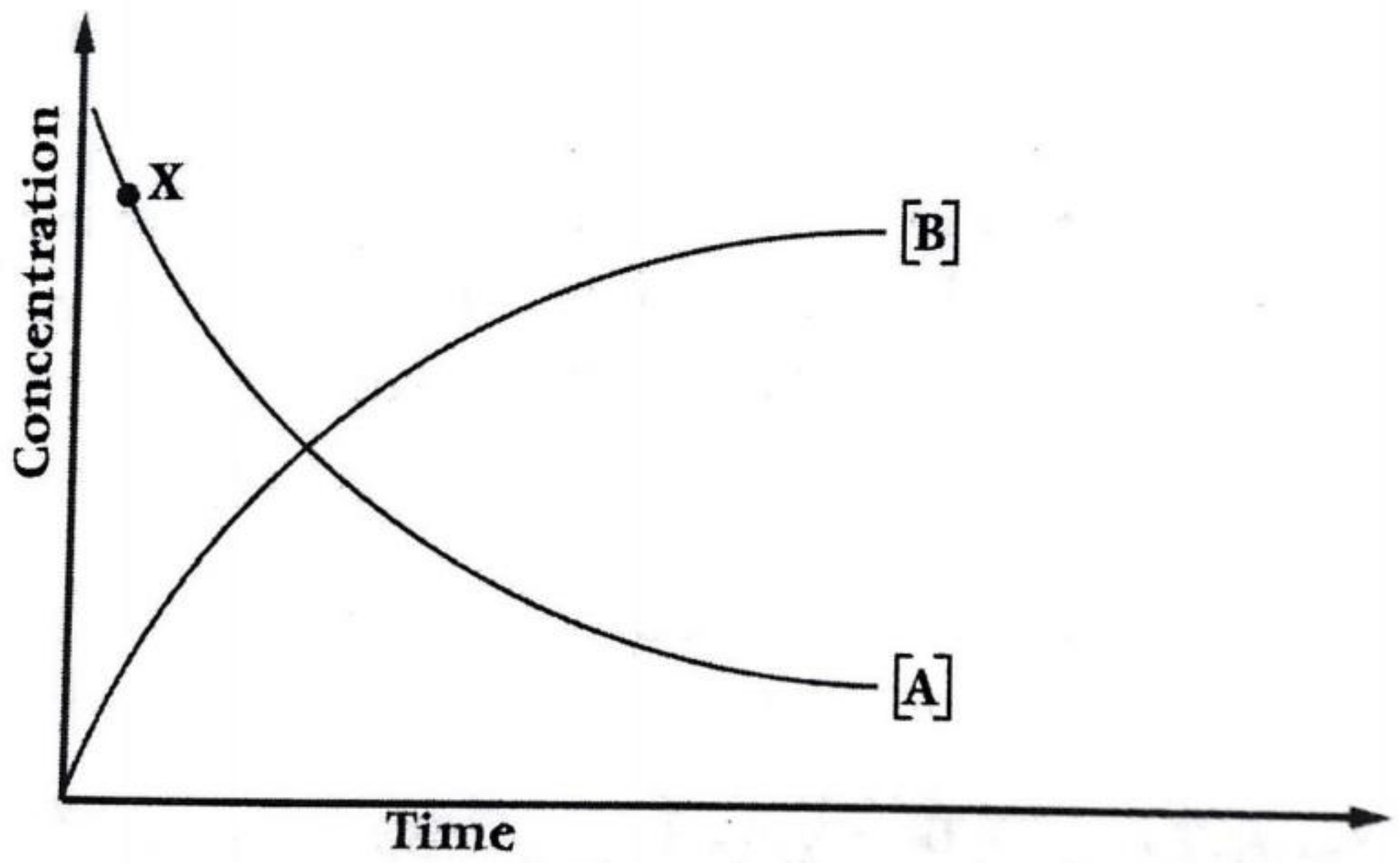


Figure 4.1 The concentration time graph for the reaction A ⟶ B

➦ The rate of the reaction is maximum around point X

➦ Some reactions are fast and others are slow.

Example

Fast reactions	Slow reaction
• Explosion of gun powder	• Rusting of iron
• Combustion of fuel	• Fermentation
• Reaction of sodium with water	• Weathering of rocks

The methods used to determine the amounts of products depends on the type of reaction. Consider some of the methods listed below

1. Mass (gain or loss of mass)
2. Volume increase or decrease in size /for gases/
3. Colour changes
4. Formation of precipitation
5. Pressure (decrease or increase in pressure for gases)

Pre – conditions for a reaction to occur.

1. Collision of reactants
   - ➦ Reactans must collide for a reaction to occur in a proper way.
2. The energy of reacting species should be greater than or equal to activation energy.

   Activation energy (Ea) is the minimum amount of energy required to initiate a chemical reaction. Lacking this energy, the molecules remain intact and no change results from the collision.
3. Proper orientation
   - ➦ The above two preconditions alone will not bring a reaction if the reactants lack proper orientation

Illustrative Examples

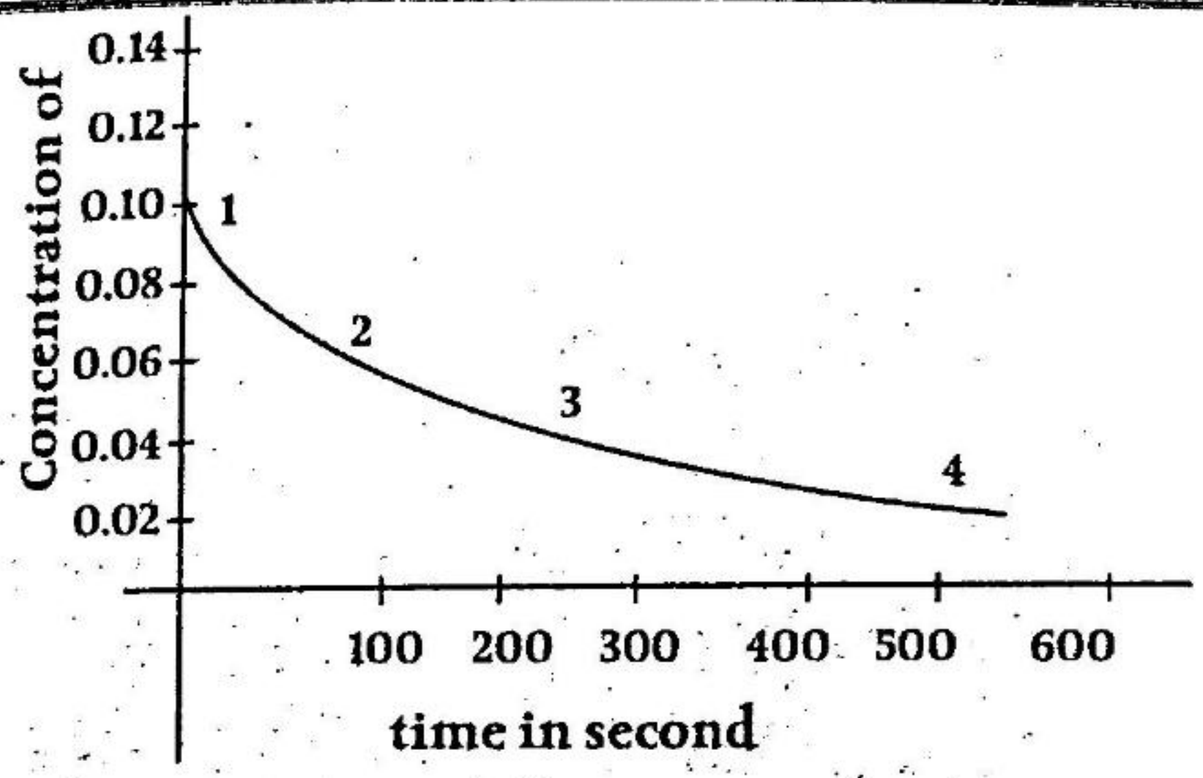
71. How does activation energy cause a chemical reaction to occur? Initially by (EGSECE. . . 2006E.C)

A. Minimizing the number of collision
B. Hindering the reverse reaction
C. Forming a new bond.
D. Breaking an original bond

Answer: D

72. The following graph shows the plot of concentration of bromine verse time represented by the reaction equation

$Br_2 + HCOOH \longrightarrow 2HBr + CO_2$



At which point on the graph the reaction will occur at fastest rate?

A. 3 B. 4 C. 1 D. 2

Answer: C

73. If colliding molecules have proper orientation, then they have:

A. Sufficient reaction speed that is neither too fast nor too slow.

B. No chance to react at all.

C. Greater chance that leads production

D. Sufficient energy for producing reaction production.

Answer: C

74. Why the rate of many chemical reactions decrease with time?

Solution

As time increases concentration of reactants decreases. As a result smaller particles collide and rates slow down.

Factors affecting the rate of a reaction

There are many factors that affect the rate of a reaction. These are:

1. Nature of reactants
2. Concentration of reactants
3. Pressure
4. Surface area
5. Catalysts

1. Nature of reactants

Reactive reactants – fast rate. Less reactive reactants – slow rate.

Example

➦ $2K + 2H_2O \longrightarrow 2KOH + H_2$ . . . fast reaction

➦ $Ca + 2H_2O \longrightarrow Ca(OH)_2 + H_2$ . . . slow reaction

➦ K is more reactive than Ca

2. Concentration of reactants

➦ As concentration of reactants increases, rate also increases. Because rate is directly proportion to concentration.

➦ Concentration mainly affects the rate of **homogenous reaction**, that is reaction in the same phase.

Example

$HCl(1M) + NaOH\ (1M) \longrightarrow NaCl\ (aq) + H_2O(1)$

Slow reaction

$HCl(3M) + NaOH\ (1M) \longrightarrow NaCl\ (aq) + H_2O(1)$

Fast reaction

3. Pressure

➦ Pressure affects the rate of reactions involving gases.

➦ Increasing pressure of gaseous reactants increases the rate of the reaction.

4. Surface area

➦ Surface area affects the rate of **heterogeneous reactions** that is reactions involving solid reactants

Example

$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2\ (aq) + CO_2(g) + H_2O(1)$

➥ Powdering $CaCO_3$ increases the rate of the reaction.

5. Catalysts

➥ Catalysts are substances that increases the rate of a reaction by lowering activation energy

Catalyst increases a rate of reaction without itself being consumed.

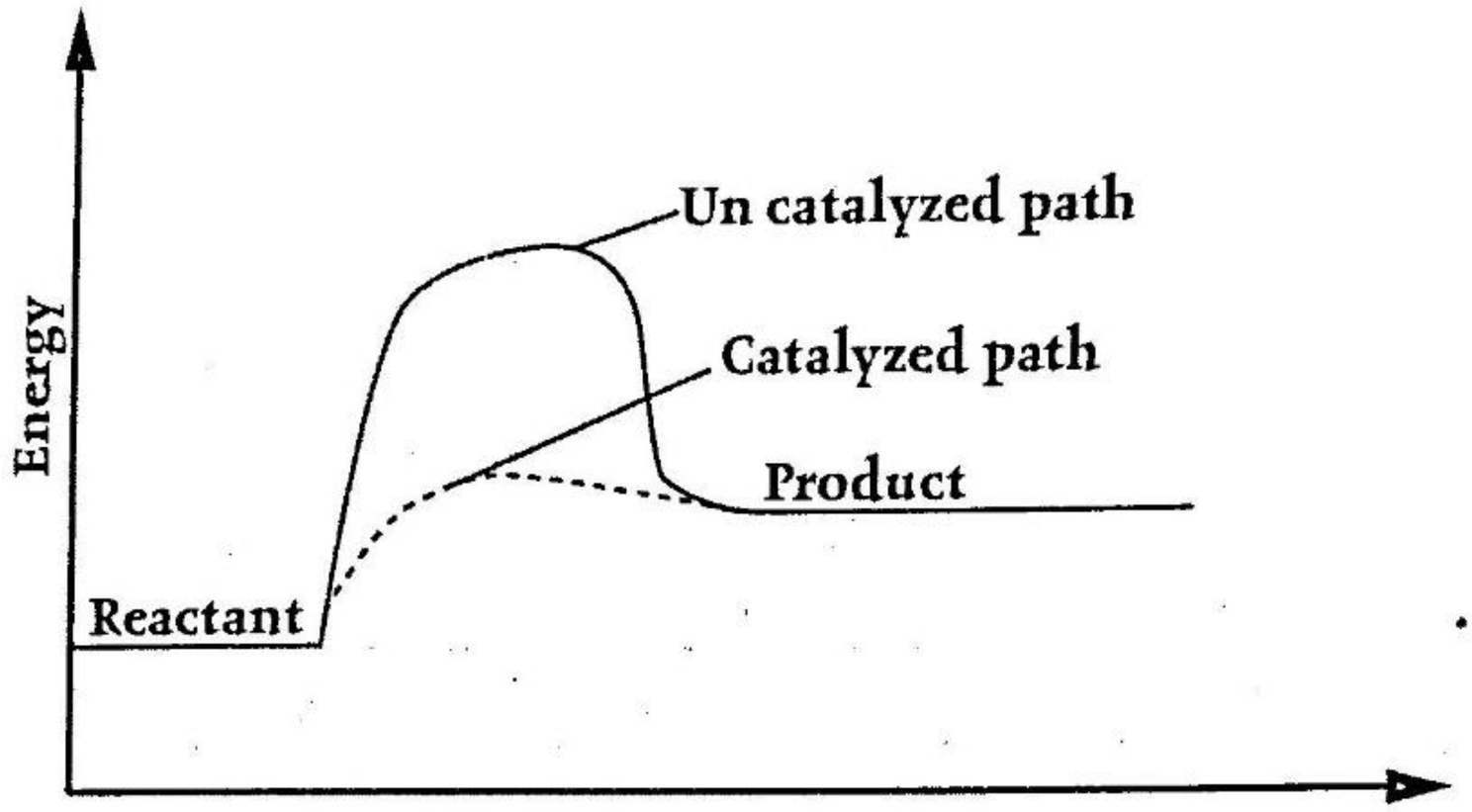


Figure 4.2 Effect of Catalyst on rate of reaction

- Catalysts based on their role are classified as:

i Positive catalysts

These type of catalysts increases the rate of a reaction.

ii Negative catalysts (inhibitors)

These catalysts slow down the rate of a reaction

- Catalysts based on their state classified as

i Homogeneous catalyst

➥ It is a catalyst that has similar state with reactants and products.

Example

$$CH_3COOH(l) + CH_3OH(l) \xrightarrow{H_2SO_4(l)} CH_3COOCH_3(l) + H_2O(l)$$

➥ H_2SO_4 is a catalyst

ii Heterogeneous catalyst

It is a catalyst that has no similar state with reactants and products

Example

$N_2(g) + 3H_2(g) \xrightarrow{Fe(s)} 2NH_3(g)$

➦ Fe is a catalyst

Illustrative Examples

75. During a laboratory activity of investigation reaction rate, a student reacts 1.0gam sample of zinc with 10.0mL of samples of HCl (aq). The table below shows information about the variables in five experiments. The student performed at a constant temperature.

Experiment	Description of Zinc sample	Molar concentration of HCl (aq)
1	lumps	0.10
2	Powder	0.10
3	lumps	0.10
4	Lumps	1.0
5	Powder	1.0

Which experiments can be used to investigate the effect of concentration on the reaction rate? (EGSECE . . . 2007E.C)

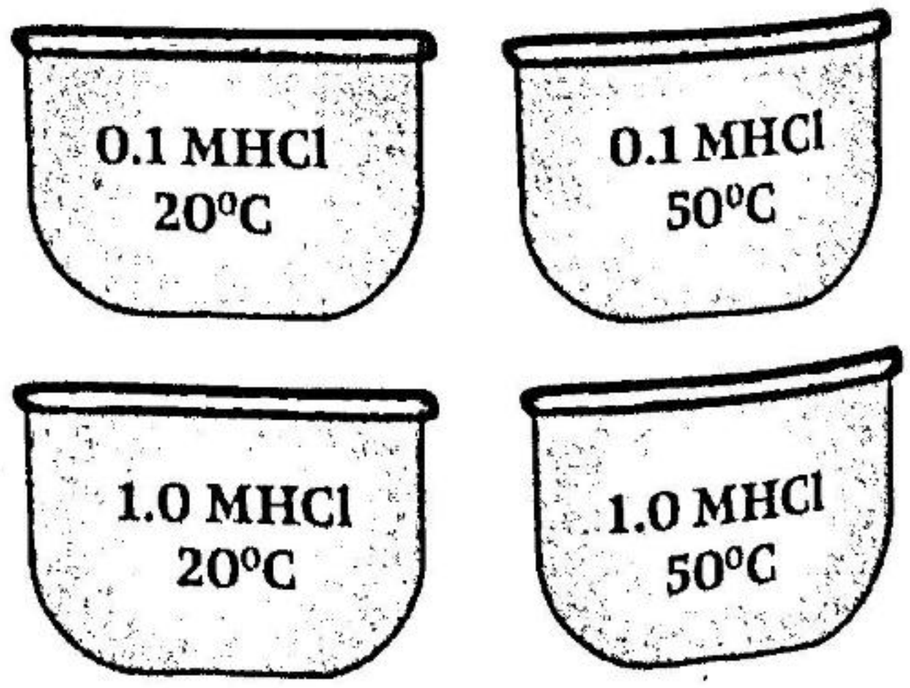
A. (2,3) and (1,3)
C. (1,2) and (4,5)
B. (1,4) and (2,5)
D. (2,4) and (1,5)

Explanation:

➦ Concentration effect can be investigated when the surface area of zinc remain same i.e as lumps or as powdered form.

Answer: B

76. In each of the four beakers shown below, a 2.0cm strip of magnesium ribbon reacts with 100mL of aqueous HCl under the given condition shown



In which beaker will the reaction occur at the fastest rate? In the beaker that contains:

A. 0.1M HCl at 50°C
B. 1.0M HCl at 50°C
C. 0.1MHCl at 20°c
D. 1.0MH Cl at 20°c

Explanation: In general increasing temperature and concentration increases the rate of a reaction.

Answer: B

77. All of the following affects the rate of a reaction. EXCEPT

A. concentration
B. surface area
C. heat of reaction
D. temperature

Answer: C

78. $1cm^3$ cube of sodium reacts more readily in $1cm^3$ cube of calcium does at the same temperature. This difference in the rate of the reaction is most likely

A. The concentration of the metals
B. The density of the metals
C. The nature of the metals
D. Surface area of the metals

Explanation: Sodium is more reactive than calcium due to their nature.

Answer: C

79. What happens when surface area of solid reactants increases?

Solution

As surface area increases number of collision between reacting species increases. This leads to increase rate of reaction.

80. How catalysts increase the rate of a reaction?

Solution

Catalysts increase the rate of reaction by lowering activation energy.

81. The rate of what type of reaction decrease with increasing temperature?

Answer: Exothermic reaction

4.8 Chemical Equilibrium

➥ Chemical reaction can be classified into two. These are:

1. Irreversible reaction

➥ It is a one way reaction that proceeds from left to right

Reactants ⟶ products

➥ It proceeds to completion

➥ Reactants are quantitatively converted into products

➥ The amount of limiting reactant that remains in negligible.

2. Reversible reaction

➥ It is a reaction that proceeds into two directions.

$A \rightleftarrows B$

➥ It does not go to completion.

➥ The reaction that proceeds from left to right is called **forward reaction**

A ⟶ B

➥ The reaction that proceeds from right to left is termed as **backward or reverse reaction**

B ⟶ A

➥ When the rates of forward and reverse reactions are equal and the concentration of the reactants and products no longer change with time, **chemical equilibrium is reached.**

➥ Chemical equilibrium is a dynamic process

➥ If the forward reaction is endothermic the reverse must be exothermic.

➥ Temperature, concentration and pressure should be constant at equilibrium.

➥ Concentration of reactants is not necessary equal to concentration of products at equilibrium.

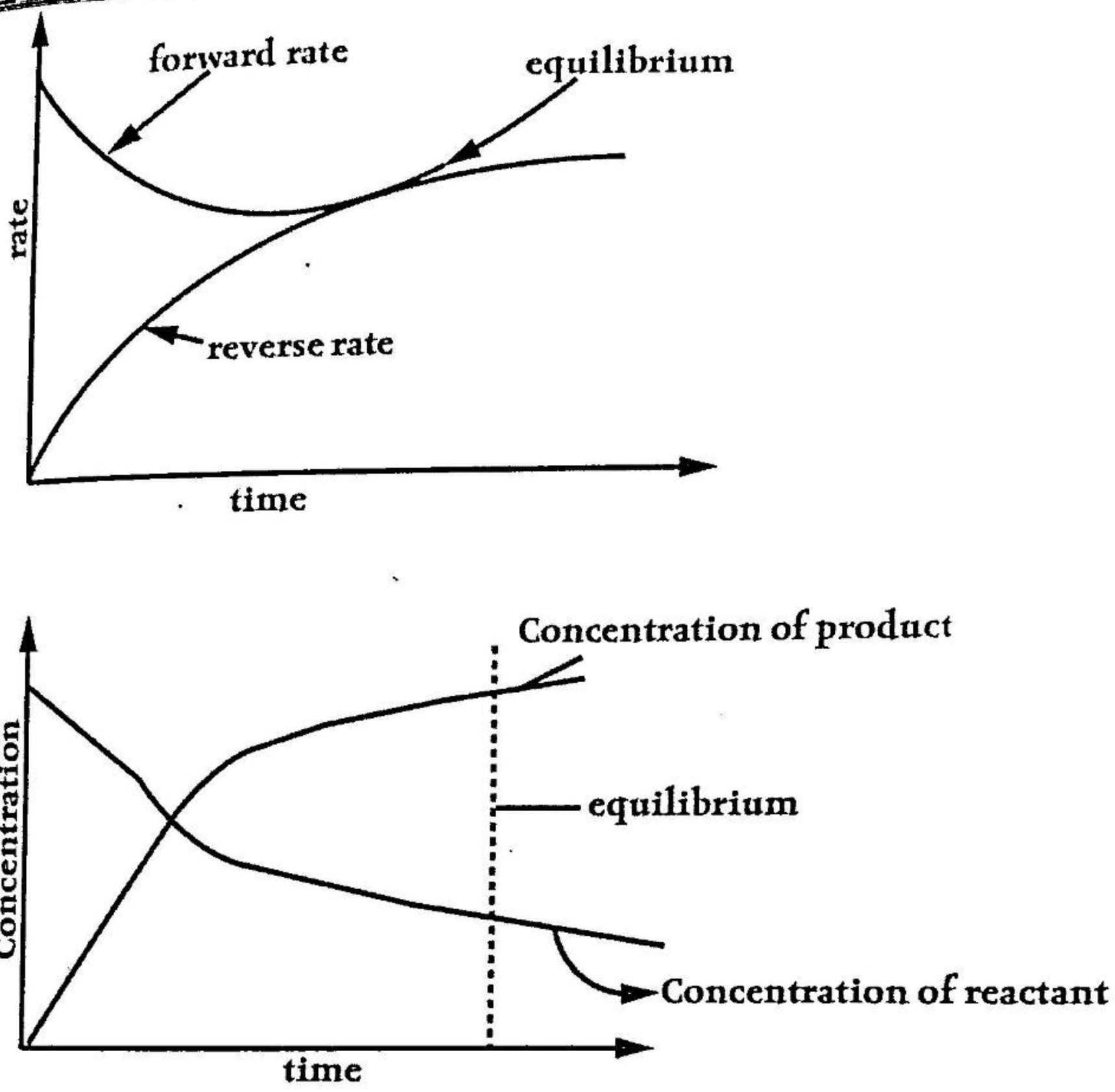


Figure 4.3 Concentration time graph for a reversible reaction.

Chemical equilibrium constant expression

The law of mass action or law of concentration action states that "for a reversible reaction at equilibrium at a given temperature equilibrium constant, Keq is equal to amount of products divided by that of reactants each raised to the power of its coefficients in the balanced equation."

Consider a general reaction:

$$aA + bB \rightleftharpoons cC + dD$$

According to law of mass action

$$r_f \alpha [A]^a [B]^b \Rightarrow r_f = k_f [A]^a [B]^b$$

$$r_b \alpha [A]^c [O]^d \Rightarrow r_b = k_b [C]^c [D]^d$$

At equilibrium, $r_f = r_b$

$\Rightarrow k_f [A]^a [B]^b = k_b [C]^c [D]^d$

After rearranging the equation:

$$\frac{k_f}{k_b} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

➦ The ratio of k_f to k_b is called **equilibrium constant (Keq)**

r_f - rate of forward reaction

k_f – rate constant for the forward reaction

r_b – rate of backward reaction

k_b – rate constant of backward reaction

➦ Keq = K_C, if the amount of products and reactants expressed in molarity

➦ Keq = K_p, if the amount of products and reactants expressed in partial pressure

- $K_C = \frac{[C]^c [D]^d}{[A]^a [B]^b}$
- $K_p = \frac{P_C^c \cdot P_D^d}{P_A^a \cdot P_B^b}$

➦ K_p is applied only for gaseous substances.

Note:

1. Equilibrium constant is affected by temperature.
2. When we write equilibrium constant expression, pure solids and solvents or liquids are not included in the equilibrium constant expression because their amount is constant.
3. Homogeneous equilibrium is an equilibrium with same state of reactants and products.

   Example: $2NO_2(g) \rightleftharpoons N_2O_4(g)$

4. Heterogeneous equilibrium is an equilibrium that contains different states of products and reactants.

   Example: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

Illustrative Examples

82. Which of the following is characteristic of a reaction at equilibrium? (EGSECE 2004 E.C)

A. Rate of forward and reverse reactions are equal

B. Concentration of reactants and products is equal

C. If it is exothermic in one direction, it also exothermic in another direction.

D. Appearance of reactants and disapperance

Answer: A

83. Which of the following is the correct Kc expression for:

$PCl_5(s) \rightleftharpoons PCl_3(s) + Cl_2(g)$?

A. $\frac{[PCl_3][Cl_2]}{[PCl_5]}$

B. $\frac{[PCl_5]}{[PCl_3][Cl_2]}$

C. $[Cl_2]$

D. $\frac{1}{[Cl_2]}$

Explanation:

Solids are not included in the equilibrium constant expression

➦ $Kc = [Cl_2]$

Answer: C

84. Write Kp expression for

a. $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$

b. $2SO_2(g) + O_2 \rightleftharpoons 2SO_3(g)$

c. $2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$

d. $FeO(s) + CO(g) \rightleftharpoons Fe(s) + CO_2(g)$

Answer:

a. $K_p = \frac{P_{CO}^{\ 2}}{P_{CO_2}}$

b. $K_p = \frac{P^2_{SO_3}}{P^2_{SO_2} \cdot P_{O_2}}$

c. $K_p = \frac{P_{NO}^{\ 2} \cdot P_{O_2}}{P_{NO_2}}$

d. $K_p = \frac{P_{CO_2}}{PCO}$

85. Write Kc expression for

a. $AgCl(s) \rightleftharpoons Ag^{+}(aq) + Cl^{-}(aq)$

b. $CO_2(g) + MgO(s) \rightleftharpoons MgCO_3(s)$

c. $HF(g) + H_2O(l) \rightleftharpoons H_3O^{+}(aq) + F^{-}(aq)$

d. $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$

Answer

a. $Kc = [Ag^{+}]\,[Cl^{-}]$

b. $K_c = \dfrac{1}{[CO_2]}$

c. $K_c = \dfrac{[H_3O^{+}][F^{-}]}{[HF]}$

d. $K_c = \dfrac{[NO]^4[H_2O]^6}{[NH_3]^4[O_2]^5}$

86. Calculate the value of equilibrium constant for the reaction: $2A + B \rightleftharpoons 3C + 2D$, If at equilibrium there are 1 mol of A, 2 mole of B, 3mol of C and 4 mol of D in a 1.0L vessel

Solution:

First find molarity of reactants and products. S

$$[A] = \frac{n_A}{v} = \frac{1mol}{1L} = 1M$$

$$[B] = \frac{n_B}{v} = \frac{2mol}{1L} = 2M$$

$$[C] = \frac{n_c}{v} = \frac{3mol}{1L} = 3M$$

$$[D] = \frac{n_D}{v} = \frac{4mol}{1L} = 4M$$

$$K_C \frac{[C]^3[D]^2}{[A]^2[B]}$$

$$K_c = \frac{[3]^3[4]^2}{[1]^2 \times [2]^1} = 216$$

87. Given the reaction: $A + B \rightleftharpoons 2C$, $Kc = 100$ At equilibrium, $[A] = 0.25M$, $[B] = 1M$. Find concentration of C at equilibrium.

Solution

$$K_c = \frac{[C]^2}{[A]^2[B]}$$

$[C]^2 = Kc\,[A]\,[B]$

$$[C] = \sqrt{K_c[A][B]}$$

$$= \sqrt{100 \times 0.25 \times 1} = 5M$$

88. Classify the following as homogeneous or heterogeneous equilibrium.
    a. $HCl\,(aq) \rightleftharpoons H^+(aq) + Cl^-(aq)$
    b. $2ZnS(s) + 3O_2(g) \rightleftharpoons 2ZnO(s) + 2SO_2(g)$
    c. $4HCl(g) + O_2(g) \rightleftharpoons 2H_2O(g) + 2Cl_2(g)$

Solution

- a and c are homogeneous equilibria
- b is heterogeneous equilibrium.

Factors affecting the position of equilibrium

Le chatelier's principle provides a means of predicting the influence of disturbing factors on equilibrium systems. This principle states that "if a system is subjected to a stress, the equilibrium is shifted in the direction that relieves the stress." The stress could be concentration, temperature or pressure

1. Effect of concentration

- Decreasing concentration of products or increasing concentration of reactants shifts the equilibrium to the right.
- Increasing concentration of products or removing reactants shifts the equilibrium to the left.

Example

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

- removing NH_3 or adding N_2 and H_2 shifts the equilibrium to the right.

2. Effect of pressure

- Pressure has effect when:
  i. The reaction contain gases.
  ii. Number of moles of reactants and products are not equal.
- Increasing pressure shifts the equilibrium toward the **smaller** number of molecules.
- Decreasing pressure shifts the equilibrium toward the **larger** number of molecules

Example

$$\underbrace{2SO_2(g) + O_2}_{3moles} \rightleftharpoons \underbrace{2SO_3(g)}_{2moles}$$

- Increasing pressure shifts the equilibrium to the right
- Decreasing pressure shifts the equilibrium to the left.

3. Effect of temperature

- Increasing temperature shifts the equilibrium toward the **endothermic side.**
- Lowering temperature shifts to the **exothermic side.**

Example

$4HCl(g) + O_2(g) \rightleftharpoons 2H_2O(g) + 2Cl_2(g) + 114.4kJ$

- The forward reaction is exothermic, so the reverse reaction must be endothermic
- Lowering temperature shifts the equilibrium toward the right while increasing temperature to the left.

4. Effect of catalysts

Catalysts do not affect the position of equilibrium, but they alter the time at which equilibrium is established.

Illustrative Examples

89. Given the system at equilibrium $N_2O_4(g)+58.1kJ \rightleftharpoons 2NO_2(g)$ What will be the result of an increase in temperature at constant pressure? (EGSECE. . . 2007E.C)

A. The equilibrium will shift to the left and concentration of NO_2 will increase

B. The equilibrium will shift to the left and concentration of NO_2 will decrease

C. The equilibrium will shift to the right and concentration of NO_2 will increase

D. The equilibrium will shift to the right and concentration of NO_2 will decrease.

Explanation: The forward reaction is endothermic. As temperature increases the equilibrium will shift to the right. As a result, concentration of NO_2 increases

Answer: C

90. Given the reaction at equilibrium $N_2(g) + O_2(g) + \text{energy} \rightleftharpoons 2NO(g)$, which change will result in a decrease in the amount of NO(g)?

A. Decreasing concentration of $N_2(g)$

B. Increasing the temperature

C. Decreasing pressure

D. Increasing concentration of $O_2(g)$

Explanation: The concentration of NO(g) decreases when

➥ N_2 or O_2 are removed

- ➦ Temperature decreases
- ➦ Pressure and volume have no confect because number of moles of reactants and product is equal.

Answer: A

91. Given the equation representing a system at equilibrium: $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ what occurs when the concentration of $O_2(g)$ is increased?

A. The equilibrium shift to the left and concentration of $SO_2(g)$ decreases.

B. The equilibrium shifts to the right and concentration of $SO_2(g)$ decreases.

C. The equilibrium shifts to the left and concentration of $SO_2(g)$ increases

D. The equilibrium shifts to the right and concentration of $SO_2(g)$ increases

Answer: B

92. Given the reaction at equilibrium $X_2(g) + 2Y_2(g) \rightleftharpoons 2XY_2(g) +$ 80Cal How does the equilibrium shift to the right of the equation? (EGSECE. . . 1993 E.C)

A. By increasing temperature and pressure

B. By increasing pressure and decreasing temperature

C. By decreasing pressure and increasing temperature

D. By decreasing pressure and temperature

Answer: B

93. Which of the following reactions at equilibrium is not affected by pressure?

A. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

B. $2HF(g) \rightleftharpoons H_2(g) + F_2(g)$

C. $2N_2O_5(g) \rightleftharpoons 4NO_2(g) + O_2$

D. $N_2(g) + 2H_2O(g) \rightleftharpoons 2NO(g) + 2H_2(g)$

Explanation: When number of moles of reactants and products is equal pressure has no effect.

Answer: B

94. Given the reaction at equilibrium

$$2NaHCO_3(s) \rightleftharpoons Na_2CO_3(s) + CO_2(g) + H_2O(g),\ \Delta H = +128kJ$$

explain the effect of the following on the position of equilibrium.

a. lowering pressure
b. adding $NaHCO_3(s)$
c. increasing temperature
d. removing $CO_2(g)$
e. increasing concentration of $H_2O(g)$
f. adding catalyst
g. increasing temperature and pressure
h. increasing concentration of $CO_2(g)$ and pressure.

Solution:

a. shifts to the right
b. no effect because it is solid reactant.
c. Shifts to the right
d. Shift to the right
e. Shifts to the left
f. No effect
g. Unpredictable because increasing temperature shifts to the right but increasing pressure shifts to the left
h. Shifts to the left because both factors shift in the same direction.

95. Write two applications of chemical equilibrium.

Solution

1. Production of ammonia by Haber process

$$N_2(g) + 3H_2(g) \rightleftharpoons NH_3(g) + 92kJ$$

➦ Formation of $NH_3(g)$ is favored at low temperature and high pressure.

2. Production of sulphuric acid by contact process.

Steps in contact process

- $S(s) + O_2(g) \longrightarrow SO_2(g)$
- $2SO_2(g) + O_2(g) \underset{}{\overset{V_2O_5(s)}{\rightleftharpoons}} 2SO_3(g) + 196kJ$

➡ Formation of $SO_3(g)$ is favoured at low temperature and high pressure

- $SO_3(g) + H_2SO_4(1) \longrightarrow H_2S_2O_7(1)$
- $H_2S_2O_7(1) + H_2O \longrightarrow 2H_2SO_4(1)$

Solved Problems

96. When 6.5g of zinc were added to solution of copper (II) sulphate, 6.4g of copper were obtained.

a. Write balanced chemical equation

b. What type of chemical reaction is it?

c. How many zinc atoms were used?

d. How many grams of copper (II) sulphate were consumed?

e. How many grams of zinc sulphate were obtained?

Solution

a. $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$

b. It is single displacement reaction

c. $number\ of\ Zn\ atoms\ used = \frac{mzn}{Mzn} \times 6.02 \times 10^{23}\ atoms$

$= \frac{6.5g}{65} \times 6.02 \times 10^{23}\ atoms$

$= 0.1 \times 6.02 \times 10^{23}$ atoms

$= 6.02 \times 10^{22}$ atoms of zinc were used

d. $\overset{6.5g}{Zn} + \overset{x}{CuSO_4} \rightarrow ZnSO_4 + Cu$

$\underset{1\cdot65}{}\quad\underset{1\cdot160}{}$

$\frac{6.5g}{65} = \frac{x}{160}$

X = 16.0g of $CuSO_4$ were consumed

e. $\underset{1\times65}{\overset{6.5g}{Zn}} + CuSO_4 \rightarrow \underset{1\times161}{\overset{x}{ZnSO_4}} + Cu$

$$\frac{6.5g}{65} = \frac{x}{161}$$

X = 16.1g of $ZnSO_4$ were obtained.

97. Given the reaction: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ identify

a. reduced substance
b. oxidized substance
c. oxidizing agent
d. reducing agent

Solution:

$$\overset{+3}{Fe_2}\overset{}{O_3} + \overset{+3}{3CO} \rightarrow \overset{0}{2Fe} + \overset{+4}{3CO_2}$$

Solution:

a. reduced substance is Fe because the oxidation number of Fe reduced from +3 to 0
b. oxidized substance is C since is oxidation state increased from +2 to +4.
c. Oxidizing agent contains reduced substance. Therefore, Fe_2O_3 is oxidizing agent.
d. Reducing agent is CO because it contains oxidized substance.

98. A 1.256g sample of elemental sulphur is combined with fluorine to given a compound with the formula SF_x. If you isolate 5.722g of SF_x, what is the value of X? (At mass, = S = 32, F = 19)

Solution:

$$S + F_2 \longrightarrow SF_x$$
$$1.256g \quad 5.722g$$

by using percentage composition, it is possible to get the value of x

i $\%S = \frac{m_s}{m_{SFX}} \times 100$, by using given mass

ii $\%S = \frac{m_s}{m_{SFX}} \times 100$, by using molar mass

In both eases the percentage of sulfur is same

$$\rightarrow \quad \frac{m_s}{m_{SFX}} \times 100 = \frac{M_s}{M_{SFX}} \times 100$$

$$\rightarrow \quad \frac{1.256g}{5.722g} \times 100 = \frac{32g/mol}{(32+19x)g/mol} \times 100$$

$$\rightarrow \quad \Rightarrow \frac{1.256g}{5.722g} = \frac{32}{32+19x}$$

Solving for x, $x = 5.9886 \approx 6$

$\therefore$ The compound isolated is SF_6.

99. At 22^0c, a 30L flask contains 7 mol HI, 4 mole H_2 and 0.6mol I_2 in equilibrium. What is the value of K_C at 22^0C?

$H_2(g) + I_2(g) \rightarrow 2HI(g)$

Solution:

$$[H_2] = \frac{nH_2}{V} = \frac{4mol}{3L} = \frac{4}{3}M$$

$$[I_2] = \frac{nI_2}{V} = \frac{0.6mol}{3L} = 0.2M$$

$$[HI] = \frac{nHI}{V} = \frac{7mol}{3L} = \frac{7}{3}M$$

$$H_C = \frac{[HI]}{[H_2][I_2]} = \frac{\left(\frac{7}{3}\right)^2}{\left(\frac{4}{3}\right)(0.2)} = 20.4$$

100. What is the mass of SO_3 produced when 48g of SO_2 and 12g of O_2 are reacted?

Solution:

First write balanced chemical equation and identify the limiting reaction.

$2SO_2 + O_2 \longrightarrow 2SO_3$

$$n_{SO_2} = \frac{m_{SO_2}}{M_{SO_2}} = \frac{48g}{64g/mol} = 0.75mol$$

$$n_{O_2} = \frac{m_{O_2}}{M_{O_2}} = \frac{12g}{32g/mol} = 0.375mol$$

To identify the limiting reactant, divide number of mole to coefficient.

$$n_{SO_2} = \frac{0.75mol}{2} = 0.375mol$$

$$n_{SO_2} = \frac{0.375mol}{1} = 0.375mol$$

$$\Rightarrow n_{SO_2} = n_{O_2} = 0.375mol$$

➦ Both are limiting reactants consumed at the same time. The amount of SO_3 produced can be determined by using either of the reactants.

$$m_{SO_3} = \frac{coeff\ SO_3 \times M_{SO_3} \times m_{O_2}}{coeff.O_2 \times M_{O_2}}$$

$$= \frac{2 \times 80g/mol \times 12g}{1 \times 32gmol} = 60g$$

$$m_{SO_3} = \frac{coeff.SO_3 \times M_{SO_3} \times m_{SO_2}}{coeff.SO_2 \times M_{SO_2}}$$

$$= \frac{2 \times 80g/mol \times 48g}{2 \times 64gmol} = 60g$$

101. A1.871g sample of unknown metallic carbonate is decomposed by heating to from the metallic oxide and 0.656g of carbon dioxide according to the equation.

$MCO_3 \longrightarrow MO + CO_2$

What is the metal?

Solution:

In order to know the metal, we have to find the atomic mass of the metal M.

$$MCO_3 \longrightarrow MO + CO_2$$

1.871g ? 0.656g

mass of MO = 1.871 – 0.656 = 1.215g

by using mass – mass relationship

$$\frac{mass\,MO}{molar\,mass\,MO \times coeff} = \frac{mass\,CO_2}{molar\,mass\,CO_2 \times coeff\,CO_2}$$

Let the atomic mass of M be X

$$\Rightarrow \frac{1.215}{(x+16)\times 1} = \frac{0.656}{(44)\times 1} \Rightarrow \frac{1.215}{x+16} = \frac{0.656}{44}$$

$$\Rightarrow 53.46 = 0.656x + 10.496$$

$x = 65g/mol$ The metal with atomic mass 65 is zinc.

102. What are the number of moles and mass of calcium in a sample containing 12.04×10^{23} atoms?

Solution:

1 mole Ca = 6.02×10^{23} atoms

? = 12.04×10^{23} atoms

$$n_{ca} = \frac{1mole Ca \times 12.04 \times 10^{23}\,atom}{6.02 \times 10^{23}\,atoms}$$

n_{Ca} = 2 mol Ca

$m_{Ca} = n_{Ca} M_{Ca}$ =2mol × 40g/mol= <u>80g</u>

103. Given the reaction:

$$3NaBH_4 + 4BF_3 \longrightarrow 3NaBF_4 + 2B_2H_6.$$

If the reaction has a percentage yield of 70%, how many moles of $NaBH_4$ should be used with excess BF_3 in order to obtain 0.2mol of B_2H_6?

Solution:

Given: Percentage yield = 70%

Actual yield = 0.02 mol B_2H_6

$$m_{B_2H_6} = M_{B_2H_6} = 0.2mol \times (2 \times 11 + 6 \times 1)g/mol = 5.6g$$

$$Percentage\ yield = \frac{Actual\ yield}{Theoretical\ yield} \times 100$$

$$= \frac{5.6g}{70} \times 100 = 8g\ B_2H_6 = 0.286mol$$

Based on theoretical yield:

$$n_{NaBH_4} = \frac{naBH_4 \times coeff.NaBH_4}{coeff.B_2H_6} = \frac{0.286mol \times 3}{2} = 0.429mol$$

or By using number of mole

$$\%\ yield \frac{Actual\ yield}{Theoretical\ yield} \times 100$$

$$Theoretical\ yield = \frac{0.2mol}{70} = 0.286mol\ B_2H_6$$

$$\rightarrow\ n_{NaBH_4} = 3 \times 0.286mol = 0.42g/mol$$

104. Write equilibrium constant expression for

a. $2ICl(s) = I_2(s) + Cl_2(g)$

b. $CO(g) + \frac{1}{2} O_2 = CO_2(g)$

c. $2SO_3(g) = 2SO_2(g) + O_2(g)$

Solution:

a. $K_c = [Cl_2]$. $Kp = PCl_2$

b. $K_c = \frac{(CO_2]}{CO]^1 O_2]^{\frac{1}{2}}}, Kp = \frac{P_{CO_2}}{P_{CO} \cdot P_{O_2}^{\frac{1}{2}}}$

c. $K_c = \frac{(SO_2]^2 [O_2]}{SO_3]^2}, Kp = \frac{P^2_{SO_2} \cdot P_{O_2}}{P^2_{SO_3}}$

105. Arrange the following in increasing order of number of molecules

a. 1g of CH_4

b. 1g of O_2

c. 1g of H_2

d. 1g of H_2O

Solution:

First find number of molecules

Number of molecules $\frac{m}{M}\times 6.02\times 10^{23}$

M = Molecular mass

no. of molecules of $CH_4 = \frac{1}{16}\times 6.02\times 10^{23}$

no. of molecules of $O_2 = \frac{1}{32}\times 6.02\times 10^{23}$

no. of molecules of $H_2 = \frac{1}{2}\times 6.02\times 10^{23}$

no. of molecules of $H_2 = \frac{1}{18}\times 6.02\times 10^{23}$

$\therefore b < d < a < c$

Note:

If the given mass of different molecules is the same, the smaller the molecular mass the higher is the number of molecules.

106. A sample of sucrose, $C_{12}H_{22}O_{11}$ contains 2×10^{20} atoms of oxygen. How many atoms of hydrogen does it contain?

Solution:

Use subscript ratio to get the answer easily

H : O

22 = 11

$? = 2\times 10^{20}$ atoms

number of hydrogen atoms= $\frac{22\times2\times10^{20}\,atoms}{11}=4\times10^{20}\,atoms$

107. Balance the following chemical equation

$Fe_2S_3 + H_2O + O_2 \longrightarrow Fe(OH)_3 + S$

Solution:

By using LCM method

$$\overset{6}{Fe_2}\overset{6}{S_3}+\overset{2}{H_2}O+\overset{4}{O_2}\rightarrow\overset{3}{Fe}\left(\overset{3}{OH}\right)_3+\overset{2}{S}$$

➦ $2Fe_2S_3 + 6H_2O + 3O_2 \longrightarrow 4Fe(OH)_3 + 6S$

108. What mass of magnesium contains the same number of atoms as 5g of calcium?

Solution:

40g Ca = 6.02×10^{23}atoms

$$atoms\,of\,Ca=\frac{5g\times6.02\times10^{23}}{40g}=7.525\times10^{22}\,atoms$$

atoms of Ca = atoms of Mg = 7.525×10^{22} atoms

➦ 24g = 6.02×10^{23} atoms

? = 7.525×10^{23} atoms

$$mass\,Mg=\frac{24g\times7.525\times10^{22}}{6.02\times10^{23}}=3.01g$$

109. How many grams of solid KO_2 is needed to react with 50L of CO_2 at STP according to the reaction:

$4KO_2(s) + 2CO_2(g) \longrightarrow 3O_2(g)+2K_2CO_3$

$$m_{KO_2}=\frac{coeff.KO_2\times M_{KO_2}\times V_{CO_2}}{coeff.CO_2\times22.4L/mol}=\frac{4\times71g/mol}{2\times22.4L/mol}$$

$= \underline{316.96g}$

110. 60g of a metal M (M = 60) combine with 24g of oxygen (O = 16) to form an oxide. What is the formula of the oxide?

Solution:

M	O
60	24
(1	1.5) × 2

∴ The formula of the oxide is M_2O_3

111. **What is the oxidation state of nitrogen in NH_4Br?**

Solution:

The oxidation state of hydrogen is + 1 except in metal hydrides. Which is – 1.

$\overset{x}{N}H_4\overset{-1}{Br}$

$x + 4(+1) + (-1) = 0$

$x + 3 = 0$

$x = -3$

➥ **The oxidation state of nitrogen is NH_4Br is – 3**

112. **Ten grams of Calcium carbonate was decomposed to give 3.96g of carbon dioxide according to an equation representing the reaction.**

$CaCO_3 \longrightarrow CaO + CO_2$

What is the percentage yield of this reaction?

Solution:

Determine the theoretical yield (Ty) from the balanced chemical equation Molar mass of $CaCO_3$ = 100 and CO_2 = 44

$$\underset{100g}{\overset{10g}{CaCO_3}} \rightarrow \underset{44g}{\overset{x}{CO_2}}$$

Set a proportion

$$\frac{10}{100g} = \frac{x}{44g}$$

x = 4.4g = Ty

Actual yield (Ay) is given i.e., 3.96g

$$\text{Percentage yield} = \frac{Ay}{Ty} \times 100$$

$$\text{Percentage yield} = \frac{3.96g}{4.4g} \times 100 = 90\%$$

113. When the following equation is correctly balanced, what is the sum of the balancing coefficients?

$Na_2SO_4 + BaCl_2 \longrightarrow NaCl + BaSO_4$

Solution:

Use inspection method to balance simple chemical equation.

Place 2 as a coefficient for NaCl to balance Na and Cl.

➥ $Na_2SO_4 + BaCl_2 \longrightarrow NaCl + BaSO_4$

Sum of coefficient = 1 + 1 + 2 + 1 = 5

114. What is the mass of one molecule of ethane (C_2H_6) gas?

Solution:

molecular mass of ethane = 6.02×10^{23} molecules

➥ 30g C_2H_6 = 6.02×10^{23} molecules

? = one molecules

$$\text{Mass of one molecules of } C_2H_6 = \frac{30gC_2H_6 \times 1 molecules}{6.02 \times 10^{23} molecules}$$

Mass of one molecule of C_2H_6 = 4.98×10^{-23}g

115. What is the number of chlorine atoms in 1.0×10^{-5} mol of CCl_4?

Solution:

Number of atoms of an element in a compound = number of mole of compound × subscript of the element × 6.02×10^{23}

Number of Cl atoms in

$$CCl_4 = 1.0 \times 10^{-5}\ mol \times 4 \times 6.02 \times 10^{23} \frac{atoms}{mol}$$

Number of Cl in $CCl_4 = 2.408 \times 10^{19}$ atoms

116. How many grams of $Fe(OH)_3$ can be produced by reacting 1.0 mole Fe_2S_3, 2.0moles of H_2O and 3.0 moles of O_2 according to the following chemical equation?

$$2Fe_2S_3 + 6H_2O + 3O_2 \longrightarrow 4Fe(OH)_3 + 6S$$

Solution:

The amount of product formed is determined by the limiting reactant. Divided the given number of mole to the coefficient to identify the limiting reactant.

➦ $n_{Fe_2S_3} = \frac{1mol}{2} = 0.5mol$

$n_{H_2O} = \frac{2\,mol}{6} = 0.33mol$

$n_{O_2} = \frac{3mol}{3} = 1mol$

➦ H_2O is limiting reactant. Find the number of moles of $Fe(OH)_3$ by mole – mole relationship

$$2Fe_2S_3 + 6\overset{mol}{H_2O} + 3O_2 \rightarrow 4\overset{x}{Fe}_4(OH)_3 + 6S$$

$$\frac{2mol}{6} = \frac{x}{4}$$

$$x = \frac{8mol}{6} = 1.33mol$$

$$M_{Fe(OH)_3} = 56 + 3 \times 17 = 107g/mol$$

$$n = \frac{m}{M}$$

$$m = n \times M$$

$$m = 1.33 \text{ mol} \times 107 \text{g/mol} = 142.31 \text{g}$$

117. 2 grams of a mixture of copper (I) oxide (Cu_2O) and copper (II) oxide (CuO) was reduced to 0.8g of metallic copper. What was the weight in grams of CuO in the original sample?

Solution:

Mass of mixture of Cu_2O and CuO = 2g

Formula mass of Cu_2O and CuO = formula mass of Cu_2O + formula mass of CuO

Formula mass of Cu_2O and CuO = (2×64×1×16)+ (1×64×1×16) = 224g

$$Mass\ of\ CuO \text{ in the original sample} = \frac{formula\ mass\ of\ CuO}{formula\ mass\ of\ mixture} \times mass\ of\ the\ mixture$$

$$Mass\ of\ CuO\ in\ the\ original\ sample = \frac{80g}{224g} \times 2g = 0.714g$$

118. A sample of a pure metal M weighing 2.8g was converted quantitatively to 3.2g of the oxide MO. what is the atomic mass of M? (Atomic mass: O = 16)

Solution:

Write a balanced chemical equation and determine the atomic mass of the metal, M. Let the atomic mass

$$\underset{2x}{\overset{2.8g}{2M}} + O_2 \rightarrow \underset{2(x+16)}{\overset{3.2g}{2MO}}$$

Set a proportion

$$\frac{2.8g}{2x} \times \frac{3.2g}{2(x+16)}$$

$$3.2x = 2.8\,x + 44.8$$

$3.2x - 2.8x = 44.8$

$0.4x = 44.8$

$$x = \frac{44.8}{0.4} = 112g$$

The atomic mass of the metal, M is 112g.

119. Given the reaction:

$$2HNO_3 + 3SO_2 + 2H_2O \rightarrow 3H_2SO_4 + 2NO$$

Identify:

a. Reduced substance
b. Oxidized substance
c. Reducing agent
d. Oxidizing agent

Solution:

a. Nitrogen is reduced, its oxidation state reduced from +5 to +2.
b. Sulphur is oxidated, its oxidation state increased from +4 to + 6
c. SO_2 is reducing agent
d. HNO_3 is oxidizing agent

120. What is the mass of 1.5×10^{23} atoms of magnesium? (atomic mass: Mg = 24)

Solution:

$24g = 6.02 \times 10^{23}$ atoms

$? = 1.5 \times 10^{23}$ atoms

$$? = \frac{24g \times 1.5 \times 10^{23}\ atoms}{6.02 \times 10^{23}\ atoms} = 5.98g$$

➦ The mass of 1.5×10^{23} atoms of magnesium is 5.98g.

121. How much Zinc will be required to produce 100 liters of hydrogen gas when it reacts with express hydrochloric acid at STP? (Atomic mass: Zn = 65,H = 1)

Solution:

Write balanced chemical equation and find the answer by mass – volume relationship.

The volume of one mole of any gas at STP is 22.4L.

$\underset{65g}{\overset{x}{Zn}} + 2HCl \rightarrow ZnCl_2 + \underset{22.4L}{\overset{100L}{H_2}}$ Set a proportion:

$$\frac{x}{65g} = \frac{100L}{22.4L}; x = 290.2g$$

The mass of zinc required to react with excess hydrochloric acid to form 100L of hydrogen gas is 290.2g.

122. How many liter will 3.5g of nitrogen, N_2, occupy at standard temperature and pressure?

Solution:

1 mole of N_2 = 28g of N_2 = 22.4L of N_2 at STP

$\Rightarrow$ 28g of N_2 = 22.4L of N_2

3.5g N_2 = ?

$$? = \frac{3.5gN_2 \times 22.4LN_2}{28gof\ N_2}$$

? = 2.8L of N_2

$\Rightarrow$ 3.5g of N_2 will occupy 2.8L at STP

123. How many liter of carbon dioxide, CO_2 at STP will be produced according to the reaction: $2CO + O_2 \rightarrow 2CO_2$ from 11.2 liters of oxygen and excess carbon monoxide, CO?

Solution:

$2CO(g) + \underset{1}{\overset{11.2L}{O_2}}(g) \rightarrow \underset{2}{\overset{x}{2CO_2}}(g)$ Set a proportion

$$\frac{11.2L}{1} = \frac{x}{2}$$

x = 22.4L, of CO_2

$\Rightarrow$ 22.4L of CO_2 will be produced from 11.2L of O_2 and excess CO.

124. How many grams of barium chloride will be required to react completely with 50g of iron (III) sulphate?

Solution:

Write balanced chemical equation

$$\underset{3\times 208g}{\overset{x}{3BaCl_2}} + \underset{1\times 400g^3}{\overset{50g}{Fe_2(SO_4)}} \rightarrow 3BaSO_4 + 2FeCl_3$$

Set a proportion

$$\frac{x}{3\times 208g} = \frac{50g}{400g}$$

x = 78g

➦ 78 grams of barium chloride will be required to react with 50g of iron (III) sulphate.

125. What is the coefficient of HCl when the

$KMnO_4 + HCl \longrightarrow KCl + MnCl_2 + Cl_2 + H_2O$

Use algebraic method to balance more complicated chemical equation.

$aKMnO_4 + bHCl \longrightarrow cKCl + dMnCl_2 + eCl_2 + fH_2O$

K: a = c

Mn: a = d

O: 4a = f

H: b = 2f

Cl: b = c + 2d + 2e

Assume a = 1

➦ $a = c = d = 1$

$4a = f$ $\quad a = 1$

$4 \times 1 = f$ $\quad b = 8$

➦ $f = 4$ $\quad c = 1$

➦ $\Rightarrow b = 2f$ $\quad d = 1$

$b = 2 \times 4$ $\quad e = 5/2$

$b = 8$ $\quad f = 4$

➦ $b = c + 2d + 2e$ $\quad a = 2 \times 1 = 2$

$8 = 1 \times 2 \times 1 + 2e$ $\quad b = 8 \times 2 = 16$

$8 = 3 + 2e$ $\quad c = 1 \times 2 = 2$

$2e = 8 - 3$ $\quad d = 1 \times 2 = 2$

$e = 5/2$ $\quad e = 5/2 \times 2 = 5$

$f = 4 \times 2 = 8$

➦ $2KMnO_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 5Cl_2 + 8H_2O$

➦ The coefficient of HCl is 16.

126. What is the molecular mass of a gas if 150 mL of the gas measured at STP, weighs 0.228g?

Solution:

mole of the gas at STP = 22400mL

1 mole of the gas = Molecular mass of the gas (M)

➦ $M = 22400mL$

$0.228g = 150mL$

$$M = \frac{0.228g \times 22400mL}{150mL}$$

$M = 34g$

127. How many liters of ammonia are formed when. 36 liters of the reacts with 13 liters of N_2?

Solution:

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

Identify the limiting reactant by dividing the given volume to its coefficient.

$$V_{N_2} = \frac{13L}{1} = 13L$$

$$V_{H_2} = \frac{36L}{3} = 12L$$

➥ H_2 is limiting reactant

$$\therefore N_2(g) + \overset{36L}{3}H_2(9) \rightarrow 2N\overset{x}{H}_3(g)$$

Set a proportion

$$\frac{36L}{3} = \frac{x}{2}$$

$$x = \frac{2 \times 36L}{3} = 24L of\ NH_3$$

24L of NH_3 will be formed

128. In 17g of ammonia there are 14g nitrogen and 3g of hydrogen. The law which governs this statement is _____.

Solution:

In ordinary chemical reaction, the total mass of reactants is equal to the total mass of products formed. This statement is referred as the law of conservation of mass or atom or the law of conservation of matter.

129. If the equilibrium concentrations found in the reaction

$2A(g) + B(g) \rightleftharpoons C(g)$ are $[A]$ 2.4×10^{-3} M,

$[B] = 4.6 \times 10^{-3}$ M and $[C] = 6.2 \times 10^{-3}$ M, then what is the value of Kc?

Solution:

$$K_c = \frac{[C]}{[A]^2[B]}$$

$$K_c = \frac{6.2\times10^{-3}\,M}{\left(2.4\times10^{-2}\,M\right)^2\left(4.6\times10^{-3}\,M\right)}$$

$$K_c = 2.34 \times 10^3\ M^{-2}$$

130. Find the number of grams left over of the reagent in excess for the following equation, if the reaction began with 5g of chromium (III) oxide, Cr_2O_3, and 12g of carbon tetra chloride CCl_4?

$$Cr_2O_3 + 3CCl_4 \rightarrow 2CrCl_3 + 3COCl_2$$

Solution:

First identify the limiting reactant:

$$n_{Cr_2O_3} = \frac{m_{Cr_2O_3}}{coeff\ M_{Cr_2O_3}} = \frac{5g}{1\times152g/mol} = 0.033mol$$

$$n_{CCl_4} = \frac{M_{CCl_4}}{coeff\ M_{CCl_4}} = \frac{12g}{3\times154g/mol} = 0.026mol$$

➦ CCl_4 is limiting reactant. Then find the mass of Cr_2O_3 that is consumed.

$$\underset{1\times152g}{\overset{x}{Cr_2O_3}} + \underset{3\times154g}{\overset{12g}{3CCl_4}} \rightarrow 2CrCl_3 + 3COCl_2$$

Set a proportion

$$\frac{x}{152g} = \frac{12g}{3\times154g}$$

x = mass of Cr_2O_3 consumed = 3.95g

➦ Mass of Cr_2O_3 left = 5g – 3.95g = 1.05g

131. How many chloride ions are there in 2 moles of $MgCl_2$?

Solution:

Number of ions in ionic compound = Number of moles of compound × subscripts of the ion × 6.02×10^{23}

Number of Cl ions in 2 mol of $MgCl_2$ =

$2mol \times 2 \times 6.02 \times 10^{13}$ ion/mol = 2.408×10^{24} Cl^- ions

132. How many atoms are present in 64g of SO_3?

Solution:

$$Total\ number\ of\ atoms\ in\ a\ given\ molecule = \frac{given\ mass}{molecular\ mass} \times sum\ of\ subscripts \times 6.02 \times 10^{23}$$

Number of atoms present in 64g of SO_3 = 1.93×10^{24} atoms

133. 3 moles of $(NH_4)_2HPO_4$ contains _____ moles of hydrogen atoms

Solution:

Number of moles of atoms in a compound = Number of moles of compound × subscript of the atom

Number of moles of hydrogen atoms =3mole×9 = 27 moles

134. How many grams of Zinc (Zn) are present in 2.5 mol of Zn?

Solution:

$n = \frac{m}{M}$, M = 65g/mol

m = n × M = 2.5mol × 65 g/mol = 162.5g

135. Calculate the mass percent of $MgSO_4$ in $MgSO_4 . 7H_2O$.

Solution:

$$mass\ oercent\ of\ MgSO_4 = \frac{mass\ of\ MgSO_4}{mass\ of\ MgSO_4} \times 100$$

Mass of $MgSO_4$ = 24 + 32 + 4 × 14 = 120g

Mass of $MgSO_4.7H_2O = 120g + 7\times\ 18g$

Mass of $MgSO_4.7H_2O = 246g$

➦ $Mass\,percent\,of\,MgSO_4 = \frac{120g}{246g}\times 100$

Mass percent of $MgSO_4 = 48.78\%$

136. How many moles of water will be produced from the complete combustion of 2 moles of benzene, C_6H_6?

Solution:

First, write balanced chemical equation and determine the answer by mole – mole relationship.

➦ $\overset{2mol}{2C_6H_6} + 15O_2 \rightarrow 6\overset{x}{H_2O} + 12CO_2$

Set a proportion:

$$\frac{2mol}{2} = \frac{x}{6}$$

x = 6 moles of H_2O

Thus, 6 moles of water will be produced

137. How many grams of CO_2 would be formed from the complete combustion of 3 moles of ethane, C_2H_6?

Solution:

$$\overset{3mol}{2C_2H_6} + 7O_2 \rightarrow \overset{x}{4CO_2} + 6H_2O$$
$$\underset{2mol}{} \qquad\qquad \underset{4\times 44g}{}$$

$$\frac{3mol}{2mol} = \frac{x}{4\times 44g}$$

x = 264g of C_2H_6

Thus, 246 grams of CO_2 would be formed.

138. A chemist wants to make 20 kilograms of methane, CH_4. How many molecules is that?

Solution:

Molar mass of CH_4 = 16g/mol

16g of CH_4 contains $6.02 \times 10^{23} CH_4$ molecules

20kg = 20,000g

Number of CH_4 molecules =

$$20{,}000g\,CH_4 \times \left(\frac{1mol}{16g}\right) \times \left(6.02 \times 10^{23}\,\frac{molecule}{mol}\right) = 7.525 \times 10^{26}\ molecules$$

$= 7.525 \times 10^{26}$ molecules

139. How many moles of hydrogen bromide will be produced from 5.6 liters of hydrogen gas reacts with excess bromine, at STP.

Answer: 0.5mol

140. What volume of nitrogen, N_2 will be produced when 80L of ammonia, NH_3 reacts with Copper (II) oxide, CuO.

$3CuO + 2NH_3 \rightarrow 3Cu + N_2 + 3H_2O$

Answer: 40L

141. 64g of sulphrdioxide, SO_2, is burned in air to produce sulphur trioxide, SO_3, If the percentage yield of SO_3 is 90% find the actual yield of SO_3.

Solution:

Find the theoretical yield from the balanced

$$chemical\,equation: \underset{2\times 64g}{\overset{64g}{2SO_2}} + O_2 \rightarrow \underset{2\times 80g}{\overset{x}{2SO_3}}$$

$$\frac{64g}{2\times 64g} = \frac{x}{2\times 80g}$$

$x = 80g$ of SO_3

➥ The theoretical yield = 80g

Actual yield = ?

Percentage yield = 90%

$$Percentage\ yield = \frac{Actual\ yield}{Theoretical\ yield} \times 100$$

$$Actual\ yield = \frac{Percentage\ yield \cdot Theoretical\ yield}{100}$$

$$Actual\ yield = \frac{90 \cdot 80g}{100}$$

Actual yield = 72g of SO

142. Iron, Fe, can be produced from its ore, iron (III) oxide, Fe_2O_3, by reacting with coke, C, at high temperature according to the equation: $Fe_2O_3(s) + 3C(s) \rightarrow 2Fe(l) + 3CO(g)$ In a certain operation

2.86×10^4 kg of Fe_2O_3 is reacted with 9.82×10^3 kg of C.

a. Calculate the theoretical yield by identifying the limiting reactant.

b. Calculate the percentage yield of iron if 1.52×10^3 kg is actually obtained.

Solution:

a. $$n_{Fe_2O_3} = \frac{m}{coeff\ M} = \frac{2.86 \times 10^1\ kg}{1 \times 160g/mol} = 178.75kmol$$

$$n_C = \frac{m}{coeff\ M} = \frac{9.82 \times 10^3\ kg}{3 \times 12g/mol} = 272.78kmol$$

- Fe_2O_3 is limiting reactant. By mass – mass relationship find the theoretical yield.

$$\overset{2.86 \cdot 10^4\ kg}{Fe_2O_3} + 3C \rightarrow \overset{x}{2Fe} + 3CO$$
$$\underset{1 \cdot 160g}{} \qquad \underset{2 \cdot 56g}{}$$

$$\frac{2.86 \times 10^4\ kg}{160g} = \frac{x}{112g}$$

$x = 2.002 \times 10^4$kg = Theoretical yield

b. $$Percentage\ yield = \frac{Actual\ yield}{Theoretical\ yield} \times 100$$

$$Percentage\ yield = \frac{1.52 \times 10^4\ kg}{2.002 \times 10^4\ kg} \times 100 = 76\%$$

143. Find the oxidation state of:

a. Br in $HBrO_4$

b. S in $Fe_2(SO_3)_3$

c. N in N_2O_5

d. P in PH_4^+

Solution:

a. $\overset{+1}{H}\overset{x}{Br}O_4^{-2}; 1\times1+x+4(-2)=0$

$x - 8 + 1 = 0$

$x = +7$

The oxidation state of Br in $HBrO_4$ is +7.

b. $\overset{+3}{Fe}_2\left(\overset{x-2}{SO_3}\right)_3$

$2(3) + 3(x) + 9(-2) = 0$

$6 + 3x - 18 = 0$

$3x - 12 = 0$

$x = +4$

The oxidation state of S in $Fe_2(SO_3)$ is +4.

c. $\overset{x}{N}_2\overset{-2}{O}_5$

$2(x) + 5(-2) = 0$

$2x - 10 = 0$

$2x = 10$

$x = +5$

The oxidation state of N in N_2O_5 is +5.

d. $\overset{x}{P}\overset{-2}{H}_4$

$x + 4(+1) = +1$

$x + 4 = +1$

$x = -3$

The oxidation state of P in PH^+_4 is -3.

144. The percentage of water in $Na_2CO_3.xH_2O$ is 62.9%. What is the value of x?

Solution:

$$\%\,\text{water} = \frac{mass\,water}{mass\,Na_2CO_3\,XH_2O} \times 100$$

$$62.9 = \frac{18x}{106+18x} \times 100$$

$$66.7 + 11.322x = 18x$$

$$66.7 = 18x - 11.322x$$

$$66.7 = 6.678x$$

$$x = 9.99 = 10$$

The value of x in $Na_2CO_3.XH_2O$ is 10. Thus, the formula is $Na_2CO_3.10H_2O$

145. If the equilibrium constant for the reaction: $A + B \rightleftharpoons 2D$ is 10 then k_c for $2D \rightleftharpoons A + B$ is

Solution:

If we reverse a given reversible reaction, the new equilibrium constant is the reciprocal of the original equilibrium constant,

$$k_{c(new)} = \frac{1}{k_{c(original)}}$$

$$k_{c(new)} = \frac{1}{10} = 0.1$$

146. Given the reaction, what mass of gaseous carbon dioxide, CO_2 can be absorbed by 2kg of lithium hydroxide, LiOH?

$$2LiOH + CO_2 \longrightarrow Li_2CO_3 + H_2O$$

Solution:

2Kg = 2000g

$$\overset{2000g}{\underset{2\times24g}{2LiOH}} + \overset{x}{\underset{1\times44g}{CO_2}} \rightarrow Li_2CO_3 + H_2O$$

$$\frac{2000g}{48g} = \frac{x}{44g}$$

$$x = \frac{44g \times 2000g}{48g} = 1833.3g$$

Hence, 1833.3g of CO_2 is consumed.

147. Write the chemical formula for:

a. Iron (II) nitrite

b. Molybdenum (VI) sulfate

c. Lead (II) sulphide

d. Copper (II) phosphate

Solution:

a. $Fe(NO_2)_2$

b. $Mo(SO_4)_3$

c. PbS

d. $Cu_3(PO_4)_2$

148. For the equilibrium: $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$, $k_p = 16.75$ atm at 500°C. What is the partial pressure of CO_2 that is in equilibrium with CO whose partial pressure is 0.50 atom?

Solution:

The equilibrium constant expression in terms of partial pressure is represented by k_p.

$$k_p = \frac{P_{co}^2}{P_{co_2}}$$

$$P_{CO_2} = \frac{P_{co}^{\ 2}}{k_p} \frac{(0.5atm)^2}{16.75atm}$$

$$P_{CO_2} = 0.015atm$$

149. What volume of 2.5moles of oxygen gas would occupy at STP?

Solution:

n = 2.5mol

At STP, T = 273K, P = 1atm

PV = nRT

$$v = \frac{nRT}{P}$$

$$V = \frac{2.5mol \times 0.082 \frac{atmL}{molk} \times 273k}{1atm}$$

$V = 55.965\text{L} \approx 56\text{L}$

or

At STP:

1 mol O_2 = 22.4L

2.5mol O_2 = ?

$$? = \frac{2.5mol\,O_2 \times 22.4L}{1atmO_2}$$

? = 56L

➥ 2.5 mol of O_2 occupy 56L at STP

150. A gas at STP which contain 1.204×10^{24} atoms and forms a diatomic molecules will occupy a volume of _____

Solution:

1 mole of gas = 6.02×10^{23} molecules

For a di atomic molecules, number of molecules = number of atoms divided by two

➥ number of molecules of the gas = $\frac{1.204}{2} \times 10^{24}$

$= 6.02 \times 10^{23}$ molecules

➥ At STP, 6.02×10^{23} molecules occupy 22.4L

151. How many moles of water is formed from the complete combustion of 5 moles of hexane (C_6H_{14})?

Solution:

Write the balanced chemical equation

$$\overset{5mol}{\underset{2}{2C_6H_{14}}} + 19O_2 \rightarrow 12CO_2 + \overset{x}{\underset{14}{14H_2O}}$$

$$\frac{5mol}{2} = \frac{x}{14}$$

x = 35 mol of H_2O

➥ 35 mol of H_2O is formed from the complete combustion of hexane.

152. What is the number of moles of one liter of PH_3 at STP?

Solution:

v = 1L

At STP, P = 1 atm, T = 273k

Pv = nRT

$$n = \frac{Pv}{RT}$$

$$= \frac{1atm \times 1L}{0.082L \times 273k} = 0.045mol$$

or At STP 1 mol of PH_3 = 22.4L

? = 1L

$$? = \frac{1mol\,PH_3 \times 1L}{22.4L} = 0.045mol$$

➥ The number of mole of one liter of PH_3 at STP is 0.045 mol.

153. If 100mL of SO_2 and 100 mL of O_2 are reacted to form SO_3:

a. What volume of SO_3 is formed?

b. How many mL of excess reactant left?

Solution:

a. Write the balanced chemical reaction:

$$2SO_2 + O_2 \rightarrow 2SO_3$$

Divide the given volume to its coefficient to identify the limiting reactant.

➦ $V_{SO_2} = \frac{100mL}{2} = 50mL$

$$V_{O_2} = \frac{100mL}{1} = 100mL$$

➦ SO_2 is limiting reactant

b. $\overset{100ml}{2SO_2} + O_2 \rightarrow \overset{x}{2SO_3}$

$$\frac{100mL}{2} = \frac{x}{1}$$

x = 50mL

➦ 50mL of O_2 is required to react with 100 mL of SO_2

V_{O_2} left = 100mL – 50mL

V_{O_2} left = 50mL

➦ 50 mL of O_2 is left.

154. A sample of potassium chlorate is decomposed by heat, giving 1.4g of a solid residue and 0.6g of oxygen gas. What is the percentage of oxygen in potassium chlorate?

Solution:

$2KClO_3 \rightarrow 2KCl + 3O_2$

x 1.49g 0.96g

According to the law of conservation of mass, mass of reactant is equal to mass of product.

➦ x = 1.49g + 0.96g

x = 2.45g

➦ mass of $KClO_3$ = 2.45g

$$\%O_2 = \frac{mass\ of\ O_2}{mass\ of\ KClO_3} \times 100$$

$$\%O_2 = \frac{0.96g}{2.45g} \times 100$$

$$\%O_2 = 39.2\%$$

155. Mercuric oxide contains 92.6% mercury by mass. Calculate the mass of mercuric oxide which contains 10g of oxygen.

Solution:

% oxygen = 100 - % mercury

% oxygen = 100 – 92.6

% oxygen = 7.4%

$$\%oxygen = \frac{mass\,of\,oxygen}{mass\,of\,mercuri\,oxide} \times 100$$

$$Mass\,of\,mercuric\,oxide = \frac{mass\,of\,oxygen}{mass\,of\,mercuric\,oxide} \times 100$$

$$Mass\,of\,mercuric\,oxide = \frac{10g}{7.4g} \times 100$$

Mass of mercuric oxide = 135.14g

156. Find the formula of a hydrate with compositions: Calcium = 18.2% Chlorine = 32.42% water = 49.31% and a molar mass of 219.9 g/mol.

Solution:

Write the symbols and formula. Then place their percentage below the symbols and formula. Find the number of mole for each. Divide each number of mole to the smaller one. If fractions are available, convert into simple whole numbers.

Ca	Cl	H_2O
18.27g	32.42g	49.31g
$\frac{18.27g}{40g/mol}$	$\frac{32.42g}{35.5g/mol}$	49.31g / 18g/mol
= 0.45675mol	= 0.91324mol	= 2.7394 mol
$\frac{0.45675mol}{0.45675\ mol}=1$	$\frac{0.91324mol}{0.45675\ mol}=2$	$\frac{2.7394\ mol}{0.45675\ mol}=6$

The formula of the hydrate is $CaCl_2.\ 6H_2O$

157. Substitute symbols and formulas for names and balance the equation representing the following reactions.

a. Copper (II) carbonate decompose to copper (II) oxide and a gas

b. Barium hydroxide reacts with carbon dioxide to form barium carbonate and water.

c. Calcium carbonate reacts with hydrochloric acid to form calcium chloride, water and a gas.

d. Copper combines with sulphur to form copper (I) sulphide

Solution:

a. $CuCO_3 \longrightarrow CuO + CO_2$

b. $Ba(OH)_2 + CO_2 \longrightarrow BaCO_3 + H_2O$

c. $CaCO_3 + 22HCl \longrightarrow CaCl_2 + CO_2 + H_2O$

d. $2Cu + S \longrightarrow Cu_2S$

158. Make the following conversations.

a. 5.8×10^{24} formula units of $CaCO_3$ to moles and grams.

b. 20g of HCl to moles and molecules.

c. 0.25 mol caffeine, $C_8H_{10}O_2N_4$, to total number of atoms and mass in grams.

Solution:

a. 6.02×10^{24} formula units = 1 mol $CaCO_3$

5.8×10^{24} formula units = ?

$$? = \frac{5.8\times10^{24}\ formula units \times 1 mol\ CaCo_3}{6.02\times10^{23}\ formula units}$$

$? = 9.6$ mol $CaCO_3$

➥ 5.8×10^{24} formula units of $CaCO_3$ is equal to 9.6mol of $CaCO_3$.

$$n = \frac{m}{M},$$

$M_{CaCO_2} = 40 + 12 + 3 \times 16 = 100$g/mol

$m = n \times M$

$m = 9.6$mol $\times$ 100g/mol $= 960$g

➥ 5.8×10^{24} formula units of $CaCO_3$ is equal to 960g of $CaCO_3$

b. $n = \frac{m}{M}$, $M_{HCl} = 1 + 35.5 = 35.5$g/mol

$$n = \frac{26g}{36.5g/mol} = 0.71mol$$

➥ 26g of HCl = 0.71mol of HCl

1 mol HCl = 36.5g HCl = 6.02 $\times 10^{23}$ molecules HC

0.71mol HCl = ?

$$? = \frac{0.71mol\ HCl \times 6.02\times10^{23}\ Molecules\ HCl}{1mol\ HCl}$$

$? = 3.612 \times 10^{23}$ atoms

c. 0.25 mol $C_8H_{10}O_2N_4 = 3.612 \times 10^{23}$ atoms.

$m = n \times M, M = 8 \times 12 + 10 \times 1 + 2 \times 16 + 4 \times 14$

M = 194g/mol

m = 0.25 mol $\times$ 194g/mol

m = 48.5g

Supplementary exercise on Unit - 4

I. Choose the correct answer from the given alternatives.

1. All of the following are chemical reactions except
   A. Souring of milk
   B. Fermenting tella
   C. Heating of sugar
   D. Formation of cloud

2. Which of the following represents the chemical equation for burning of magnesium in air?
   A. $Mg + O \rightarrow MgO$
   B. $2Mg + O_2 \rightarrow 2MgO$
   C. $Mn + O \rightarrow MnO$
   D. $4Mn + 3O_2 \rightarrow 2Mn_2O_3$

3. Which of the following is single displacement reaction?
   A. $2CO + O_2 \rightarrow 2CO_2$
   B. $BaCO3 \rightarrow BaO + CO_2$
   C. $Mg + 2HI \rightarrow MgI_2 + H_2$
   D. $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + CO_2 + H_2O$

4. The equilibrium constant (K_c) expression for
   $AgCl\,(s) \rightarrow Ag^+\,(aq) + Cl^-(aq)$ is
   A. $\frac{[Ag^+][Cl^-]}{[AgCl]}$
   B. $[Ag^+][Cl^-]$
   C. $[Ag^+][Cl^-]$
   D. $\frac{1}{[Ag^+][Cl^-]}$

5. How many moles of Al_2O_3 can be formed when a mixture of 0.36 moles of Al and 0.36 mol of O_2 gas is ignited?
   A. 0.72mol
   B. 0.12 mol
   C. 0.46 mol
   D. 0.18mol

6. What is the oxidation number of chlorine in ClO_4^- ?
   A. +1
   B. +3
   C. +5
   D. +7

7. Which of the following is powerful oxidizing agent?
   A. Chlorine
   B. Fluorine
   C. iodine
   D. nitrogen

8. How many grams of H_2O are produced in burning 2.25 liters of $H_2(g)$ at STP?

   A. 40.5g
   C. 1.8g
   B. 81g
   D. 18g

9. Which one of the following contains the largest number of molecules? (At. Mass: C = 12. H = 1, N = 14, 0 = 16)

   A. 1.0g CH_4
   C. 1.0g H_2O
   B. 1.0g CO_2
   D. 1.0g N_2O_4

10. The percent by mass of water in $CuSO_4.5H_2O$ is

    A. 7.2%
    C. 36.1%
    B. 14%
    D. 56.4%

11. Which of the following statements is not true for the following reaction?

$$Fe^{2+} + MnO_4^- + H^+ \rightarrow Fe^{3+} + Mn^{2+} + H_2O$$

    A. Fe^{2+} loses electrons
    C. MnO_4^- is the oxidizing agent
    B. Fe^{2+} is the reducing agent
    D. H^+ gains electrons

12. How many grams of phosphorus are there in 20g of $Ca_3(PO_4)_2$?

    A. 62g
    C. 31g
    B. 4g
    D. 15.5g

13. In which of the following species does nitrogen have the lowest oxidation number?

    A. N_2H_4
    C. NH_3
    B. N_2O_4
    D. N_2O_3

14. Which of the following has the highest percentage by mass of nitrogen?

    A. N_2O
    C. NO_2
    B. NO
    D. N_2O_3

15. Which of the following is false about 2 moles of $(NH_4)_2Cr_2O_7$?

    A. 4 moles of nitrogen atoms
    C. 2 moles of chromium atoms
    B. 16 moles of hydrogen atoms
    D. 14 moles of oxygen atoms

16. When the following equation is balanced:
$MnO_2 + KOH + O_2 \rightarrow K_2MnO_4 + H_2O$, the coefficient of oxygen (O_2) is

A. 5 B. 4 C. 2 D. 3

17. A sample of pure metal X weighting 2.70g was converted to 3.76g of the oxide XO_2. What is the atomic weight of X?

A. 81.50 C. 56.25

B. 40.75 D. 86.4

18. The value of heat of reaction (ΔH) for the forward reaction is – 10kJ. What will be the value of ΔH for the reverse reaction?

$A + B \rightleftharpoons C + D$

A. -10KJ C. -20KJ

B. +10kJ D. +2-KJ

19. If a molecule of CX_2 has a relative mass of 3.67 times that of a carbon atom, then the atomic mass of X would be

A. 44 C. 32

B. 16 D. 22

20. Which of the following affects the value of equilibrium constant?

A. Addition of a catalyst C. Change in total pressure

B. Change in concentration of the reactant D. Change in temperature

21. Given the equation: $2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$ At STP, how many liters of hydrogen will react with 20 liters of oxygen?

A. 40L C. 10L

B. 20L D. 22.4L

22. Which of the following is wrong regarding the rate of a chemical reaction?

A. A catalyst increases the rate of a reaction by lowering activation energy.

B. The rate of reactions between ions in general is fast

C. The rate of gaseous reaction decreases by decreasing volume.

D. Rate of reaction depends on the nature of reactants.

23. The gaseous reaction $A + 2B \rightleftharpoons C$ is exothermic. Which of the following factors favors forward reaction?

A. Low pressure
B. Low temperature
C. Introducing C
D. Presence of catalyst

24. Which of the following changes will always increase the amount of products at equilibrium?

A. An increase in pressure
B. An increase in temperature
C. Increase the reactant concentration
D. Addition of catalyst

25. The formula mass of $Al_2(SO_4)_3$ is

A. 342
B. 171
C. 180
D. 310

26. For a chemical reaction to takes place, one of the following preconditions is not necessarily important. Which one is it?

A. Reactant particles must have sufficient energy of collision
B. Reactant particles should come in contact to one another
C. Reactant particles should have similar sizes
D. Reactant particles should have proper orientation in space.

27. Which one of the following statements is true about a chemical reaction?

A. Reactants combine in certain specific proportions.
B. The amount of products of a chemical reaction depends
C. on the excess reactant.
D. It is usually common to get 100 percent yield in chemical reaction.
E. If the reactants are impure, the actual yield exceeds the theoretical yield.

II. Give short answer

28. An element X forms both a dichloride (XCl_2) and a tetrachloride (XCl_4). Treatment of 10g XCl_2 with excess chlorine forms 12.55g of XCl_4. What is the atomic mass of X.

29. A sample of glucose, $C_6H_{12}O_6$ contains 4.0×10^{22} atoms of carbon. How many atom of hydrogen dose it contain?

30. In an experiment to prepare bromobenzene according to the equation, $C_6H_6 + Br_2 \rightarrow C_6H_5Br + HBr$. A student reacted 20g of C_sH_5 with 0.310 mol of Br_2. If 28g of C_6H_5 Br was obtained

    a. What is the limiting reactant?

    b. How many grams of the excess reactant left over?

    c. What was the percentage yield?

31. Answer the following questions. Based on the diagram below

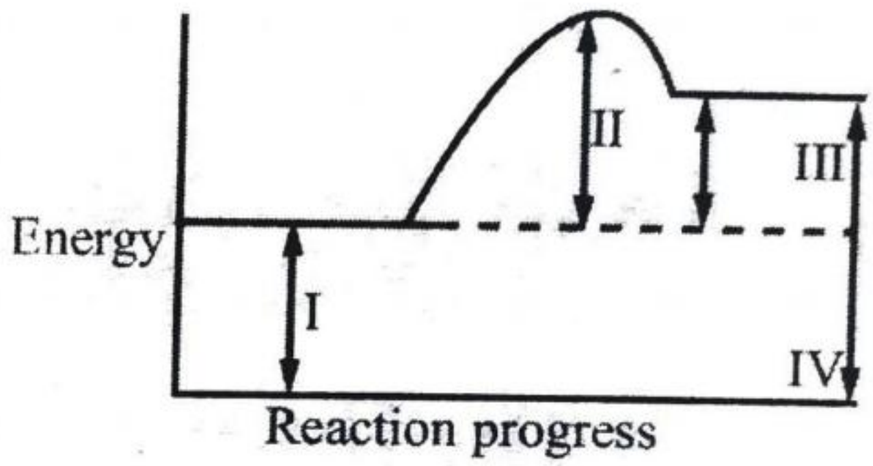
    a. Enthalpy change of the reaction _____

    b. Heat content of reactant _____

    c. Heat content of product _____

    d. Activation energy of the reaction _____

    e. The diagram represents _____ reaction.



32. Change the following chemical equations to word equations

    a. $Fe + H_2SO_4 \rightarrow FeSO_4 + H_2$

    b. $P_4 + 5O_2 \rightarrow 2P_2O_5$

33. Write the uses of chemical changes.

34. Complete and balance the following equation

    a. $Zn + CuSO_4 \rightarrow$

    b. $Na_2CO_3 + H_2S \rightarrow$

35. When 56.0g of calcium and excess amount nitrogen gas undergo a reaction that has a 90% yield, what is the actual yield?

36. Define homogeneous equilibrium and heterogeneous equilibrium and give example for each.

37. Classify the following as physical or chemical properties of substance

    a. Odor
    b. Fermentation
    c. Density
    d. Rusting
    e. Respiration
    f. Malleability
    g. contraction
    h. Freezing point
    i. Reactivity
    j. hardness

UNIT 5

Physical states of Matter

Unit Outline:

5.1 Kinetic theory of Matter

5.2 The kinetic theory of gases

5.3 Gas laws

5.4 The liquid State

5.5 The Solid State

Introduction:

Every material around our environment is matter.

Matter is anything that has mass and occupies space.

Example

The salt we eat, the water we drink and the air we breath in and out are matter.

- All matter, at least in principle, can exist in three states: **solid, liquid and gas**

1. **Solids** are rigid objects with definite shapes and volumes.
2. **Liquids** are less rigid than solids and are fluid. They are able to flow and assume the shape of their containers.
3. **Gases** are fluids but unlike liquids they can expand indefinitely

- These three states of matter are inter convertable.
- In addition to three states of matter, there is also a fourth states of matter, called the **plasma state**. This state of matter exist at extremely high temperature. No material exists in the plasma state.

5.1 Kinetic theory of matter

- All matter is composed of particles which are in constant motion
- The particles have kinetic and potential energy.
- The difference between the three states of matter is because of their energy content and the motion of particles

Properties of matter

1. Properties of solids

- They have definite shape and volume.
- They are incompressible because the particles are very close each other.

- They have higher boiling and melting points than liquids and gases.
- They are not fluid
- They do not diffuse
- They exist at low temperature and high pressure
- Their kinetic energy is much less than attraction force of particles.
- They are incompressible

2. Properties of liquids

- They have definite volume but no definite shape
- They are slightly compressible
- They are fluids
- They have lower boiling point than solids.
- Their particles have more restricted motion than gas.
- They are slightly compressible.
- They exist at intermediate temperature and pressure
- Their kinetic energy is almost equal to attraction force of particles.

3. Properties of gases

- They have neither define shape nor definite volume.
- They have lower boiling point than liquids and solids.
- They are highly compressible because particles are very far apart.
- They exert pressure in all direction.
- They diffuse spontaneously into each other
- They exist at high temperature and low pressure.
- Their kinetic energy is much greater than the attraction force between particles.

5.2 The kinetic theory of gases

- Gas molecules are very far apart and have high freedom of motion.
- Collisions between molecules of gases is perfectly elastic.

- The attraction force between gas molecules is almost negligible.
- The volume of the gas is negligible compared to the total volume of the gas
- The average kinetic energy of the gas is directly proportional to the absolute temperature of gas.

Illustrative Examples

1. According to kinetic theory of matter: (EGSECE...2008E.C)
   A. All matter is made up of small particles that has space between them.
   B. Particles in a liquid have less energy that in a solid.
   C. Particles in gases are held together by strong attractive forces.
   D. Particles in solid move freely than in a liquid.

Explanation:

- Energy of particles: gas > liquid > solid
- Freedom of motion: gas > liquid > solid
- Compressibility: gas > liquid > solid
- Attraction force: solid > liquid > gas

Answer: A

2. According to the kinetic molecular theory of gases,
   A. The particles in gases have more energy than in liquids
   B. The distance between the particles in gases is small as compared to liquids and solids.
   C. The particles in gases move about in one direction at high sped
   D. Particles are able to vibrate but not translate.

Answer: A

3. Which of the following is not true about gases?
   A. Particles of gases are highly disorded
   B. Gases posses definite shape or volume
   C. Gases diffuse spontaneously into each other
   D. Gases are lighter than liquids and solids.

Answer: B

4. States of matter that exist at very high temperature is

Answer: Plasma state

5. Arrange the three states of matter in increasing order of

a. compressibility.

Answer: solid < liquid < gas.

b. strength of force that holds particles together.

Answer: gas < liquid < solid

6. Why gases and liquids are termed as fluids?

Answer: Because liquids and gases flow easily.

7. At a given temperature, all gases have the same kinetic energy. Why?

Explanation: Temperature and kinetic energy of a gas are directly related.

8. From ice, water, and water vapor, which one has:

a. Highest kinetic energy

b. Greatest density

c. Regular arrangement of molecules.

Solution:

a. Water vapor

b. Liquid water

c. Ice

5.3 Gas Laws

➦ Gas laws relate volume, pressure, number of mole, and temperature.

1. Volume (V)

➦ It is the space occupied by an object.

➦ The SI unit of volume is cubic meter (m^3)

$1m^3 = 1000L = 1000dm^3 = 10^6cm^3 = 10^6mL$

$1L = 1dm^3 = 1000mL = 1000cm^3$

➦ $1mL = 1cm^3 = 0.001L$

2. Pressure (P)

➦ It is force divided by area

$$\text{Pressure} = \frac{concentration}{time}$$

$$P = \frac{F}{A}$$

➦ The SI unit of pressure is pascal (pas)

$1Pa = 1N/m$

➦ Other units of pressure are, atmosphere (atm), torr, millimeter of mercury (mmHg), centimeter of mercury (cmHg)

1 atm = 760torr = 760mmHg = 101.3 kpa = 101325pas

1 torr = 1mmHg.

3. Temperature (T)

➦ Temperature is the measure of the average kinetic energy of particles.

➦ Its SI unit is Kelvin.

➦ The lowest attainable temperature is – **273.15^{0}c or OK**

➦ The Kelvin and degree celsium are related by the formula:

$K = {}^oC + 273.15$

$^oC = K - 273.15$

➦ Other unit of temperature is degree Fahrenheit (^oF)

Relationship between ^{o}C and ^{o}F

➦ The freezing and boiling points of water are 0^{o}C (32^{o}F) and 100^{0}c (212^{0}F) respectively. Substract the smallest value from both sides and by ratio method find the relationship.

$$\frac{\text{Freezing point}}{^{\circ}C-0} = \frac{\text{Boiling point}}{100-0}$$

$$\frac{^{\circ}C-0}{^{\circ}F-32} = \frac{100-0}{212-32}$$

$$\frac{^{\circ}C}{^{\circ}F-32} = \frac{100}{180}$$

$$^{\circ}C = \frac{5}{9}(^{\circ}F-32)$$

$$^{\circ}F = \frac{9}{5}.^{\circ}C+32$$

Mole

- It is the amount of a substance that contains as many elementary entities (atoms, molecules, or other particles) as there are atoms.

Exactly equal to 12 grams of the carbon – 12 isotope,

1mole $=6.02\times10^{23}$ particles (atoms, molecules formula units)

1 mole = 1gram atom = atomic mass of an element

Example:

1 mole Na = 1gram – atom Na = 23 Na

1mole = 1 gram – molecule = molecular mass of molecule

Example:

1mol H_2O = 1gram – molecules H_2O = 18g of H_2O

1mol = 1 gram – formula = formula mass of compound

Example:

1 molNaCl = 1gram – formula NaCl = 58.5gNaCl.

Illustrative Examples

9. All of the following quantities represent the same amount of component particles EXCEPT (EGSECE…2000E.C)

A. A mole
B. A gram – molecule
C. a formula unit
D. a gram atom

Explanation:

A mole = a gram atom = a gram molecule = 6.02×10^{23}.

A formula unit is $\frac{1}{6.02\times10^{23}}$

Answer: C

10. -40^0F ——— K

A. 200
B. 300
C. 273
D. 233

Solution:

$-40^0F = -40^0C$

$K = {}^0C + 273 = -40 + 273 = 233K.$

Answer: D

11. Which of the following is not equal to $0.1dm^3$?

A. 0.1L
B. 100mL
C. $0.1m^3$
D. $100cm^3$

Solution:

$0.1dm^3 = 0.1L = 100mL = 100cm^3 = 0.0001m^3$

Answer: C

12. Convert the following into atmosphere.

a. 1520torr
b. 380mmHg
c. 100cmHg

Solution:

a. 1 atm = 760 torr

$$1520\text{torr} = \frac{1520atm}{760} = 2 \text{ atm}$$

b. 1 atm = 760mmHg

➦ $380\text{mmHg} = \frac{380atm}{760} = 0.5\text{atm}$

c. 1 atm = 76cmHg

$$100\text{cmHg} = \frac{100atm}{76} = 1.32\text{atm}$$

13. Express your body temperature (37^0c) in

a. K

b. ^oF

Solution:

a. $K = {}^oC + 273 = 37 + 273 = 310k$

b. $^oF = 98.6^oF$

14. Convert the following in to Kelvin.

a. -173^0c b. 212^0F

Solution:

a. $K = {}^oC + 273 = -173 + 273 = 100K$

b. $212^oF = 100^0C$

➦ $K = {}^0c + 273 = 100 + 273 = 373K$

1. Boyle's law: The pressure–volume relationship

Boyle's law states that "the volume of a fixed amount of gas maintained at constant temperature is inversely proportional to the gas pressure."

Mathematically: $V \propto \frac{1}{P}$ (at constant temperature)

➦ $V = K.\ \frac{1}{P}$

$Vp = k$, where, k is constant

$V_1P_1 = K_1$, $V_2P_2 = k_2$,

But $K_1 = K_2 \Rightarrow$ $P_1V_1 = P_2V_2$

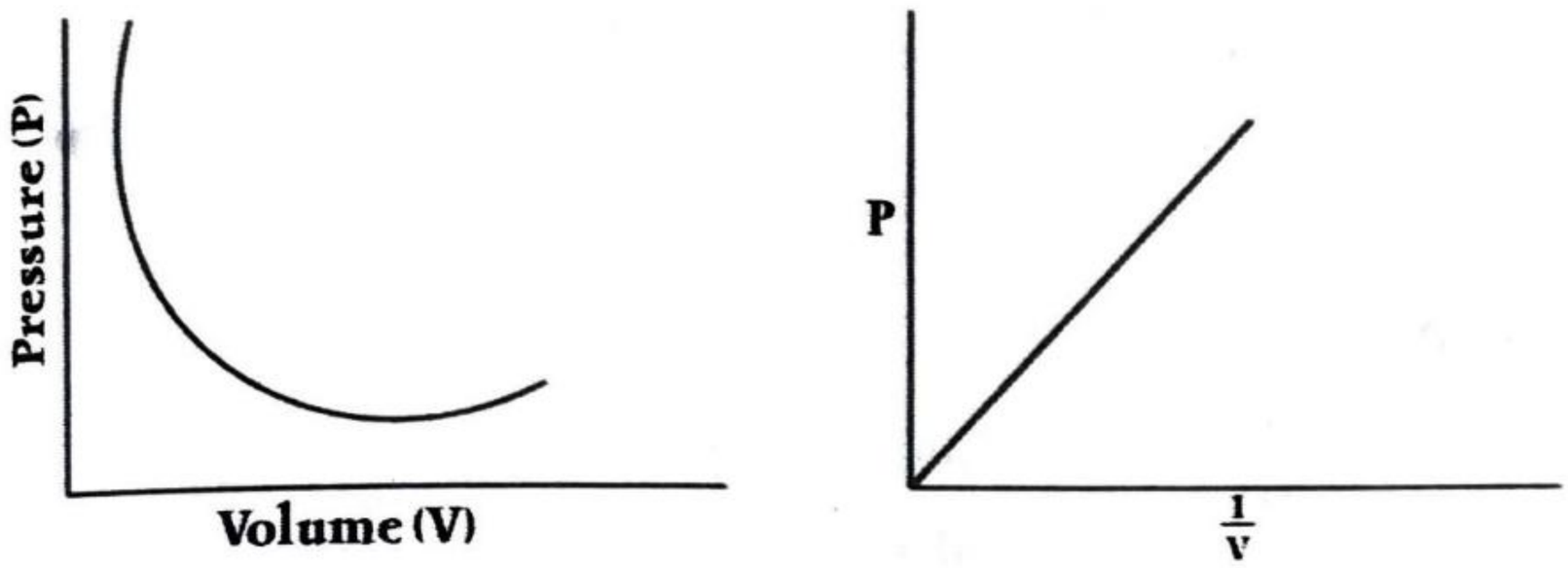


Figure 5.1 graphs showing variation of the volume of a gas sample with the pressure exerted at constant temperature.

If the volume of gas doubles the pressure halves at constant temperature

Illustrative Examples

15. For a fixed mass of gas, Boyle's law states that: (EGSECE)

A. At constant temperature, the volume is inversely proportional to the pressure.

B. At constant temperature, the volume is inversely proportional to the temperature

C. At constant pressure, the volume is directly proportional to the Kelvin temperature

D. Equal volume of all gases at same temperature and pressure have the same number of molecules.

Explanation: Boyle's law states that $V \propto \frac{1}{P}$ at constant temperature:

Answer: A

16. A sample of gas has a volume of $183cm^3$ and a pressure of 740mmHg. If the volume is changed to $116cm^3$. Assuming the temperature is held constant, what will be the new pressure. (EGSECE)

A. 856mmHg
B. 469mmHg
C. 28.7mmHg
D. 1167mmHg

Solution:

$P_1 = 740mmHg$ $V_2 = 116cm^3$

$V_1 = 183cm^3$ $P_2 = ?$

$\rightarrow P_1V_1 = P_2V_2$

$$P_2 = \frac{P_1V_1}{V_2} = 1167mmHg$$

Answer: D

17. By what factor must the pressure be increased to compress 8L of nitrogen to 2.5L?

Solution:

$V_1 = 8L$, $V_2 = 2.5L$, $P_2 = ?$ P_1

$$P_2 = \frac{8}{3}$$

18. A gas in a ballon occupies 10L at a pressure of atm. If the temperature is kept constant, at what pressure will the volume of the gas double?

Solution:

$V_1 = 10L$, $V_2 = 2V_1$, $P_1 = 1$ atm

$$P_2 = \frac{1}{2}P_1 = 0.5\ atm$$

19. A gas occupies a volume of 5L at 300 torr. The pressure increases to 400 torr at constant temperature, what is the new volume?

Solution:

Find by your own: **Answer: $V_2 = 3.75L$**

2. Charles'law: The temperature–volume relationship

➦ Charles' law states that "the volume of a fixed amount of gas maintained at constant pressure is directly proportional to the absolute temperature (temperature in Kelvin) of the gas."

Mathematically: $V \alpha T$ (at constant pressure)

$$\frac{V}{T} = K(\text{constant})$$

$$K_1 = \frac{V_1}{T_1}, \frac{V_2}{T_2} = K_2$$

But $K_1 = K_2 \Rightarrow \boxed{\frac{V_1}{T_1} = \frac{V_2}{T_2}}$

If the volume of a gas doubles, the Kelvin temperature also doubles

3. Gay Lussac's Law: The temperature – pressure relationship.

➦ Gay Lussac's law states that "the pressure of a fixed amount of gas maintained at constant volume is directly proportional to Kelvin temperature."

Mathematically = $P \alpha T$ (at constant volume)

➦ $\boxed{\frac{P_1}{T_1} = \frac{P_2}{T_2}}$

Note:

Gay - Lussac's law is simply charles' law.

Illustrative Examples

20. If you heat a 5L balloon from a temperature of 25^0c to 50^0c, its new volume will be: (EGSECE. . . 2001E.C)

A. 4.61L
B. 2.5L
C. 10L
D. 5.42L

Solution:

$V_1 = 5L, V_2 = ?$

$T_1 = 25^0c = 50 + 273 = 298K$

$T_2 = 50^0c = 50 + 273 = 323K$

$V_2T_1 = T_2V_1$

$$V_2 = \frac{323K \times 5L}{298K} = 5.42L$$

Answer: D

21. A sample of gas is sealed in a 150mL flask at 127^0c and 380mmHg. What will be its pressure if the gas is cooled to 27^0c at constant volume? (EGSECE…2000E.C)

A. 280mmHg
B. 180mmHg
C. 305mmHg
D. 608mmHg

Solution:

$P_1 = 380mmHg, \ T_1 = 127^0c = 127 + 273 = 400K$

$P_2 = ? \qquad T_2 = 27^0c = 27 + 273 = 300K$

$P_1T_2 = P_2T_1$

$$P_2 = \frac{380 \times 300}{400}$$

$P_2 = 285mmHg$

Answer: A

22. At constant pressure, by what factor of its volume will a quantity of a gas change if the temperature changes from -73^0c to 127^0c?

Solution:

$T_1 = -73^0c = 200K$

$T_2 = 127^0c = 400K$

➦ $V_2 = 2V_1$

23. A given mass of oxygen occupies $20dm^3$ at 47^0c. Determine its temperature in ^0C when the gas occupies $8dm^3$ of volume at constant pressure

Answer: -145^0c

24. At what temperature will the volume of the gas be

a. doubled?

b. tripled?

c. halved if the initial temperature is 327^0c?

Solution:

$T_1 = 327^0c = 600K$

a. $V_2 = 2V_1$

$$T_2 = \frac{V_2T_1}{V_1} = \frac{2V_1 \times 600K}{V_1} = 1200K$$

b. $V_2 = 3V_1$

$$T_2 = \frac{V_2T_1}{V_1} = \frac{3V_1 600K}{V_1} = 1800K$$

c. $V_2 = \frac{1}{2}V_1$

$$T_2 = \frac{1}{2} \times 600K = 300K$$

25. What would be the pressure of the gas, originally at 100Kpa? If the temperature is lowered from 30^0c to 20^0c at constant volume.

Answer: 96.7KPa

26. The pressure of a confined gas is doubled. Calculate the temperature if the original temperature was 0^0c. What is the change in temperature?

Solution:

$P_2 = 2P_1$

$T_1 = 0^0c = 273K$

$T_2 = 2T_1 = 546K$

$$\Delta T = T_2 - T_1 = 546 - 273 = 273K$$

or

$$\Delta T = T_2 - T_2 = 273 - 0 = 273^0c$$

The value of change in temperature is same in Kelvin and degree Celsius scales

4. The combined gas law

Temperature, volume – pressure relationship

➦ This law is the combination of Boyle's and Charles' laws

$V \propto \frac{1}{P}$ (Boyle's law)

$V \alpha T$ (Charles' law) $\Rightarrow \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$V \propto \frac{T}{P}$ (Combined gas law)

$V_1P_1T_2 = V_2P_2T_1$

Illustrative Examples

27. If the conditions on $166cm^3$ of a gas at 93000KPa and 15^0c are increased by 7kpa and 5^0c, what will be the new volume of the gas in liters? (EGSECE. . . 2008E.C)

A. 0.160L B. 160L C. 206L D. 20.6L

Solution:

$V_1 = 166cm^3 = 0.166L, V_2 = ?$

$T_1 = 15^0c = 288K$

$T_2 = 15 + 5 = 200c = 293K$

$P_1 = 93000kpa, P_2 = 93007Kpa$

$P_1V_1T_2 = V_2P_2T_1$

$V_2 = 0.169L$

Answer: A

28. A quantity of methane gas has a volume of 100L at 27^0c and 700mHg of pressure. What will be its volume at 0^0c and 760mmHg? (EGSECE)

Answer: $V_2 = 83.8L$

29. The volume of a gas at 17^0c and 1500 torr is 500mL. Calculate its volume at STP.

Solution:

$V_1 = 500mL$ $T_1 = 17^0c = 290K$

$P_1 = 1500torr$ $V_2 = ?$

at STP $T_2 = 0^0C = 273K, P_2 = 760$ torr

$V_2 = 929mL$

30. A sample of gas kept in a 2.0L container at 50^0c exerts a pressure of 700torr. If the volume is halved and temperature is tripled, what will be the final pressure of the gas?

Solution:

$V_1 = 2L, V_2 = V_1 = 1L$

$T_1 = 50^0c = 323K, T_2 = 3T_1 = 969K$

$$P_2 = \frac{V_1 P_1 T_2}{V_2 T_1} = \frac{2L \times 700 \text{ torr} \times 969K}{1L \times 323K}$$

$P_2 = 4200$ torr

31. A gas collected at 227^0c has a volume of $20dm^3$ and a pressure of 800 torr. What would the temperature of the gas be if the volume is changed to 500mL at standard pressure.

Solution:

At standard condition, $P_2 = 760$ torr, $P_1 = 800$torr

$V_1 = 20dm^3, V_2 = 500mL = 0.5dm^3$

$T_1 = 227^0c = 500K$, $T_2 = ?$

$$T_2 = \frac{V_2 P_2 T_1}{V_1 P_1}$$

$$T_2 = \frac{500mL \times 760torr \times 500K}{20,000mL \times 800torr}$$

$$T_2 = 11.88K$$

32. A certain sample of gas occupies 44.8L at 298K and 210KPa. Calculate its volume at STP.

Solution:

V1 = 44.8L. T1 = 298K, P1 = 210KPa

V_2 = ? , at STP, T_2 = 273K, P_2 = 1atm = 101.3KPas

$$V_2 = \frac{V_1 P_1 T_2}{P_2 T_1}$$

$$V_2 = \frac{44.8 \times 210 \times 243K}{101.3 \times 298K}$$

$$V_2 = 85.1L$$

5. Avogadro's Law: The volume – amount relationship

This law states that "At the same temperature and pressure, equal volumes of different gases contain the same number of molecules."

Mathematically: V α n (at constant temperature and pressure)

$$V = Kn$$

$$\boxed{\frac{V_1}{n_1} = \frac{V_2}{n_2}}$$

Example

A 5L container held 2 moles of a gas. What volume will be needed to hold 4 moles of that gas at the same temperature and pressure?

Solution:

$V_1 = 5L$ $n_1 = 2mol,$ $n_2 = 4mol,$ $V_2 = ?$

$$V_2 = \frac{4mol \times 5L}{2mol}$$

$V_2 = 10L$

6. The ideal gas Equation

➦ Let us summarize the gas laws we have seen so far:

Boyle's law: $V \propto \frac{1}{P}$ (at constant T and n)

Charles' law: V α T (at constant P and n)

Avogadro's law: V α n (at constant T and P)

➦ We can combine all three expressions to form a single master equation for the behavior of gases.

$$V \propto \frac{nT}{P}$$

$VP = nRT$ . . . . ideal gas equation

➦ Where R, the proportionality constant, is called the universal gas constant.

Standard temperature and pressure (STP)

1mole of ideal gas at STP occupies 22.4L.

➦ The value of R can be determined as follow:

$$R = \frac{PV}{nT} = \frac{1atm \times 22.4L}{1mol \times 273K} = 0.0820 \frac{atm.L}{mol.k}$$

➦ **An ideal gas** is a hypothetical gas whose pressure – volume – temperature behavior can be completely accounted for by ideal gas equation.

- The molecules of ideal gas do not attract or repel one another and their volume is negligible compared with the volume of the container.
- The ideal gas equation can be applied to determine density or molar mass of gaseous substances. $\boxed{P = \frac{dRT}{M}}$

d - density

M - molar mass

Illustrative Examples

34. Which of the following statements is true about an ideal gas? It is a hypothetical gas that obeys the gas laws (EGSECE…2008E.C)

A. Closely at high temperature and low pressure

B. Closely at high temperature and high pressure

C. At low temperature and low pressure

D. At low temperature and high pressure

Answer: A

35. What is the volume of 2.8grams of carbonmonoxide at 27^0c and 0.821atm?

($R = 0.082\ \frac{atm.L}{mol.K}$ molecular mass of CO = 28g/mol)

A. 3L

B. 30L

C. 0.27L

D. 27L

Solution:

$$n = \frac{m}{M} = \frac{2.8}{28} = 0.10\,mol$$

$$T = 27^0c = 300K$$

$$PV = nRT \Rightarrow V = \frac{0.1 \times 0.082 \times 300K}{0.821} = 3L$$

Answer: A

36. Equal volume of different gases under the same conditions of temperature and pressure have equal ___ (EGSECE…2006E.C)

A. number of atoms

B. density

C. number of molecule

D. molar mass

Answer: C

37. A 60L balloon is required to filled with helium gas at 37^0c and a pressure of 8.2 atm. What mass of helium gas is needed?

Solution:

V = 60L, T = 27^0c = 300K, P = 8.2atm, molar mass = 4g/mol

m = ?

$$m = \frac{PVM}{RT}$$

$$m = 8.2\,atm \times 60L \times 4\frac{g}{mol}$$

$$m = 0.082\frac{atm.L}{mol.k} \times 300K = 80g$$

38. Unknown gas has a density of 1.173g/L at STP. Determine its molar mass.

Solution:

At STP, T = 273K, P = 1atm

$\Rightarrow$ PV = nRT, but PM = dRT

$$M = \frac{dRT}{P}$$

$$= \frac{1.173 \times 0.082 \times 273}{1}$$

$$= 26 \frac{g}{mol}$$

39. Calculate the volume of 8g of CH_4 at 227°c and under a pressure of 760torr.

Solution:

$$V = \frac{8 \times 0.082 \times 500}{16 \times 1}$$

$$V = 20.5L$$

40. Calculate the volume of 2.5g of oxygen at STP. Find the answer by your own:

Answer: 1.75L

41. Compute the approximate density of methane, CH_4, at 20°c and 6 atm.

Answer: 4g/L

42. An electric vacuum tube was sealed off during manufacture at a pressure of 2×10^{-3} torr at 25°c. Its volume is 596mL, calculate the number of gas molecules remaining in the tube.

Answer: 3.85×10^{16} molcules

43. What is the molecular mass of a gas whose density is 3.5g/L at 27°c and a pressure of 2 atom? (EGSECE)

Solution:

$$M = \frac{dRT}{P}$$

$$= \frac{3.5 \times 0.082 \times 300}{2\ atm}$$

$$M = 44 \frac{g}{mol}$$

44. Calculate the volume occupied by 7.40g of CO_2 at STP.

Answer: 3.77L

45. A chemist has synthesized a greenish – yellow gaseous compound of chlorine and oxygen and finds that its density is 7.71g/l, at 36°c and 2.88atm. calculate the molar mass of the compound and determine its molecular formula.

Answer: 67.9g/mol, ClO_2.

7. Graham's law of diffusion

Rate of diffusion – Molar mass or density relationship.

Diffusion is the tendency of a gas to mix spontaneously with and speed through out another gas.

Graham's law of diffusion states that the rate of diffusion of a gas is inversely proportional to the square root of its density or the square root of its molar mass at constant temperature and pressure.

➦ Consider gas A

$$r_A \propto \sqrt{\frac{1}{dA}} \text{ or } r_A \propto \sqrt{\frac{1}{MA}}$$

➦ For two gases A and B

$$\frac{r_A}{r_B} = \sqrt{\frac{d_B}{d_A}} = \sqrt{\frac{M_B}{M_A}}$$

r_A - rate of gas A

dA - density of gas A

M_A - molar mass of gas A

dB – density of gas B

MB – molar mass of gas B

When gases contained in a vessel are allowed to escape through small hole, it is called **Effusion.**

The smaller the molar mass or density, the higher is the rate of diffusion or effusion.

Example:

$r_{H_2} > r_{He}$ because the molar mass of H_2 is 2g/mol while that of He is 4g/mol

Illustrative Examples

46. If an unknown gas diffuses 1.73 times faster than krypton, what is the molar mass of the unknown gas? (Molar mass of krypton = 83.80g/mol) (EGSECE. . .2008E.C)

A. 28.0g/mol
B. 63.7g/mol
C. 110g/mol
D. 251g/mol

Solution:

Let the unknown gas be X

$r_x = 1.73\, r_{kr}$ . . . . given , r_x – rate of x

, r_{kr} – rate of Kr

$$\frac{r_x}{r_{kr}} = \sqrt{\frac{M_{kr}}{M_x}}$$

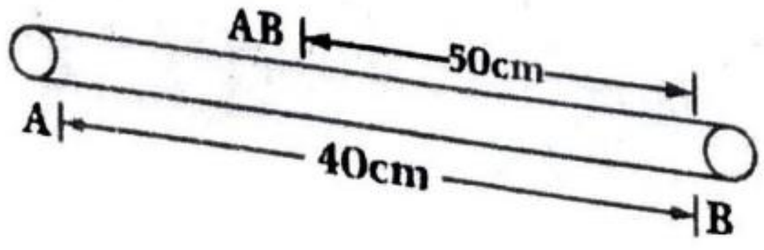
$$\Rightarrow (1.73)^2 = \left(\sqrt{\frac{83.3}{M_x}}\right)^2$$

$M_x = 28$g/mol

Answer: A

47. Two gases A and B were applied to the two ends of a 40cm long – glass tube. The two gases met at a distance of 30cm the side where B is applied, which statement is true about those gases?
    A. The molar mass of A is bigger than the molar mass of B.
    B. The rate of diffusion of A is bigger than the rate of diffusion of B
    C. The density of A is smaller than the density of B.
    D. The rate of effusion of B is smaller than the rate of diffusion of A

Explanation:



- The rate of diffusion of gas B is higher than that of gas A.
- The density and molar mass of B are smaller than the density and molar mass of A

Answer: A

48. Which one has the higher molecular speed?

A. O_2 B. N_2 C. CO_2 D. CH_4

Answer: D

49. If $140cm^3$ of methane diffuses into air in 72 seconds, how long will it take $210cm^3$ of sulphurdioxide (SO_2) to diffuse under the same condition?

Solution:

$$\frac{\frac{V_{CH_4}}{t_{CH_4}}}{\frac{V_{SO_2}}{t_{SO_2}}} = \sqrt{\frac{M_{CH_4}}{M_{SO_2}}}$$

$$\frac{\frac{140}{72}}{\frac{210}{t}} = \sqrt{\frac{16}{64}}$$

$t = 216$ sec

50. Gas A diffuses twice as fast as gas B. Gas B has a molecular weight of 60. What is the molar mass of gas A?

Answer: $M_A = 15$ g/mol

51. Helium and argon are monoatomic and their masses are 4 and 40 respectively. What is the rate of diffusion of helium?

Answer: $r_{He} = 3.16\, r_{Ar}$

52. The pressure in a vessel that contained pure oxygen dropped from 1000Kpa to 500 kpa in 40min as the oxygen leaked through a small hole in to a vacuum when the same verses was filled with another gas, the pressure dropped from 1000kpa to 500kpa in 70 min. what is the molecular weight of the second gas?

Solution: $M_{unknown\ gas} = 98\ g/mol$

$M_A = 98g/mol$

53. What is the relative rate of diffusion of H_2 and O_2?

Solution: $\Rightarrow rH_2 = 4rO_2$

54. Arrange the following gases in increasing order of rate diffusion. SO_3, CO_2, N_2, CH_4. NH_3

Answer:

$SO_3 < CO_2 < N_2 < NH_3 < CH_4$

55. A light gas diffuses faster than a heavy gas. Why?

Explanation:

This is because rate of diffusion and weight are inversely related.

5.4 The liquid state

➦ The liquid state exists at intermediate temperature and pressure. Attraction force between molecules of liquids is almost equal to kinetic energy of molecules.

Phase changes in liquids

➦ Liquids undergo two types of phase changes. These are evaporation and condensation

Evaporation/ vaporization/ is the conversion of liquid to gas It is endothermic process

Condensation is the reverse of evaporation. It is exothermic process

Heat of vaporization (ΔH_{vap}) and heat of condensation (ΔH_{cond}) are equal in magnitude but opposite in sign.

$$\Delta H_{vap} = -\Delta H_{cond}$$

The liquid and gaseous phases are in dynamic equilibrium if the rate of evaporation is equal to the rate of condensation.

Rate of evaporation is the speed at which liquids vaporize.

It is affected by:

- ➦ Temperature
- ➦ Surface area
- ➦ Intermolecular forces
- ➦ Types of liquids

1. **Temperature**: As temperature increases, the rate of evaporation also increases.
2. **Surface area**. Rate of evaporation and surface area are directly related.
3. **Intermolecular forces**. As the strength of intermolecular force increases rate of evaporation decreases.
4. **Types of liquids** – liquids are classified into two based on their tendency to evaporate.

   i **Volatile liquids** – are liquids that evaporate readily at room temperature

   - They have weak inter molecular forces between their particles.
   - They have high vapour pressure.

Example:

benzene, ethanol, diethyl ether etc

ii **Non – volatile liquids**

- They have a low tendency to evaporate at room temperature.
- They have relatively strong intermolecular forces between their particles.
- They have low vapour pressure.

Example:

Sulphuric acid, molten ionic compounds, water, mercury, etc

➦ **Vapour pressure** is the pressure exerted by vapor.

- **Boiling** occurs when a liquid heated in an open container starts to vaporize from every parts of its body. It is the conversion of liquid within as well as at the surface of liquid. However, during **evaporation**, only molecules at the surface escapes in to the vapour state.
- **Boiling point** is the temperature at which the vapour pressure of the liquid is equal to the external atmospheric pressure,
- **Normal boiling point** is the temperature at which the external pressure equal to one atmosphere.

Illustrative Examples

56. Which of the following statements explains why water vapour condenses as temperature decreases? (EGSECE. . . .2007 E.C)

A. Vapor molecules are pulled together by the force of gravity.

B. Attractive forces between vapor molecules overcome molecular motion.

C. Collisions between vapor molecules become elastic

D. Vapor molecules chemically react when they collide

Explanation:

- As vapour condenses, temperature decreases and attraction forces between particles overcome kinetic energy.

Answer: B

57. A substance has a freezing point of -38^0c and a boiling point of 356^0c. At what temperature would this substance be in its liquid state?

A. 80^0c B. -50^0c C. 375^0c D. -100^0c

Explanation: The substance exists in liquid state between its freezing and boiling points. Above its boiling point exists in gas phase but below its freezing point occurs in solid state.

Answer: A

58. Which sample of carbondioxides has a definite volume but no definite shape?

A. $CO_2(g)$ B. $CO_2(1)$ C. $CO_2(s)$ D. $CO_2(aq)$

Answer: B

59. Which of the following factors does not have an effect on the rate of evaporation of a liquid?

A. Surface area
B. Volume of liquid
C. temperature
D. intermolecular force

Answer: B

60. Among the following liquids which is expected to have the largest vapour pressure?

A. H_2O
B. CH_3CH_2OH
C. CH_3OH
D. CCl_4

Explanation: Non – polar covalent compounds with smaller size have high tendency to evaporate at low temperature. As a result, they have large vapour pressure.

Answer: D

61. Which of the following takes place at the boiling point of a liquid? (EGSECE)

A. The chemical bonds between atoms of liquids will be broken down.
B. The intermolecular forces of attraction in the liquid increase.
C. The temperature of the liquid increases
D. The vapor pressure over the liquid surface equals to atmospheric pressure.

Answer: D

62. The boiling points of three liquid A,B and C are 50^0c, 78^0c and 100^0c respectively. Which of the following is true about these liquids under the same conditions of pressure and temperature?

A. "C" exists in solid state at 105^0c
B. "B" is more volatile than "A".
C. "C" has the weakest intermolecular force of attraction.
D. "A" has the highest vapor pressure.

Explanation: As boiling point of a liquid increases, volatility decreases

Answer: D

63. The vapour pressure of benzene is greater than the vapor pressure of water under the same pressure and temperature. Why?

Answer: This is because the intermolecular force in benzene is weaker than it is in water.

64. Arrange the following liquids in increasing order of rate of evaporation. H_2O, C_6H_6, CH_3OH, C_2H_5OH

Answer: $H_2O < C_2H_5OH < CH_3OH < C_6H_6$

65. Why polar liquids have lower vapour pressure than non – polar liquids?

Answer: The strength of intermolecular force in polar liquids is higher than in non – polar liquids

5.5 The solid state

➦ The solid state exists at low temperature and high pressure. The attraction force between particles is higher than kinetic energy of particles.

Phase changes in solid

The change of solid to liquid is called **melting** or **fusion** and the reverse process is **Freezing or solidification**

$$\text{Solid} \underset{\text{Freezing}}{\overset{\text{Melting}}{\rightleftharpoons}} \text{Liquid}$$

The direct change of solid to gas without melting is called **sublimation.** The reverse of sublimation is called **re -sublimation or deposition.**

$$\text{Solid} \underset{\text{Deposition}}{\overset{\text{Sublimation}}{\rightleftharpoons}} \text{Liquid}$$

Melting and sublimation are endothermic processes while Freezing and deposition are exothermic processes.

Heat of fusion (ΔH_{fus}) is the amount of heat required to convert solid in to liquid.

Heat of solidification or crystallization (ΔH_{cry}) is the amount of energy released when liquid converted in to solid.

$\Delta H_{fus} = -\Delta H_{fus} + \Delta H_{cry.}$

Phase and energy changes in solids

- **Phase** is a homogeneous part of a system in contact with other parts of the system but separated from them by a well – defined boundary.
- **Phase changes** are transformations from one phase to another, occur when energy (usually in the form of heat) is added or removed.

Phase changes are physical changes.

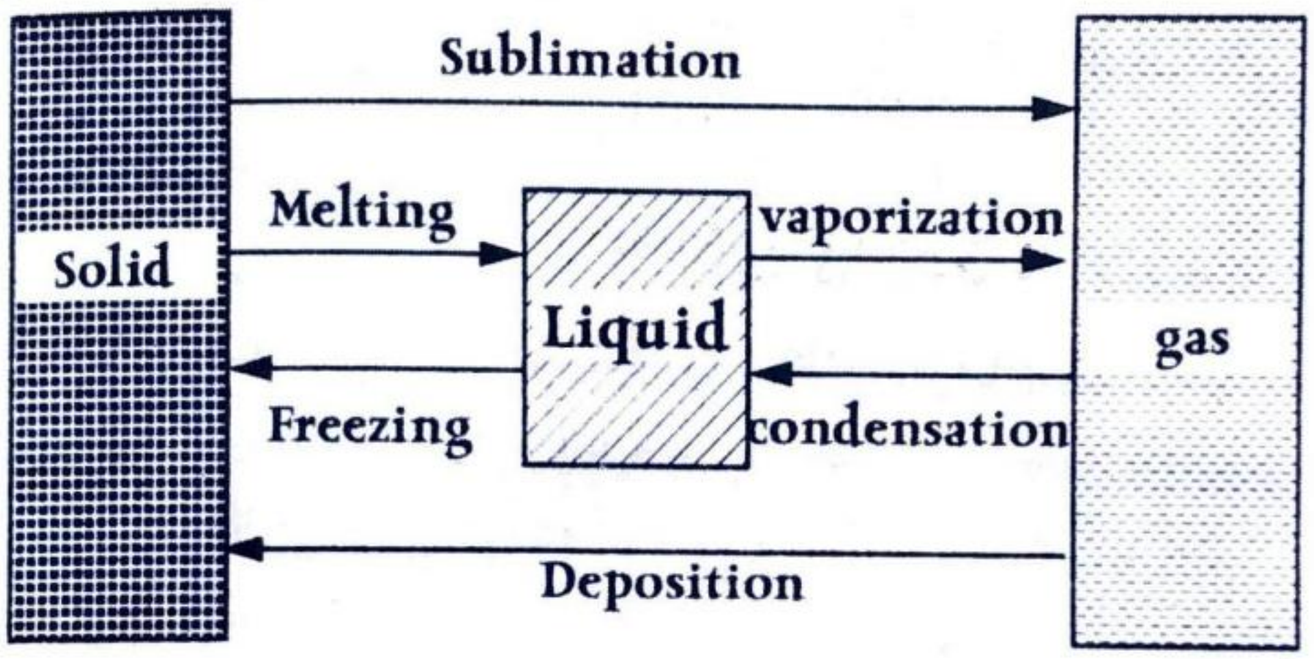
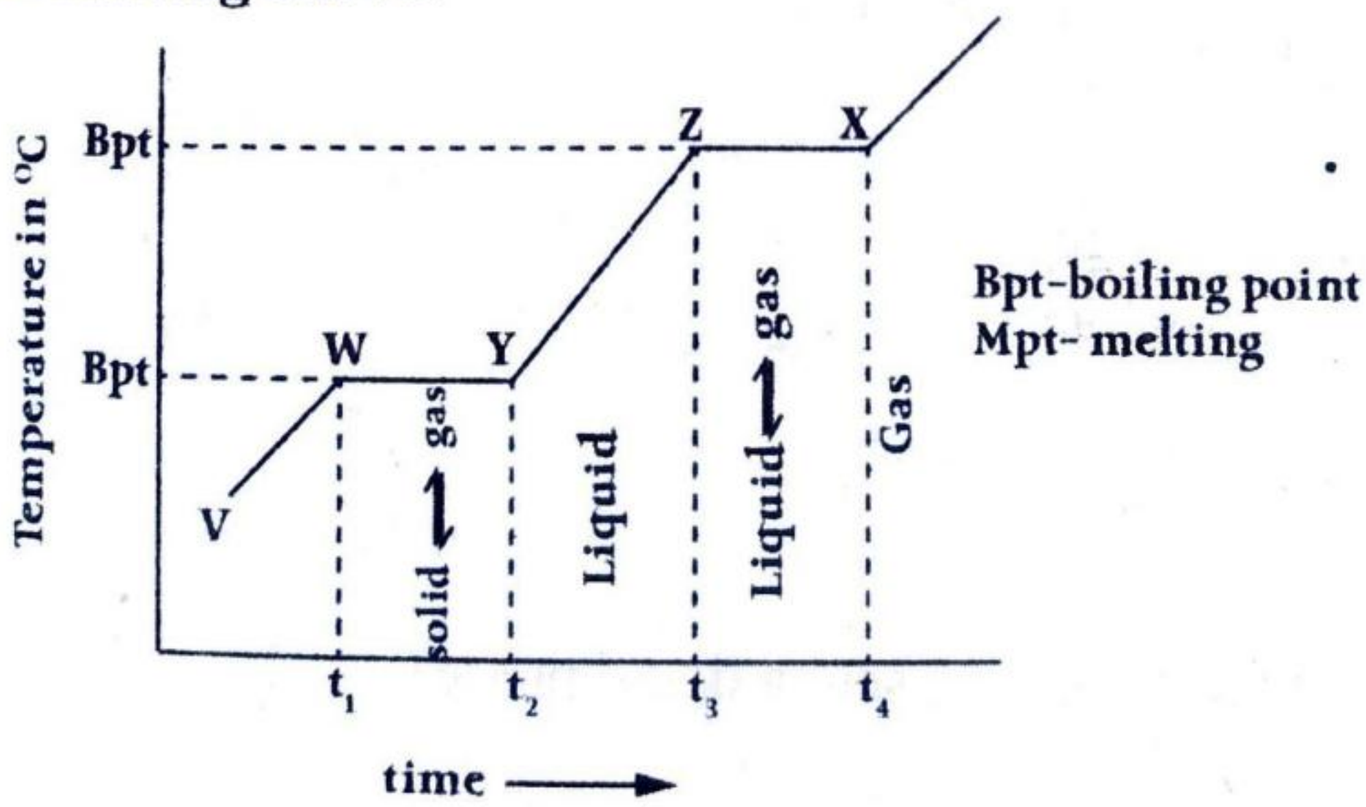


Figure 5.2 The various phase changes that a substance undergo.

The changes that take place in the physical state of a substance as the temperature is varied can be represented by a plot of temperature versus time curve called **heating curve**.



- Region V to W – temperature of solid rises
- Region W to Y – temperature remains constant (melting point)
- Region Y to Z – temperature of liquid rises.

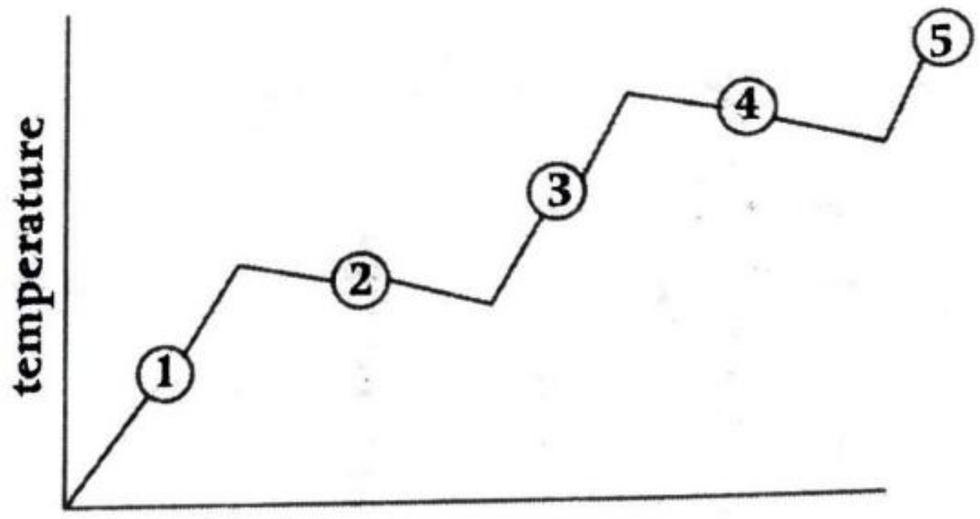
- Region Z to X – temperature remains constant (boiling point) phases
- Above region X – temperature of gases rises.
- **For conversion of**: solid to liquid and to gas
  - Temperature increases
  - Order of particles decreases
  - Kinetic energy of particles increases
  - Strength of attraction force between particles decreases.
- **For conversion of**: gas → liquid →solid.
  - temperature decreases
  - order of particle increases
  - kinetic energy decreases
  - strength of intermolecular force of attraction increases

The following exist at the same temperature

- melting point and freezing point
- boiling point and condensation point
- sublimation point and deposition point

Illustrative Examples

66. Given the following heating curve:



Which part of a curve represent a liquid phase?

A. 1 B. 2 C. 3 D. 4

Explanation:

- Part 1 – solid phase
- Part 2 – solid and liquid phase at equilibrium

- Part 3 – liquid phase
- Part 4 – liquid and gas phases at equilibrium
- Part 5 – phase of gas

Answer: C

67. The amount of heat released when one mole of liquid is converted to a solid at its freezing point is

A. Molar heat of vaporization
B. Molar heat of sublimation
C. Molar heat of solidification
D. Molar heat of fusion.

Answer: C

68. Write three solids that vaporize without melting. Iodine, solid carbondiodixe, ammoniumchloride, naphthalene.

Solved Problems

69. Explaxin why a helium weather ballon expands as it rises in the air. Assume that the temperature remains constant.

Explanation:

A helium weather balloon expands as it rises in the air because of decrease in pressure.

70. Convert – 225^0c to Kelvin

Answer: 48K

71. A sample of nitrogen gas kept in a container of volume 2.3L and at a temperature of 32^0c exerts a pressure of 4.7atm. calculate the number of moles of gas present.

Solution: V = 2.3L, T = 32^0c = 305K, P = 4.7atm

$$n = \frac{PV}{RT} = \frac{4.7\ atm \times 2.3L}{0.082 \frac{atm.L}{mol.K} \times 305K}$$

n = 0.43mol

72. What volume would 5.6moles of sulfur hexafluoride (SF_6) gas occupy if the temperature and pressure of the gas are 128^0c and 9.4atm?

Answer: 19.6L

73. Calculate the density of air at 27^0c and 745torr if its molar mass is 30g/mol.

Solution:

$T = 27^0c = 300K$, $P = 745torr = 0.98atm$

$$d = \frac{PM}{RT} = \frac{0.98 \times 30}{0.082 \times 300} = 1.2g/L$$

74. A sample of gas occupies 500mL at 40^0c and a pressure of 250mmHg. What is the volume if the temperature is raised to 60^0c and pressure is raised to 300mmHg.

Answer: $V_2 = 443.3mL$

75. A gas whose molar mass you wish to know effuses through an opening at a rate only one third as great as the effusion rate of helium. What is the molar mass of the unknown gas?

Solution:

Let the unknown gas be x

$$\frac{r_x}{r_{He}} = \sqrt{\frac{M_{He}}{M_x}}$$

$$\left(\frac{1}{3}\right)^2 = \frac{4}{M_x}$$

$$M_x = 36\frac{g}{mol}$$

76. Calculate the number of molecules of hydrogen gas in a 1.0L sample of gas at STP.

Answer: 2.5×10^{22} molecules of H_2

77. Determine the volume occupied by 4g of oxygen at STP.

Solution:

$$V = \frac{4 \times 0.082 \times 273}{32 \times 1}$$

$$V = 2.8L$$

78. What happens to each of the following during the phase transition of: Gas → liquid → solid?

a. distance between particles
b. intermolecular force of attraction.
c. order of particles.
d. Speed of particles.

Answer:

a. decreases
b. increases
c. increases
d. decreases

79. What is the difference between evaporation and boiling?

Explanation:

- Evaporation takes place at any temperature above the melting point on the surface of liquid while boiling occurs throughout the liquid.

80. Arrange the following gases in decreasing order of rate of diffusion.
NO_2, NO, N_2O, N_2O_3, N_2O_5

Answer:

$NO > N_2O > NO_2 > N_2O_3 > N_2O_5$

81. What gas law related number of moles with volume of gas at constant temperature and pressure?

Answer: Avogadro's law

82. A sample of oxygen gas escapes through a tiny hole in 64sec. and an unknown gas requires 16 sec under the same conditions. What is the molar mass of the unknown gas.

Answer: 2 g/mol

83. A sample of methane gas escapes through a tiny hole in 72 sec. How long would it take for a sample of sulphurdioxide to effuse?

Answer: t_{SO_2} = 144 sec

84. The density of carbondioxide is 1.96g/dm^3 at 0°c and 1 atm, Determine its density at 86.7 kpa and 25°c

Solution:

$$d = \frac{PM}{RT} = \frac{0.86 \times 44}{0.082 \times 298}$$

$d = 1.54$ g/dm^3

Supplementary Exercise on Unit Five

I. Match the gas laws stated on the left with the scientists stated on the right

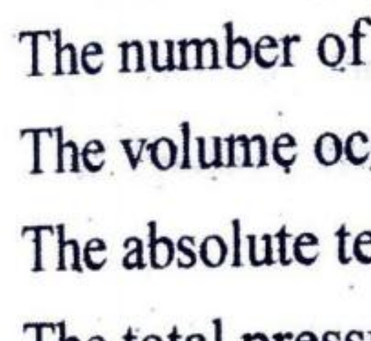
A	B
1. $V \propto \frac{1}{P}$	A. Charles' law
2. $V \alpha T$	B. Boyle's law
3. $P \alpha T$	C. Graham's law
4. $V \alpha n$	D. Gay – Lussac's law
5. $V \propto \frac{T}{P}$	E. Avogadro's law
6. $V \propto \frac{nT}{P}$	F. Combined gas law
7. $R \propto \sqrt{\frac{1}{d}}$	G. Ideal gas law
8. $P_T = P_1 + P_2 + P_3 + \ldots$	H. Dalton's law

II. Choose the best answer from the given alternatives.

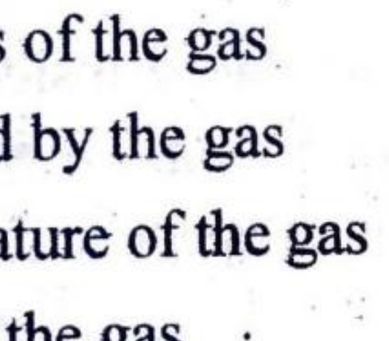
9. Which one has no definite shape and volume?
   A. H_2O C. NaCl
   B. O_2 D. Fe
10. In which of the following order density decrease?
   A. Solid, liquid, gas C. Liquid, gas, solid
   B. Solid, gas, liquid D. Gas, liquid, solid.
11. The lowest attainable temperature is
   A. O°c C. -273.15°c
   B. 273K D. 100K

12. If the temperature in Kelvin of a confined gas sample is doubled while the volume is held constant, what will happen to the pressure?
    A. It will decrease by half
    B. It will be doubled
    C. It will increase by four folds
    D. It will remain the same
13. What is the volume at STP of a gas whose volume is $200cm^3$ at 27^0c and 700mmHg pressure?
    A. $203cm^3$
    B. $167.6cm^3$
    C. $230cm^3$
    D. $181.3cm^3$
14. If 150mL of a gas at STP, weigh 0.228g, what is its molecular weight?
    A. 34 B. 17 C. 16 D. 32
15. A bottle of cold beverage taken only of a refrigerator often covered with drops of water. The process that leads to the formation of those drops of water is
    A. Boiling
    B. Condensation
    C. freezing
    D. sublimation
16. The average kinetic energy of gas molecules is directly proportional to:
    A. The number of moles of the gas
    B. The volume occupied by the gas
    C. The absolute temperature of the gas
    D. The total pressure of the gas
17. Which of the following gives a compressive explanation about physical states of substances?
    A. Kinetic theory of matter
    B. Le – chatler's principle
    C. Avogadro's law
    D. Dalton's atomic theory.
18. Which of the following graph shows the Charles' Law?

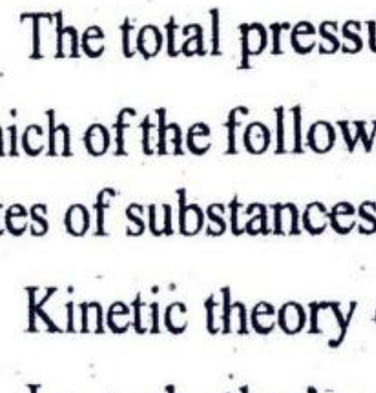
A.



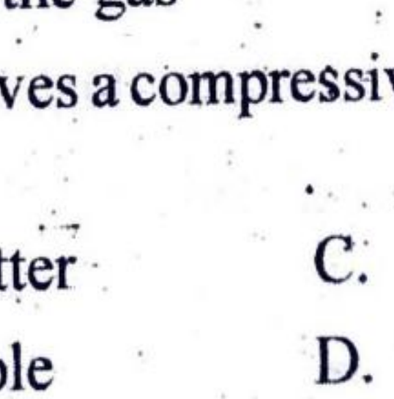
B.



C.



D.



19. At what temperature in degree celius, does 3L of gaseous oxygen weigh 12g if the pressure is 1520 torr?

A. -127 B. -203 C. -146 D. -78

20. Which of the following vaporize without melting?

A. Molten NaCl
B. Sugar
C. ammonium chloride
D. sulphuric acid

21. All of the following are same processes. EXCEPT

A. Condensation
B. Freezing
C. crystallization
D. solidification

22. The direct change of gas to solid is

A. sublimation
B. deposition
C. evaporation
D. condensation

23. Which two states of matter are fluids?

A. Solid and liquid
B. Solid and gas
C. liquid and gas
D. plasma and liquid

24. If 3moles of oxygen gas was placed inside a 500mL container, how many moles of oxygen can be placed in 125mL vessel assuming pressure and temperature remain same?

A. 0.75mol
B. 0.25mol
C. 0.5mol
D. 1.25mol

25. Which of the following is commonly known as dry ice?

A. Solid water
B. Solid carbondioxide
C. solid oxygen
D. solid nitrogen

26. Which one has the least rate of diffusion?

A. $H_2(g)$ B. $CH_4(g)$ C. $N_2(g)$ D. $SO_3(g)$

27. Identify the correct statement about gases.

A. Gas particles are highly disorded when compared with that of liquids and solids.
B. Energy if absorbed when a gas changes to a liquid.
C. The closer the particles in gases are the smaller is the force of attraction between them
D. Particles in gases vibrate about a fixed position.

28. If you heat a 5L balloon from a temperature of 25°c to 50°c, its new volume will be

A. 10L
C. 4.61L
B. 5.42L
D. 2.5L

29. By what factor must the pressure be increased to compress 8 liters of nitrogen to 4 liters?

A. 2
B. 4
C. 16
D. 8

30. During melting of a solid, all are true. EXCEPT

A. intermolecular force of attraction remain same
B. two phases ------- in the equilibrium
C. temperature remains same in the process.
D. energy change is involved in the process.

III. Give short answer

31. Express the following in atmosphere

a. 300cmHg
b. 380torr
c. 1 bar

32. Calculate the volume which 4.0L of gas at 0°c will occupy at 100°c and the same pressure.

33. Sample of gas at 1.20atm and 27°c is heated at constant pressure to 57°c. Its final volume is 4.75L. What was its original volume?

34. When 2.0L of a gas at 760 torr and 27°c is expanded to 3.0L at 77°c. What will the final pressure be?

35. A mass of gas at 80°c and 785 torr occupies 350mL. What volume will the gas occupy at STP?

36. Calculate the mass of 6.0L of ammonia gas (NH_3) at STP.

37. Determine the volume occupied by 4.0g of oxygen at STP.

38. A quantity of hydrogen gas occupies a volume of 30.0mL at a certain temperature and pressure. What volume would half mass of hydrogen occupy and triple the absolute temperature if the pressure were one ninth that of the original gas?

39. Determine the approximate molecular weight of a gas if 560mL weighs 1.80g at STP.

40. The volume of one mole of gas at STP is called

UNIT 6

Introduction to Organic Chemistry

Unit Outline:

6.1 Classification of Organic Compunds

6.2 Open Chain Hydrocarbons

6.3 Cyclic hydrocarbons

6.4 Aromatic hydrocarbons

6.5 Natural Sources of hydrocarbons

6.6 Alcohols

6.7 Aldehydes and Ketones

6.8 Carboxylic acids and Ketones

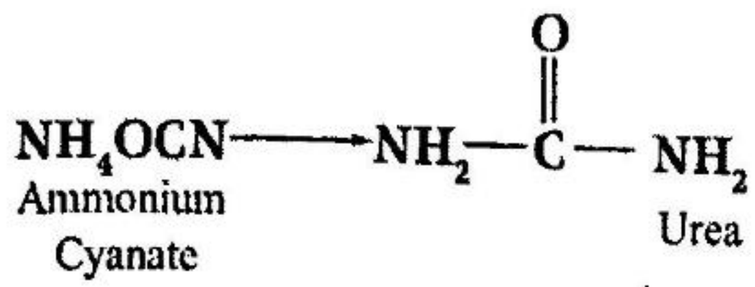
6.9 Industrial and Agricultural applications of Orga
Compounds

Introduction

In the early time, organic chemistry deals with substances that were produced by living organisms with the help of "vital force" that could not be manufactured in the laboratory.

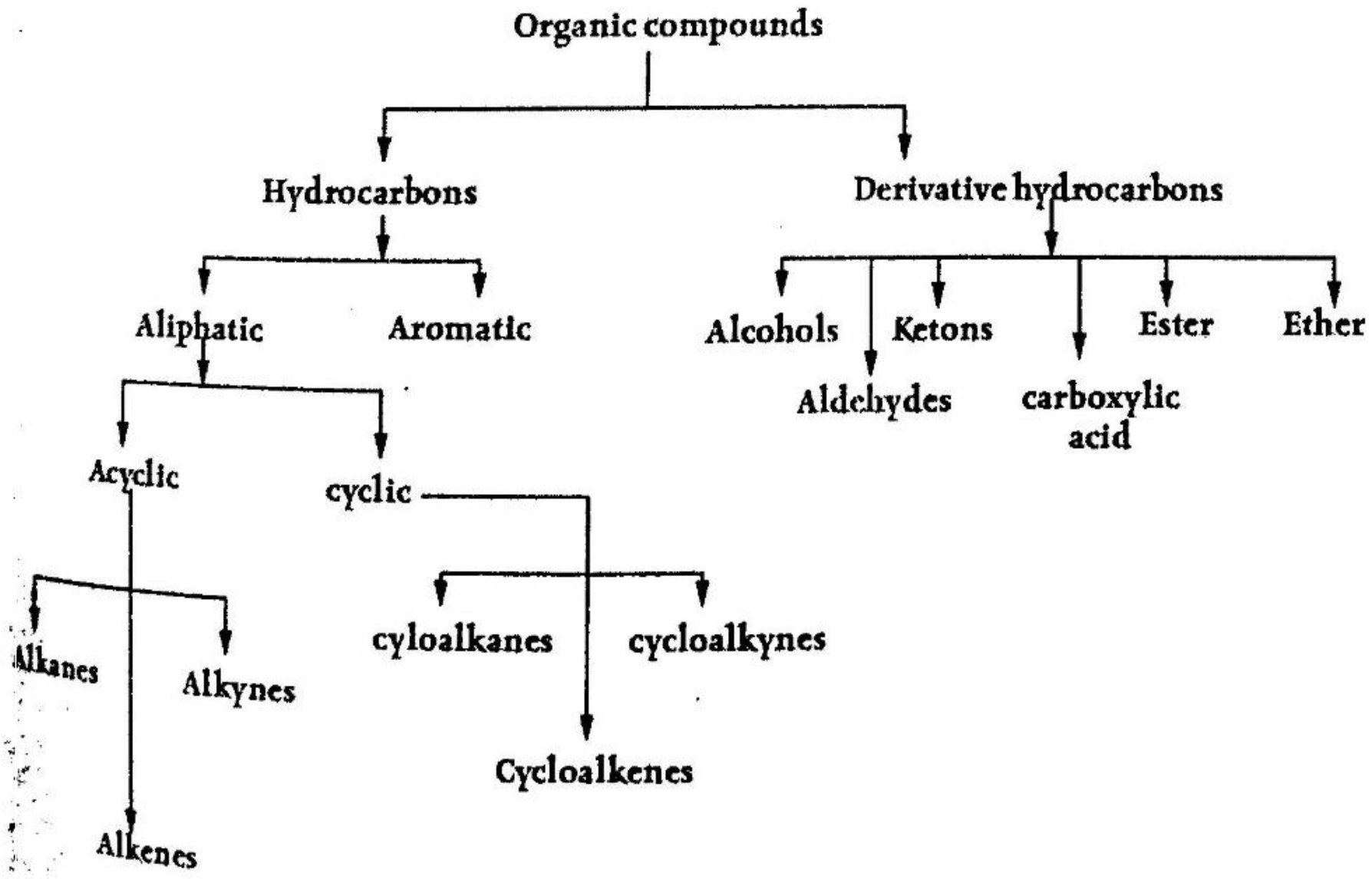
- Friedrich Wohler disproved the idea of vital force theory in 1828 by manufacturing the first synthetic organic compound (urea) in the laboratory.

$$NaOCN + NH_4Cl \longrightarrow NH_4OCN + NaCl$$



6.1 Classification of Organic Compounds

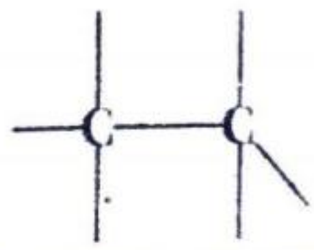
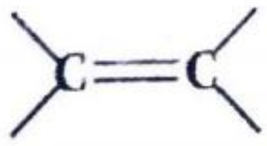
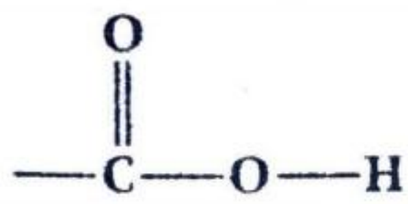
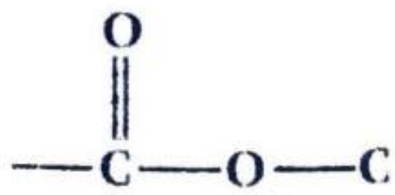
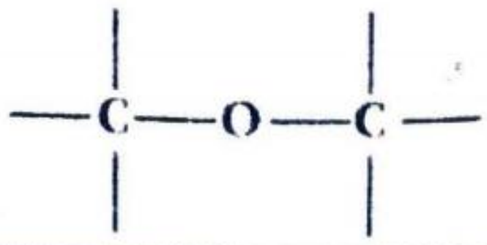
- Organic chemistry is the chemistry of hydrocarbons and their derivatives.
- Hydrocarbons are compounds of hydrogen and carbon only.
- Derivative hydrocarbons are hydrocarbons in which one more hydrogen replaced by other atom such as oxygen.



- Carbon atom is capable of forming straight chains, branched chains, cyclic compounds and other complicated structure of hydrocarbons. This property of carbon is called **catenation**.

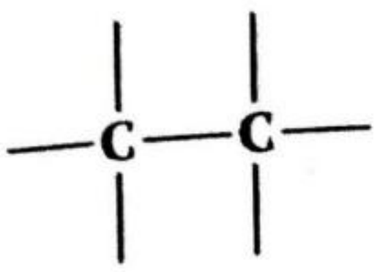
- **Functional group** is part of the molecule or compound that determines the chemical properties of that molecules or compounds.

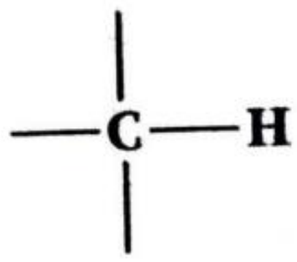
Table 6.1. Some organic compounds and their functional groups

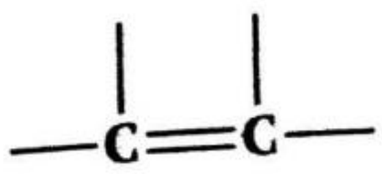
Organic compounds	Functional
Alkanes	
Alkenes	
Alkynes	
Aromatics	
Alcohols	
Aldehydes	
Ketones	
Carboxylic acids	
Ester	
Ether	

Illustrative Examples

1. One of the following is the functional group of alkene. Which one is it?. . . (EGSECE 2004E.C)

A. 

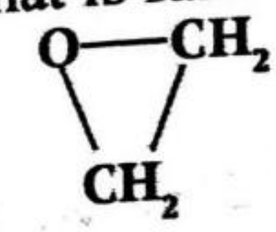
C. 

B. 

D.

Answer: B

2. What is functional group present in the following organic compound?



A. Ether
B. Ester
C. Aldehyde
D. Ketone

Solution:

The presence of C – O – C bonds represent an ether.

Answer: A

3. Which of the following is the first organic compound synthesized from inorganic source?

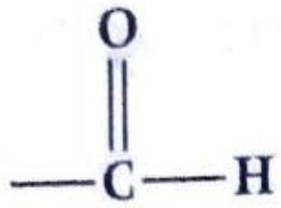
A. NH_4Cl
B. NH_4CNO
C. $(NH_2)_2CO$
D. AgCNO

Solution:

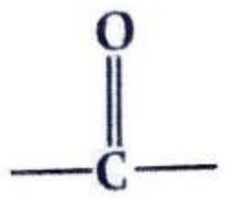
The 1st synthetic organic compound produced in the laboratory was urea. The condensed formula of urea is (NH_2) CO or N_2H_4CO

Answer: C

4. Esters contain the following functional groups.

A. 

C. —OH

B. 

D. —COO—

Answer: D

6.2 Open chain hydrocarbons

Alkanes

- are also called paraffins
- are saturated hydrocarbons (contain carbon-carbon single bond)
- have a general formula C_nH_{2n+2}, n = 1, 2 . . .

Example

4. Find the formula of an alkane with the value of n equals to:
   a) n = 1 b) n = 6 c) c = 10 d) n = 18

Solution

a) n = 1

$C_nH_{2n+2} = C_1H_2(1) + 2$

$= C_1H_{2+2}$

$= CH_4$

b) n = 6

$C_nH_{2n+2} = C_6H_{2(6)+2}$

$= C_6H_{12+2}$

$= C_6H_{14}$

c) c = 10

$C_nH_{2n+2} = C_{10}H_{2(10)+2}$

$= C_{10}H_{20+2} = C_{10}H_{22}$

d) c = 18

$C_nH_{2n+2} = C_{18}H_{2(18)+2}$

$= C_{18}H_{36+2} = C_{18}H_{38}$

In a given alkane, number of carbon atoms equal to n but number of hydrogen atom equal to 2n + 2.

5. Find the number of carbon atoms of alkanes with hydrogen atoms equal to:

a. 14 b. 32

Solution:

a. Number of hydrogen atoms = 2n + 2

$\Rightarrow 2n + 2 = 14$

$2n = 14 - 2$

$2n = 12$

$n = 6$

Therefore, number of carbon atoms = n = 6

b. $2n + 2 = 32$

$2n = 32 - 2$

$2n = 32$

$n = 15$

Therefore, number of carbon atoms = 15

6. Find the number of hydrogen atoms of an alkane with carbon atoms equal to:

a. 12 b. 7

Solution:

a. Carbon atoms n

Hydrogen atoms = 2n + 2

When n = 12

Number of hydrogen atoms = 2(12) + 2 = 26

b. n = 7, number of hydrogen atoms = 2n + 2

= 2(7) + 2 = 16

7. An alkane with 39 carbon atoms would have the formula: EGSEC

A. $C_{39}H_{76}$

B. $C_{39}H_{78}$

C. $C_{39}H_{79}$

D. $C_{39}H_{80}$

Solution:

$$C_nH_{2n+2} = C_{39}H_{2(39)+2} = C_{39}H_{80}$$

Answer: D

- Alkanes form homologous series.
- **Homologous series** is a series in which one member differs from the next by the same repeating cluster of atoms. Consider the first four alkanes:

CH_4, C_2H_6,C_3H_8,C_4H_{10}, that differ by methylene group ($-CH_2-$). The individual members are called **homologues**.

Illustrative Examples

8. Which sets of compounds represent a homologous series? (EGSECE)

A. C_2H_4, C_3H_8, C_4H_8

B. C_2H_2, C_3H_6, C_4H_8

C. CH_4, C_2H_6, C_3H_8

D. CH_4, C_2H_4, C_3H_6

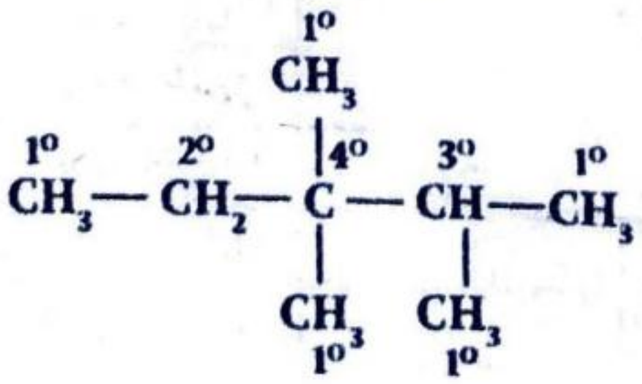
Solution:

Each member differs from the next by $-CH_2-$ is CH_4, C_2H_6, C_3H_8

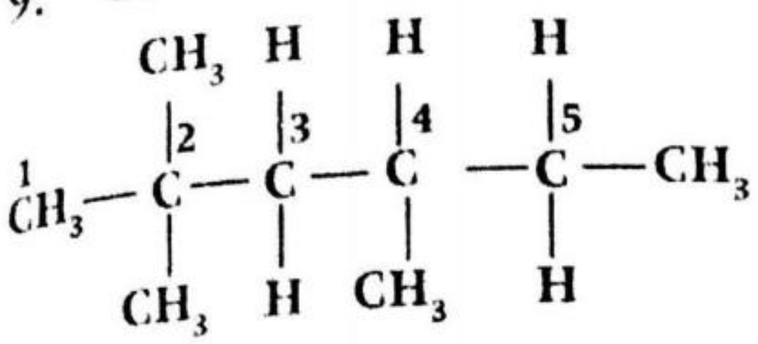
Answer: C

The carbon atoms bounded to one carbon, two carbons, three carbons and four carbons are called primary (1^o). Secondary (2^o). tertiary (3^o) and quaternary carbonates (4^o) respectively.

Example



9. Given the structure



A tertiary carbon atom is show by the number:

Answer: C – 4

➦ The hydrogen atom attached to 1^0,2^0 and 3^0 carbon atoms is termed as primary, secondary and tertiary hydrogen atom respectively.

There is no quaternary hydrogen because carbon atom never forms more than four bonds in any compound.

Nomenclature of Alkanes

The name of alkanes end with the suffix "-ane"

Table 6.2. Prefixes, carbon number, molecular formula and name

Prefixes	Carbon number	Molecular formula	Name
meth-	1	CH_4	Methane
eth-	2	C_2H_6	Ethane
prop-	3	C_3H_8	Propane
but-	4	C_4H_{10}	Butane
pent-	5	C_5H_{12}	Pentane
hex-	6	C_6H_{14}	Hexane
hept-	7	C_7H_{16}	Heptane
oct-	8	C_8H_{18}	Octane
non-	9	C_9H_{20}	Nonane
dec-	10	$C_{10}H_{22}$	Decane
undec-	11	$C_{11}H_{24}$	Udecane
dodec-	12	$C_{12}H_{26}$	Dodecane
tridec-	13	$C_{13}H_{28}$	Tridecane
tetradec-	14	$C_{14}H_{30}$	Tetradecane
pentadec-	15	$C_{15}H_{32}$	Pentadecane
hexdec-	16	$C_{16}H_{34}$	Hexadecane

heptadec-	17	$C_{17}H_{36}$	Heptadecane
octadoc-	18	$C_{18}H_{38}$	Octandecane
nonadec-	19	$C_{19}H_{40}$	Nonadecane
eicos-	20	$C_{20}H_{42}$	Eicosane

Methane is the simplest alkane, without functional group

Alkyl group (R) – are radicals obtained from alkane by reducing one hydrogen atom.

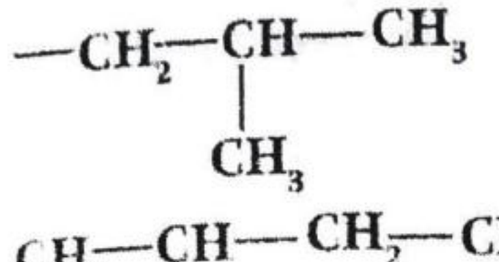
➡ Have general formula C_nH_{2n+1}, n = 1,2,... .

Table 6.3. Shows molecular formula of alkane, molecular and condensed formula of alkyl group and common name of alkayl groups.

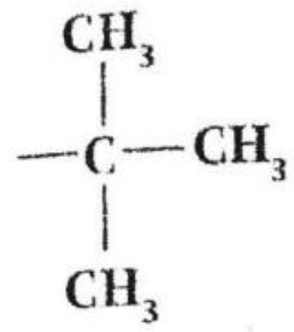
Molecular formula or alkane	Molecular formula of alkyl group	Condensed formula of alkyl group	Common name
CH_4	CH_3—	CH_3—	Methyl
C_2H_6	C_2H_5—	CH_3CH_2—	Ethyl
C_3H_8	C_3H_7—	$CH_3CH_2CH_2$—	n-propyl
		$CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}-$	Isopropyl
C_4H_{10}	C_4H_9—	$CH_3CH_2CH_2CH_2$—	n-butyl
		$CH_3\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}CH_2$—	Isobutyl
		$CH_3\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}CH_2CH_3$	Sec-butyl(s-butyl)
		$CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{\displaystyle |}{\overset{\displaystyle |}{C}}}}-$	Tert-butyl (t-butyl)
C_5H_{12}	C_5H_{11}	$CH_3CH_2CH_2CH_2CH_2$—	n-pentyl (n – amyl)

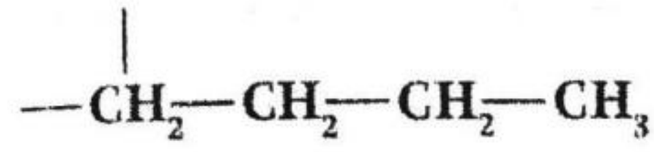
Illustrative Examples

10. Which one of the following structures represents an iso – butyl alkyl group? (EGSECE 2002E.C)

A. 

B.

C. 

D. 

Answer: A

11. Find the molecular formula of the alkyl group with carbon

a. 6

b. 8

Solution:

a. $C_nH_{2n+1} = C_6H_{2(6)+1} = C_6H_{13}$

b. $C_nH_{2n+1} = C_8H_{2(8)+1} = C_8H_{17}$

1. Common names of alkanes

Common name of alkanes take some prefixes to identify one from other.

i **Normal (n-):** to name an alkane without any branch.

Example

$CH_3CH_2CH_2CH_3$

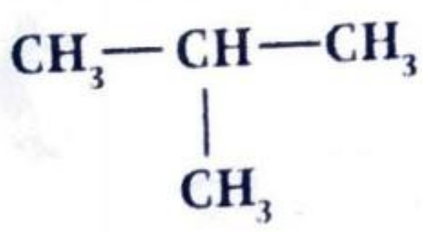
n – pentane

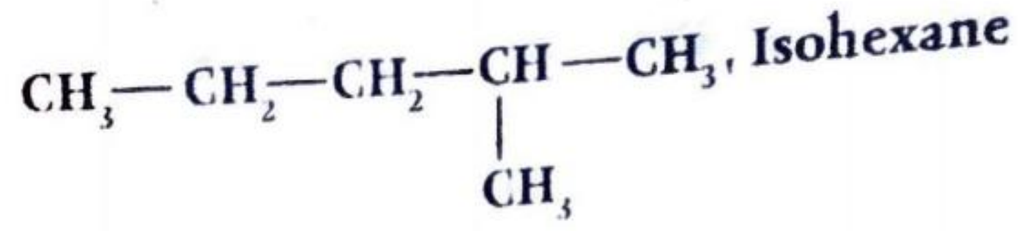
$CH_3CH_2CH_2\,CH_2CH_2\,CH_2CH_3$

n-heptane

ii **Iso:** to name alkanes that contain one methyl on the second carbon.

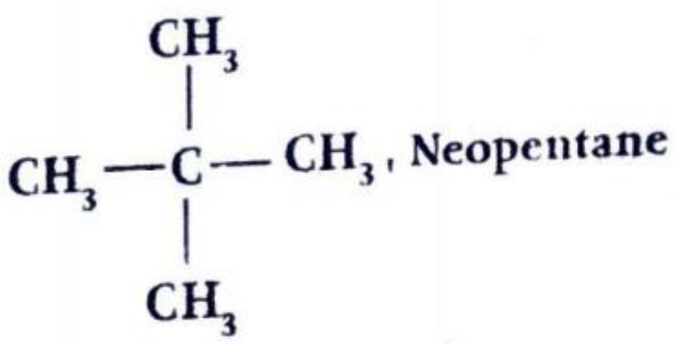
Example

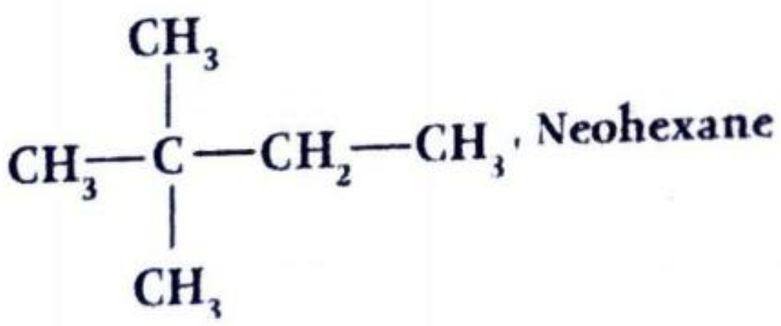




iii **Neo:** is used for naming alkanes when the central carbon atoms attached to other four carbon atoms.

Example





2. IUPA name of alkanes

- IUPAC – international Union Pure and Applied chemistry.
- It is very important to name branched chain hydrocarbons.

Rules

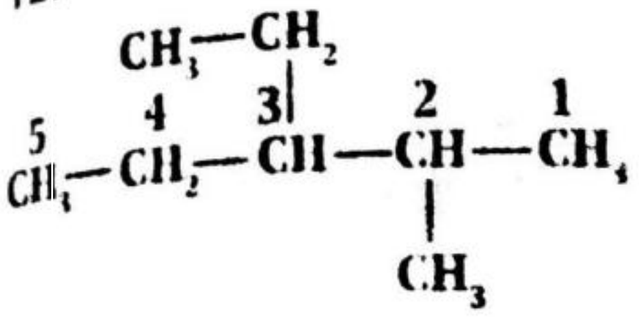
1. Identify the longest continuous chain.
2. Count from the end nearer to the side chains.
3. It the same side chain appears twice, three times, four times, use the prefix di-, tri-, tetra- respectively.
4. Number and number separated by comma (,).

5. Number and letter separated by hyphen(-)

6. Arrange the side chains alphabetically.

Illustrative Examples

12. Given the IUPAC name for the following alkanes



Solution:

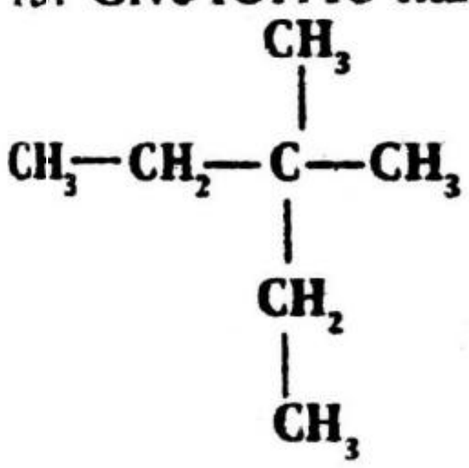
Count from right to left which is nearer to the side chain.

Answer: 3 – Ethyl – 2 – Methyl pentane

Note:

If there are two ways to identify the longest chain select the one with more side chains or branches.

13. Give IUPAC name

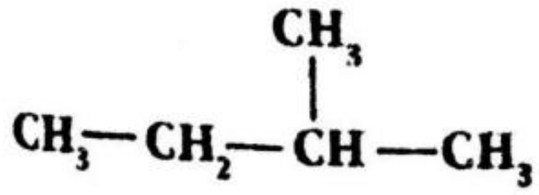


Solution:

Follow the zigzag structure to known the longest chain.

Answer: 3,3—dimethylpentane

14.

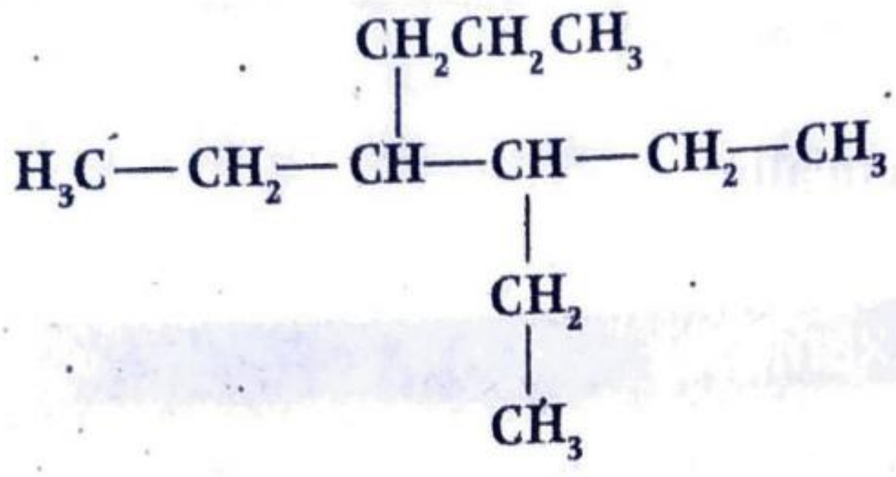


Solution:

start counting from the right to the left.

Answer: 2 – methylbutane

15.

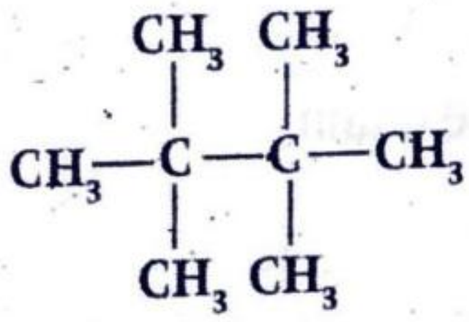


Solution:

Start from left and follow the top zigzag continuous chain to identify the longest chain. The parent name is hexane.

Answer: 3,4 - diethyl heptane

16.

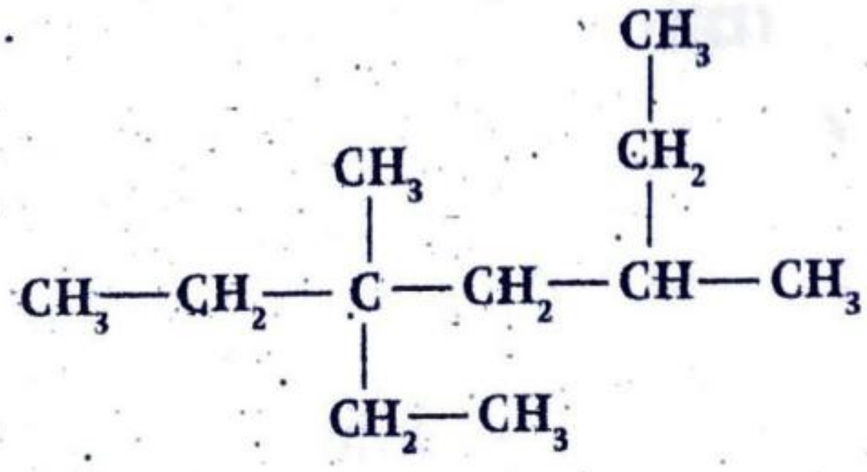


Solution:

If side chains appear on the second carbon on both sides, give priority for the carbon that contains more side chains. So, for the above problem start from the left.

Answer: 2, 2, 3, 3 – tetramethylbutane

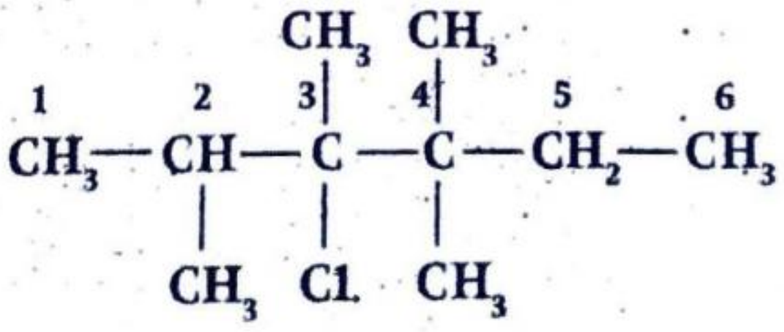
17.



Solution:

Answer: 3 – ethyl – 3,5 – dimethyl heptane

18.

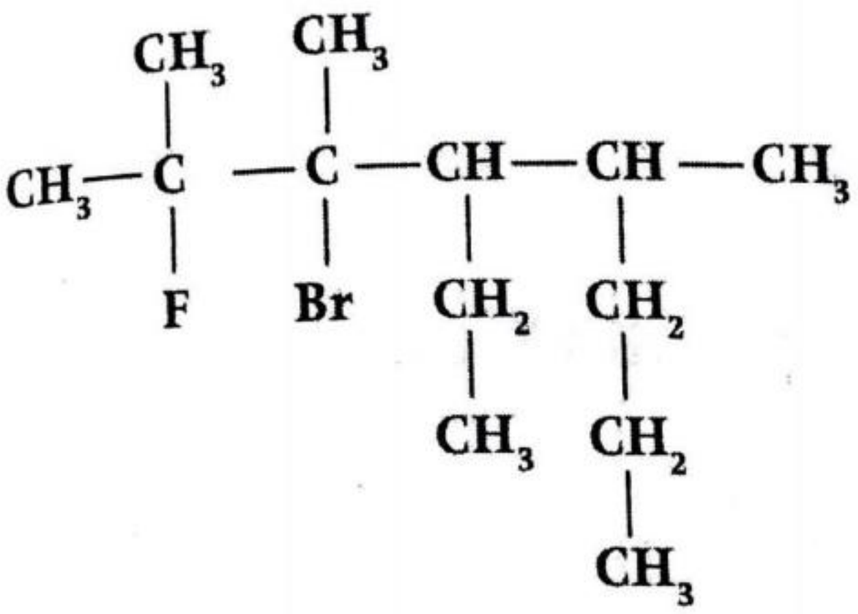


Solution:

Start from left to right and arrange the side chains alphabetically.

Answer: 3 – chloro – 2, 3, 4, 4 – tetramethyl hexane

19.



Solution:

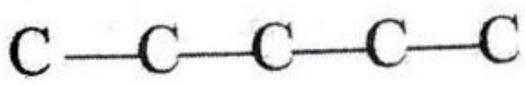
Count starting from the left. The parent name is octane.

Answer: 3 – Bromo – 4 – ethyl – 2 – flouro – 2, 3, 5 – trimethyl – octane.

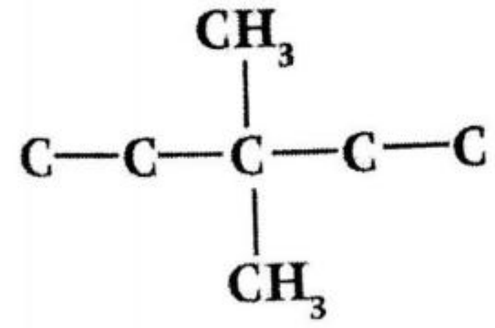
20. Write the structural formula for 2,2 – dimethyl pentane.

Solution:

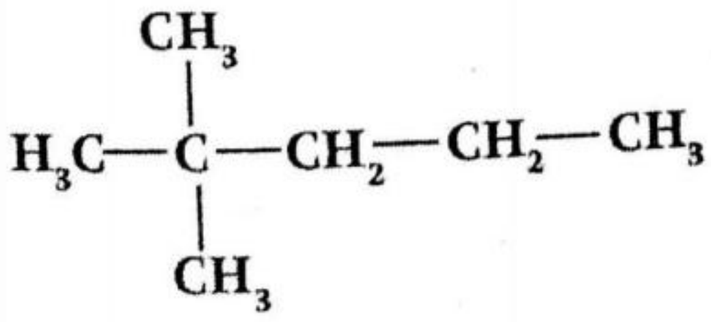
i Based on the parent name, draw skeletal structure.



ii From any side you like, place the side chains on the given position.

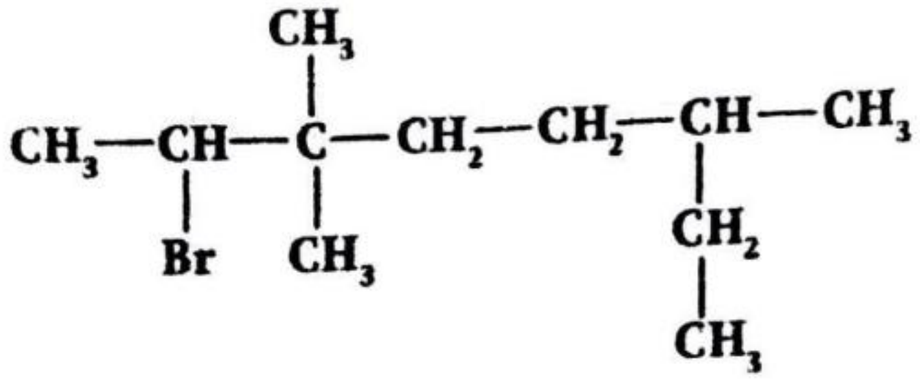


iii Give the required number of hydrogen atoms for each carbon.



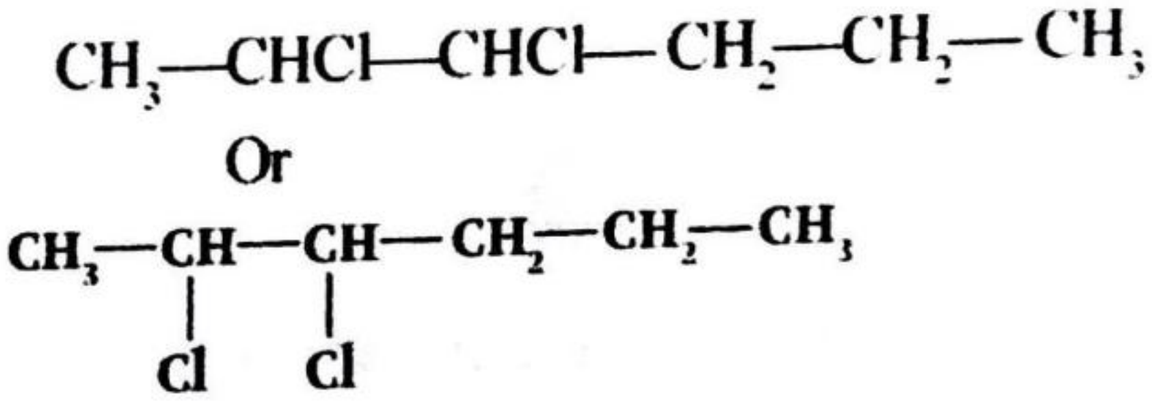
21. Draw the structure of 2 – Bromo – 3, 3, 6 – trimethiloctane

Answer:



22. What is the correct structure for 2,3 – dichloro hexane?

Answer:



Isomerism

Isomerism is the occurrence of two or more compounds with the same molecular formula but different structural formula. The compounds are called **isomers**.

Methane(CH_4), Ethane (C_3H_6) and propane (C_3H_8) do not have isomers.

Alkanes show structural isomerism.

Table 6.4. Isomers of butane and pentane

Alkane	Structural formula	Molecular formula	Number of isomers
Butane	$CH_3—CH_2—CH_2—CH_3$	C_4H_{10}	2
	$CH_3—CH(CH_3)—CH_3$	C_4H_{10}	

Pentane	$CH_3—CH_2—CH_2—CH_2—CH_3$	C_5H_{12}	3
	$CH_3—CH(CH_3)—CH_2—CH_3$	C_5H_{12}	
	$CH_3—C(CH_3)(CH_3)—CH_3$	C_5H_{12}	

As carbon number increases, number of isomers also increase

Illustrative Examples

23. Which type of isomers are common in alkanes?

A. Position isomers
B. Optical isomers
C. Geometrical isomers
D. Structural isomers

Answer: D

24. Isomers are different in all of the following EXCEPT. . . .

A. Chemical property
B. Structure
C. Physical property
D. Molecular mass

Answer: D

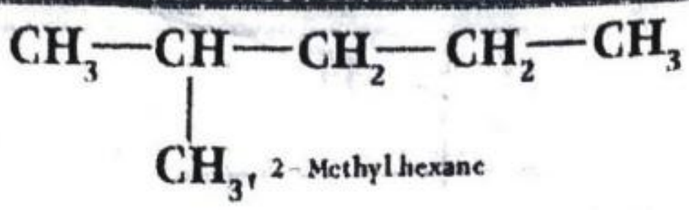
25. How many different structural arrangements are possible for heptane?.... EGSECE

A. 3
B. 5
C. 9
D. 7

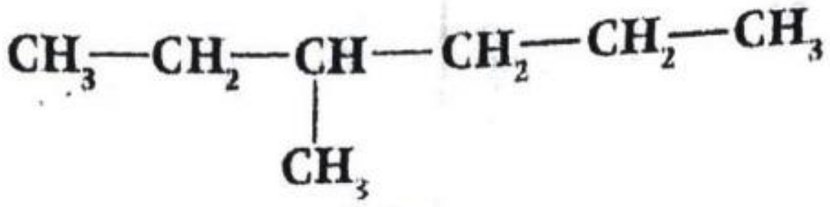
Solution:

Isomers of heptanes are:

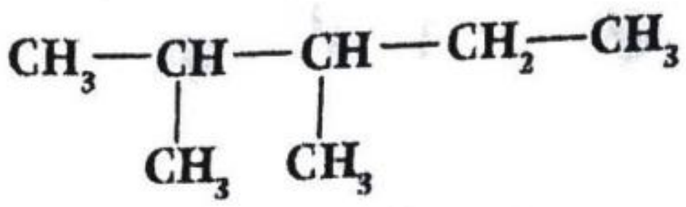
i $CH_3—CH_2—CH_2—CH_2—CH_2—CH_2—CH_3$, n-heptane

ii 

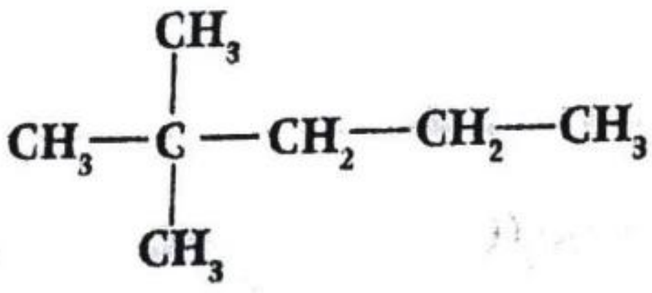
2 - Methyl hexane

iii 

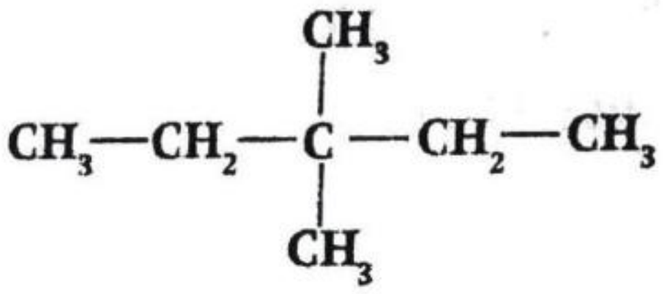
3 - Methyl hexane

iv 

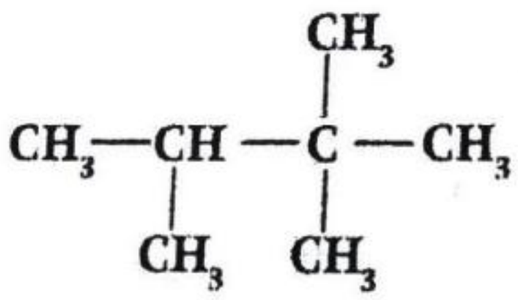
2, 3- di methyl pentane

v 

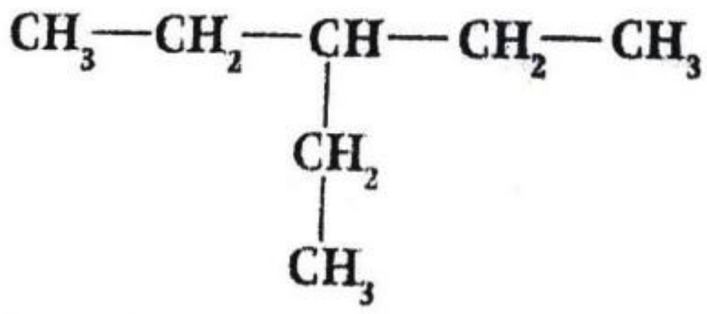
2,2 – di methyl pentane

vi 

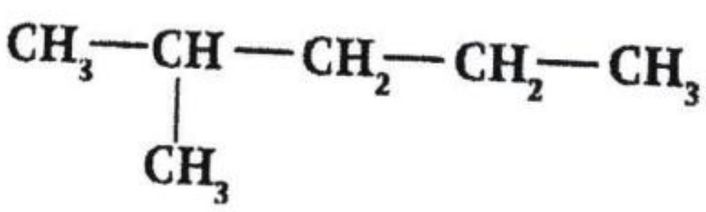
3,3 – dimethyl pentane

vii 

2,2,3 – tri methyl butane

viii 

3 – ethyl pentane

ix 

2, 4 – dimethyl pentane

All of these nine isomers have same molecular formula (C_7H_{16})

Answer: C

To find the number of isomers of simple alkanes use the formula $2^{n-4}+1$, $n = 4,5, \ldots$

Examples: Butane (C_4H_{10})

Number of isomers $= 2^4 - 4 + 1 = 2^0 + 1 = 1 + 1 = 2$

Hexane (C_6H_{14}) number of isomers $= 2^{6-4} + 1 = 2^2 + 1 = 4 + 1 = 5$

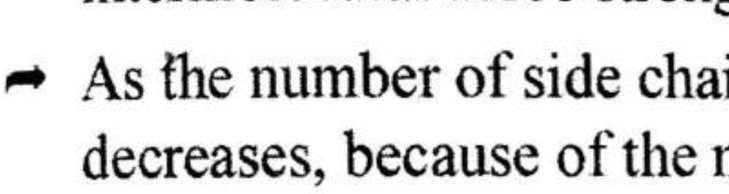
Physical Properties of alkanes

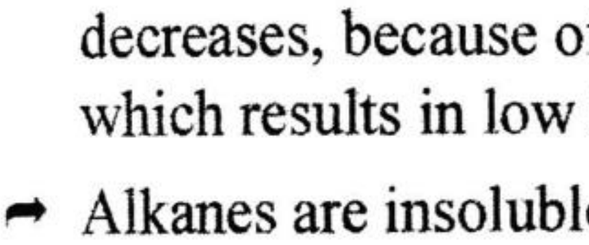
- Alkanes are colorless hydrocarbons.
- The first four ($C_1 - C_4$) alkanes are gases.
- From pentane up to heptadecane (C_{17})exist in liquid state.
- From octadecane (C_{18}) and above exist in solid state at room temperature.
- As carbon number increases in the homologous series., boiling point, melting point and density also increase because intermolecular force strength increase.
- As the number of side chains (branches) increase boiling point decreases, because of the molecules approaches that of sphere which results in low inter molecular force of attraction.
- Alkanes are insoluble in water because they are non – polar covalent molecules.

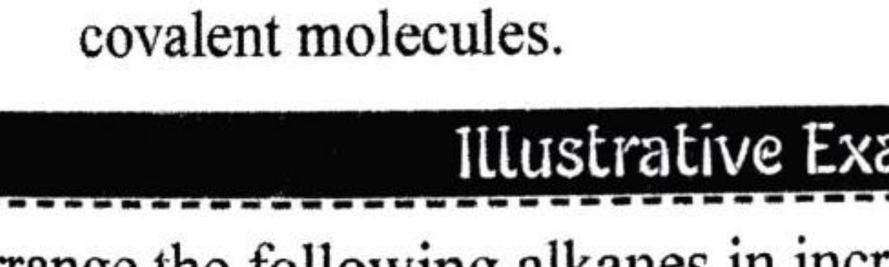
Illustrative Examples

26. Arrange the following alkanes in increasing order of boiling points

a. $CH_3—CH_2—CH_2—CH_2—CH_2—CH_3$

b. 

c. 

d. 

Solution:

c< b < a < d. since the first three contain the same carbon number as branching increase boiling point decrease but in (d) there are 7 carbon atoms, as a result, it has the highest boiling point.

27. Which of the following compounds would be in the liquid state at room temperature?

A. CH_4

B. C_6H_{14}

C. C_3H_8

D. $C_{25}H_{52}$

Solution:

$C_5 - C_{17}$ exist in liquid phase.

Answer: B

28. Which of the following is the physical property of alkanes?

A. Their densities decreases with increasing molecular mass along the homologous series.

B. Their boiling points decrease with carbon number.

C. The first four n-alkanes are solids at room temperature.

D. They are colorless.

Answer: D

29. Alkanes are insoluble in water because

A. They are solid substances.

B. They have strong intermolecular attraction forces.

C. They are non – polar substances.

D. They posses carbon – carbon single bond

Solution:

Due to the "Like dissolve like" principle alkanes are non-polar while water is polar.

Answer: C

30. Which of the following physical properties decrease as the number of carbon atom increases in a homologous series of alkanes?

A. Intermolecular force
B. Density
C. Melting point
D. solubility

Solution:

In general alkanes are insoluble in water, comparatively, as carbon number increases solubility decreases

Answer: D

Chemical Properties of Alkanes

1. **Substitution reaction** is the typical reaction of alkanes.
   - ➦ Halogenations of alkanes is a substitution reaction.
   - ➦ The reaction of bromine and chlorine with alkanes occur in the presence of sunlight. Such type of reaction is called photo chemical reaction.

Bromination of methane may produce different products depending on the amount of bromine.

$$CH_4 + Br_2 \xrightarrow{light} CH_3Br + HBr$$

$$CH_3Br + Br_2 \longrightarrow CH_2Br_2 + HBr$$

$$CH_2Br + Br_2 \longrightarrow CHBr_3 + HBr$$

$$CHBr_3 + Br_2 \longrightarrow CBr_4 + HBr$$

Over all reaction

$$CH_4 + 4Br_2 \longrightarrow CBr_4 + HBr$$

2. **Elimination Reaction**

Alkanes undergo elimination reaction at relatively high temperature.

Example:

$$CH_2CH_2CH_3 \xrightarrow{heat} CH_2 = CH_2 + CH_4$$

3. Combustion reactions

Alkanes + excess air ⟶ carbondioxide + water + heat

Example:

$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O + heat$

Illustrative Examples

31. Which of the following is an elimination reaction? (EGSECE)

A. $CH_3CH_2CH_3 + 5O_2 \longrightarrow 3CO_2 + 4H_2O$

B. $CH_3CH = CH_2 \rightarrow CH_3CH_2 + 4H_2O$

C. $CH_3CH_2CH_3 \xrightarrow{heat} CH_3CH = CH_2 + H_2$

D. $CH_3CH_2CH_3 + Cl_2 \xrightarrow{UV} CH_3CH_2CH_2Cl + HCl$

Answer: C

32. The process of converting higher alkanes into smaller alkanes and alkenes to increase the gasoline yield is known as.

A. Alkylation
C. Cracking
B. Polymerization
D. saponification

Solution:

Cracking is a special type of elimination reaction that converts large alkanes in to smaller alkenes and alkanes.

Answer: C

33. The possible reason why alkanes are relatively stable to most common laboratory reagents at room temperature is due to ….. (EGSECE 2003 E.C)

A. High strength of carbon to carbon and hydrogen to carbon bonds
B. Their strong intermolecular forces
C. Their insoluibility in water
D. Their molecular masses

Solution:

Alkanes are non-polar saturated hydrocarbons and inert towards many laboratory reagents such as acids, bases, oxidizing agents and reducing agents.

Answer: C

34. Which of the following is the structure for methylchloride?

A. CH_3Cl

B. CH_2CH_2

C. $CHCl_3$

D. CCl_4

Solution:

Methane undergoes chlorination reaction

$$CH_4 + Cl_2 \xrightarrow{heat} CH_3Cl + HCl$$

Methyl chloride (chloromethane) } CH_3Cl

$$CH_3Cl_4 + Cl_2 \longrightarrow CH_2Cl_2 + HCl$$

Methylene chloride (Dichloromethane) } CH_2Cl_2

$$CH_2Cl_2 + Cl_2 \longrightarrow CHCl_3 + HCl$$

Chloroform (Trichloromethane) } $CHCl_3$

$$CHCl_3 + Cl_2 \longrightarrow CCl + HCl$$

Carbon tetrachloride (Tetrachloromethane) } CCl_4

Answer: A

Preparation of Alkanes

The most important method to manufacture alkanes is **Decarboxylation method.**

$$RCOONa + 2NaOH \longrightarrow R - H + Na_2CO_3$$

Sodium carboxylate Alkane

Example

$$CH_3CH_2COONa + 2NaOH \longrightarrow CH_3CH_3 + Na_2CO_3$$

Sodium Propanoate — Ethane

Uses of Alkanes

- Alkanes are largely used as a fuel. Methane is main constituent of natural gas and biogas.Butane and kerosene are examples of alkanes which are used as fuels in our day to day life.
- Alkanes are also used as solvent, raw materials in plastics, soaps, alcohols and other chemical industries.

Illustrative Examples

35. All of the following reactions produce an alkane except

A. $H_3COONa + NaOH \longrightarrow$

B. $CH_2 = CH_2 + H_2 \xrightarrow[catalyst]{Heat}$

C. $2CH_3I + 2Na \longrightarrow$

D. $CH_3 - CH = CH_2 + H_2O \xrightarrow{H_2SO_4}$

Solution:

$H_3COONa + NaOH \rightarrow CH_4 + Na_2CO_2$, decarboxylation

$CH_2 = CH_2 + H_2 \xrightarrow[catalyst]{Heat} CH_3 - CH_3$, hydrogenation reaction

$2CH_3I + 2Na \xrightarrow[manufacture]{} CH_3CH_3 + 2NaI$, Wurtz reaction used to manufacture alkanes.

$$CH_3 - CH = CH_2 + H_2O \longrightarrow CH_3 - \underset{\displaystyle OH}{\underset{|}{CH}} - CH_3$$

, Hydration reaction

Answer: D

36. Alkanes are primary used as (EGSECE)

A. Fuels
B. Organic solvents
C. raw materials to make soaps
D. raw materials to make plastics

Answer: A

37. Which of the following reactions could produce alkanes in the laboratory? (EGSECE ... 2009 E.C)

A. $R-CH=CH_2 + H_2 \xrightarrow[heat]{Pt} \rightarrow$

B. $R-CH_2-CH_2-OH \xrightarrow[heat]{H_2SO_4} \rightarrow$

C. $R-CH_2-CH_2-X + KOH \longrightarrow$

D. $R-CH=CH_2 + H_2O \longrightarrow$

Answer: A

Alkenes

- Alkenes are also called **olefins**
- They are unsaturated hydrocarbons containing carbon-carbon double bond.
- They have a general formula C_nH_{2n}, n = 2,3. . .
- Form homologous series.
- Have two fewer hydrogen atoms than the corresponding alkanes.

Illustrative Examples

38. Find the molecular formula of an alkene with carbon number.

a. 10 b. 22

Solution:

a. $C_nH_{2n} = C_{10}H_{2(10)} = C_{10}H_{20}$

b. $C_nH_{2n} = C_{22}H_{2\times22} = C_{22}H_{44}$

39. Which of the following is the third alkene?

A. ethene C. butene

B. propene D. pentene

Solution:

The simplest alkene has two carbon atoms

First alkene, C_2H_4 = ethene

Second alkene, C_3H_6 = propene

Third alkene, C_4H_8 = butene, etc

Answer: C

40. Which of the following general formula represents the group of olefins?

A. C_nH_{2n+2}

B. C_nH_{2n-2}

C. C_nH_{2n}

D. C_nH_{2n+1}

Answer: C

Nomeclature of alkenes

A. Common name

Table 6.5. Shows some alkenes and their common names

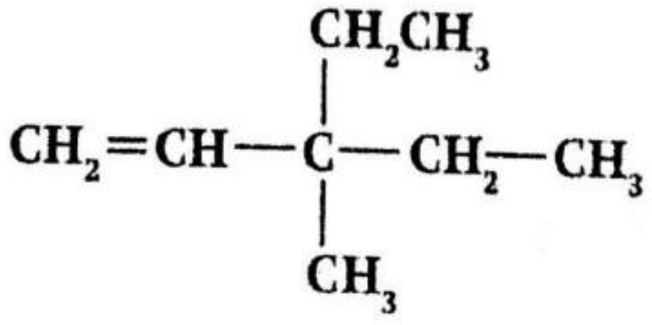
Alkene	Common name
C_2H_4	Ethylene
C_3H_6	Propylene
C_4H_8	Butylene
C_5H_{10}	Pentylene

B. IUPAC name

In the IUPAC system of naming alkenes, identify the longest chain by including the double bond. Start from the end nearer to the double bond and indicate the position of the double bond by number in the name of the alkenes. The rest rules are similar with that of alkanes.

Illustrative Examples

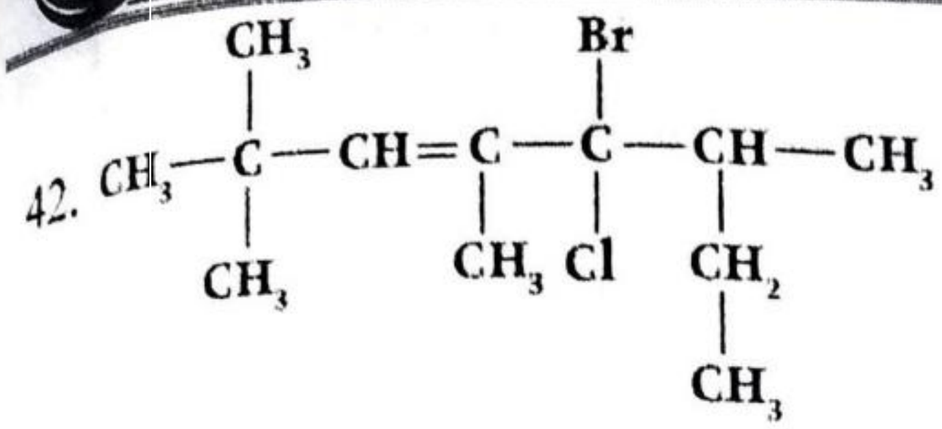
41. Give the IUPAC name of the following alkenes.



Solution:

Start from the end nearer to the double bond. The parent

Answer: 3 – ethyl – 3 – methyl – 1 – pentene or
3 –ethyl – 3 – methyl pent – 1 – ene

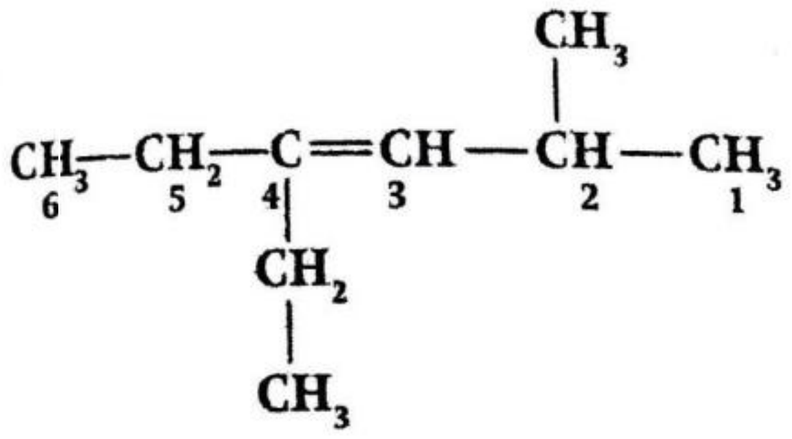
42. 

Solution:

Start counting from the left. Parent name is octane.

Answer: 5–bromo–5–chloro–2, 2, 4, 6 – tetramethyl – 3 – octene

(5– bromo –5– chloro – 2,2,4,6–tetramethyloct –3– ene)

43. 

Solution:

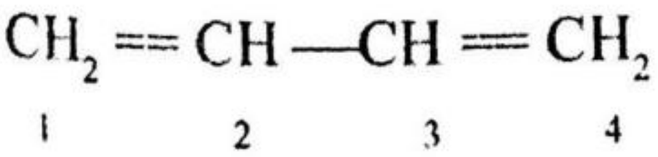
If the double bond appears in the same carbon number both sides, start from the end earer to the side chain. For the above example start from the right.

Answer: 4 – ethyl – 2 – methyl – 3 – hexene (4 – ethyl – 2 – methyl hex – 3 – ene)

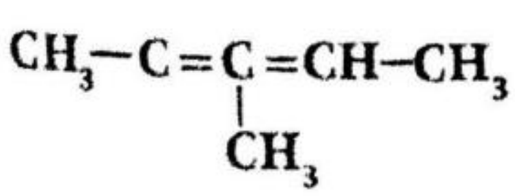
Dienes are unsaturated hydrocarbons thatposses two double bonds. Their general formula is C_nH_{2n-2}, n = 3, 4, 5. . . .

Examples:

45.

a. 

1, 3 – butadiene

b. 

Answer: 2 – methyl – 2, 3 – pentadiene

46. Draw the structure of

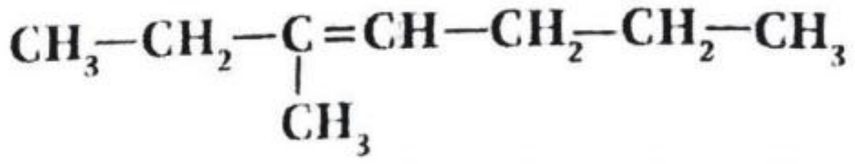
a. 3 – methyl – 3 – heptene

Solution:

➦ draw the structure of the parent chain

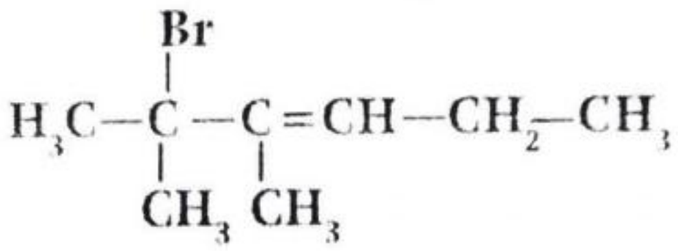
CCC = C CCC

➦ place the side chain and fulfill the octet of each carbon by providing the necessary hydrogen.



b. 2 –Bromo – 2, 3 – dimethyl hex – 3 – ene

Solution:



Isomerism in Alkenes

Ethene and propene do not have isomers because they have only one possible structure.

Alkenes show the following three isomerism

1. Structural isomerism
2. Position isomerism
3. Geometric isomerism (cis – trans isomerism)

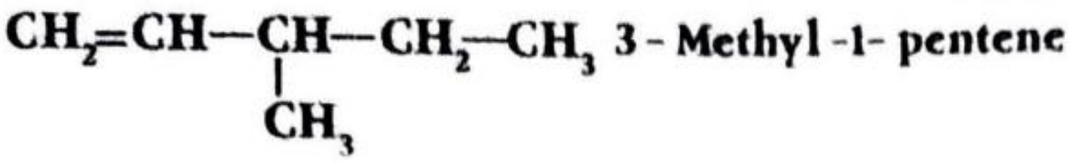
Example

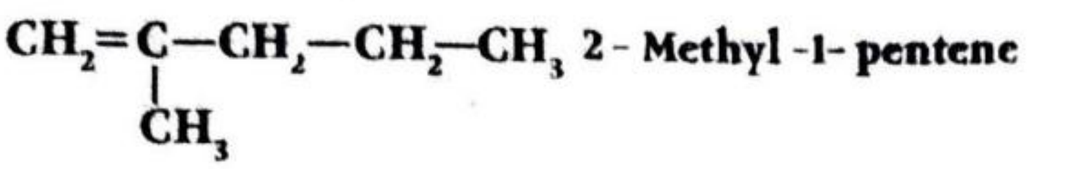
1. Write the structural isomers, position isomers and geometric isomers of hexane (C_6H_{12}).

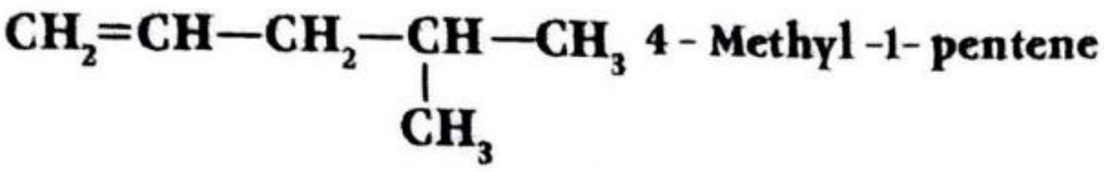
Solution:

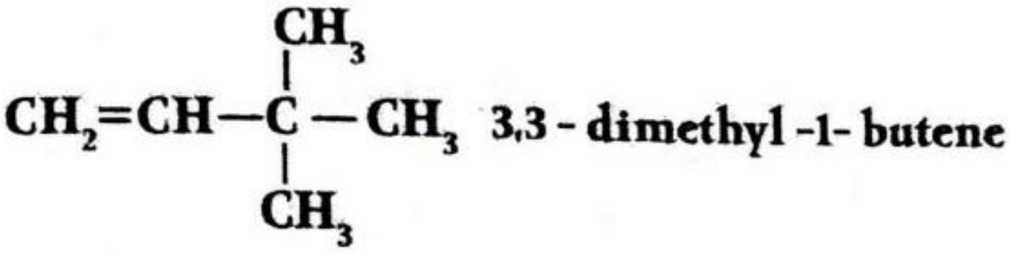
i Structural isomers due the difference in the arrangement of carbon atoms.

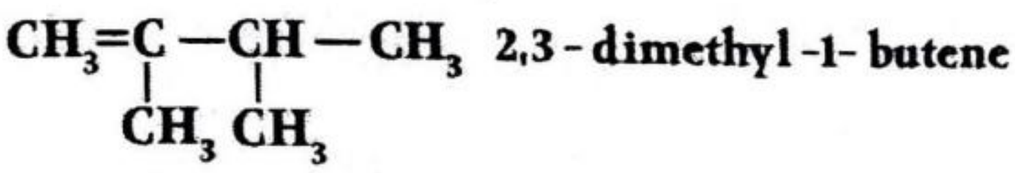
$CH_2 = CH — CH_2 — CH_2 — CH_2 — CH_3$, 1 – Hexene


3 - Methyl -1- pentene

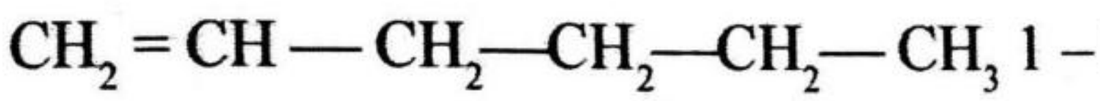

2 - Methyl -1- pentene

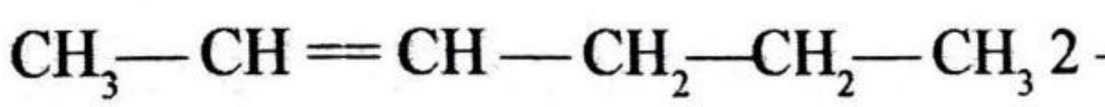

4 - Methyl -1- pentene

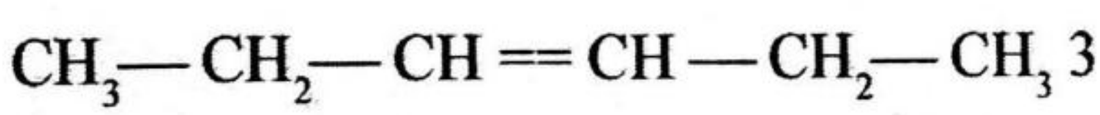

3,3 - dimethyl -1- butene


2,3 - dimethyl -1- butene

ii Position isomer is due to difference in the location of double bond.


1 – hexene


2 – hexene

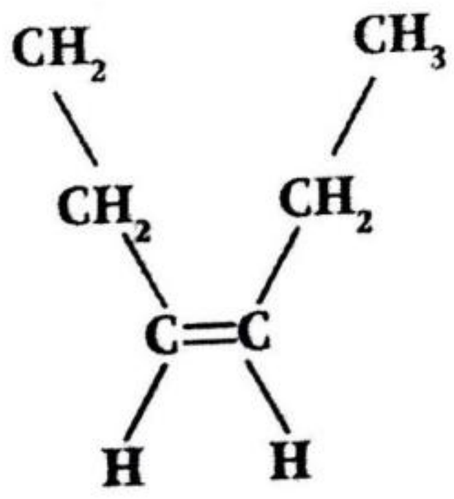

3 – hexene

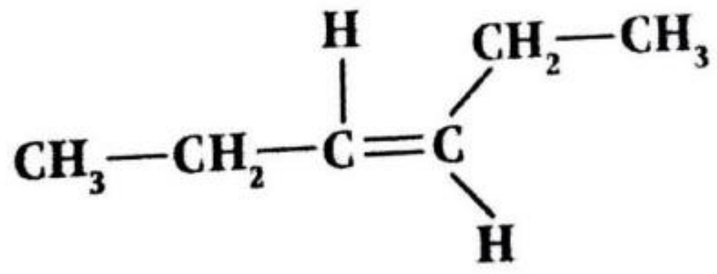
iii Geometric (cis-trans) isomers formed due arrangement of atoms or groups about the double bond.

Cis - two similar groups lie on the same side of the double bond.

Trans – two similar groups lie on the opposite side of the double bond.

Cis – trans isomerism of the carbon – carbon double bond cannot occur when one carbon of the double bond is bonded to two identical groups.


Cis – 3 - hexene


trans – 3 – hexene

48. Which of the following organic compounds shows geometric (C istrans) isomerism? (EGSECE 2004 E.C)

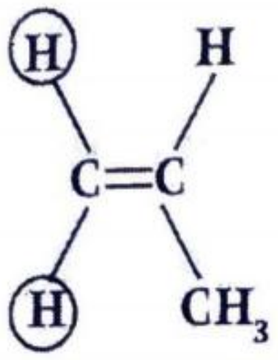
A. $CH_3—CH=CH—CH_3$
B. $CH_2—CH=CH_2$
C. $CH_3—CCl=CH_2$
D. $CH_2=CH_2$

Answer: A

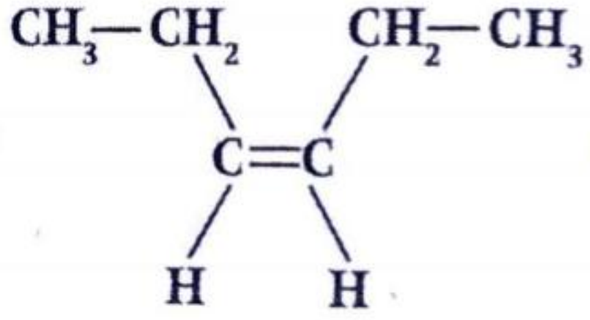
49. Isomerism of the following hydrocarbons exhibit a cis – trans isomerism? …(EGSECE)

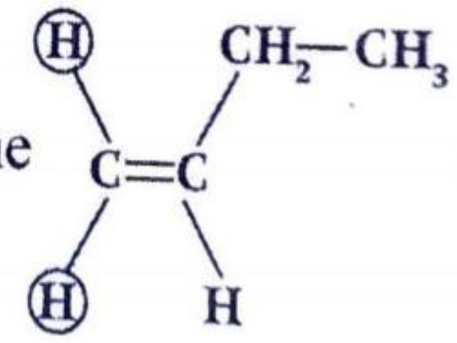
A. 1 – propene
B. 2 – pentyne
C. 3 – hexene
D. 1 – butene

Solution:

A.  doesn't show

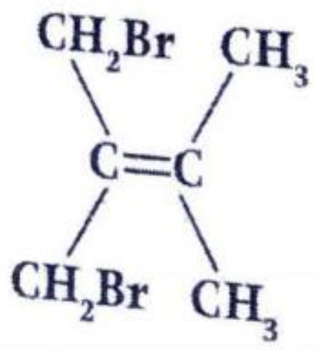
B. Alkynes do not show C is trans isomerism.

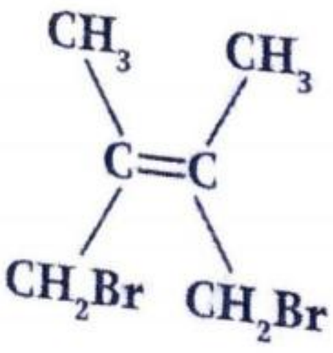
C. 3 - hexene  shows c is trans isomerism

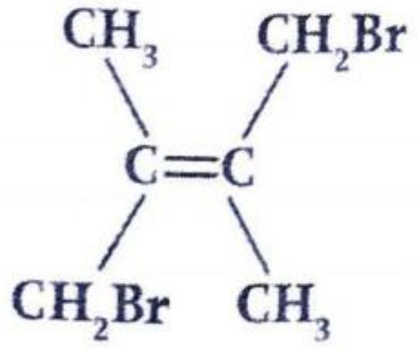
D. 1 – butene  doesn't show

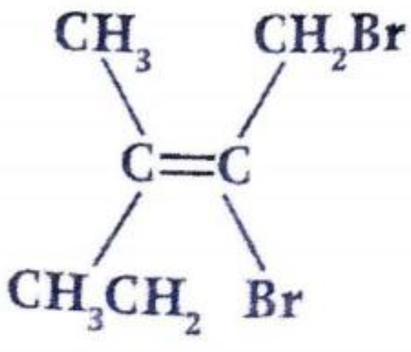
Answer: C

50. Among the following isomeric substances, which one cis – 1, 4 - dibromo – 2, 3 – dimethyl – 2 – butane?

A. 

B. 

C. 

D. 

Solution:

A and D don't exhibit cis-trans insomerism because they have the same group in one carbon.

cis means same side. Therefore the answer is B.

Physical Properties of Alkenes

- $C_2 - C_4$ exist in gaseous state
- $C_5 - C_{17}$ occur in liquid state
- greater than or equal to C_{18} are found in solid state
- are insoluble in water
- as carbon number increases, boiling point, melting point and density increase.

Chemical properties of Alkenes

1. **Addition reaction** is the most common reaction of alkenes

a. Addition of hydrogen (hydrogenation)

$$\underset{\text{Ethene}}{CH_2 = CH_2} + H_2 \xrightarrow[Ni]{heat} \underset{\text{Ethane}}{CH_3 - CH_3}$$

Hydrogenation of alkenes is used to make alkanes

b. Addition of halogens (Halogenations)

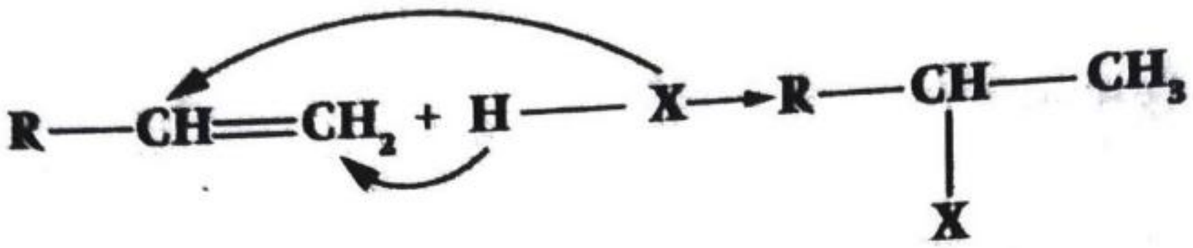
$$\underset{\text{Alkene}}{R - CH = CH_2} + X_2 \rightarrow \underset{\text{Dihaloalkane}}{R{-}\overset{X}{\overset{|}{C}}H{-}\overset{X}{\overset{|}{C}}H_2} \quad (X_2 = Cl_2, Br_2)$$

$$\underset{\text{Ethene}}{CH_2 = CH_2} + Cl_2 \rightarrow \underset{\text{1,2 – di chloroethane}}{ClCH_2 - CH_2Cl}$$

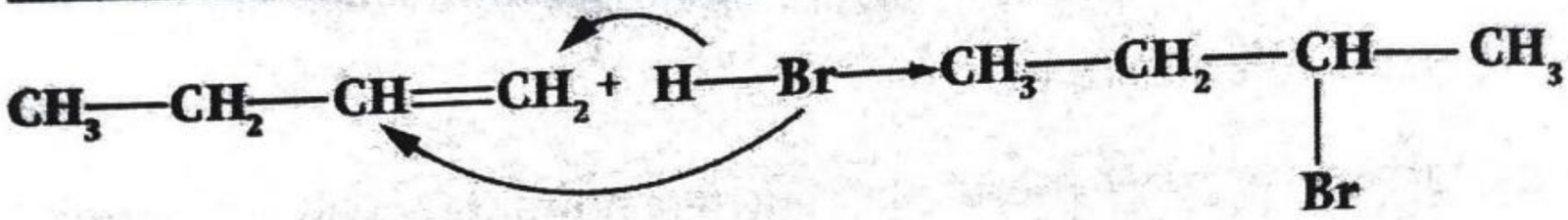
c. Addition of hydrogen halide (Hydrohalogenation)

The product is determined by Marko Vinkov's rule.

This rule states that "Those that have get more" meaning the hydrogen move to the carbon that contains more hydrogen while the halogen toward the carbon that contains contains less hydrogen.

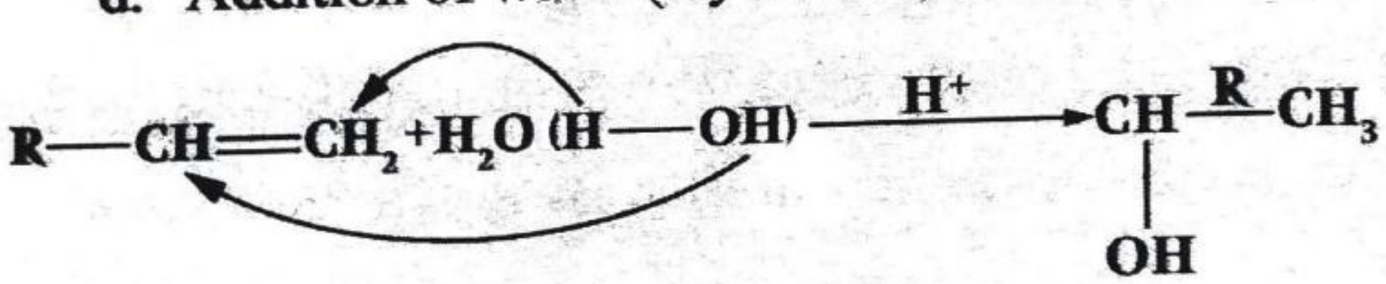


Example



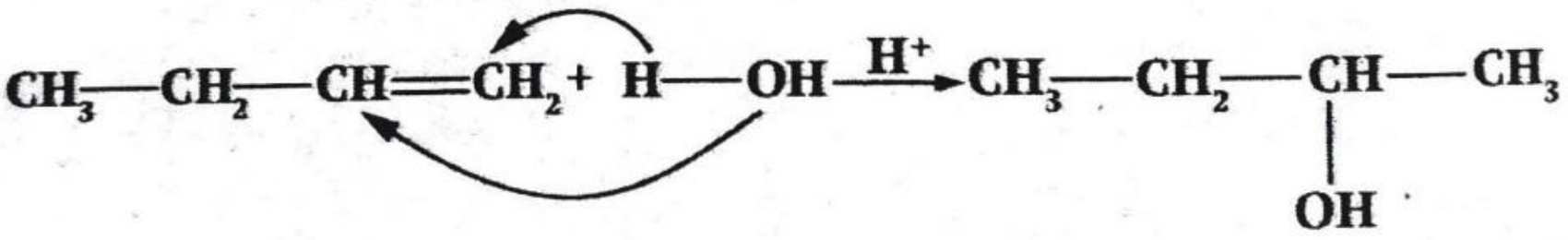
1-Butene **2-Bromobutane**

d. Addition of water (Hydration) is followed by Markovinikov's rule.



Aleken **Alcohol**

Example



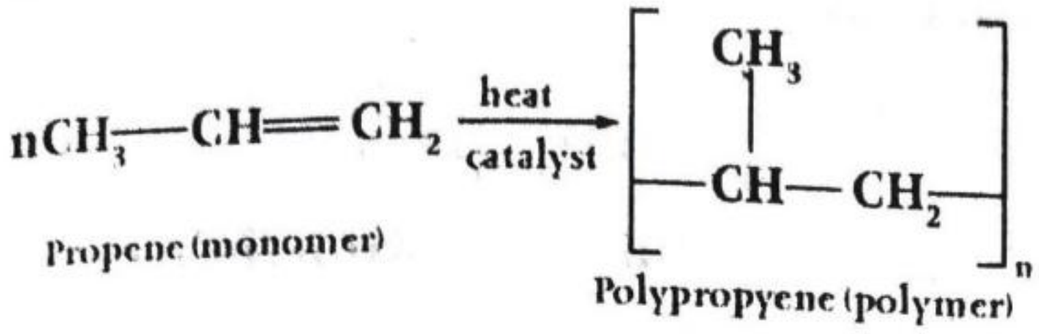
1-Butene **2-Butanol**

Note:

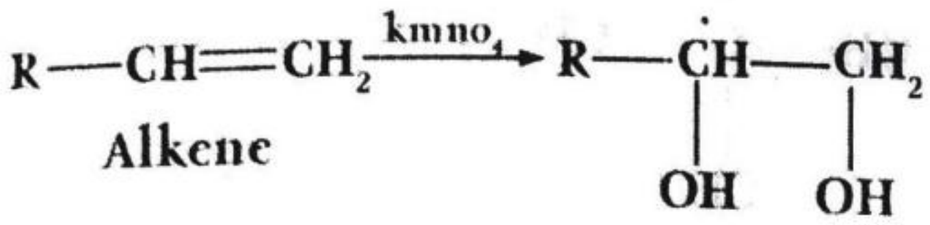
The most important acid used as catalyst in organic chemistry is sulfuric acid.

a. Polymerization (self-addition) is formation of large molecules (polymers) from smaller molecules (Monomers)

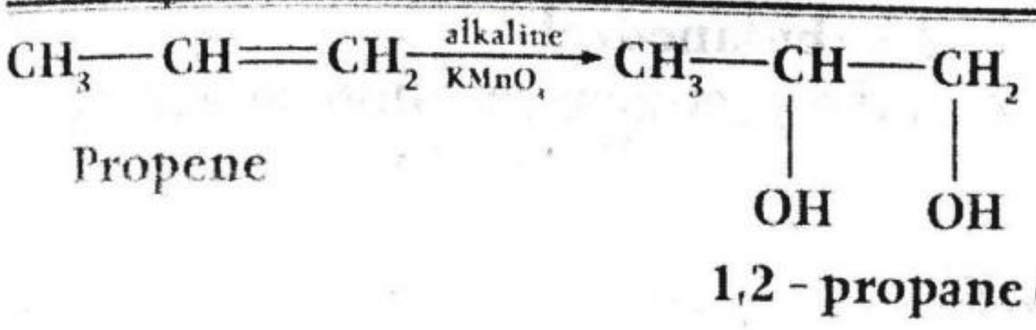
Example



Oxidation of alkenes by strong oxidizing agent potassium permanganate



Example



Illustrative Examples

51. Complete each of the following and name the product formed.

a. $CH_3—CH_2—C(CH_3)=CH_2 + HBr \longrightarrow$

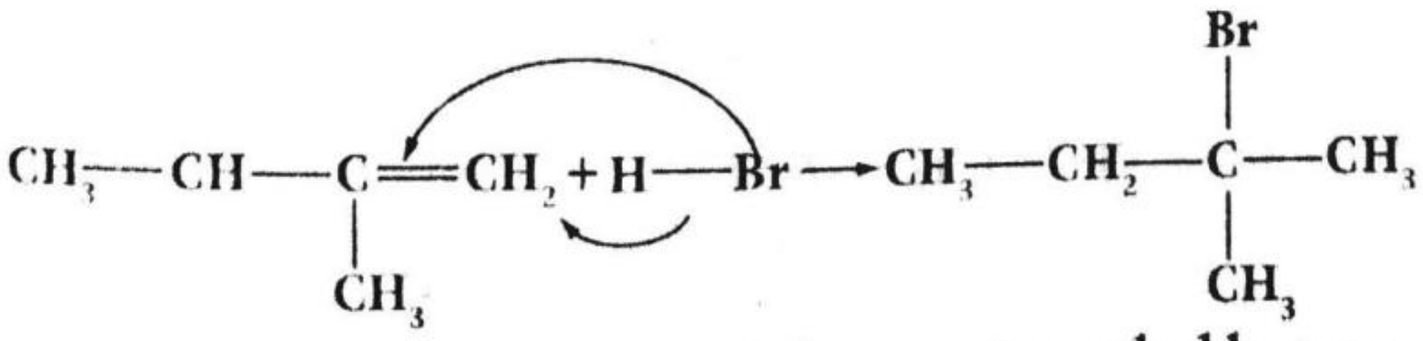
b. $CH_3—C(CH_3)=C(CH_3)—CH_3 + Br_2 \longrightarrow$

c. $CH_3—C(CH_3)=CH_2 + H_2O \longrightarrow$

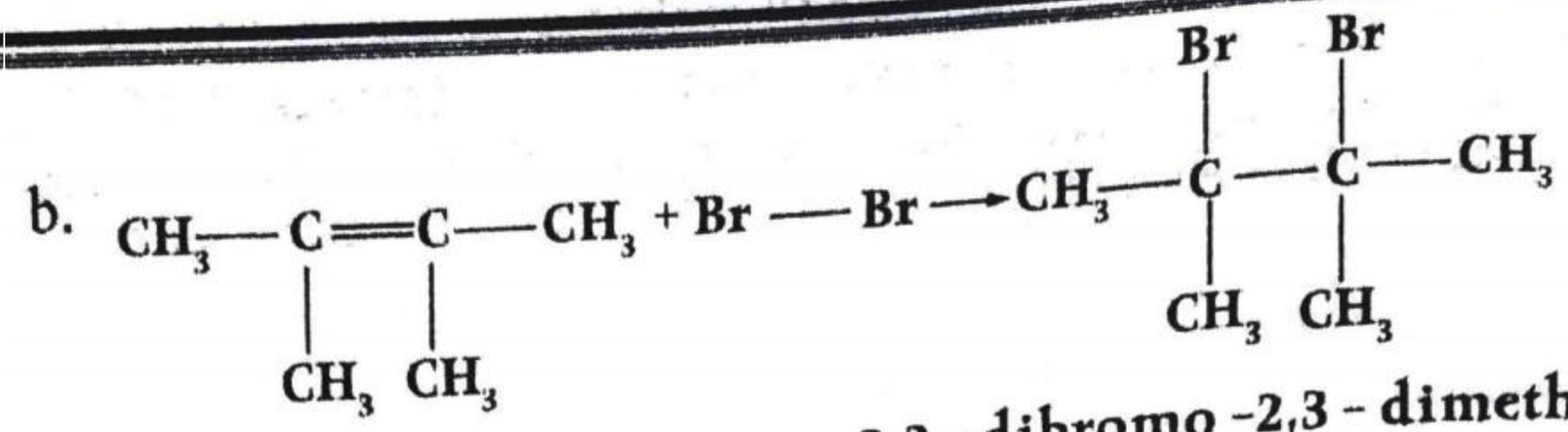
d. $CH_3—C(H)=CH—CH_3 \longrightarrow$

Solution:

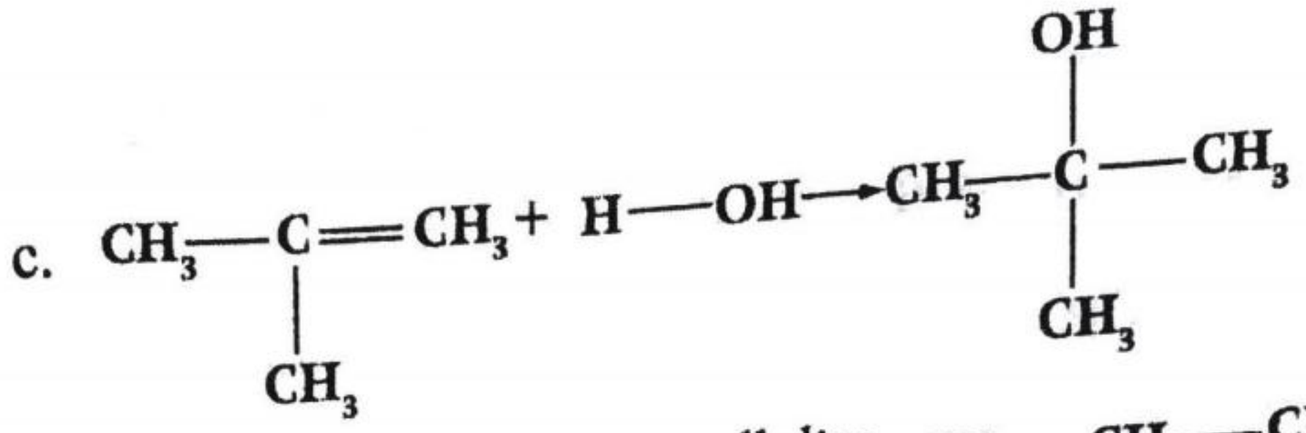
a.

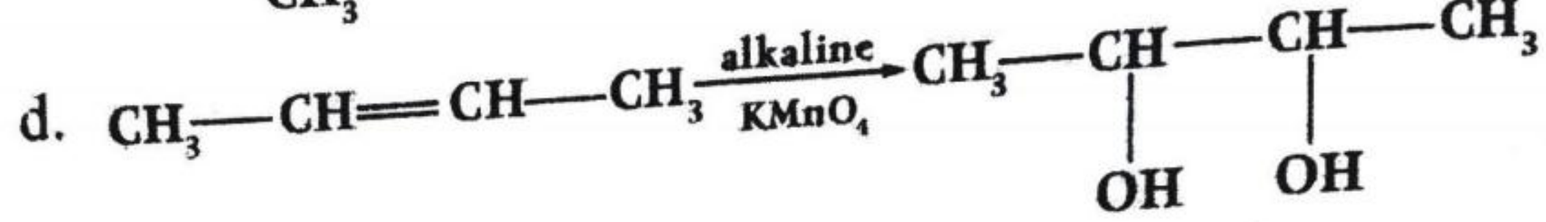


2-bromo-2-methyl butane

b. 

2,3 - dibromo -2,3 - dimethyl butane

c. 

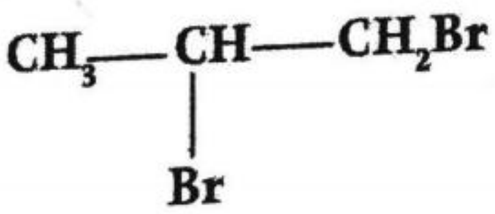
d. 

2,3 - butanediol

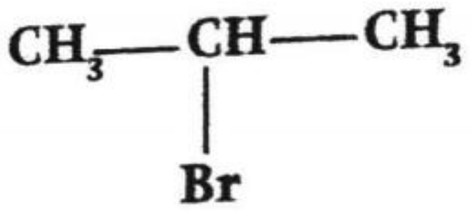
52. What would be the product X in the chemical reaction shown below?
(EGSECE 2000 E.C)

$CH_3—CH=CH_2 + HBr \rightarrow X$`

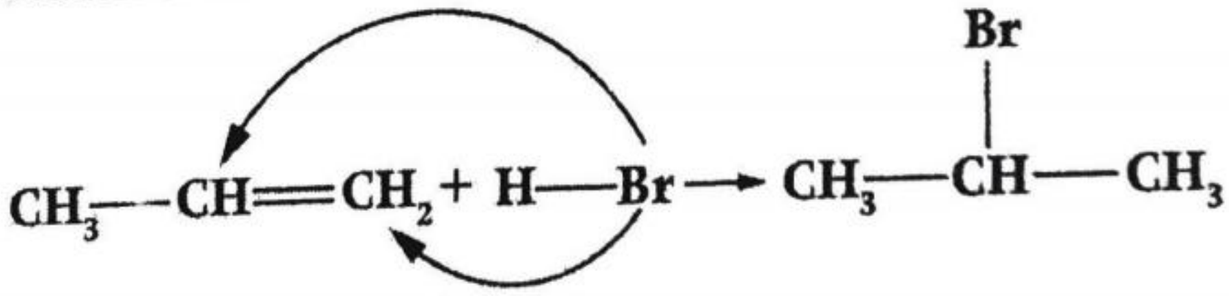
A. $CH_3—CBr_2—CH_3$

B. 

C. $CH_3—CH_2—CH_2—Br$

D. 

Solution:



Answer: D

53. Which of the following chemical reactions is a confirmatory test for un-saturation?

A. $CH_2 = CH_2 + 3O_2 \rightarrow 2CO_2 + 2H_2O$

B. $CH_2 = CH_2 + Br_2 \rightarrow CH_2Br - CH_2Br$

C. $CH_2 = CH_2 + H_2 \rightarrow CH_3 - CH_3$

D. $nCH_2—CH_2 \rightarrow [—CH_2—CH_2—]_n$

Solution:

Bromination is used to test un-saturation (presence of double or triple bond)

Answer: B

54. Given the following reaction

$$R{-}CH{=}CH_2 + X{-}Y \rightarrow R{-}\underset{\underset{OH}{|}}{CH}{-}CH_3$$

Which of the following can correctly represent X and Y respectively.

A. H_2O and H_2SO_4

B. O_2 and Ni

C. Ag^+ and H_2O

D. OH^- and yeast

Solution:

An alcohol is obtained by acid catalyzed hydration of alkene. Therefore, X represents water while Y represents sulfuric acid.

Answer: A

55. Which of the following compounds can undergo bromination reaction readily?

A. C_3H_8

B. C_4H_8

C. C_5H_{12}

D. C_6H_{14}

Solution:

Alkenes undergo bromination reaction more readily. A,C and D are alkanes that fulfill the general formula C_nH_{2n+2}.

Answer: B

56. If $CH_3 - CH = CH_2$ is treated with HCl, the product of the reaction will be

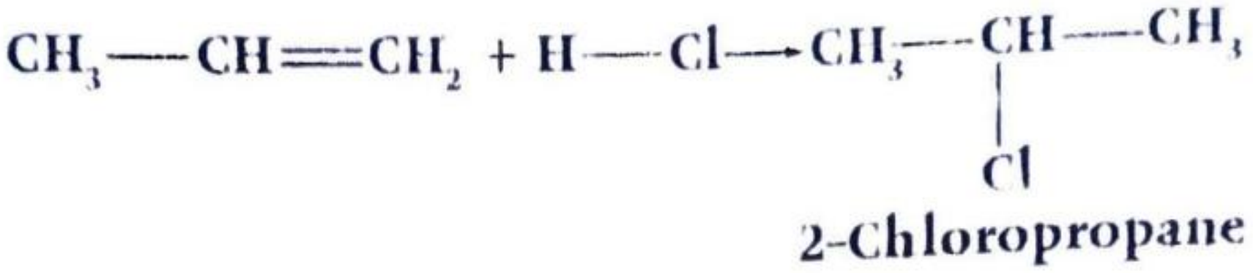
A. Propane

B. 1-chloropropene

C. 1 – chloro propane

D. 2 – chloropropane

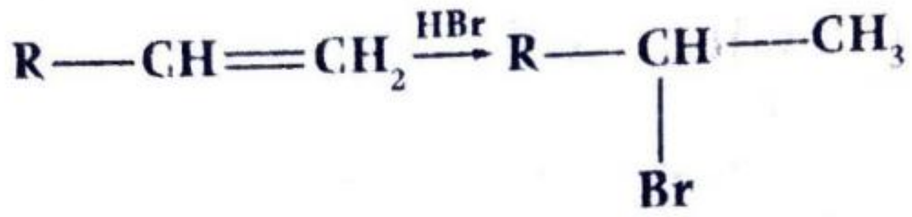
Solution:

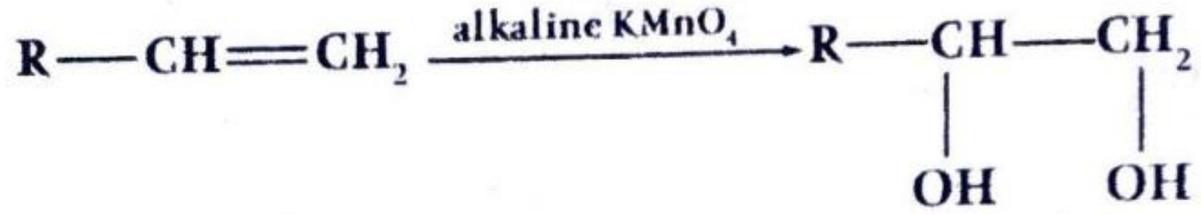


2-Chloropropane

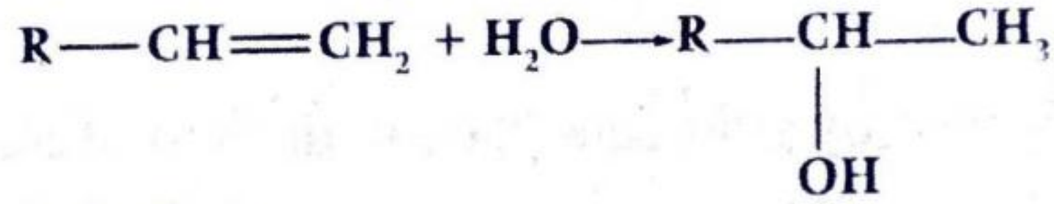
Answer: D

57. One of the following reactions is used to test for the presence of multiple bonds. Which one is it?

A. 

B. 

C. $R—CH == CH_2 + H_2 \longrightarrow R - CH_2 - CH_3$

D. 

Solution:

In addition to bromination, treating with alkaline potassium permanganate used to test un-saturation.

Answer: B

2. **Combustion reaction** is burning of alkenes with oxygen to form carbondioxide and water.

The heat released when 1 mol of alkene is burnt is greater that of alkane of similar mass due to the presence of double bond.

Example:

$2C_3H_6 + 9O_2 \longrightarrow 6CO_2 + 6H_2O + heat$

$C_2H_4 + 3O_2 \longrightarrow 2CO_2 + 2H_2O + heat$

Preparation of alkenes

1. **Dehydration of alcohols** – is removal of water from compound by using dehydrating agent (H_2SO_4).

$$R-CH_2-CH_2OH \xrightarrow[heat]{H_2SO_4} R-CH=CH_2 + H_2O$$

Alcohol — Alkene

Example

$$CH_3-CH_2OH \xrightarrow[heat]{H_2SO_4} CH_2=CH_2 + H_2O$$

Ethanol — Ethene

2. **Dehydro halogenation of alkyl halides with base** – is the removal of hydrogen halide from a compound

$$R-CH_2-CH_2-X + KOH \longrightarrow R-CH=CH_2 + KX + H_2O$$

Alkyl halide — Alkene

Example

$$CH_3-CH_2-CH_2Cl + KOH \rightarrow CH_3-CH=CH_2 + KCl + H_2O$$

Illustrative Example

58. Complete and balance the following reactions.

a. $CH_2-CH_2-CH_2-CH_3OH \xrightarrow[heat]{H_2SO_4}$

b. $CH_3-CH_2-CH_2Br + KOH \longrightarrow$

Solution:

a. $CH_3-CH_2-CH_2-CH_2-OH \xrightarrow[heat]{H_2SO_4} CH_3-CH_2-CH=CH_2 + H_2O$

b. $CH_3-CH_2-CH_2-Br + KOH \longrightarrow CH_3-CH=CH_2 + KBr + H_2O$

59. Write the alkyl halide used to make ethene.

Solution:

$CH_3-CH_2X + KOH \rightarrow CH_2=CH_2 + KX + H_2O$

Haloethane (X = Cl or Br)

Uses of Alkenes

➥ Alkenes are mainly used asraw materials to manufacture polymers.

Example:

$$nCH_2=CH_2 \longrightarrow [-CH_2-CH_2-]_n$$

$$nCH_2 = CH - CH_3 \rightarrow \left[\begin{matrix} CH_3 \\ | \\ CH_2 - CH \end{matrix} \right]_n$$

Propene Polypropene

- Ethene is also used for the preservation and artificial ripening of fruits.
- Alkenes undergo acid – catalyzed hydration reaction to form alcohols.
- Alkenes are also used to prepare alkyl halides

Illustrative Example

60. Which of the following is the monomer of PVC?

A. C_2H_4 C. C_2H_3Cl

B. C_3H_5 D. C_2H_6Cl

Solution:

The monomer of polyvinyl chloride (PVC) is vinyl chloride

($CH_2 = CHCl$ or C_2H_3Cl

Answer: C

61. The most important industrial application of alkenes is

A. as fuel C. in polymer production

B. as solvent D. in insecticide production

Answer: C

Alkynes

- alkynes are unsaturated hydrocarbons containing carbon-carbon triple bond.
- Their general formula is C_nH_{2n-2}, n = 2,3,....
- The alkynes homologous series is also called acetylene series.

Example

C_2H_2, C_3H_4, C_4H_6,...

Find the molecular formula of the first four homologous series of alkynes.

Solution:

First alkyne, n = 2

$$C_nH_{2n-2} = C_2H_{2\times3-2} = C_2H_2$$

Second alkyne, n = 3

$$C_nH_{2n-2} = C_3H_{2\times3-2} = C_3H_4$$

Third alkyne, n = 4

$$C_nH_{2n-4} = C_4H_{2\times4-2} = C_4H_{6.}$$

Fourth alkayne, n = 5

$$C_nH_{2n-2} = C_5H_{2\times5-2} = C_5H_{8.}$$

Nomenclature of alkynes

A. Common name

Alkynes are commonly named as derivative of acetylene

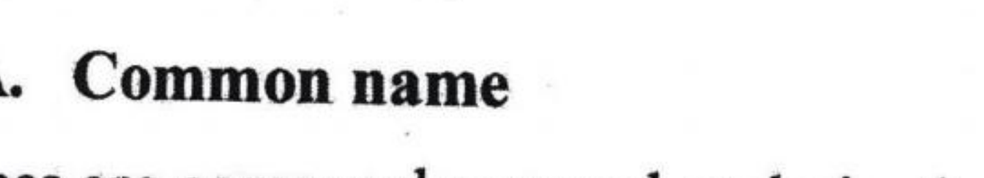
Example

a. $H-C\equiv C-H$, acetylene

b. $H-C\equiv C-CH_3$, methyl acetylene

c. $CH_3-C\equiv C-CH_3$, dimethyl acetylene

d. $CH_3-CH_2-C\equiv C-CH_3$, ethyl methyl acetylene

e. 

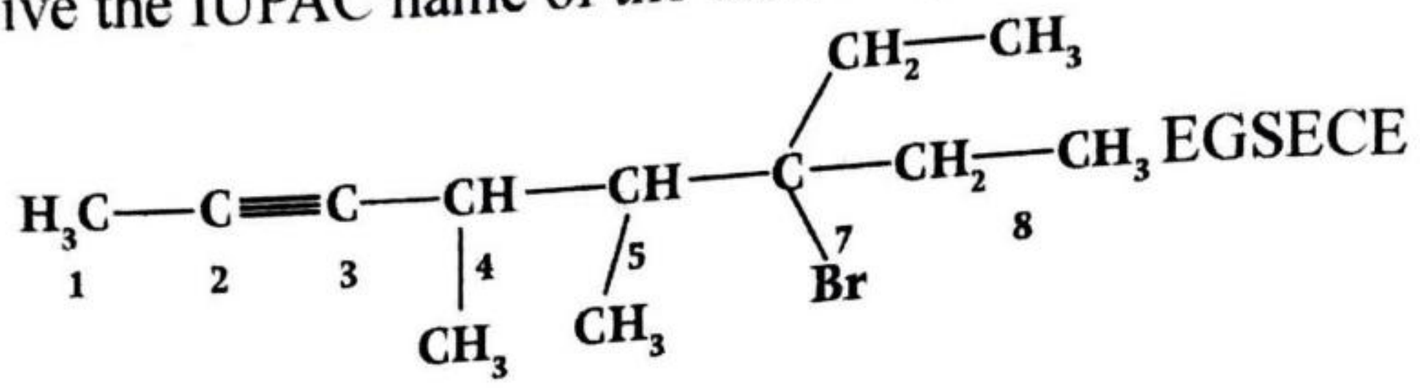
$CH_3-CH(CH_3)-C\equiv CH$, isopropylacetylene

B. IUPAC name of alkynes

Identify the longest chain including the triple bond and count from the end nearer to the triple bond.

Illustrative Example

62. Give the IUPAC name of the following alkynes.

a. 

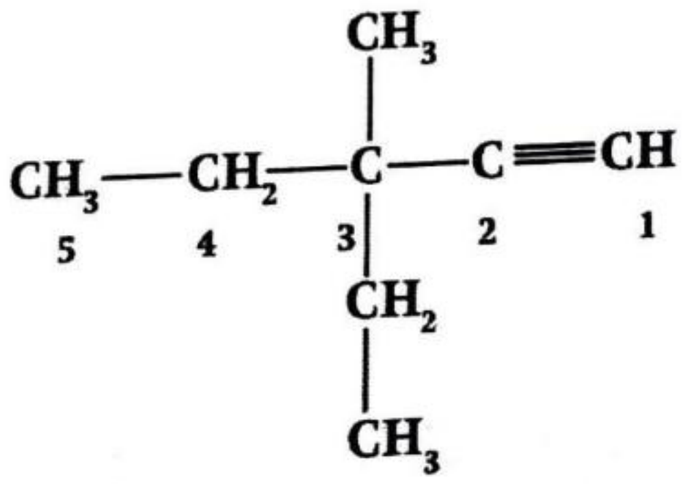
Solution:

Start counting from the left side. The parent name is octyne.

Answer:

6 – Bromo – 6 – ethyl – 4, 5 – dimethyl – 2 – octlyne

(6 – Bromo – 6 – ethyl – 4, 5 – dimethyl oct – 2 – yne)

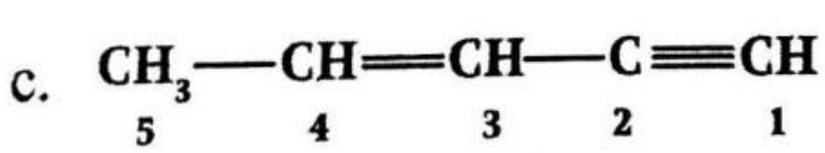
b. 

Solution:

Start from the right side to know the longest chain. The parent name is pentyne.

Answer: 3 – ethyl – 3 – methyl – 1 – pentyne

(3 – ethyl – 3 – methyl pent – 1 – yne)

c. 

Solution:

If the double bond and triple bond appear in the same structure start counting from the end nearer to the triple bond.

Answer: 3 – penten – 1 – yne

63. Draw the structure of

a. 1 – chloro – 2 – butyne

Solution:

Draw the skeletal structure of the parent structure

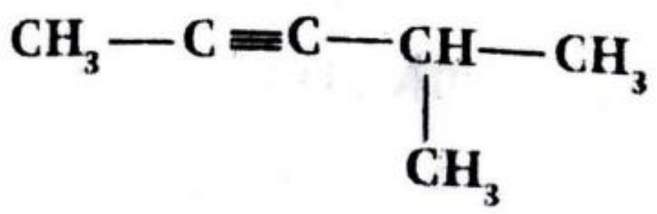
C – C ≡C – C

Place the side chain and fulfill the octet of carbon.

$H_3C - C \equiv C - CH_2Cl$ or $ClH_2C - C \equiv C - CH_3$

b. 4 – methyl – 2 – pentyne

Solution:



Isomerism in Alkynes

Alkynes exhibit structural and position isomerism but does not show cis trans isomerism.

Ethyne and propyne do not show any isomerism. As carbon number increases number of isomers increases.

Example

1. Write the position isomers of butyne (C_4H_6)

Solution:

a) $CH_3 - CH_2 - C \equiv CH$ b) $CH_3 - C \equiv C - CH_3$

1 – butyne 2 – butyne

Physical properties of alkynes

- Ethyne, propyne and butyne are gases.
- $C_5 - C_{17}$ exist in liquids state.

- Alkynes are insoluble in water because they are non polar.
- Alkynes with carbon number greater than or equal to 18 are solids at room temperature.

64. Which one has the highest boiling point?

A. C_4H_{10} B. C_4H_8 C. C_4H_6 D. CH_4

Solution:

The boiling point of alkynes is greater than alkenes and alkanes of same carbon number due to the presence of triple bond.

Answer: C

65. Which one of the following has the least boiling point?

A. 1 – hexyne
B. 2 – Hexyne
C. 3 – Methyl – 1 – pentyne
D. 3, 3 – dimethyl – 1 – butyne

Solution:

All have the same carbon number, as branching increase boiling point decrease.

Answer: D

Chemical properties of alkynes

Alkynes are the most reactive aliphatic hydrocarbons due to the presence of carbon – carbon triple bond.

1. Addition reaction is the most common reaction of alkynes.

A. Addition of hydrogen

$$C_nH_{2n-2} + H_2 \xrightarrow[\text{catalyst}]{\text{heat}} C_nH_{2n} + H_2 \xrightarrow[\text{catalyst}]{\text{heat}} C_nH_{2n+2}$$

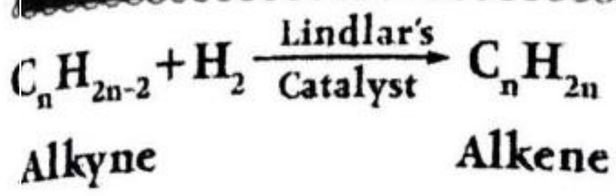
Alkyne Alkene Alkane

Example

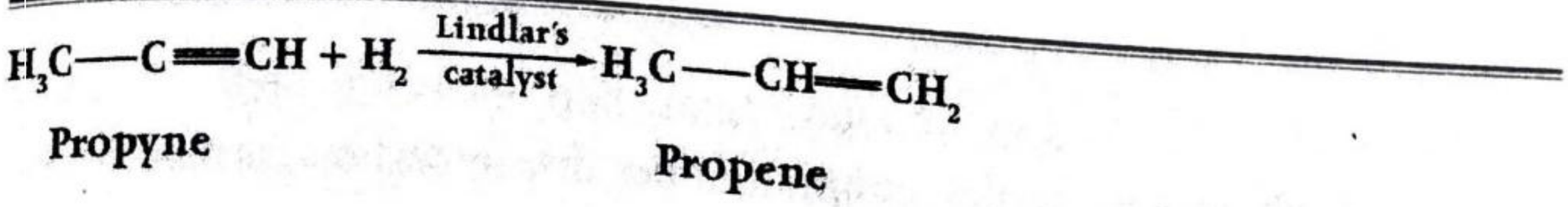
$$H-C\equiv C-CH_3 + H_2 \xrightarrow[\text{heat}]{Ni} CH_3-CH=CH_2 \xrightarrow[\text{heat}]{Ni} H_3C-CH_2-CH_3$$

Propyne Propene Propane

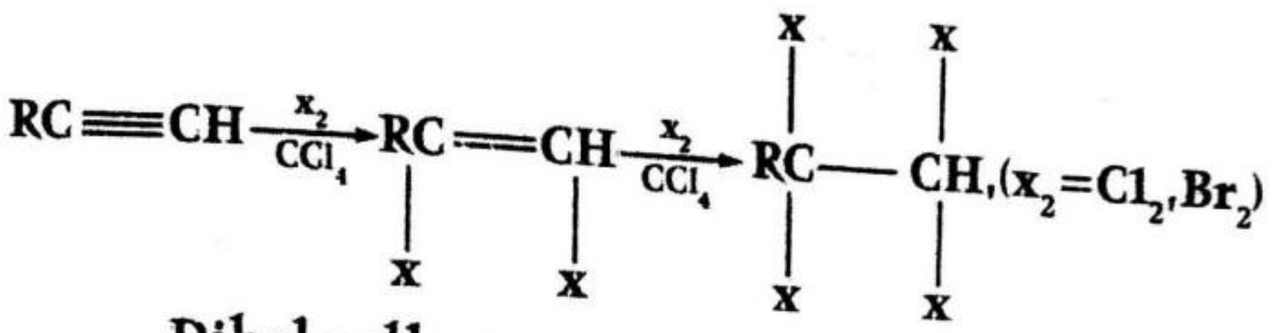
Partial hydrogenation can be carried out by using a special catalyst called Lindlar catalyst.



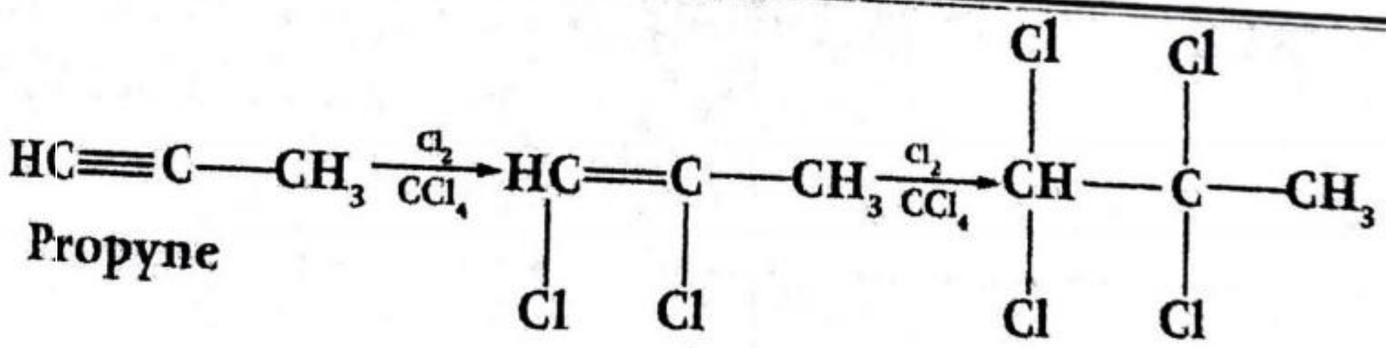
Example



B. Addition of halogen

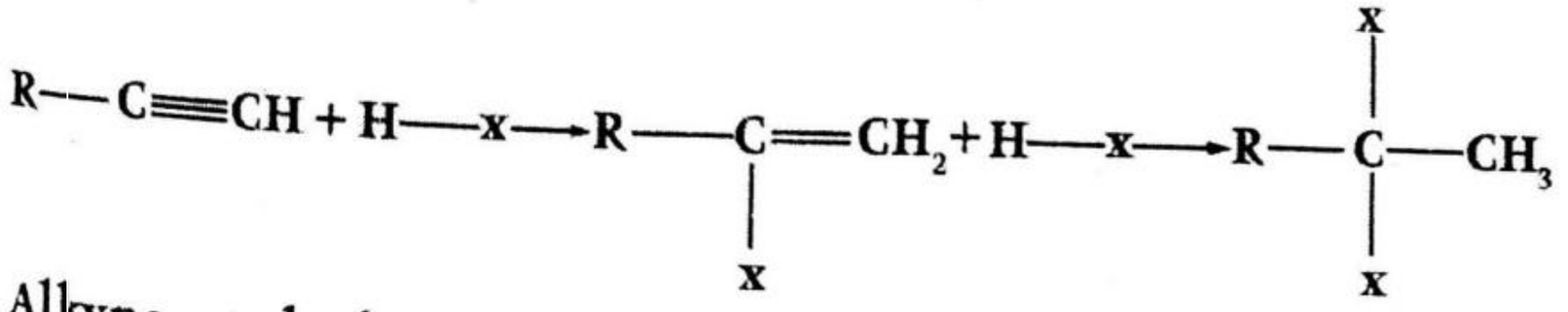


Example

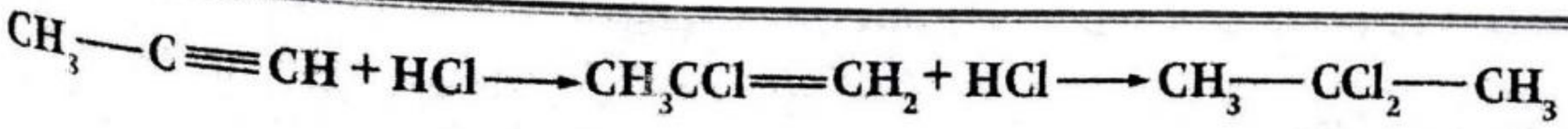


1,2-dichloropropene 1,1,2,2,-tetrachloropropane

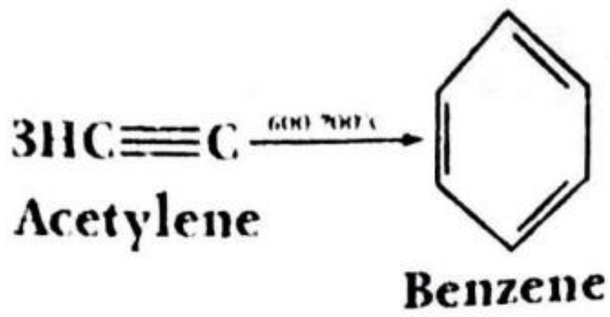
C. Addition of hydrogen halide



Example



D. Trimerization of acetylene yields benzene



Combustion of alkynes

Alkynes burn with smoky luminous flame and release greater quantity of heat than alkenes of similar carbon number due to carbon-carbon triple bond.

Example

$2C_4H_6 + 11O_2 \rightarrow 8CO_2 + 6H_2O + heat$

Illustrative Example

66. Identification of the unknown hydrocarbons of equal number of moles has shown the following experimental results in the laboratory.

Hydrocarbon	Addition of Br_2 water	Amount of Br_2 consumed
A	decolorized	1 volume
B	No change	None
C	Decolorized	2 volumes

The three hydrocarbons could respectively be (EGSECE 2004 E.C)

A. Alkane, alkene, alkyne
B. Alkene, alkyne, alkane
C. Alkyne, alkene, alkane
D. alkene, alkane, alkyne

Solution:

Bromination is used to test un-saturation. Bromine water decolorized in akene after one volume is consumed while in alkyne two volumes of Br_2 is consumed.

Answer: D

67. What are the products 'X' and 'Y' in the reaction below respectively? (EGSECE)

$CH_3 - C \equiv C - CH_3 + Cl_2 \rightarrow X + H_2 \rightarrow Y$

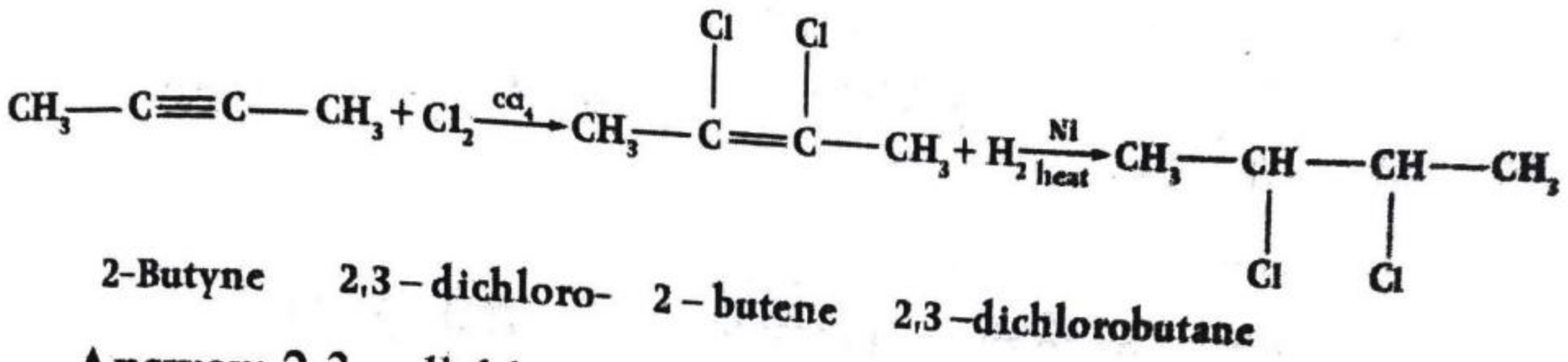
A. 2,3 – dichlorobutene and 2,3 – dichlorobutane

B. 2,2 – dichlorobutene and 2,2 – dichlorobutane

C. 2 – butyne and 2 – butne

D. 1,2 – dichlorobutene and 1,2 – dichlorobutane

Solution:



Answer: 2,3 – dichloro – 2 – butene and 2,3 – dichlorobutane

68. When one mole of acetylene is completely reacted in excess amount of bromine

Which product is likely to occur? ..(EGSECE 2002E.C)

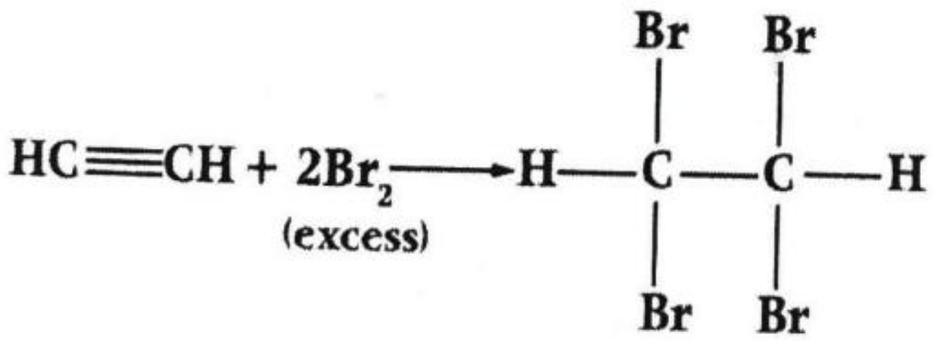
A. $CH_3 - CHBr_2$

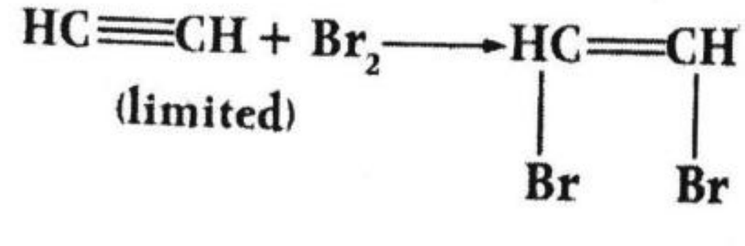
B. $CHBr_2 - CHBr_2$

C. $CHBr = CHBR$

D. $CH_3 - CH_2Br$

Solution:





Answer: B

69. What is the missing term "X" in the following reaction?

$CH_3 - C \equiv CH + 2HCl \rightarrow X$

A. $CH_3 - CCl_2 - CH_3$

B. $CH_3 - CHCl - CH_2Cl$

C. $CH_3 - CH_2 - CHCl_2$

D. $CH_3 - CCl{=}CHCl$

Solution:

$CH_3 - C \equiv CH + 2HCl \rightarrow CH_3CCl_2 - CH_3$, this reaction must follow Markovinkow's rule.

Answer: A

Preparation of alkynes

1. Dehydrohalogenation of vicinal (adjacent) dihalides with NaOH or KOH of $NaNH_2$.

$$R - \underset{|}{\overset{X}{CH}} - \underset{|}{\overset{X}{CH_2}} + 2NaOH \rightarrow \underset{\text{Alkyne}}{R - C \equiv CH} + 2NaX + 2H_2O$$

2. Alkylation of sodium accetylide with primary alkylhalide.

$$HC \equiv C^- Na^+ + R - X \rightarrow HC \equiv C - R + NaX$$

Sodium carbide 1^{st} alkyl halide alkyne

$$HC \equiv C^- Na^+ + CH_3Cl \rightarrow HC \equiv C - CH_3 + NaCl$$

Illustrative Example

70. Complete the following reactions.

a. $CH_3 - CHBr - CHBr - CH_3 + 2KOH \rightarrow$

b. $H - C \equiv C^- Na^+ + CH_3 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_2Cl \rightarrow$

Solution:

a. $CH_3-CHBr-CHBr-CH_3+2KOH \rightarrow CH_3-C\equiv C-CH_3 + 2KBr+2H_2O$

b. $H-C\equiv CNa^+ + CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2Cl \rightarrow H-C\equiv C-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3 + NaCl$

71. A reaction for the general preparation for a member of homologous series of certain hydrocarbon is given by:

$R - CHCl - CH_2Cl + 2NaOH \rightarrow X + 2NaCl + 2H_2O$

"X" in the above chemical equation represents

A. $R-C\equiv CH$

B. $R-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$

C. $R-CH=CH_2$

D. $R-CH_2-CH_3$

Solution:

The above chemical equation represents dehydrohalogenation of adjacent dihalides which is used to make alkynes.

Answer: A

Uses of alkynes

The most important alkyne is acetylene (ethyne). It is mainly used for cutting and welding metals in oxyacetylene torch. Acetylene is also used to make vinylchloride, vinylcyanide and to make a polymer acrylon.

Illustrative Example

72. Which of the following hydrocarbons has the highest heat of combustion? (EGSECE)

A. CH_4

B. C_2H_6

C. C_2H_4

D. C_2H_2

Solution:

Acetylene (C_2H_2) has the highest heat of combustion due to the carbon to carbon triple bond.

Answer: D

73. Which of the following reagents are used to prepare acetylene in the laboratory? (EGSECE)

A. Sodium acctate and soda lime

B. Alcohol and concentrated sulfuric acid

C. Calcium carbide and water

D. Benzene and nitric acid

Solution:

Acetylene is prepared in the laboratory by the reaction of calcium carbide with water.

$$CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$$

Calcium carbide acetylene calcium hydroxide

Answer: C

74. **The flame of acetylene is smokey and luminous flame. This is because of**

A. **High carbon content of the compound**

B. **High reactivity of acetylene**

C. **Saturation of the compound**

D. **Low molecular mass**

Answer: A

75. **The gas that is widely used for cutting and welding of metals has the formula**

A. C_2H_6

B. C_3H_8

C. C_2H_2

D. CH_4

Solution:

Ethyne is simplest alkyne used for cutting and welding of metals, its molecular formula is C_2H_2.

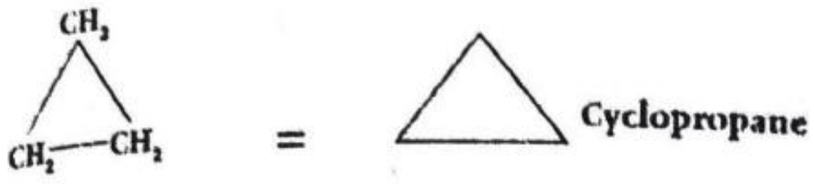
Answer: C

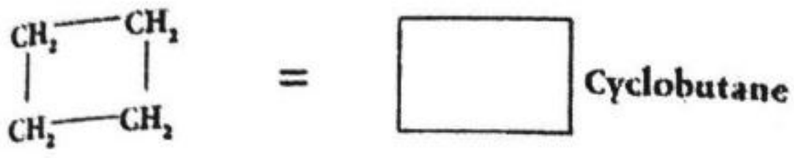
.3 Cyclic Hydrocarbons

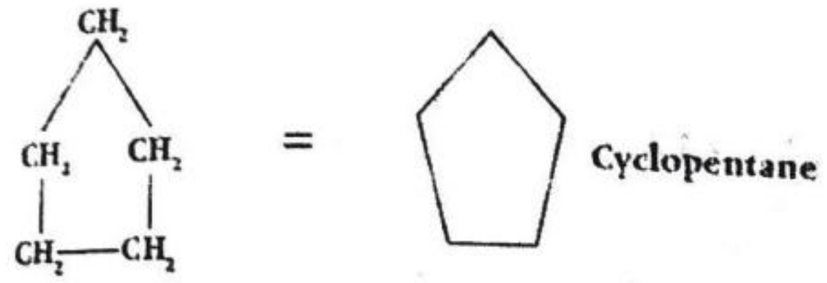
. **Cycloalkanes** have general formula C_nH_{2n}, $n \geq 3$

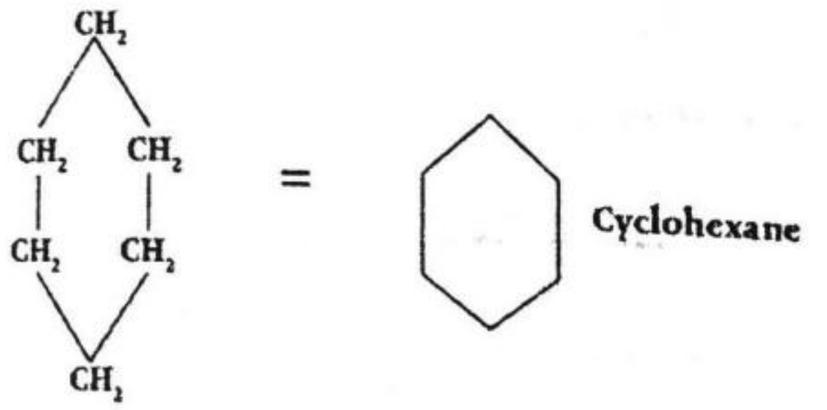
They are isomeric with open chain alkenes.

Example





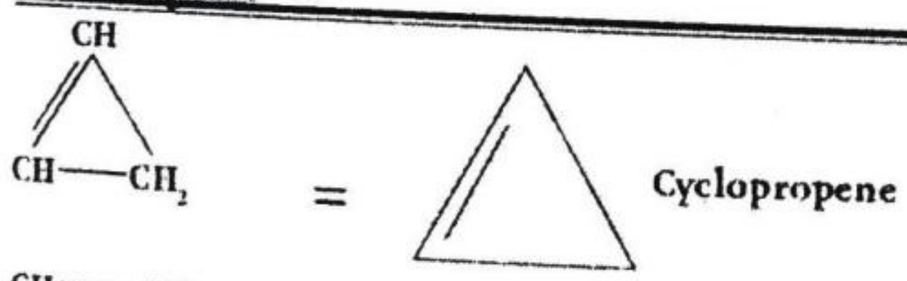


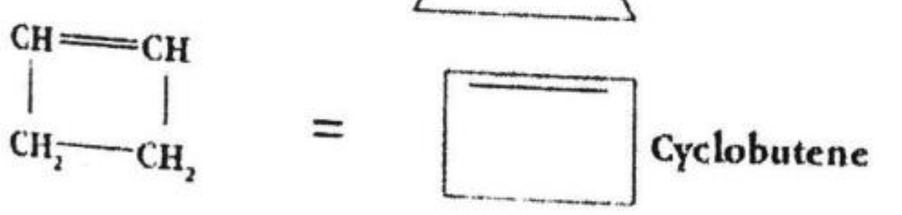


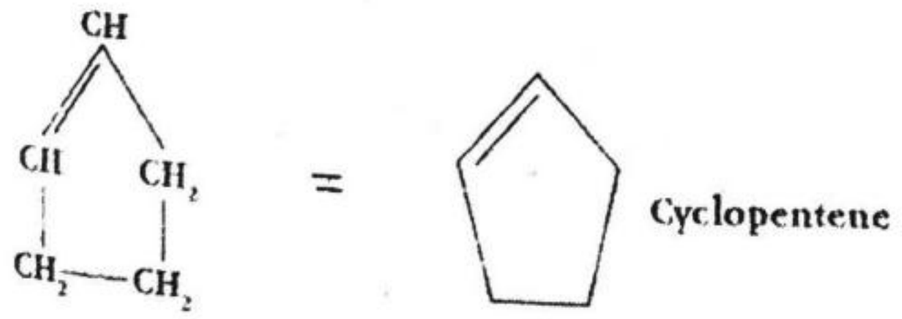
B. Cycloalkeneshave a general formula C_nH_{2n-2}, $n \geq 3$

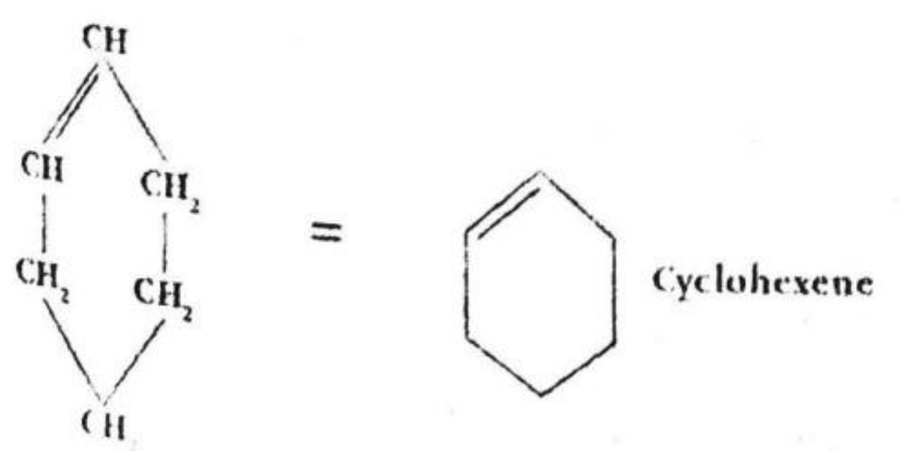
They are isomeric with open chain alkynes and dienes.

Example



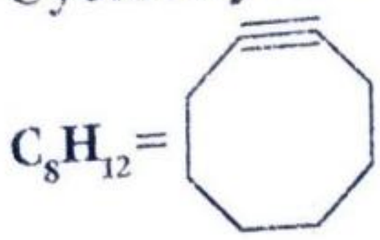






C. Cyclo alkynes have a general formula CnH2n-4. , n ≥ 8

- Because of the linear nature of the – C ≡C – alkyne unit, cycloalkynes are usually highly strained and can only exist when the number of carbon atoms in the ring is great enough to provide the flexibility necessary to accommodate this geometry.
- Cyclooctyne is the simplest cycloalkyne that is stable compound.



cyclooctyne

Illustrative Example

76. Find the molecular formula of cycloalkane, cycloalkene and cycloalkyne with carbon number 9.

Solution:

For n = 9

Cycloalkane, $C_nH_{2n} = C_9H_{2\times9} = C_9H_{18}$

Cycloalkene, $C_nH_{2n-2} = C_9H_{2\times9-2} = C_9H_{16}$

Cycloalkyne, $C_nH_{2n-4} = C_9H_{2\times9-4} = C_9H_{14}$

77. Which of the following is isomeric to cyclobutane?

A. n – butane

B. 1 – butyne

C. 1 – butene

D. cyclobutene

Solution:

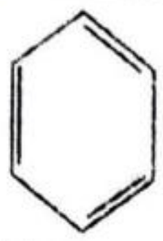
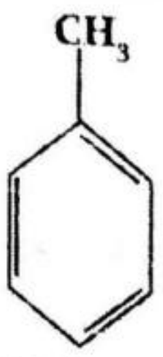
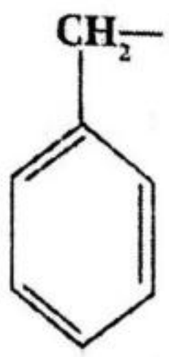
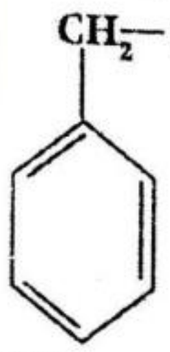
Cycloalkanes are isomeric with open chain alkenes

Answer: C

6.4 Aromatic hydrocarbons

- They are benzene and benzene derivatives.
- They are stable towards many chemical reagents.
- They have a general formula $C_nH_{2n} - 6$, $n \geq 6$

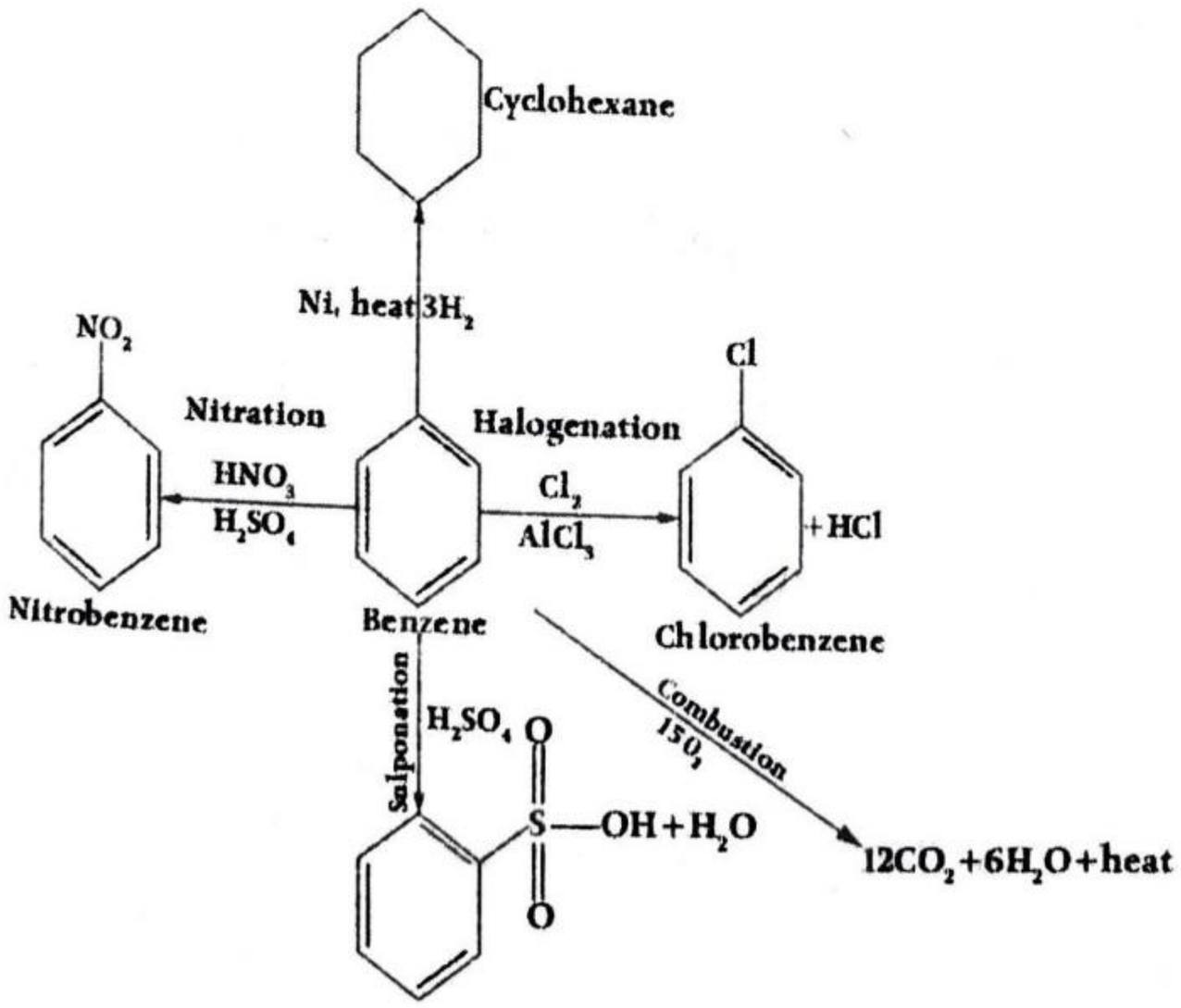
Table 6.6. Some aromatic hydrocarbons

Aromatic hydrocarbons	Structure formula	Molecular formula
Benzene		C_6H_6
Toluene (methyl benzene		C_7H_8
Ethyl benzene		C_8H_{10}
Propyl benzene		C_9H_{13}

Benzene is the simplest and important aromatic hydrocarbon.

➥ The characteristics reaction of aromatic hydrocarbons is substitution reaction.

Reactions of benzene can be summarized as follows.



Note:

Nitration, halogenation and sulphonation are substitution reaction

Uses of aromatic hydrocarbons

- They are used as organic solvents.
- Benzene is used as fuel for cars, raw materials for production of plastics, dyes, medicines, rubber and other benzene derivatives.

Illustrative Example

78. Benzene does not react with bromine or with cold potassium permanent, why?

Answer: because of the stability of aromatic ring

79. Aromatic hydrocarbons mainly react by…

A. Addition
B. Substitution
C. elimination
D. hydrogenation

Solution:

The most common reaction of aromatic hydrocarbon is substitution reaction.

Answer: B

80. Methyl benzene (toluene) is a member of the homologous series of benzene, what is its chemical formula? (EGSECE 2003E.C)

Solution:

Carbon number = 7, $C_nH_{2n-6} = C_7H_{2\times7-6} = C_7H_8$

81. Benzene burns with intense smoky flame. This is because…. (EGSECE 2000E.C)

A. It has the highest carbon content
B. It has resonance hybrid structure
C. It is highly unsaturated
D. It is resistant to oxidizing agents.

Solution:

$2C_6H_6 + 15O_2 \rightarrow 12CO_2 + 6H_2O + heat$

Answer: A This is due to high carbon content

82. Benzene is essential resonance hybrid of the two Kekule structure. The reason for this is that… (EGSECE)

A. Half the molecule corresponds to one structure and the other half the second structure

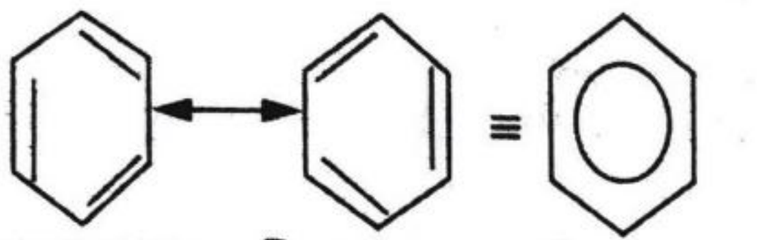
B. Individual benzene molecule changes back and forth between the two structures.

C. At low temperatures, benzene can be separated in to the two structures.

D. The two structures make equal contributions to the resonance hybrid.

Solution:

Kekule determined the structure of benzene.



Resonance Structure I Resonance Structure II Resonance hybrid Sturcture

both resonance structures have equal contribution to the hybrid structure **Answer: D**

83. What is the product formed when benzene is treated with concentrated nitric acid in the presence of sulphuric acid?.... (EGSECE 1995 E.C)

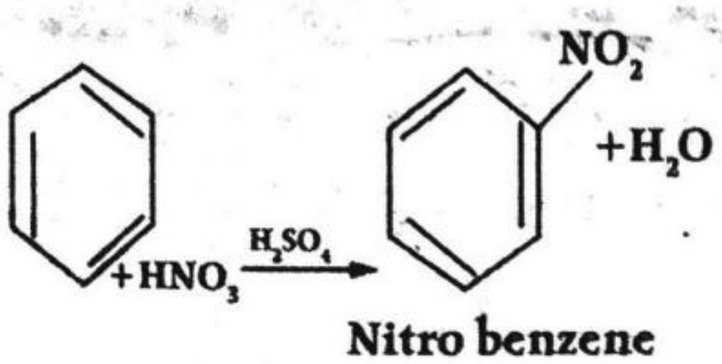
A. No reaction

B. Nitrobenzene

C. cyclohexane

D. Benzene sulphoric acid

Solution:



Nitro benzene

Answer: B

84. Which of the following properties is not appropriate to describe benzene? (EGSECE 2002 E.C)

A. It has a characteristics reactivity of alkenes

B. It has a resonance hybrid structure

C. It is an aromatic hydrocarbon

D. It is volatile liquid

Solution:

Benzene has less reactive than alkenes and alkynes due to the stability of the aromatic ring.

Answer: A

6.5 Natural sources of hydrocarbons

A. Natural gas – mainly contain methane, ethane, propane and butane. It is used as fuel and raw material for production organic substances.

B. Petroleum–contain mainly alkanes, cycloalkanes and aromatic compounds.

- ➦ It cannot be used for various purposes directly.
- ➦ Fractional distillation changes petroleum to useful products. This physical technique separates mixtures that contain different components with different boiling point.
- ➦ Cracking is the most important process that breaks long chain hydrocarbons into smaller units.

C. Coal – is an important source of aromatic hydrocarbons. Coal is heated in the absence of air and break down in to coal gas and coal tar. The coal gas used as fuel.

Illustrative Example

85. Which physical property makes it possible to separate the components of crude oil by means of fractional distillation? (EGSECE 2004 E.C)

A. Boiling point
B. Melting point
C. Conductivity
D. D. Density

Answer: A

86. Water gas is a mixture of … (EGSECE)

A. CO and H_2O
B. CO and H_2
C. H_2 and CO
D. H_2 and CH_4

Solution:

Coke is a form of carbon used to produce gaseous fuels:

- Water gas mixture of CO and H_2 in one to one ratio).
- Producer gas (mixture of CO and N_2 in one to one ratio).

Answer: C

87. Which of the following products of petroleum are arranged in the correct order of increasing boiling point?

A. Petrol, asphalt, diesel oil, kerosene

B. Asphalt, diesel oil, kerosene, petrol

C. Diesel oil, petrol, asphalt, kerosene

D. Petrol, kerosene, diesel oil, asphalt

Solution:

The common petroleum products are:

Fractions	Carbon number
Gases	$C_1 - C_4$
Petroleum ether	$C_5 - C_7$
Gasoline (petrol)	$C_5 - C_{10}$
Kerosene	$C_{10} - C_{16}$
Fuel oil (diesel oil)	$C_{15} - C_{16}$
Lubricating oil	
Grease	$> C_{19}$
Petroleum jell	
Paraffin max	
Asphalt	$> C_{20}$
Coke	

You remember as carbon number increase boiling point also increases.

Answer: D

88. The main constituent of petroleum are…(EGSECE 2000 E.C)

A. Alkenes and natural gas
B. Alkanes and cyclo alkanes
C. Alkanes and aromatics
D. Benzene and alkenes

Answer: B

89. A petroleum product commonly used for jet engines and tractors, heating and lighting purpose is

A. Petrol
B. Petroleum ether
C. Kerosene
D. Diesel fuel

Solution:

Petrol	Uses
Petrol	Motor fuel
Petroleum ether	Solvent and dry cleaning
Kerosene	Jet engine fuel and house hold fuel
Diesel fuel	Furnace fuel, diesel engine fuel

Answer: C

90. Which of the following substances is formed from the remains of plants and animals?

A. Natural gas
B. Petroleum
C. Coal
D. Water gas

Solution:

Natural gas originated from the decay and decomposition of animal and plant remains.

Answer: A

91. Why is crude oil unusable when it is extracted?

Solution:

Because it is a mixture of compounds that must first be refined.

Oxygen containing hydrocarbons

6.6 Alcohols

- Alcohols are derivatives hydrocarbons containing hydroxyl group.
- They are represented by the general formula R – OH where R is alkyl group.
- They are commonly known as **alkanols.**

Classification of alcohols

Alcohol

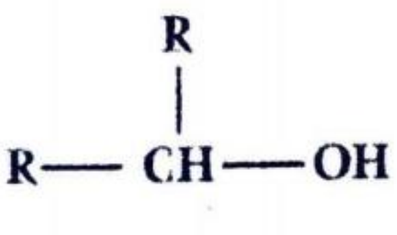
Monohydric (With one OH)

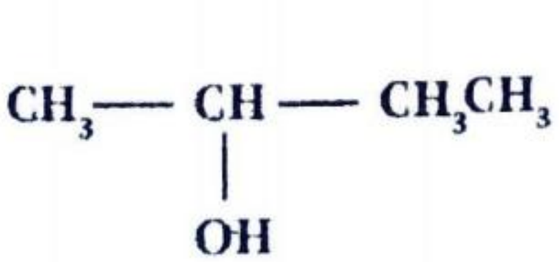
Primary
OH attached to primary carbon

RCH_2OH

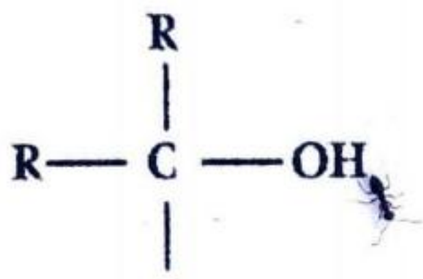
$CH_3—CH_2OH$

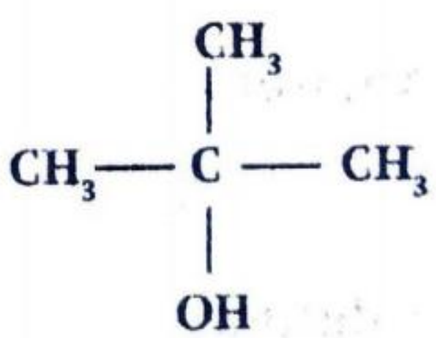
Secondary
OH attached to secondary carbon





Tertiary
OH attached to tertairy carbon





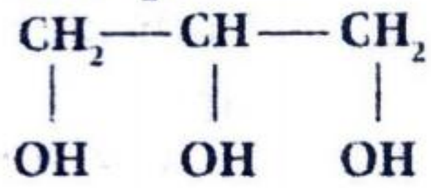
Polyhydric (more than one OH)

Dihydric-has two OH

Example: $CH_2OH\ CH_2OH$

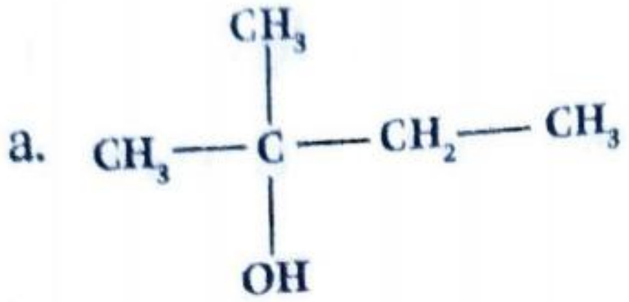
Trihydric-has three OH

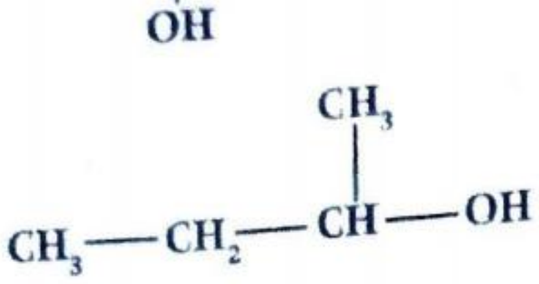
Example:

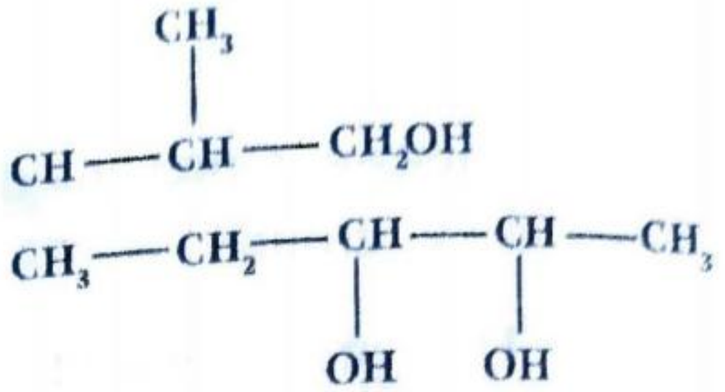


Illustrative Example

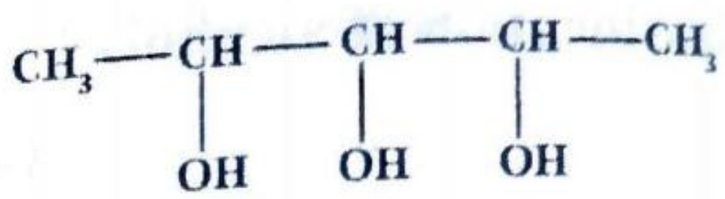
92. Classify the following alcohols as primary, secondary, tertiary, dihydric or trihydric alcohols.

a. 

b. 

c. 

d.

e. 

Solution:

a. Tertiary alcohol because hydroxyl group (OH) is bonded to tertiary carbon or the OH bearing carbon has no hydrogen.

b. Secondary alcohol since the OH is bonded to secondary carbon or the OH bearing carbon has one hydrogen.

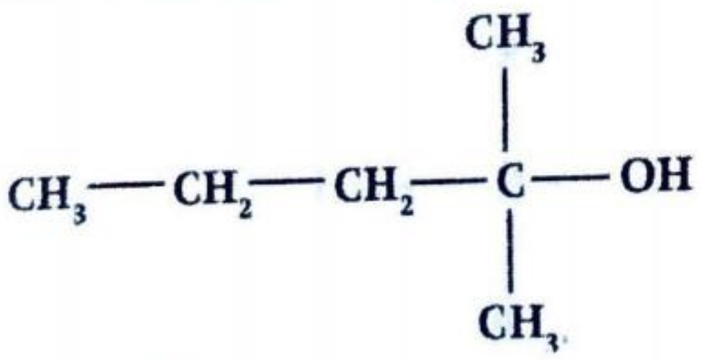
c. Primary alcohol because the OH is bonded to primary carbon or the OH containing carbon has two hydrogen atoms.

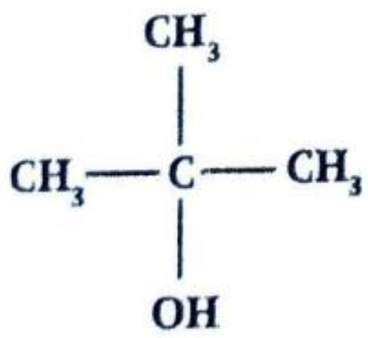
d. Dihydric alcohol contains two hydroxyl.

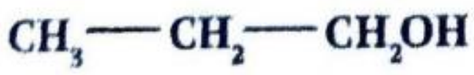
e. Trihydric alcohol has three hydroxyl

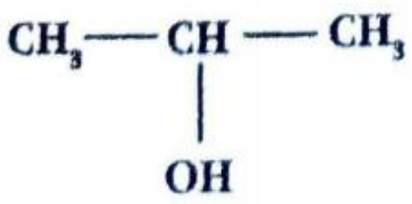
For primary alcohols, the hydroxyl group is found in the first carbon.

93. Which of the following is secondary alcohol?

A. 

B. 

C. 

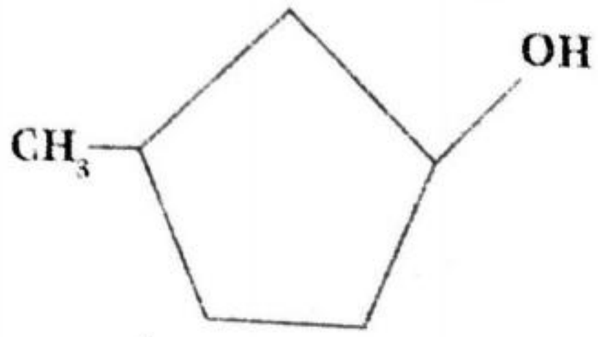
D. 

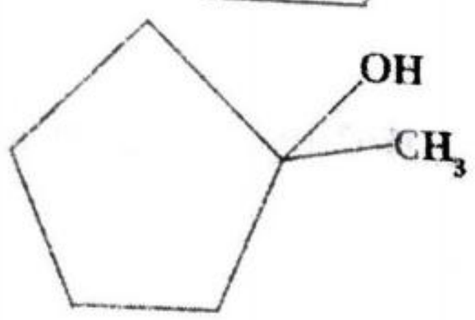
Solution:

In secondary alcohol, the OH is attached to secondary carbon.

Answer: D

94. Which of the following is tertiary alcohol

A. 

B. 

C. CH_3OH

D. $CHCH_2CH_2OH$

Answer: B

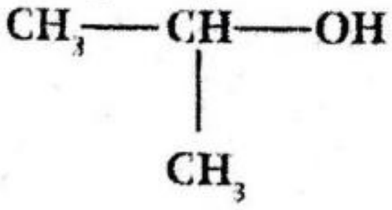
Nomenclature of alcohols

A. Common name

Common name of simple alcohols can be formed from the alkyl group name and the word alcohol.

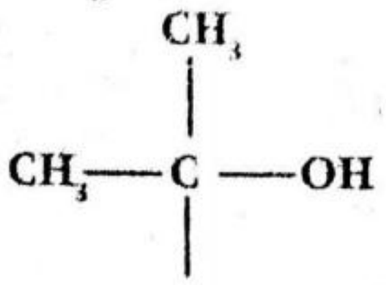
Example

- CH_3OH, Methyl alcohol
- CH_3CH_2OH, Ethyl alcohol



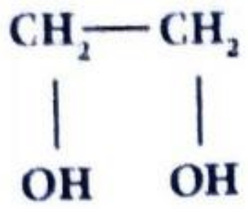
Isopropyl alcohol

- $CH_2 = CHOH$, vinyl alcohol
- $CH_2 = CH - CH_2OH$, Allyl alcohol



, t- butyl alcohol

Common names of some polyhydric alcohols



ethylene glycol

➡ $CH_3CHOHCH_2OH$ propylene glycol

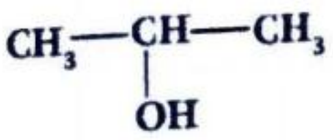
B. IUPAC name alcohols

The IUPAC name of an alcohol is formed by dropping the - e in the name of hydrocarbon and the ending – 01.

The chain is numbered to give the carbon bearing the – OH group the lowest number.

Example

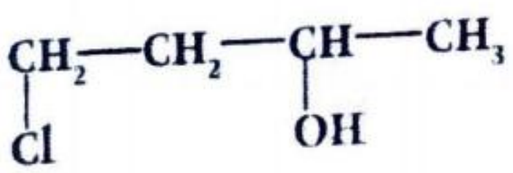
95. Give the IUPAC name of the following alcohols.

a. 

2–propanol (Propan– 2–01)

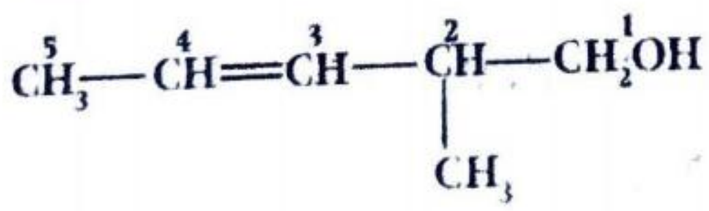
b. $ClCH_2CH_2CHOHCH_3$

Solution:



Start counting from the right.

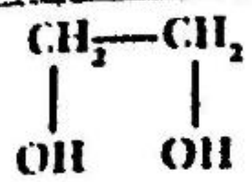
Answer: 4 – chloro – 2 – butanol (4 – chlorobutan – 2 – 01)

c. 

Solution:

Start counting from the right which is nearer to the OH bearing carbon.

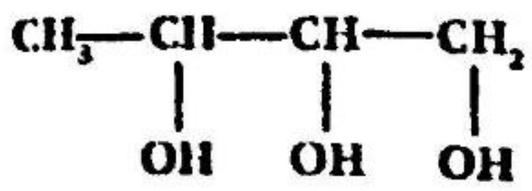
Answer:2 – Methyl – 3 – penten – 1 – 01

d. 

Solution:

For dihydric alcohol indicate the position of the hydroxyl group then the name of the alkane followed by – diol.

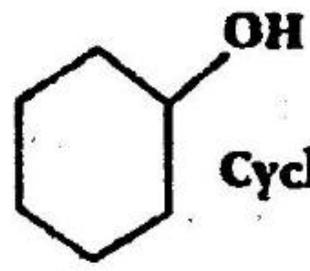
Answer: 1, 2 – ethane diol (Ethane – 1, 2 – diol)

e. 

Solution:

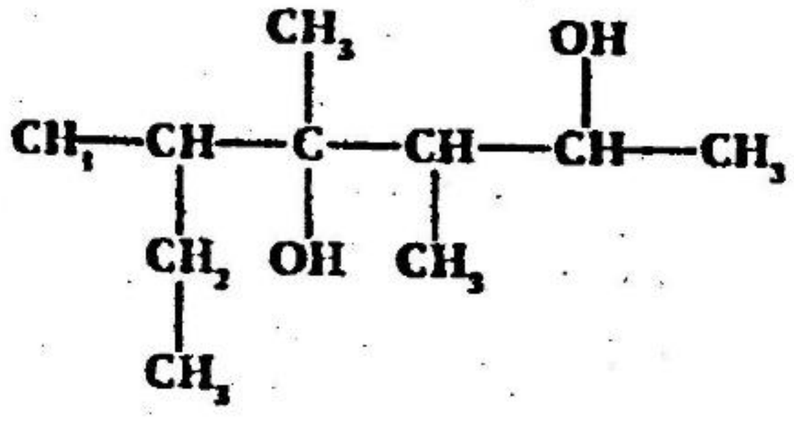
To name trihydric alcohols the name of alkane followed by – triol.

Answer: 1,2,3 – Butane triol (Butane – 1, 2, 3 – triol)

f. 

Cyclohexanol

96. What is the correct IUPAC name of the compound shown below?... (EGSECE 1994 E.C)



A. 3, 4, 5 – trimethylheptanes – 2, 4 – diol
B. 3, 4, 5 – trimethylhepatane – 4, 6 – diol
C. 5 – ethyl – 3, 4 – dimethyl hexane – 2, 4 – diol
D. 2 – ethyl – 3, 4 – dimethyl hexane – 2, 4 – diol

Solution:

Start counting from the right which is nearer to the hydroxyl group.

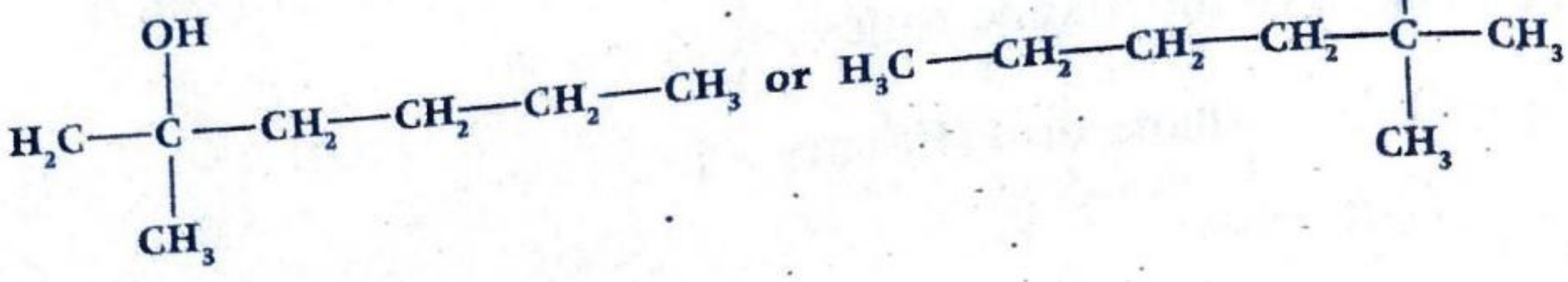
3, 4, 5 – trimethylheptanes – 2, 4 – diol

Answer: A

97. Draw the structure of

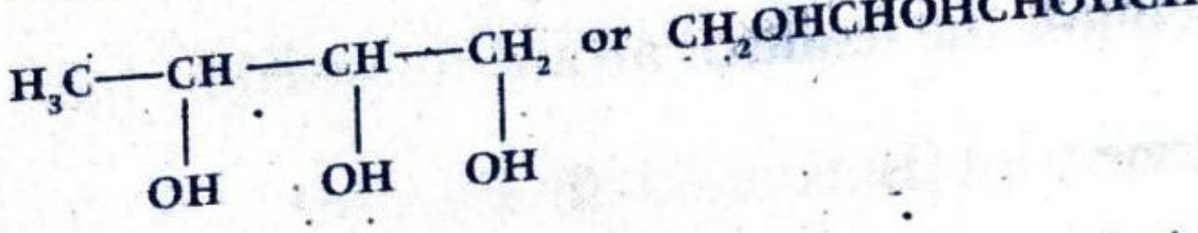
a. 2 – Methyl – 2 – hexanol

Solution:



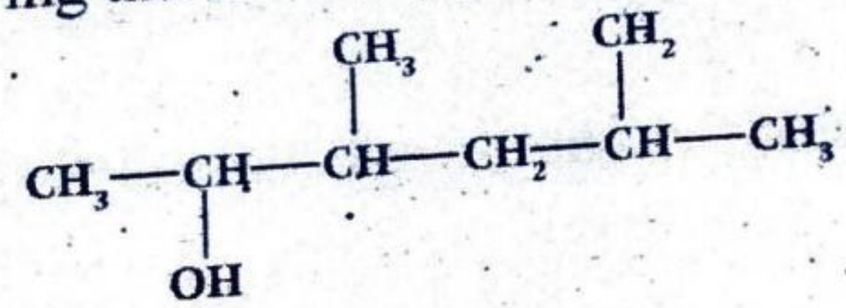
b. 1, 2, 3 – butane triol

Solution:

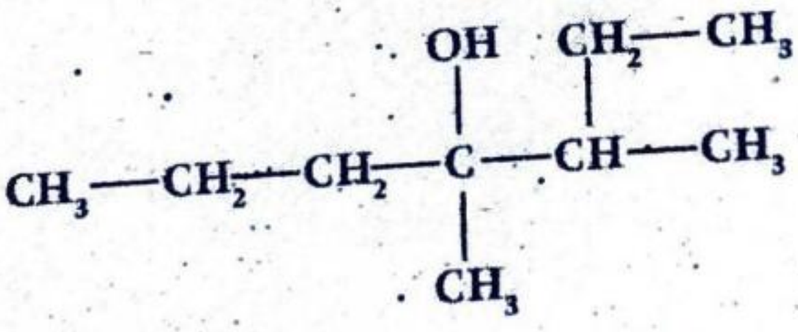


98. Which of the following is the correct structure for the hydrocarbon having the IUPAC name: 3, 5 – dimethyl – 3 – heptanol?

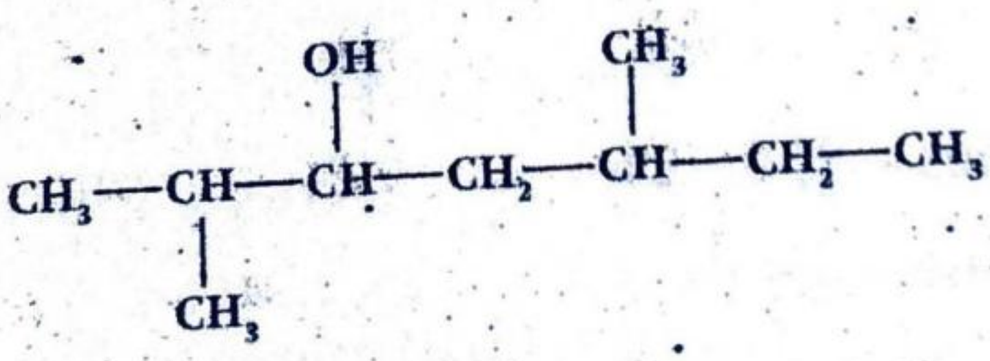
A.



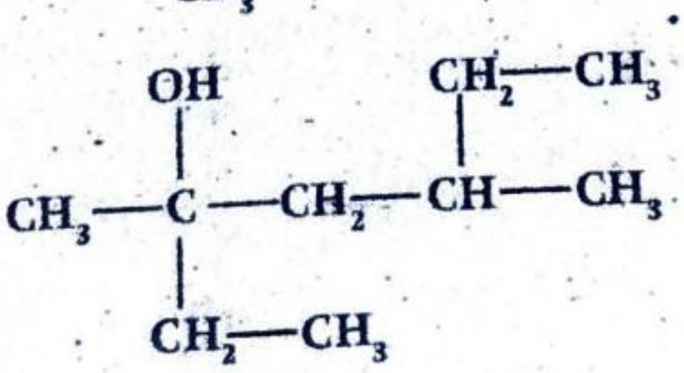
B.



C.



D.



Solution:

Start by drawing the skeletal structure of the parent chain.

Answer: D

Physical Properties of alcohols

- Alcohols have higher boiling point than hydrocarbons with similar carbon number due to hydrogen bond.
- As carbon number increases, boiling points, melting point and density increases.
- Boiling point of polyhydric > Boiling point of monohydric alcohol due to more hydrogen bonds in polyhydric alcohol.
- Lower member alcohols are completely miscible with water.
- As carbon number increases, solubility in water decreases.

Illustrative Example

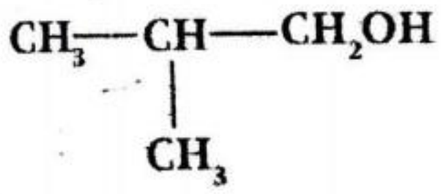
99. Arrange the following alcohols in increasing order of boiling point.

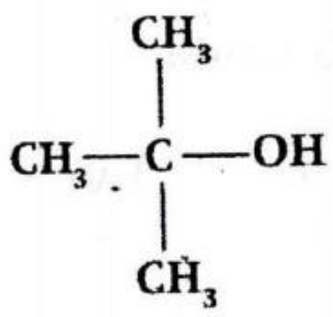
a. n–butyl alcohol b. isobutyl alcohol c. ter-butyl alcohol

Solution:

First draw the structure of each alcohol

a. $CH_3 - CH_2 - CH_2 - CH_2OH$

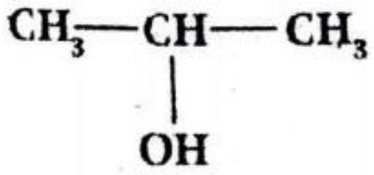
b. 

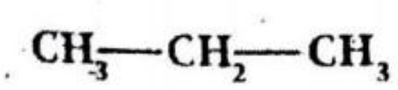
c. 

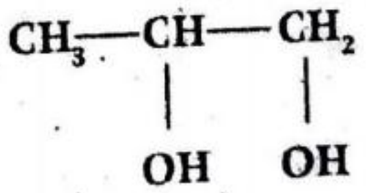
c < b < a

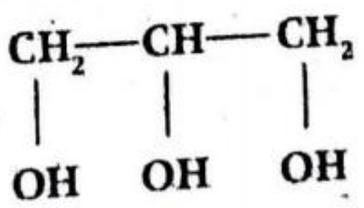
Boiling point 1° alcohol > Boiling point of 2° alcohol > Boiling point of 3° alcohol of similar carbon number.

100. Which of the following is highly soluble in water?

A. 

B. 

C. 

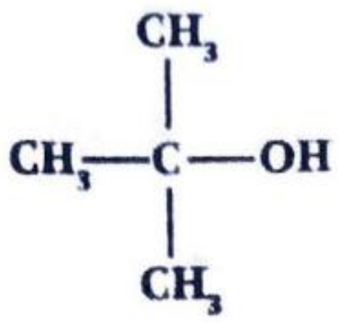
D. 

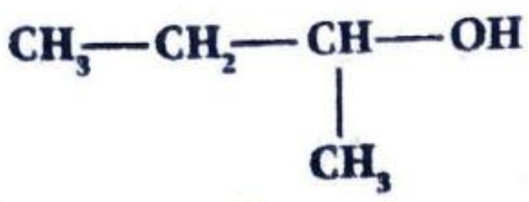
Solution:

The more the number of hydroxyl group, the higher is the solubility in water.

Answer: D

101. Among the following isomeric alcohols which one would have the highest boiling point? . . . . (EGSECE 2002 E.C)

A. 

B. 

C.

D.

Answer: C

102. Which of the following is the least dense alcohol?

A. Methanol
B. Ethanol
C. 1 – propanol
D. 1 – butanol

Solution:

As carbon number increases, density also increases.

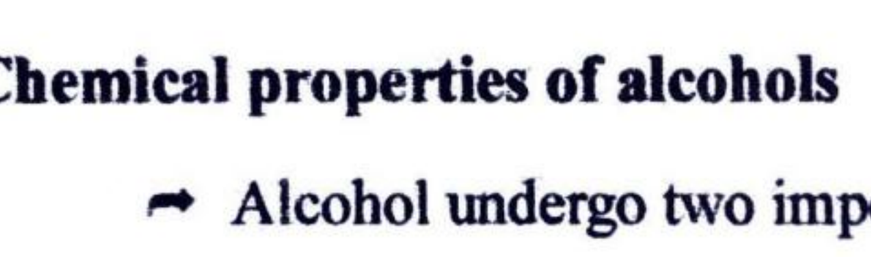
Answer: A

Chemical properties of alcohols

➦ Alcohol undergo two important types of reactions

1. Reaction of alcohol that involves oxygen – hydrogen bond cleavage.
2. Reaction of alcohols that involves carbon – oxygen bond cleavage.

Reactions of alcohols

Reactions involving O – H bond cleavage	Reactions involving C – O bond cleavage
i. Esterification 	i. Dehydration of alcohols

ii. Reaction of alcohols with active metal

$$2R - OH + 2Na \rightarrow 2RONa + H_2$$

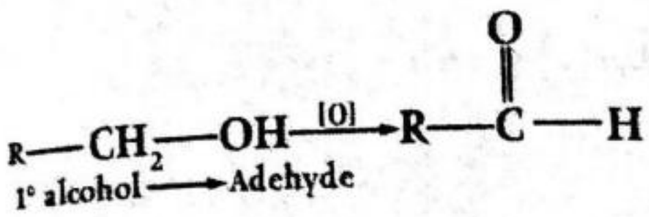
Alcohol

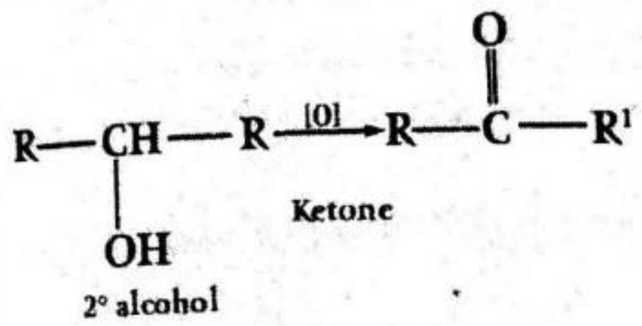
ii. Conversation of alcohols to alkyl halide

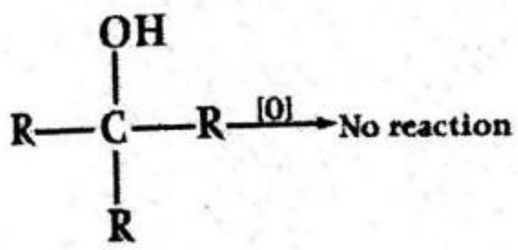
$$R{-}CH_2OH + KX \rightarrow R{-}CH_2 - X + H_2O$$

Alcohol alkyl (x = Cl – Br)

iii. Oxidation of alcohol







normal condition

3rd alcohol

iii. Reaction of alcohols with chloride of phosphorus

$$R - CH_2 - OH + PCl_5 \rightarrow R - CH_2Cl + POCl_3 + HCl$$

Alcohol

Note:

Ease of dehydration:

Tertiary alcohol > secondary alcohol > primary alcohol

Illustrative Example

103. Complete and balance the following equation.

a.

b. $CH_3 - CH_2 - CH_2OH + Na \rightarrow$

c. $CH_3 - CH_2 - OH \xrightarrow{[O]}$

d. $CH_3 - CH_2 - CH_2 - CH_2OH \xrightarrow{H_2SO_4}$

e. $CH_3OH + HCl \rightarrow$

f. $CH_3CH_2OH + PCl_5 \rightarrow$

Solution:

a. $CH_3-\overset{\overset{O}{\|}}{C}-OH + CH_3OH \rightarrow CH_3COOCH_3 + H_2O$

b. $2CH_3-CH_2-CH_2OH + 2Na \rightarrow 2CH_3-CH_2-CH_2O\,Na^+ + H_2$

c. $CH_3CH_2OH \xrightarrow{[O]} CH_3-\overset{\overset{O}{\|}}{C}-H$

d. $CH_3-CH_2-CH_2-OH \xrightarrow{H_2SO_4} CH_3-CH_2-CH=CH_2 + H_2O$

e. $CH_3CH_2OH + HCl \rightarrow CH_3Cl + H_2O$

f. $CH_3CH_2OH + PCl_5 \rightarrow CH_3CH_2Cl + POCl_3 + HCl$

104. All of the following reactions involve O – H bond cleavage except

A. $CH_3-\overset{\overset{O}{\|}}{C}-OH + CH_3OH \rightarrow$

B. $CH_3CH_2OH + Na \rightarrow$

C. $CH_3CH_2OH \xrightarrow{H_2SO_4}$

D. $CH_3CH_2OH \xrightarrow{[O]}$

Answer: C

Preparation of Alcohols

Table 6.7. preparation of alcohols

Method	General reaction	Example
1. Acid catalyzed hydration of alkenes	$R-CH=CH_2 + H_2O \xrightarrow{H^+} R-\underset{\underset{OH}{|}}{CH}-CH_3$	$CH_2=CH_2 + H_2O \xrightarrow{H^+} CH_3CH_2OH$
2. Hydrolysis of ester	$R-\overset{\overset{O}{\|}}{C}-R^1 + H_2O \xrightarrow{H^+}$ Ester $RCOOH + R^1OH$ Carboxylic acid alcohol	$CH_3COOH + H_2O \xrightarrow{H^+} CH_3CH_2OH + CH_3OH$
3. Hydrolysis of alkyl halide	$R-CH_2-X + OH^- \rightarrow R-CH_2-OH + X^-$	$CH_3-CH_2-Cl + OH^- \rightarrow CH_3-CH_2-OH + Cl^-$

Note:

Reduction of aldehyde, ketone carboxylic acid and ester by Lithium aluminum hydride ($LiAlH_4$) forms alcohols.

Uses of alcohols

- The most important alcohols are methanol and ethanol.
- Ethanol manufactured industrially by catalystic hydration of ethene and fermentation of carbohydrates.
- It is found in all alcoholic beverages and used as organic solvents
- It is used as fuel for cars mixed with benzene.

Illustrative Example

105. Complete and balance the following chemical equations.

a. $CH_3-\overset{\overset{CH_2}{\|}}{C}-CH_3 - H_2O \xrightarrow{H^+}$

b. $(CH_3)_2CH\,OOC\,CH_3 + H_2O \xrightarrow{H^+}$

c. $CH_3-\overset{\overset{CH_3}{|}}{\underset{CH_3}{|}C}-Cl + OH^- \rightarrow$

Solution:

a. $CH_3-\overset{\overset{CH_2}{\|}}{C}-CH_3 + H_2O \xrightarrow{H^+} CH_3-\overset{\overset{CH_3}{|}}{\underset{OH}{|}C}-CH_3$

2 - Methylpropene

2 - methyl - 2 - propanol

b. $CH_3-\underset{CH_3}{\underset{|}{CH}}-O-\overset{\overset{O}{\|}}{C}-CH_3 + H_2O \xrightarrow{H^+} CH_3-\underset{CH_3}{\underset{|}{CH}}-OH + CH_3-\overset{\overset{O}{\|}}{C}-OH$

c. $CH_3-\overset{\overset{CH_3}{|}}{\underset{CH_3}{|}C}-Cl + OH^- \rightarrow CH_3-\overset{\overset{CH_3}{|}}{\underset{CH_3}{|}C}-OH + Cl^-$

2-Chloro – 2 – methylpropane 2 – Methyl-2- propanol

106. The major product "X" in the reaction

$C_6H_{12}O_6 \xrightarrow{\text{Fermentation}} X + CO_2$ is… (EGSECE 2004 E.C)

A. CH_3CH_2OH

B. CH_3CH_2COOH

C. $C_5H_{12}OH$

D. H_2O

Solution:

The major product of fermentation of glucose is ethyl alcohol

$C_6H_{12}O_6 \xrightarrow{\text{Fermentation}} 2C_2H_5OH + 2CO_2$

Answer: A

107. All of the following reactions produce an alcohol EXCEPT

A. $2CH_3CH_2COOH + 2Na \rightarrow$

B. $CH_3 - CH = CH_2 + H_2O \xrightarrow{H_2SO_4}$

C. $CH_3 - CH_2 - CH - Br + OH^- \rightarrow$

D. $CH_3-\underset{\underset{O}{\|}}{C}-H + H_2 \xrightarrow{Ni}$

Solution:

A. $\rightarrow 2CH_3CH_2COONa + H_2$

B. $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$

C. $\rightarrow CH_3 - CH_2 - CH_2 - OH$

D. $\rightarrow CH_3CH_2OH$

Answer: A

108. The type of alcohol that is found in all common alcoholic beverages like "Tella", "Areqi", "Beer", etc is

A. propanol
B. butanol
C. methanol
D. ethanol

Solution:

The alcohol that is found in all alcoholic beverages is ethyl alcohol (ethanol). It is commonly known as wine of spirit.

Answer: D

109. What is the product formed when carboxylic acids react with lithium aluminum hydride?

A. Ketone
B. Ester
C. Alcohol
D. Ether

Solution:

$R-\underset{\underset{O}{\|}}{C}-OH \xrightarrow{LiAlH_4} R-CH_2OH$

Carboxylic acid → Alcohol

Answer: C

110. What is methylated spirits or denatured alcohol?

Answer: Methylated spirits is mainly ehanol mixed with a little methanol, which is poisonous. Coloring is added and a horrible taste to stop people drinking it.

111. Why wine goes 'sour' when left open in air?

Answer: Because the alcohol (ethanol) in it slowly oxidized to ethanoic acid, by bacteria in the air.

$$C_2H_5OH(\ell) + O_2(g) \rightarrow CH_3COOH\ (aq) + H_2O(\ell)$$

Ethanol Ethanoic acid

112. State the properties, preparation and uses of methanol

Answer:

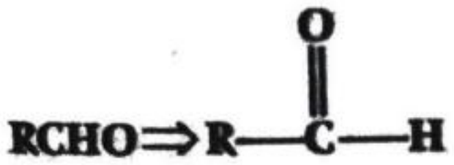
Methanol also known as wood alcohol is a deadly poison alcohol. It can cause blindness and eventually total loss of motor control and death.

$$CO + 2H_2 \longrightarrow CH_3OH$$

Methanol is used to make formaldehyde.

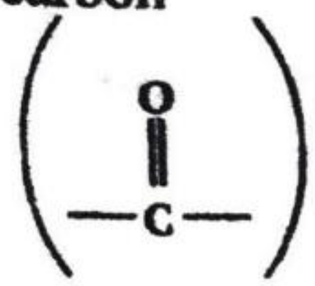
6.7 Aldehydes and Ketons

- Aldehydes are commonly known as **Alkanals**.
- Their general formula is



- Ketones are commonly known as **Alkanones**.
- Their general formula is

- Both are known as carbonyl compounds that contain carbonyl carbon



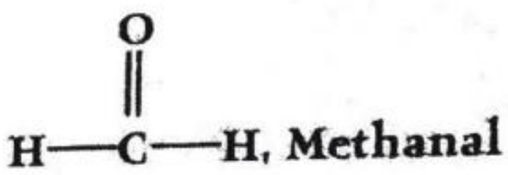
Nomenclature of aldehyde and Ketones

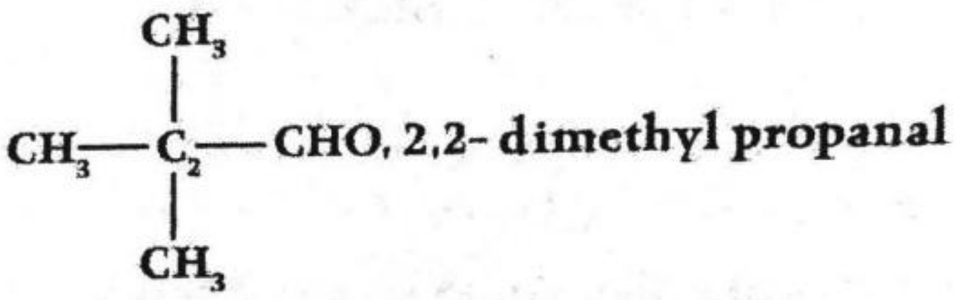
i Aldehydes

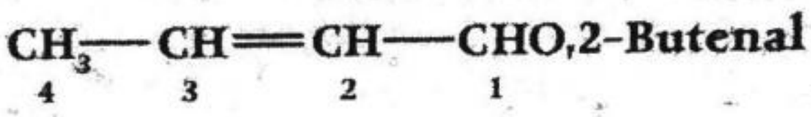
➦ Start counting from the carbonyl carbon to identify the longest chain.

Examples

Give the IUPAC name for

a. 

b. 

c. 

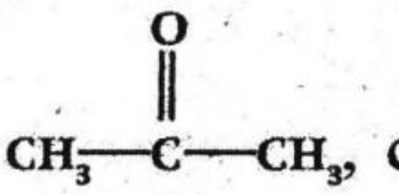
ii Ketones

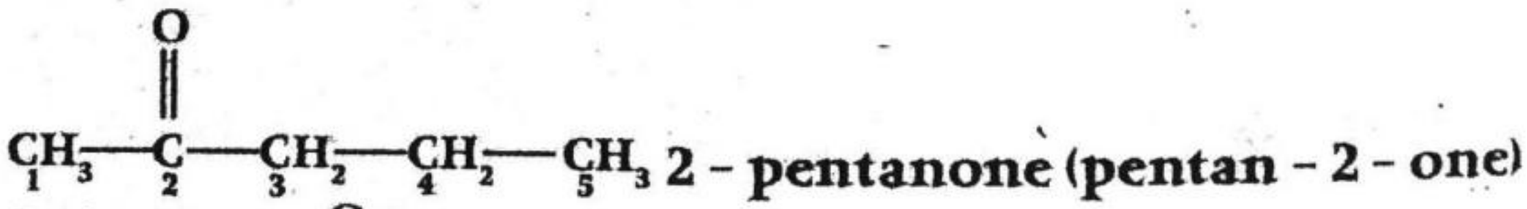
➦ In the system developed by IUPAC, ketones are named by replacing – ane with – one.

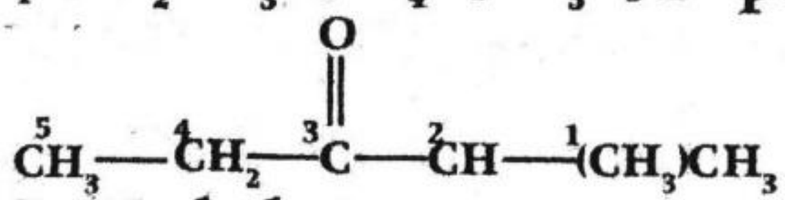
➦ The chain is numbered to give the —CO— group the lowest number

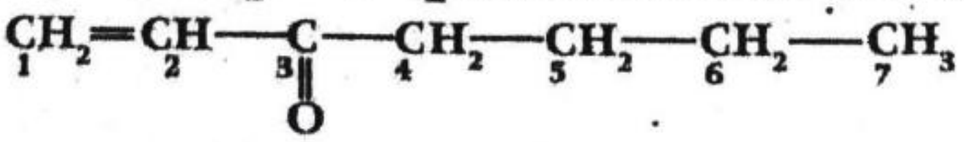
Examples

Give the name of the following ketones.

a. 

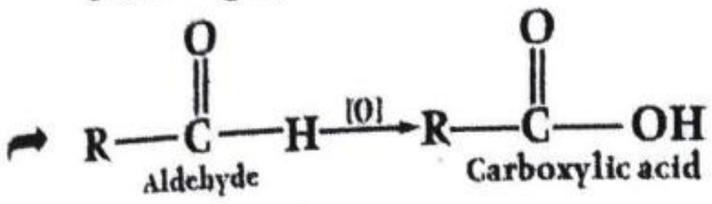
b. 

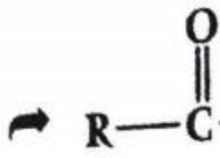
c. 

d. 

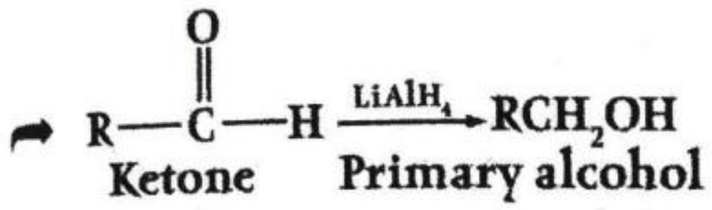
Properties of aldehydes and ketones

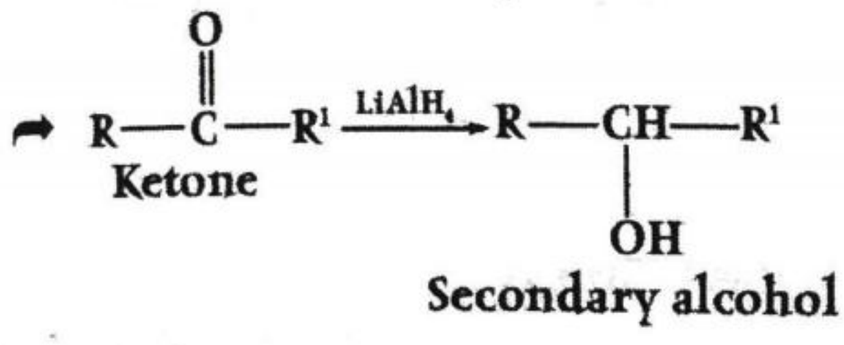
- Aldehydes and ketones have higher boiling points than hydrocarbons due to the polarity of the carbonyl carbon but lower boiling point than alcohol because of lack of hydrogen bond.





No reaction, under normal condition





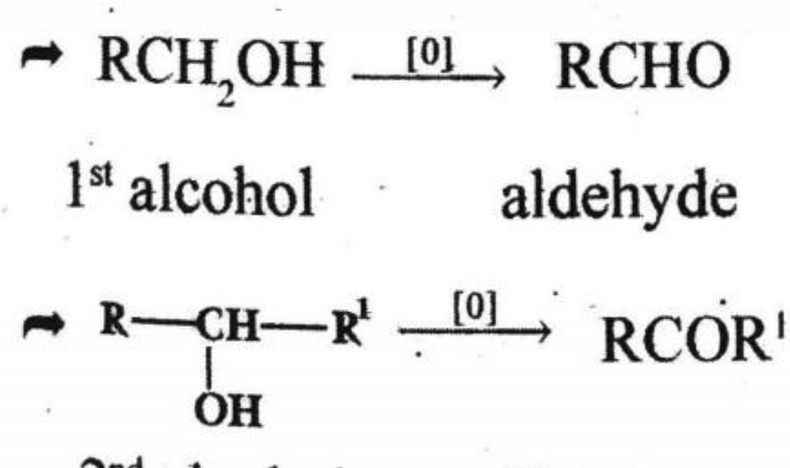
Note:

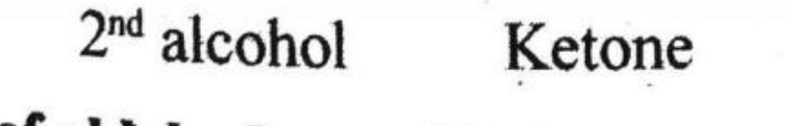
[O] - oxidizing agent

$LiAlH_4$ - reducing agent

Preparation of aldehydes and ketones

The most common method used to prepare aldehydes and ketones is oxidation of alcohols.





Uses of aldehydes and ketones

- The most important aldehyde is methanal (formaldehyde) which is used to make formalin.
- The most important ketone is acetone that is used as organic solvent.

Illustrative Example

113. Write the structure of the following compounds.

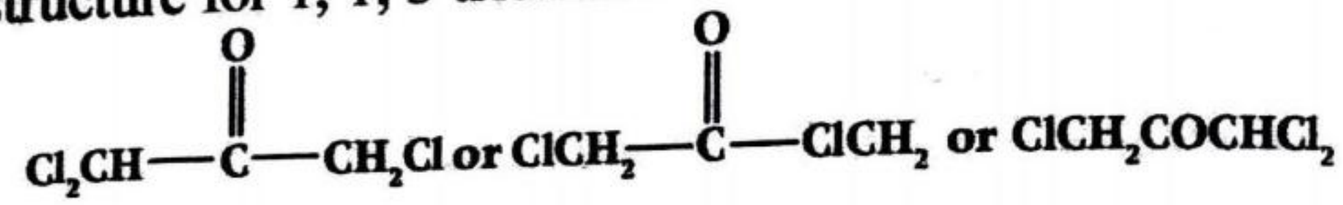
a. 1, 1, 3 – trichloro acetone

b. 3 – pentynal

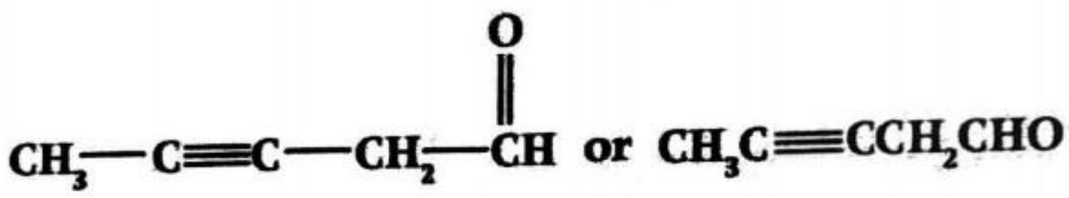
c. 2 – methyl cyclopentanone

Solution:

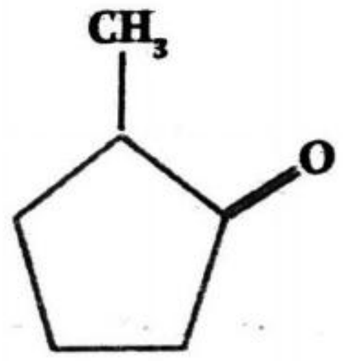
a. Acetone is the simplest ketone containing three carbon atoms. The structure for 1, 1, 3 trichloro-acetone is



b.



c.



114. Complete the following chemical equations.

a. $CH_3CHO \xrightarrow{LiAlH_4}$

b. $CH_3COCH_2CH_3 \xrightarrow{LiAlH_4}$

Solution:

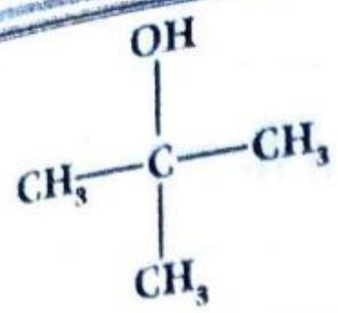
a. $CH_3CHO \xrightarrow{LiAlH_4} CH_3CH_2OH$

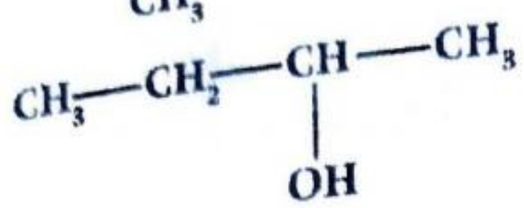
Ethanal Ethanol

b. $CH_3COCH_2CH_3 \xrightarrow{LiAlH_4} CH_3\,CHOHCH_2CH_3$

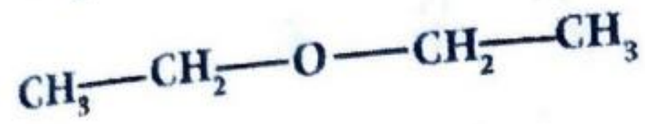
Butanone 2 – Butanol

115. An organic compound with the empirical formula $C_4H_{10}O$ was changed into a compound with empirical formula C_4H_8O under controlled oxidation. The compound found was identified to be an aldehyde. What is the structural formula of the original compound?

A. 

B. 

C.

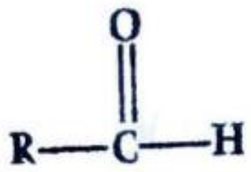
D. 

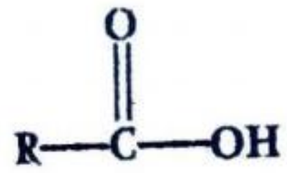
Solution:

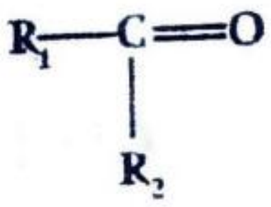
An aldehyde is formed by oxidation of primary alcohol. The original compound should be primary alcohol.

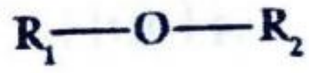
Answer: C

116. The formula of aldehyde is …(EGSECE)

A. 

B. 

C. 

D. 

Answer: A

117. Which one of the following organic compound is used to preserve dead organisms?

A. Formaldehyde
B. Acetone
C. Acetic acid
D. Ethanol

Solution:

A 40% formaldehyde is formalin that preserves dead organisms in biological laboratories.

Answer: A

118. Aldehydes usually have lower boiling points than the corresponding alcohols because

A. They are liquids at room temperature.

B. They contain carbonyl radicals in their structure

C. They have less hydrogen atoms per molecule.

D. They cannot form hydrogen bonds between their molecules.

Answer: D

119. Which of the following yields a primary alcohol upon reduction?

A. Alkene

B. Ether

C. Aldehyde

D. Ketone

Solution:

$$RCHO \xrightarrow{LiAlH_4} RCH_2OH$$

Adehyde 1^0 alcohol

Answer: C

120. Which one of the following molecules belongs to aldehydes group?

A. $CH_3CH_2CH_2CHO$

B. $CH_3CH_2CH_2OH$

C. $CH_3CH_2COCH_3$

D. $CH_3CH_2OCH_2CH_3$

Solution:

Aldehyders are represented by the general form RCHO where R is alky group or hydrogen.

Answer: A

121. Oxidation of 2 – propanol gives…(EGSECE 2003 E.C)

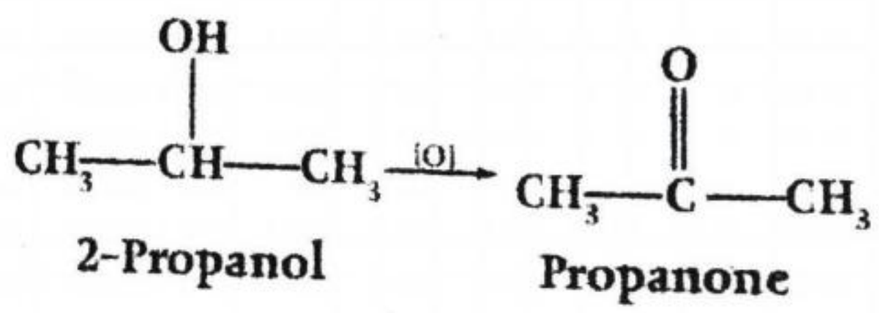
A. Propanol

B. Propyl alcohol

C. Propanone

D. Methyl propanoate

Solution:



2-Propanol Propanone

Answer: C

122. Which of the following is carbonyl compound?

A. CH_3CH_2OH

B. CH_3CHO

C. CH_3COOH

D. CH_3OCH_3

Solution:

A compound that contains either aldehyde or ketone functional group is carbonyl compound.

Answer: B

6.8 Carboxylic acids and Esters

A. Carboxylic acids

➦ Carboxylic acids are organic acids containing carboxylic group (—COOH)

➦ Their general formula is RCOOH or $R—\overset{O}{\overset{\|}{C}}—OH$

- They are commonly known as alkanoic acids.
- They have the highest boiling point than alcohols of similar molecular mass.
- They have all the properties of acids.
- They are found in living and non – living matter. For example
- Formic acid (HCOOH) – ants, bees
- Acetic acid (CH_3COOH) – vinegar
- Butyric acid ($CH_3(CH_2)_2COOH$) – rancid butter
- Valeric acid ($CH_3(CH_2)_3COOH$) – perennial herb.
- Caproic acid ($CH_3(CH_2)_4COOH$) – aroma of goat.

IUPAC name Carboxylic acids

➦ The IUPAC name of carboxylic acids is obtained by dropping the final – e of the name of alkane corresponding to the longest chain in the acid and by adding – oic acid.

➦ The carboxyl carbon atom is assigned number 1.

Example

a. HCOOH – Methanoic acid

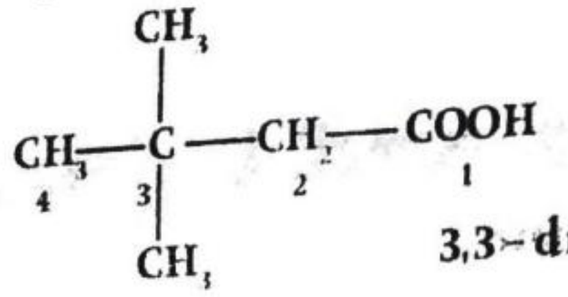
b. CH_3CCOOH – Ethanoic acid

c. CH_3CH_2COOH –Propanoic acid

d. $CH_3(CH_2)_2COOH$ – butanoic acid

e. $CH_3(CH_2)_3COOH$ – pentanoic acid

f. $CH_3(CH_2)_4COOH$ – Hexanoic acid

g. 

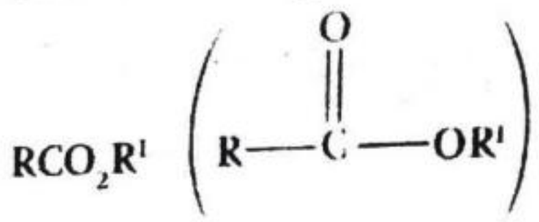
3,3– dimethyl butanoic acid

h. $CH_{3_4} - CH_3 = CH_2 - COOH_1$, 2 – butenoic acid

i. $CH_{3_4} - C \equiv C_3 - COOH_2$, 2_1 – butynoic acid

B. Esters

➦ Esters have general formula

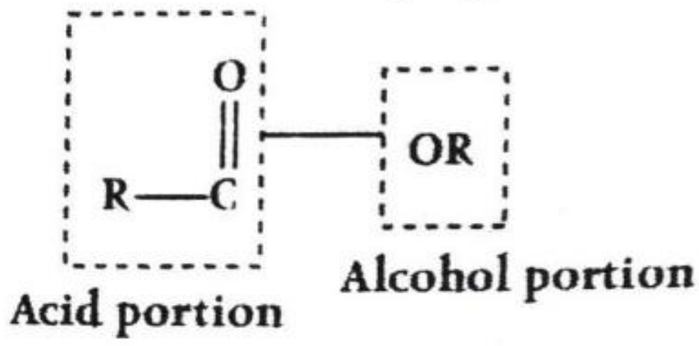


➦ Most esters have sweet, pleasant odours and esters are important constituents of fruits and flower aromas.

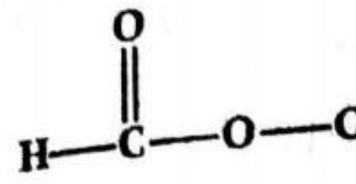
➦ Bees wax, animal fat and vegetable oil are natural esters.

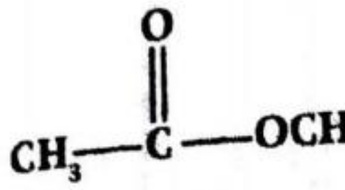
Nomenclature of esters

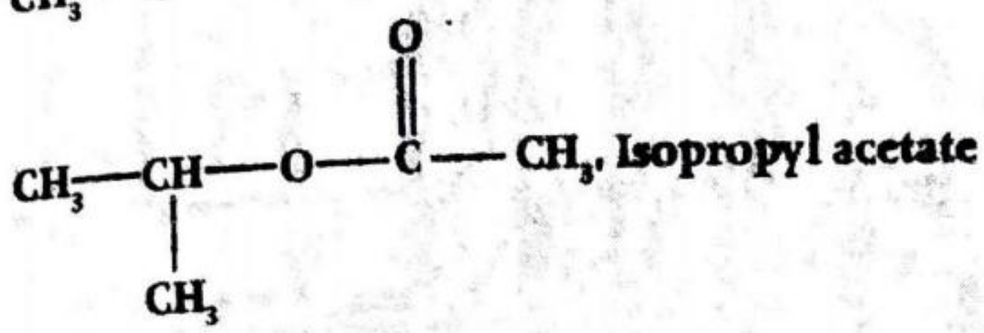
Esters are named first by naming the alkyl derived from the alcohol followed by the name of the acid changing "-ic acid" to "-ate".

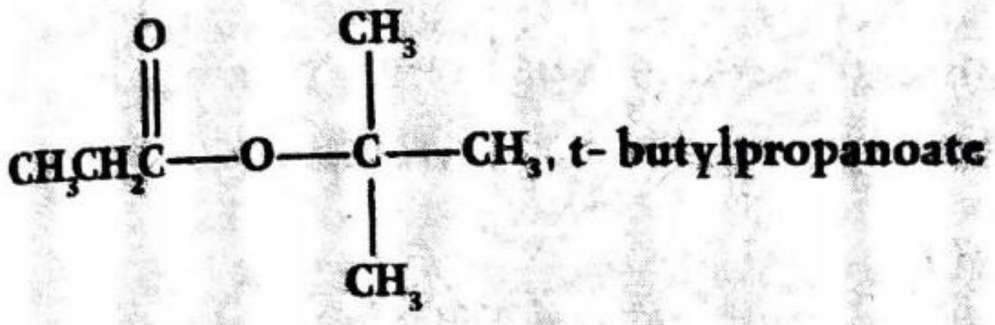


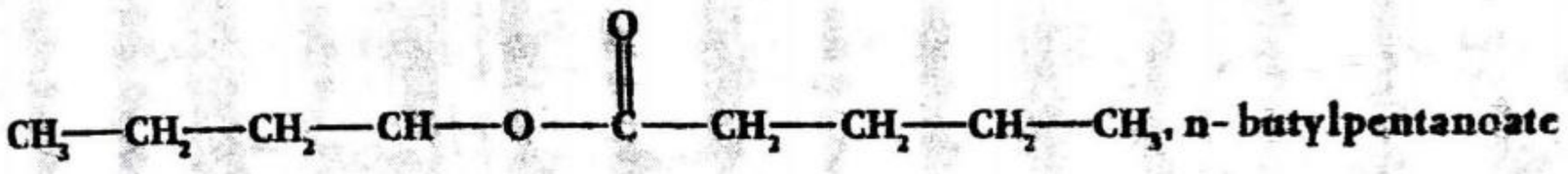
Example

a.  Methylmethanoate (methyl formate)

b.  Ethyl ethanoate (Ethyl acetate)

c.  Isopropyl acetate

d.  t- butylpropanoate

e.  n- butylpentanoate

Esters are prepared by esterification reaction

Illustrative Example

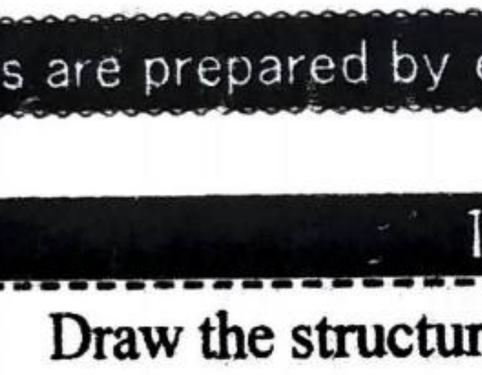
123. Draw the structure of

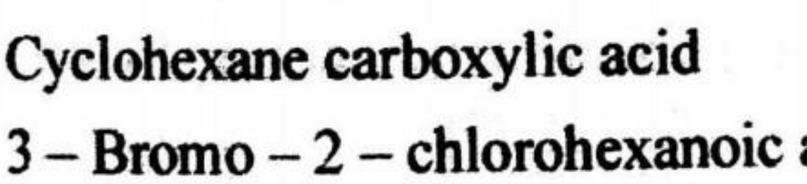
a. Cyclohexane carboxylic acid

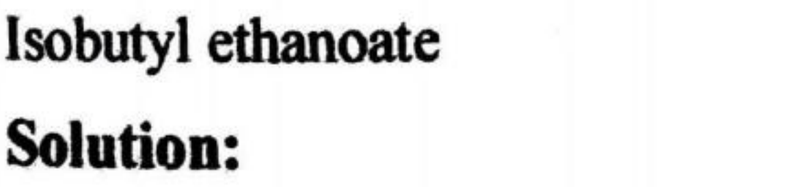
b. 3 – Bromo – 2 – chlorohexanoic acid

c. Isobutyl ethanoate

Solution:

a. 

b. 

c. 

124. The acid that is responsible for the painful feeling from the "sting of ants" is

A. Formic acid
B. Acetic acid
C. Propanoic acid
D. Butanoic acid

Solution:

The common name of carboxylic acids is derived from Latin word. For example formic acid from a Latin word formica meaning ant.

Answer: A

125. Which of the following substances would be formed when C_2H_5OH and CH_3COOH are chemically combined?.... (EGSECE 2003 E.C)

A. $CH_3OC_2H_5$
B. $C_2H_5COOCH_3$
C. $CH_3COOC_2H_5$
D. $CH_3COC_2H_5$

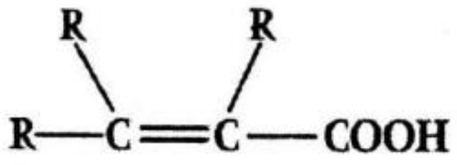
Solution:

$$C_2H_5OH + CH_3COOH \xrightarrow{H^+} C_2H_5OOCCH_3 + H_2O$$

Answer: C

Such type of reaction is called esterification reaction.

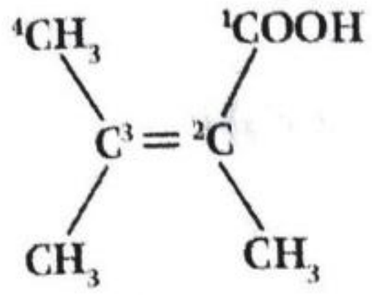
126. What is the IUPAC name of the following carboxyl compound? 'R' is the methyl group...(EGSECE 2002E.C)



A. 2, 3 – dimethyl – 2 – butenoic acid
B. 2 – hexenoic acid
C. 2,3 – dimethyl butenoic acid
D. 2, 3, 3 – trimethyl – 2 – proponoic acid

Solution:

Replace the alkyl group by methyl (–CH3)



Answer: B

2,3 - dimethyl - 2 - butenic acid

127. What are the substances that give fruits their characteristics smell and flavor?

A. Ethers
B. Esters
C. Carboxylic acids
D. Ketones

Answer: B

128. Which of the following is NOT true about carboxylic acid?

A. They react with carbonates.
B. They are weak acids.
C. They react with inert metals liberating H_2 gas.
D. They react with active metals liberating H_2 gas.

Solution:

➦ $2RCOOH + CaCO_3 \rightarrow (RCOO)_2Ca + CO_2 + H_2O$

Carboxylic acid

➦ $RCOOH + H_2O \rightarrow RCO\bar{O} + H_3O^+$,

carboxylic acids are weak acid (ionize slightly when dissolved in water).

➦ As carbon number increases acidity decreases.

➦ $2RCOOH + Mg \rightarrow (RCOO)_2Mg + H_2$

➦ $RCOOH + NaOH \rightarrow RCOONa + H_2O$

Answer: C

129. Ethyl acetate can be prepared by the reaction between ethanol and

A. CH_3CHO
B. $CH_3 - COOH$
C. HCOOH
D. CH_3COCH_3

Solution:

Esters are formed by the reaction between alcohol and carboxylic acid. To produce ethyl acetate, identify the alcohol and acid portion.

➦ The acid portion – acetic acid

- The alcohol portion – ethanol
- Draw the structure of acetic acid – CH_3COOH

Answer: B

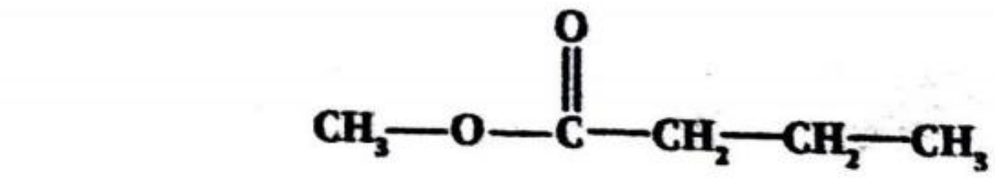
130. Substances such as vegetables oils and animal fats are examples of

A. Organic acid
B. Ester
C. Aldehydes
D. ketones

Answer: B

131. What is the structural formula of methyl butanoate?

Solution:



132. What is the molecular formula of pentanoic acid? (EGSECE ... 2009 E.C)

Solution:

The molecular formula of open chain carboxylic acid is $C_nH_{2n}O_2$ pentanoic acid, n = 5, thus its molecular formula is $C_5H_{10}O_2$.

3. Ethers

- Ethers are commonly known as **alkoxy alkanes**
- They have the general formula R – O – R^1 (ROR^1)
- Ethers can be thought of as derivatives of water in which both hydrogen atoms have been replaced by alkyl groups.

Nomenclature of ethers

- In the IUPAC system, the parent name is alkane and oxygen containing alkyl group is named as alkoxy.

Ethers	Common name	IUPAC name
$CH_3 - O - CH_3$	Dimethyl ether	Methoxy methane
$CH_3 - O - CH_2CH_3$	Ethylmethyl ether	Methoxy ethane
$CH_3 - CH(OC_2H_5) - CH_3$	Ethyl isopropyl ether	2 – ethoxyl Propane

Ethers are produced by self – condensation of alcohol.

$$R\boxed{OH+H}\ OR' \xrightarrow{H_2SO_4} R\ O\ R' + H_2O$$

$$CH_3OH + CH_3CH_2OH \xrightarrow{H_2SO_4} CH_3OCH_2CH_3 + H_2O$$

➦ The most important ether is dimehtyl ether used as solvent.

Note

1. Alcohols and ether
2. Aldehydes and ketones
3. Esters and carboxylic acids other

} are isomeric with each other.

Illustrative Example

133. Write the structural formula of

a. Ethylpropyl ether

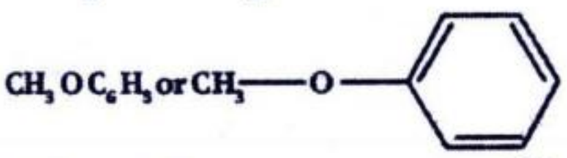
b. Methyl phenyl ether

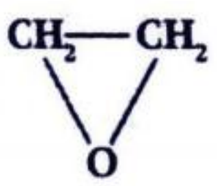
c. Ethylene oxide

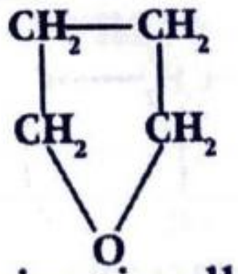
d. Tetrahydrofuran

Solution:

a. $CH_3 - CH_2 - O - CH - CH_2 - CH_3$

b. $CH_3OC_6H_5$ or 

c. 

d. 

134. Which of the following is alkoxyl alkane? (EGSECE)

A. Ester

B. Ether

C. Aldehyde

D. Ketone

Answer: B

135. Which pair of compounds are structural isomers? (EGSECE)

A. C_6H_6 and C_6H_{12}

B. CH_3CH_2OH and CH_3COOCH_3

C. C_2H_4 and C_2H_6

D. CH_3CH_2OH and CH_3OCH_3

Solution:

Alcohol and ether with similar carbon number are isomers.

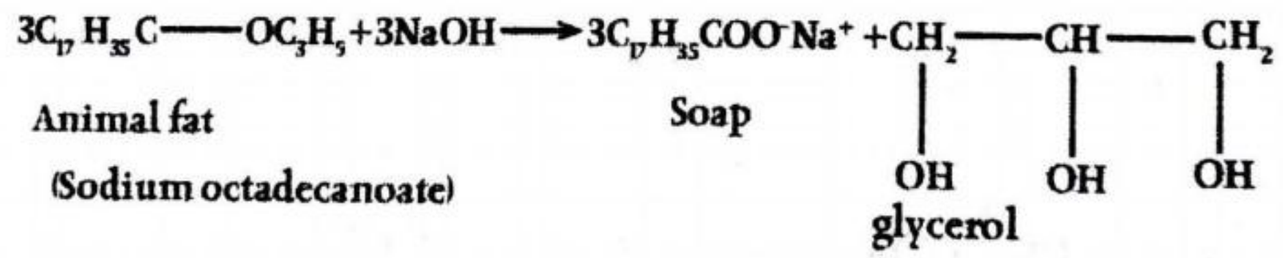
Answer: D

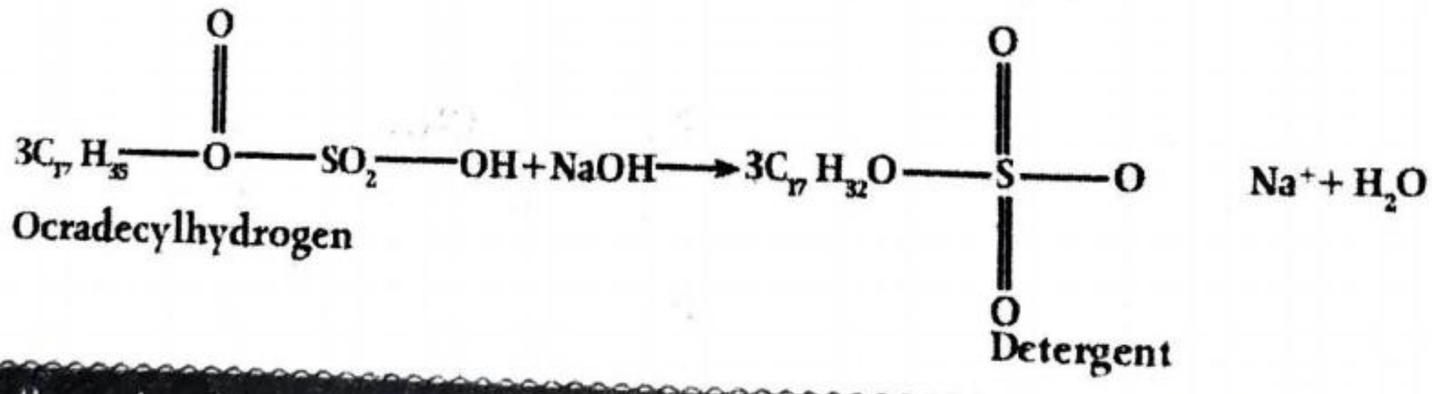
6.9 Industrial and Agricultural applications of organic compounds

A. Industrial Application

Organic chemicals are used to prepare a huge range of industrial products. These include:

1. Alcoholic beverages containing ethanol
   - ➦ Beer
   - ➦ Wine
   - ➦ Whisky
   - ➦ Gin etc
2. Pharmaceuticals
   - ➦ Analgesics
   - ➦ Anti septics
   - ➦ Disinfectants
   - ➦ Sulphonamides
   - ➦ Sedatives
3. Soaps and detergents





Sodium hydroxide (NaOH) is used to make hard soap while potassium hydroxide is used to make soft soap.

Dry cleaning

- ➦ Benzene
- ➦ Gasoline
- ➦ Carbontetrachloride
- ➦ 1, 1, 2, 2 - tetrachloroethane

4. Fuels
   - Alkanes
   - Ethanol
   - Biofuels

B. Agricultural applications

- Pesticides – kill pests
- Herbicides – weed killers
- Fertilizers – important for plant growth.

Illustrative Example

136. Which of one of the following is NOT a product of organic compounds?

A. Alcoholic beverages
B. Detergents
C. Cement
D. Pharmaceuticals

Solution:

Cement is made from inorganic compounds such as calcium carbonate, alumina and silica.

Answer: C

137. Which one of the following substances is used in dry cleaning?

A. Carbontetrachloride
B. Water
C. Vegetable oil
D. Detergent

Answer: A

138. Classify the following alcoholic beverages as undistilled or distilled alcoholic beverages.

a. Beer
b. Wine
c. Tella
d. Tej
e. Gin
f. Katicala
g. Ouzo
h. Whisky

Solution:

Undistilled alcoholic beverages	Distilled alcoholic beverages
Beer	Gin
Wine	Katicala
Tella	Whisky
Tej	Ouzo

139. The compound is $C_{17}H_{35}COONa$ (EGSECE)

A. A fiber

B. An ester

C. A soap

D. A plastic

Answer: C

Solved Problems

140. For n-hexane, Write its

a. Molecular formula

b. Condensed structural formula

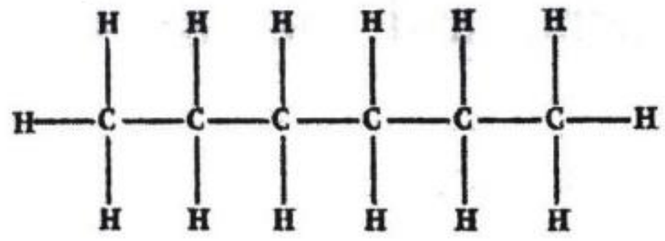
c. Detail structure

Solution:

a. $n = 6, C_nH_{2n+2} = C_6H_{2\times 6+2} = C_6H_{14}$

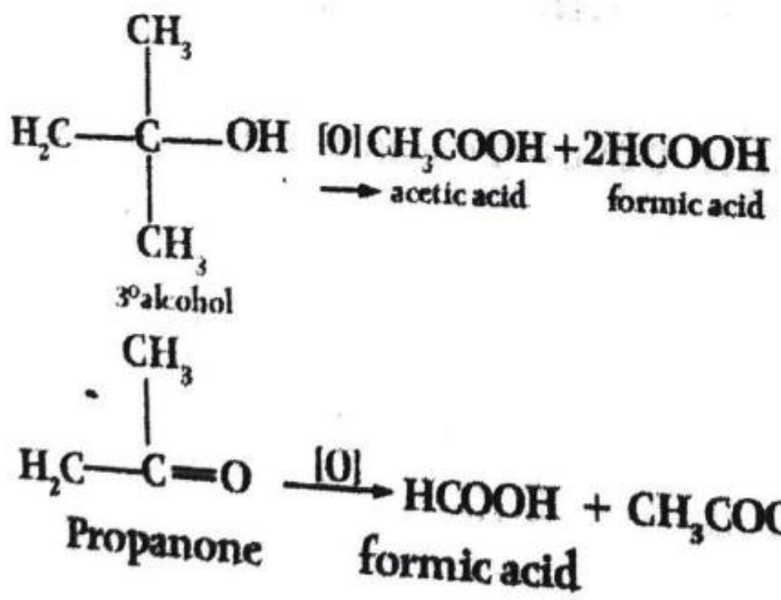
b. $CH_3CH_2CH_2CH_2CH_2CH_3$ or $CH_3(CH_2)_4CH_3$

c.



Tertairy alcohols and ketones do not undergo oxidation reaction under normal conditions. However, they under oxidation under drastic conditions. Give example:

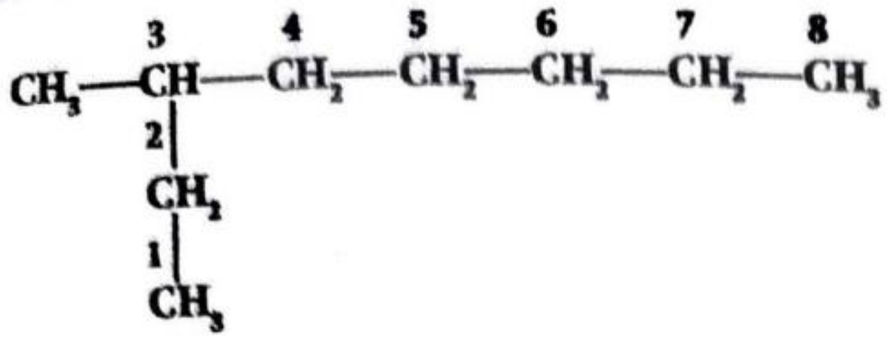
Solution:



141. Explain why 2 –ethylheptane is not standard IUPAC name.

Solution:

The longest continuous chain would include the ethyl group and the compound would have the standard name 3 – methyloctane.



142. What class of compounds is represented by the formula $RCOR^1$? Using CH_3 and C_6H_5 as the radicals, write the formula for three compounds which correspond to this type of formula.

Solution:

Ketones

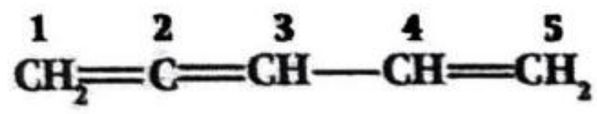
CH_3COCH_3, $C_6H_5COH_5C_6$, $C_6H_5COCH_3$

143. What is the name of

a. $CH_2 = C = CH - CH = CH_2$

b. $(CH_3CH_2)_2O$

Solution:

a.  1, 2, 4 - pentatriene

b. $(CH_3CH_2)_2O \Rightarrow CH_3CH_2OCH_2CH_3$

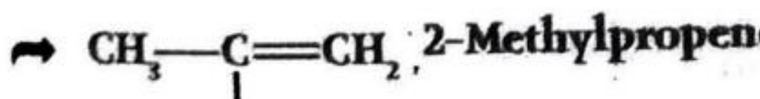
Diethylether or ethoxyethane

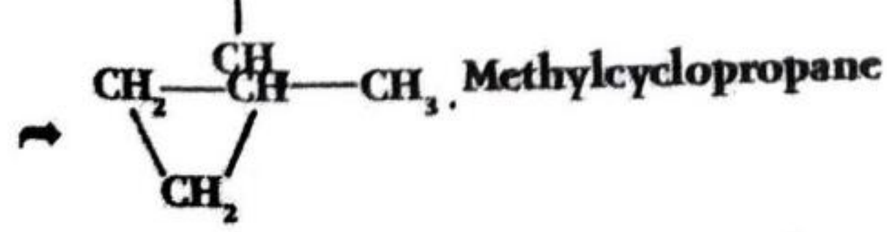
144. Write structural formula for all structural isomers of cyclobutane.

Solution:

➦ 1-butene

➦ 2-butene

➦  2-Methylpropene

➦  Methylcyclopropane

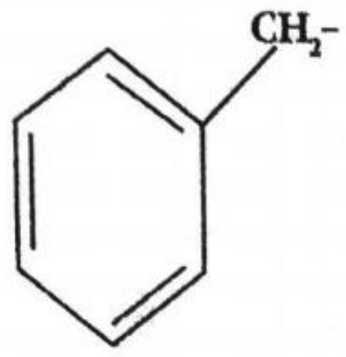
145. Write the structural formula for

a. benzyl

b. phenyl

Solution:

a.



b.

146. There are seven isomeric compounds with the formula $C_4H_{10}O$. Write their structure and give their IUPAC name.

Solution:

i $CH_3—CH_2—CH_2—CH_2—OH$, 1-butanol

ii $CH_3—CH_2—\underset{\displaystyle OH}{\underset{|}{CH}}—CH_3$,2-butanol

iii $CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_2OH$, 2-methyl - 1- propanol

iv $CH_3—\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle OH}{\overset{|}{C}}}}—CH_3$, 2-methyl - 2- propanol

v $CH_3OCH_2CH_2CH_3$, 1 - methoxypropane

vi $CH_3—\underset{\displaystyle OCH_3}{\underset{|}{CH}}—CH_3$, 2-methoxypropane

vii $CH_3CH_2OCH_2CH_3$Ethyoxy ethane

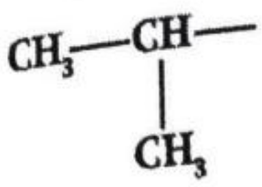
147. Write the name of alkane that contains

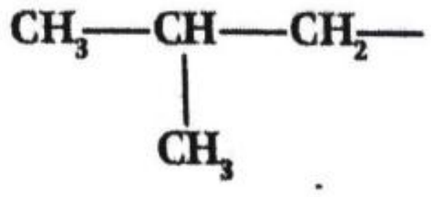
a. 30 carbon
b. 40 carbon
c. 50 carbon
d. 60 carbon
e. 70 carbon
f. 80 carbon
g. 90 carbon
h. 100 carbon

Solution:

a. Tricosane
b. Tetracosane
c. pentacosane
d. hexacosane
e. Heptacosane
f. octacosane
g. nonacosane
h. hectane

148. Give the IUPAC name of the following alkyl groups.

a. 

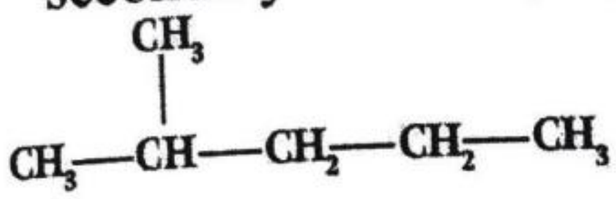
b. 

Solution:

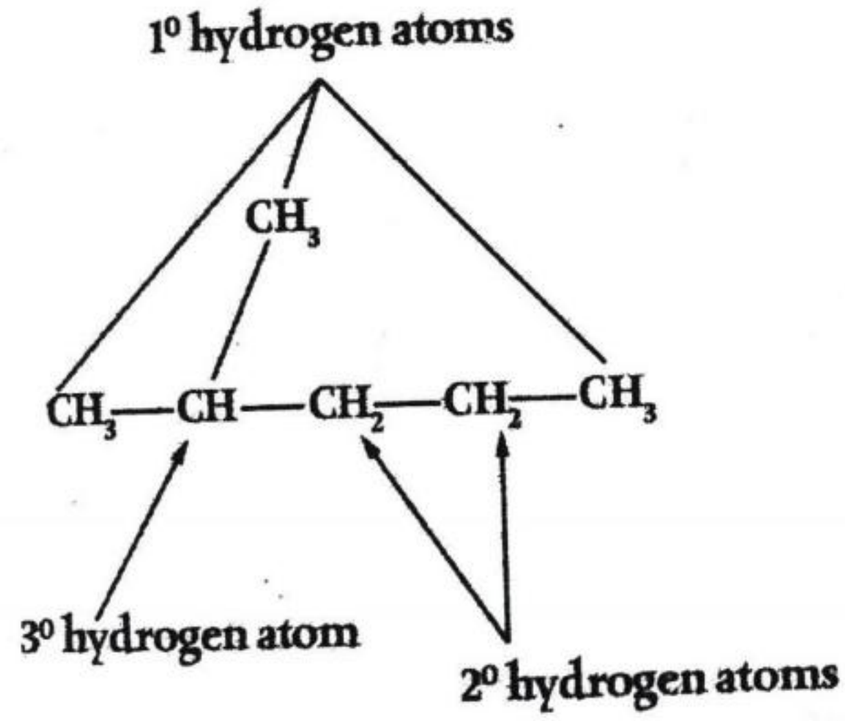
a. 1 – methylethyl

b. 2 - methylpropyl

149. Classify the hydrogen atoms in the structure below as primary, secondary or tertiary hydrogen atoms.



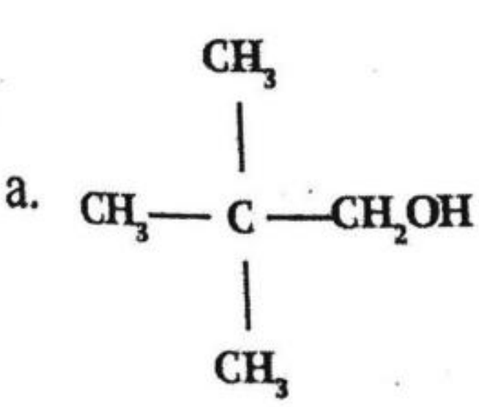
Solution:



150. Draw the structure of

a. Neopentyl alcohol b.cyclopropylcyclopropane

Solution:

a. 

b.

151. Which member has higher boiling point? Explain

a. CH_3OH or HCHO

b. CH_3COOCH_3 or $CHCH_2COOH$

c. CH_3CH_2OH or CH_3COOH

d. $HC \equiv CH$ or $CH_2 = CH_2$

Solution:

a. CH_3OH, has hydrogen bond

b. CH_3CH_2COOH, has hydrogen bond

c. CH_3COOH, can form two hydrogen bonds with its molecule.

d. $HC \equiv CH$, has triple bond

152. **Predict the products of the following reactions.**

a. $CH_3—CH_2—CH_2—OH + CH_3CH_2COOH \xrightarrow{H^+}$

b. $CH_3—\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=CH_2 + HI \longrightarrow$

c. $CH_3—CH_2—CH_2—OH \xrightarrow[heat]{KMnO_4}$

d. $CH_3—\underset{\displaystyle OH}{\overset{\displaystyle |}{CH}}—CH_2—CH_3 \xrightarrow{KMnO_4}$

e. $CH_3—CH_2—\underset{\displaystyle OH}{\overset{\displaystyle |}{CH}}—CH_2—CH_3 \xrightarrow{H_2SO_4}$

Solution:

a. $CH_3—CH_2—CH_2—O—\overset{\displaystyle O}{\overset{\displaystyle ||}{C}}—CH_2—CH_3 + H_2O$

n-propylpropanoate

b. $CH_3—\underset{\displaystyle CH_3}{\overset{\displaystyle \overset{\displaystyle I}{|}}{C}}—CH_3$

2-iodo-2-methylpropane

c. $CH_3—CH_2—COOH$

Propanoic acid

d. $CH_3—\overset{\displaystyle O}{\overset{\displaystyle ||}{C}}—CH_2—CH_3$

Butanone

$CH_3—CH_2—CH═CH_2—CH_3 + H_2O$

e. 2 - Pentene

153. Given the following table:

Components of natural gas	Boiling point at standard pressure in ^0C
Methane	-161.6
Ethane	-88.6
Propane	-41.1
Butane	-0.5

154. a. What type of physical technique is used to separate the components of natural gas?

b. Arrange the components of natural gas in increasing order of strength of inter molecular force?

Solution:

a. The physical technique used to separate components of a mixture that have different boiling points is fractional distillation.

b. The type of intermolecular force that is found in non- polar molecules is London force. As size increses bond force also increases.

⇒ Methane < Ethane < Propane < Butane

155. Identify the alcohol that does not react with acidified potassium permanganate?

a. 2 – methyl – 1 – propanol

b. 2 – methyl – 2 – propanol

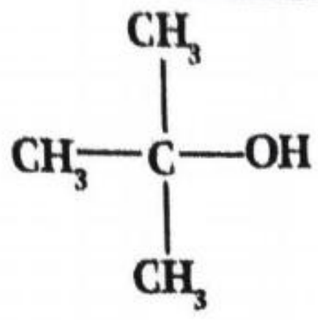
c. 1 – butanol

d. 2 – butanol

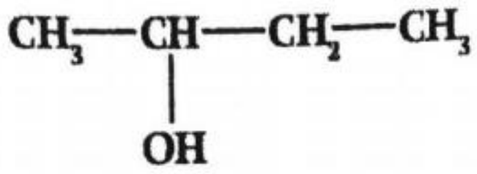
Solution:

An alcohol that does not react with acidified potassium permanganate is tertiary alcohol. Draw the structure of each alcohol and identify it.

a. $CH_3—CH—CH_2OH$, it is primary alcohol and reacts with potassium permanganate.

b.  It is tertairy alcohol and deos not reacts with potassium permanganate.

c. $CH_3—CH_2—CH_2—CH_2—OH$ It is primary alcohol and reacts with potassium permanganate.

d.  This is secondary alcohol and reacts with potassium permanganate.

156. List the methods used to prepare alkanes.

Solution:

a. Hydrogenation of alkenes with a metal catalyst:

$$R—CH=CH_2 + H_2 \xrightarrow{Pt} R—CH_2—CH_3$$

Alkene → Alkane

$$CH_3—CH=CH_2 + H_2 \xrightarrow{Pt} CH_3—CH_2—CH_3$$

b. Heating sodium salt of an organic acid with soda lime.

$$RCOONa + 2NaOH \xrightarrow[heat]{CaO} R—H + Na_2CO_3$$

Sodium salt → Alkane

Example

$$CH_3CH_2COONa + NaOH \xrightarrow[heat]{CaO} CH_3CH_3 + Na_2CO_3$$

c. Wurtz synthesis

$$R—X + 2Na \longrightarrow R—R + 2NaX$$

Halogenated alkane

$$2CH_3I + 2Na \longrightarrow CH_3—CH_3 + 2NaI$$

Methyl iodide → Ethane

157. Propyne can be prepared using the following equation representing the reaction:

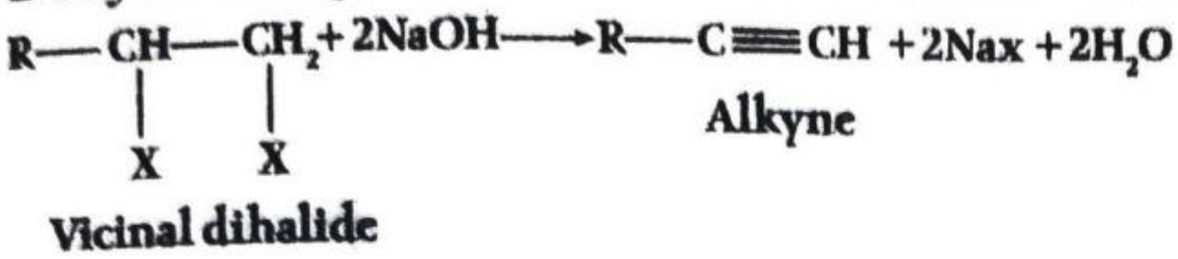
$$X + 2NaOH \rightarrow CH_3 - C \equiv CH + 2NaBr + 2H_2O$$

What is the missing structural formula "X" in the above equation?

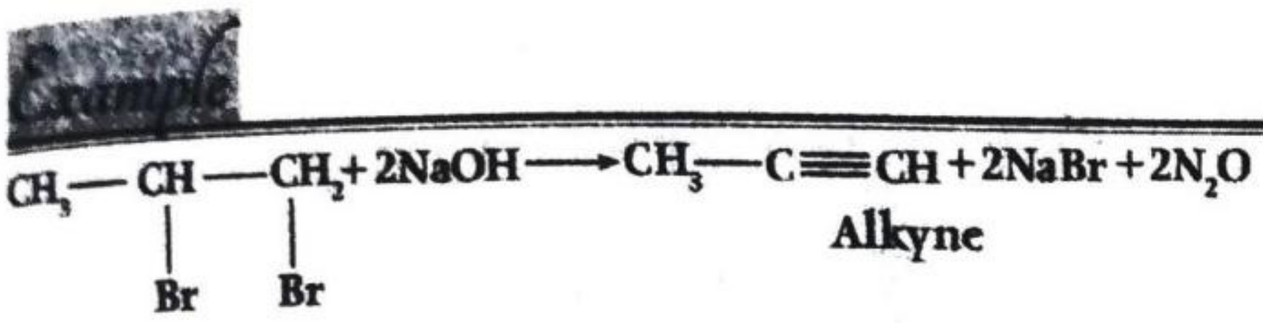
Solution:

Alkynes are prepared by several methods. Some of the general methods are:

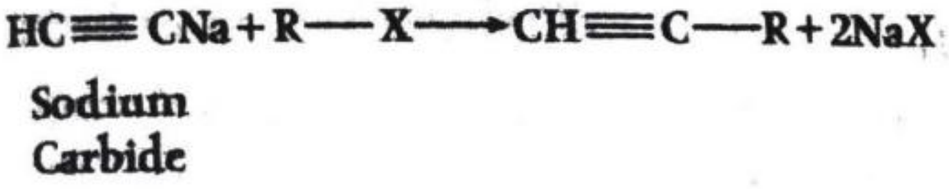
a. Dehydrohalogenation of vicinal dihalides with NaOH or KOH



Example



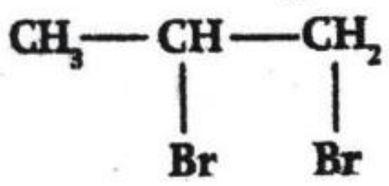
b. alkylation of sodium accetylide with a primary alkyl halide.



Example

$HC \equiv CNa + CH_3Br \rightarrow HC \equiv C - CH_3 + 2NaBr$

➦ The missing structural formula "X" in the above equation is:



158. Which two compounds are isomers of each other?

a. CH_3CH_2CHO and CH_3COCH_3

b. $CH_3CHOHCH_3$ and $CH_3CHOHCH_2OH$

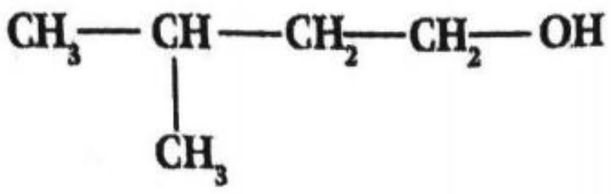
c. CH_3CH_2COOH and CH_3COOCH_3

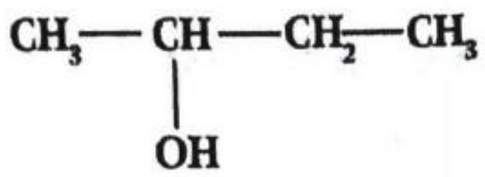
d. $CH_3CHBrCH_3$ and $CH_2BrCHBrCH_3$

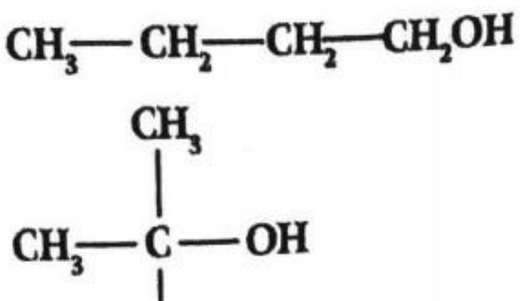
Solution:

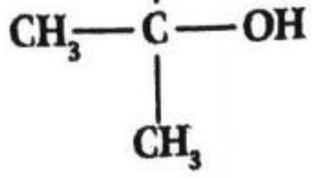
Isomers are compounds with the same molecular formula but different structural formula. ⇒a and c are isomers of each other.

159. Give the common name for:

a. 

b. 

c. 

d. 

Solution:

a. Isopentyl alcohol

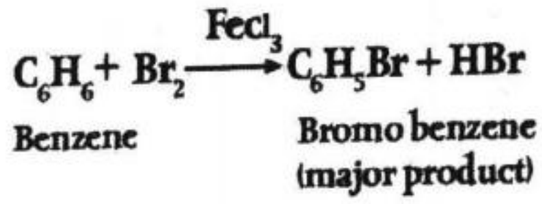
b. sec-butyl alcohol

c. n – butyl alcohol

d. tert – butyl alcohol

160. When benzene reacts with Br_2 in the presence of $FeCl_3$, write the formula of the major product formed.

Solution:



161. _____ is a group obtained by removing a hydrogen atom from an alkane.

Solution:

Alkyl group

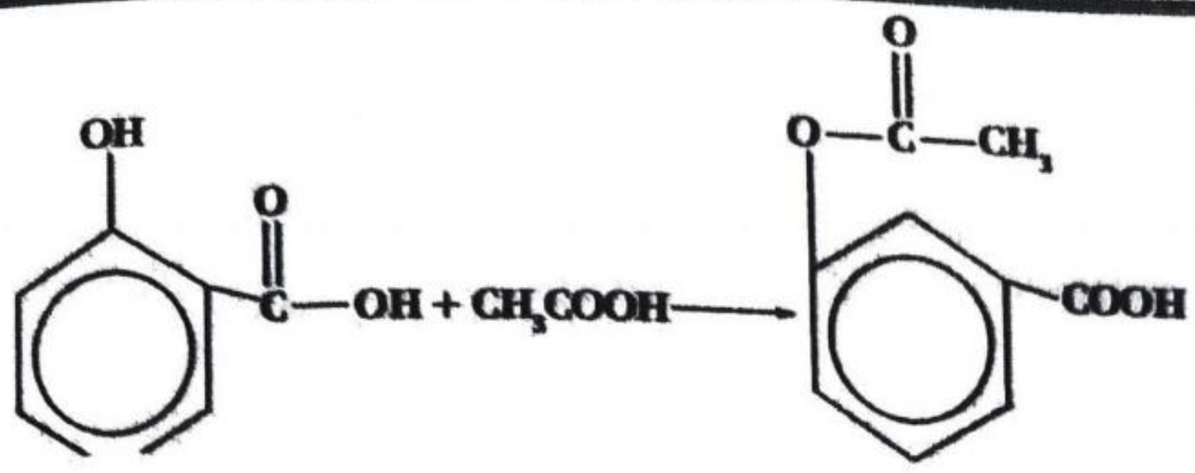
Example

$CH_4 - H \rightarrow CH_3 -$, $CH_3CH_3 - H \rightarrow CH_3\ CH_2$–Ethyl

162. What are the raw materials used to manufacture the common pain killer aspirin?

Solution:

The raw materials used to prepare aspirin are salicylic acid. (2 – hydroxyl benzoic acid) and acetic acid (Ethanoic acid).



163. **What is the difference between saturated and unsaturated hydro carbon?**

Solution:

Saturated hydrocarbons	Unsaturated hydrocarbons
➦ have all carbon – carbon single bond ➦ less reactive	➦ have carbon – carbon double or triple bond ➦ more reactive
➦ have lower boiling point ➦ do not oxidize easily ➦ release less amount of energy when burnt with oxygen	➦ have higher boiling point ➦ oxidize more easily ➦ release more amount of energy when burnt with oxygen

164. **Arrange the following compounds in increasing order of boiling point.**

a. Ethane
b. Dimethyl ether
c. Ethanal
d. ethanol
e. ethanoic acid

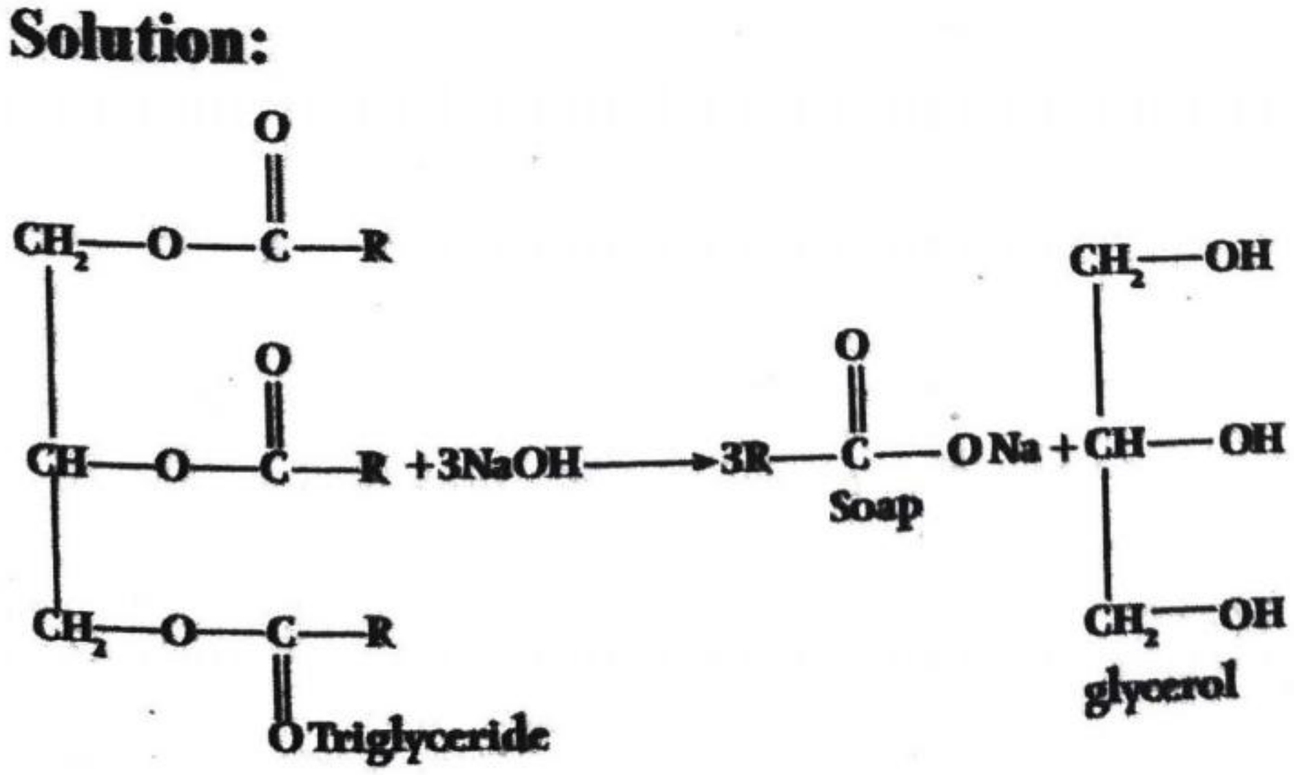
Solution:

Alkane < Alkene <ALkyne< aldehyde < ether < alcohol < carboxylic acid

➦ **ethane<ethanal< dimethyl ether < ethanol <ethanoic acid**

165. **An alcohol produced in the manufacture of soap is _____**

Solution:



➦ An alcohol formed during the production of soap is glycerol (1,2, 3, - propanetriol)

166. List some distilled and un distilled alcoholic beverages and state the approximate percentage of alcohol.

Solution:

Distilled alcohol	Source	Percentage of alcohol
Whisky	Malt of rye	35 – 40
Brandy (cognac)	Grape juice	40 – 50
Gin	Maize	50
Kati kala (araki)	Maize	70 – 75
Run	Molasses	35 – 40

Undistilled	Source	Percentage of alcohol
Beer	Malt of lops	3 – 6
Wine	Grape juice	8 – 12
Port & sherry	Grape juice	14 – 20
Tej	Honey	10 – 15
Tella	Barely, wheat	-

167. Write the molecular formula for a straight chain hydrocarbon having two carbon – carbon double bonds (dienes)

a. 7 b. 10 c. 5

Solution:

a. C_nH_{2n-2}

$C_7H_{2\times7-2}$

C_7H_{12}

b. C_nH_{2n-2}

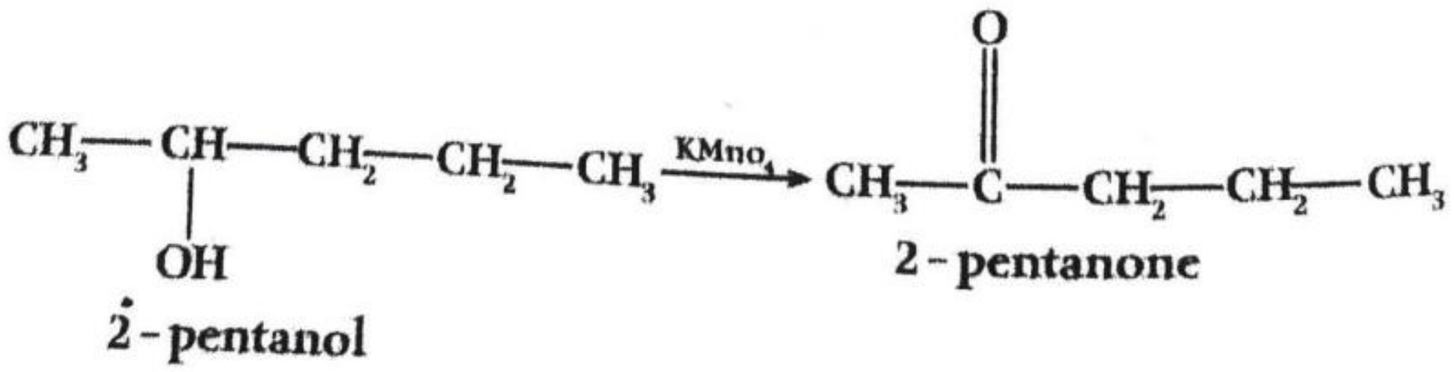
$C_{10}H_{2\times10-2}$

$= C_{10}H_{18}$

c. $C_5H_2\times5-2 = C_5H_8$

168. Write the balanced chemical equation when 2 – pentanol is oxidized with acidified potassium permanganate.

Solution:



Oxidation of primary alcohols yield aldehydes and secondary ones, ketones

169. The main combustible gas obtained from a biogas plant is ____.

Solution:

The biogas plant operates on the simple principle that when dung or any other organic matter is fermented in theabsence of air. The combustible gas produced is methane (CH_4).

170. Write the molecular formula for:

a. 2 – pentyne

b. 2 – pentene

c. n – pentane

d. cyclopentane

Solution:

a. The general formula for open cahin alkyne is C_nH_{2n-2}, n = 2,3,4,....

$\Rightarrow n = 5$

$C_5H_{2\times5-2} = C_5H_8$

b. The general formula for open chain alkene is $C_nH_{2n.}$ n = 2,3,4,....

$\Rightarrow n = 5$

$C_5H_{2\times5} = C_5H_{10}$

c. The general formula for open chain alkane is C_nH_{2n+2}

$\Rightarrow n = 5$

$C_5H_{2\times5+2} = C_5H_{12}$

d. The general formula for cyclic alkane is C_nH_{2n}, n = 3,4.5,....

$\Rightarrow n = 5$

$C_5H_{2\times5} = C_5H_{10}$

171. Arrange the products of petroleum in order of increasing boiling points.

a. Natural gas

d. lubricating oil, waxes

b. Gasoline (petrol)

c. Asphalt

e. kerosene (paraffin)

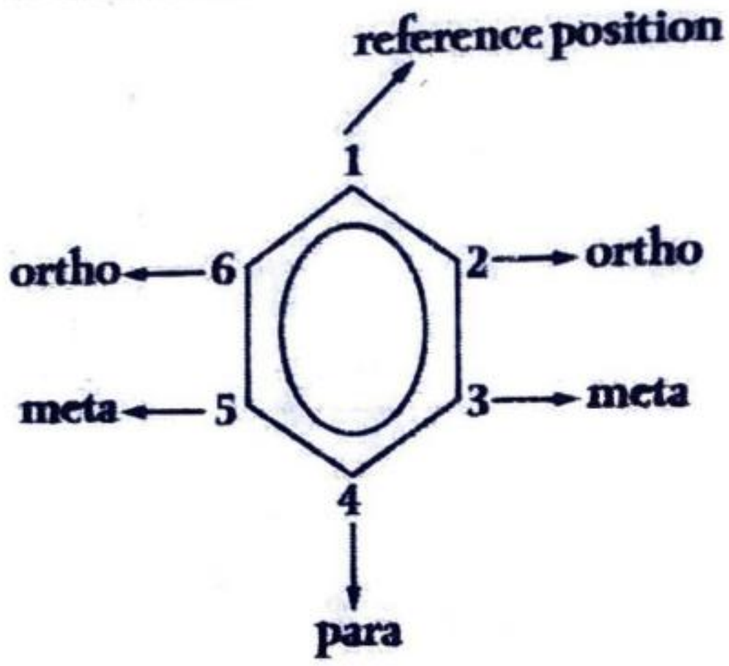
f. diesel oil (naphtha)

Solution:

a < b < e < f < d < c

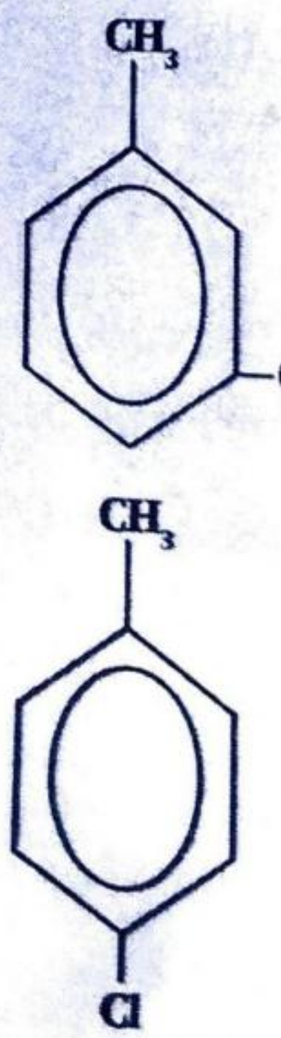
172. Represent the ortho, meta and para substituted benzene.

Solution:



Example

, orthochloromethyl benzene



, metachloromethyl benzene

, parachloromethyl benzene

173. Draw the structure for:

a. 5 – methyl – 4 – hexen – 2 – 01

b. 2,4 – hexadiene

c. Ortho – methyl benzoic acid

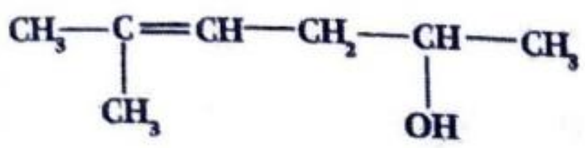
d. Cyclopentanecarbocylic acid

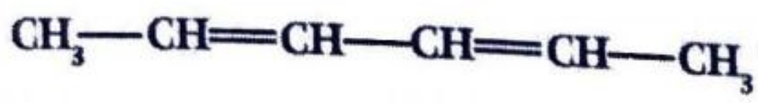
e. 1,2 – benzene di carboxylic acid

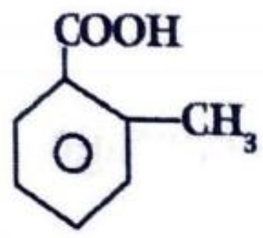
f. Citric acid

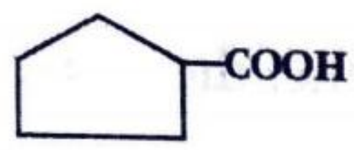
g. 2 – methylbutyl

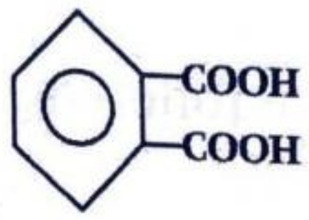
Solution:

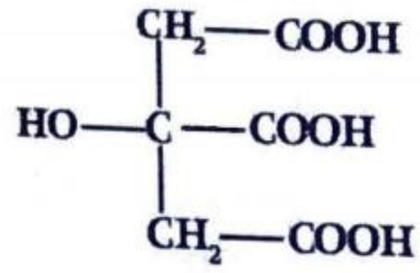
a. 

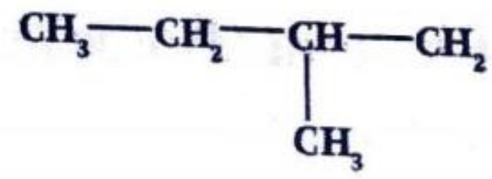
b. 

c. 

d. 

e. 

f. 

g. 

174. Explain by using chemical reaction, how ethanol manufactured by fermentation of sugar?

Solution:

Fermentation can produce an alcoholic beverage whose ethanol content is 12 – 15% only.

175. List some pharmaceuticals and give example for each.

Solution:

➥ **Analgesics** – are substances that reduce pain.

Example: aspirin, Paracetamol

➥ **Antiseptics** – are substances that help to prevent infections in wounds by killing bacteria.

Example: acriflavine

➦ **Sedatives** – are drugs that make a person go to sleep or make them feel calm and relaxed.

Example: barbiturates

➦ **Disinfectants** – are substances used to disinfect something or to kill bacteria.

Example: Dettol

➦ **Sulphonamides** – are medicines used as antibiotics.

Example: Pencillin

176. Define the following by giving example.

a. Pesticides b. Herbicides c. Fertilizers

Solution:

a. Pesticides are chemicals used for controlling the damage of fungi, insects and other pests that attack crops.

Example: Bordeaux (furoicide), DDT, aldrine, dieldrine, malathion.

b. Herbicides – are chemicals used to remove unwanted plants or weeds.

Example: Paraguat, diguat, ammonium – sulphamate, ammonium glyphosinate etc.

c. Fertilizers – are substances used to increase crop productivity.

Example: Ammonium sulphate, potassium nitrate, urea, DAP, etc.

177. Classify each compound according to its functional group.

a. $H_2C{=}CH{-}CH_3$

b. $H_3C{-}O{-}CH_3$

c. $CH_3{-}CH_2{-}CH_2{-}OH$

d. $H_3C{-}\overset{\overset{O}{\|}}{C}{-}O{-}CH_2{-}CH_3$

e. $CH_3{-}CH_2{-}CH_2{-}\overset{\overset{O}{\|}}{C}{-}OH$

f. $CH_3{-}\overset{\overset{O}{\|}}{C}{-}CH_2{-}CH$

g. $CH_3CH_2{-}\overset{\overset{O}{\|}}{C}{-}H$

h. $CH_3{-}C{\equiv}C{-}CH_3$

i. $CH_3CH_2CH_3$

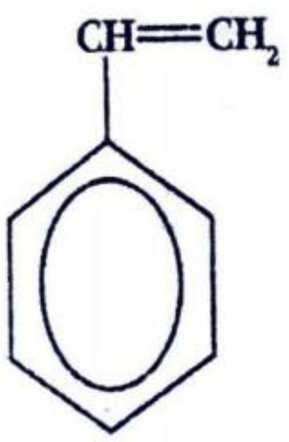
Solution:

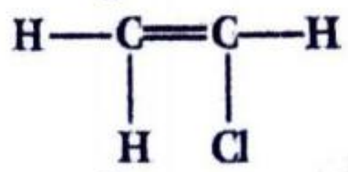
a. alkene
b. ether
c. Alcohol
d. Ester
e. Carboxylic acid
f. Ketone
g. Aldehyde
h. Alkyne
i. Alkane

178. Draw the structure of

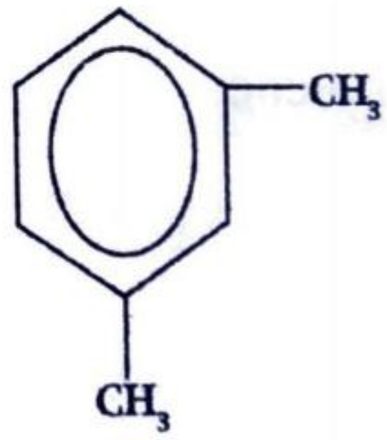
a. Styrene
b. Vinyl chloride
c. Toluene
d. m – xylene
e. phenol
f. Nitro benzene
g. O – dibromo benzene
h. 1, 3, 5 – trichloro benzene
i. 1, 2, 4 – triethylbenezene

Solution:

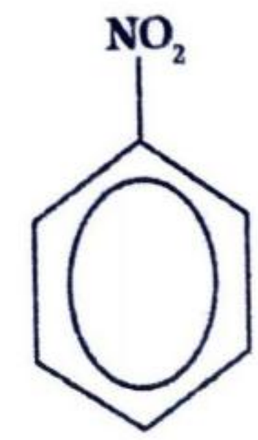
a.


b.


c.

d.


e.

f.


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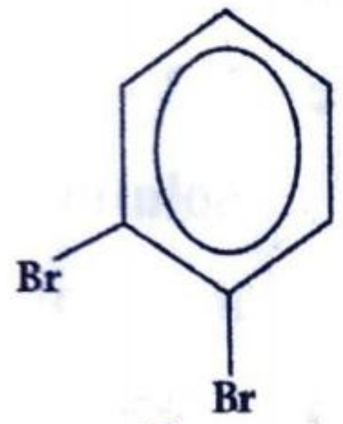
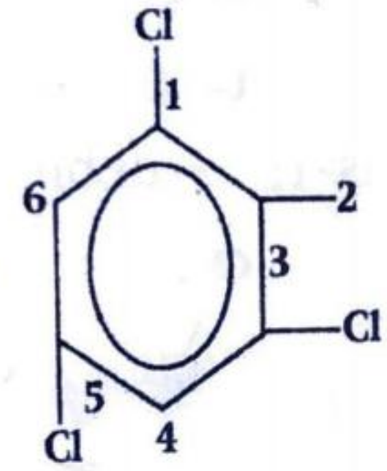
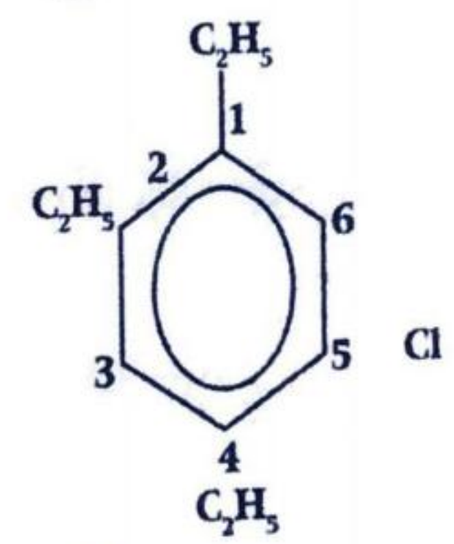
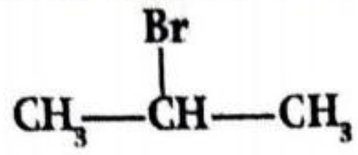


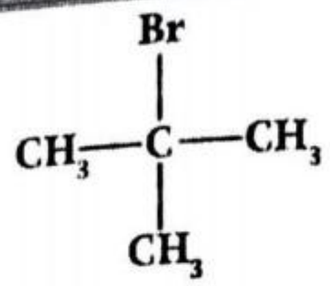
h.


i.


179. Give the systematic (IUPAC) and common name for:

a. CH_3Cl
b. CH_2Cl_2
c. $CHCl_3$
d. CCl_4
e. CH_3CH_2Cl

f. 

g. 

Solution:

Systematic name	Common name
a. Chloro methane	Methyl chloride
b. Dichloro methane	Methylene di chloride
c. Trichloro methane	Chlro form
d. Tetra chloro methane	Carbon tetra chloride
e. Chloro ethane	Ethyl chloride
f. 2 – bromo propane	Isopropyl bromide
g. 2 – Bromo 2- methyl propane	Tert – butyl bromide

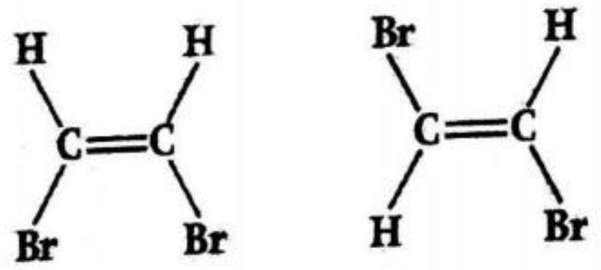
180. Draw the structures of the Cis – trans isomers of:

a. 1, 2 – di bromoethene

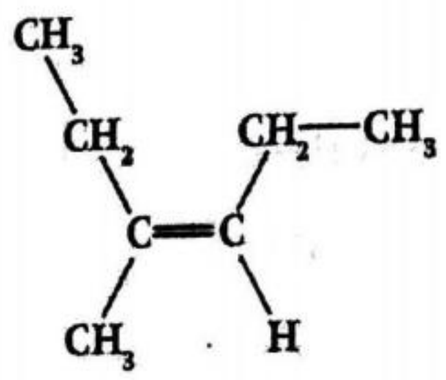
b. 3 – Methyl – 3 – hexene

c. 2, 3 – dichloro – 2 – butene

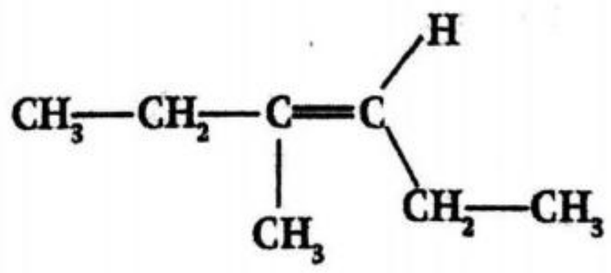
Solution:

a. 

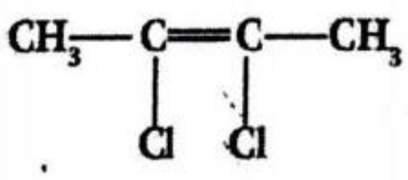
Cis-1, 2 – di bromoethene Trans – 1, 2- dibromoethene

b. 

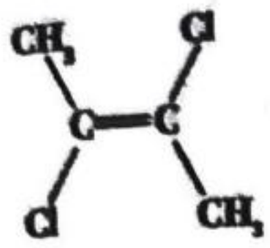
Cis – 3 – Methyl – 3 – hexene



Trans – 3 – Methyl – 2 – hexene

c. 

cis – 2, 3 – dichloro – 3 – butene



Trans -2, 3 – dichloro – 2 – butene

181. Starting with the appropriate alkene, tell how you could prepare each of the following by a single reaction.

a. 1, 2 – dibromobutane

b. 2 – bromo butane

c. 2 – butanol

Solution:

a. $CH_3-CH_2-CH=CH_2 + Br_2 \longrightarrow CH_3-CH_2-\underset{Br}{CH}-\underset{Br}{CH_2}$

1-butene — 1,2-dibromo butane

b. $CH_3-CH_2-CH=CH_2 + HBr \longrightarrow CH_3-CH_2-\underset{Br}{CH}-CH_3$

1-butene — 2-Bromobutane

c. $CH_3-CH_2-CH=CH_2 + H_2O \xrightarrow{H^+} CH_3-CH_2-\underset{OH}{CH}-CH_3$

1-butene — 2-Butanol

182. Starting with 1 – butanol, show how you could prepare 2 – butanol into two reaction steps.

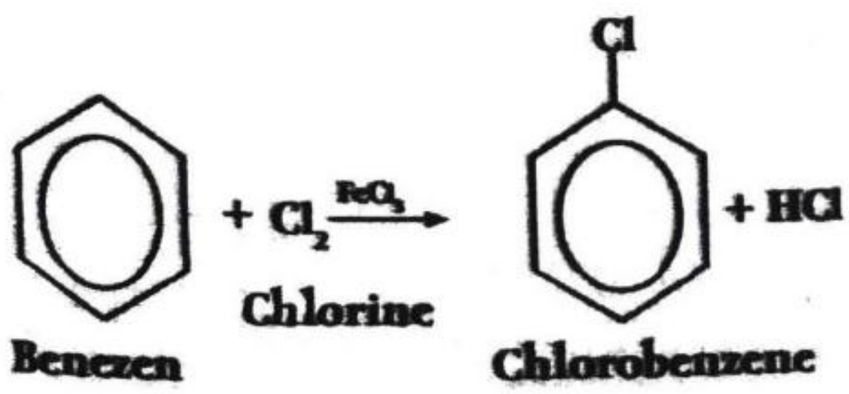
Solution:

i $CH_3-CH_2-CH_2-CH_2OH \xrightarrow{H_2SO_4} CH_3-CH_2-CH=CH_2 + H_2O$

ii $CH_3-CH_2-CH=CH_2 + H_2O \xrightarrow{H^+} CH_3-CH_2-\underset{OH}{CH}-CH_3$

183. Show how to prepare chloro benzene, in a single step from benzene and other appropriate reagents.

Solution:



Supplementary Exercise on Unit - Six

I. Choose the best answer from the given alternative

1. Acid catalyzed hydration of ethylene gives
   A. Ethyne
   B. Ethanal
   C. ethanoic acid
   D. Ethanol

2. When a piece of calcium metal is added to an alcohol, the gas that is given off is
   A. Ethane
   B. Carbondixoide
   C. Hydrogen
   D. Methane

3. The compound $CH_3COCH_2CH_3$ is
   A. Alcohol
   B. Ketone
   C. Ether
   D. Aldehyde

4. Base hydrolysis of ester is
   A. Esterification
   B. Reduction
   C. Saponification
   D. Addition

5. The common alcohol formed during production of soap is
   A. Ethanol
   B. Ethylene glycol
   C. Glycerol
   D. Propanol

6. Which of the following will not show cis-trans isomerism?
   A. 1,2 – dichloroethene
   B. 2 – butene
   C. 1,1 – dibromo propene
   D. 2 – pentene

7. Which of the addition reactions below will not proceed under ordinary circumstances?

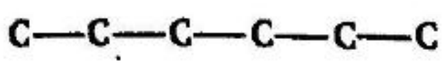
   I $CH_2 = CH_2 + HCl \rightarrow$
   II $CH_2 = CH_2 + H_2SO_4 \rightarrow$
   III $CH_2 = CH_2 + NaOH \rightarrow$
   IV $CH_2 = CH_2 + Br_2 \rightarrow$

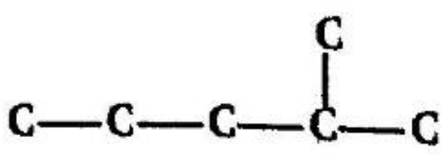
   A. III only
   B. I and II
   C. IV only
   D. I, II and III

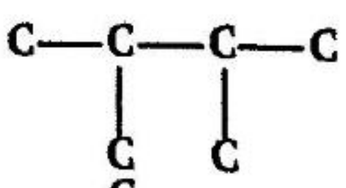
8. The isomers of butanone include all of the following except:

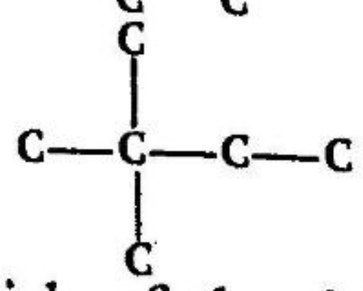
A. Butanal
B. Ethylvinyl ether
C. Tetrahydrofuran
D. Methyl isopropyl ether

9. Examine the structures below, then indicate which has the lowest boiling point.

A. 

B. 

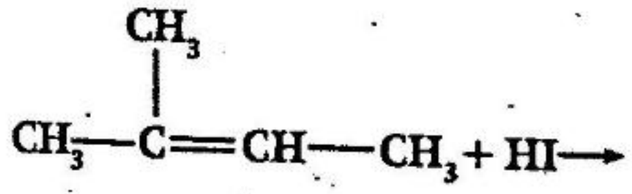
C. 

D. 

10. Which of the following react with hydrogen and nickel to form propane?

A. $CH_3CH_2CH_2OH$
B. $CH_3CH=CH_2$
C. $CH_3CHOHCH_3$
D. CH_3CH_2CHO

11. Consider the reaction below



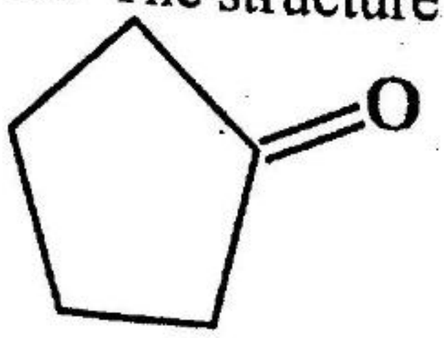
Product, the IUPAC name of the product

A. 2 – lodo – 2 – methyibutane
B. 1 – lodo – 2 – methyl butane
C. 2 – lodo – 3 – methyl butne
D. 3 – lodo –2– methyl butane

12. Which of the following has the highest boiling point?

A. CH_3CH_2COOH
B. $CH_3CH_2CH_2OH$
C. $HCOOCH_2CH_3$
D. CH_3CH_2CHO

13. The structure



A. Aldehyde
B. Ketone
C. Ether
D. Ester

14. What do $CH_3 - CH_2 - CH_2 - CH_3$ and $(CH_3)_2CHCH_3$ have exactly in common?

A. Molecular mass

B. Melting point

C. Chemical property

D. Boiling point

15. Which two compounds have the same molecular formula?

A. Propanol and ethyl propyl ether

B. Acetaldehyde and propane

C. Ethylmethyl ether and propanoic acid

D. Propanal and acetone

16. Which of the following elements is found in most organic compounds combined with carbon?

A. Nitrogen

B. Hydrogen

C. Oxygen

D. Sulfur

17. Hydrocarbon with the general formula C_nH_{2n-2}

A. Alkanes

B. Alkenes

C. Alkynes

D. Aromatics

18. In the reaction: $CaC_2 + 2H_2O \rightarrow X + Ca(OH)_2$, X represents.

A. C_2H_2 B. C_2H_6 C. CH_4 D. C_2H_4

19. Which of the following alcohol undergoes dehydration reaction most readily?

A. 3 – methyl – 1 – butanol

B. 3 – methyl – 2 – butanol

C. 2 – methyl – 2 – butanol

D. 2 – methyl – 1 – butanol

20. Which of the following compounds react with active metals to give salt?

A. CH_2CO

B. CHCOOH

C. CH_3CH_2OH

D. CH_3COOCH_3

21. Which of the following has the highest melting point?

A. Propene

B. Pentene

C. Ethene

D. Butnene

22. All of the following is true about biogas except
    A. It is cleaner and more readily available than other fuel resources.
    B. Biogas usage would greatly reduce deforestation.
    C. Biogas gas technology has no negative effect on environment.
    D. Biogas mainly contains ethanol.
23. Which of the following mismatched with its definition?
    A. Chain – propagation step: a reaction in which free radicals are consumed and generated.
    B. Photo chemical reaction: reaction caused by the presence of light.
    C. Free radical: atom or group of atoms possessing unpaired electrons.
    D. Combustion: endothermic reaction that forms carboxylic and water.
24. Which of the following is not true about benzene?
    A. It is liquid at room temperature.
    B. It is volatile substance.
    C. It is a non – polar compound that is immiscible in water.
    D. It freezes at $0^{o}C$ and boils at $100^{o}c$
25. The physical characteristics that identifies esters is
    A. Texture
    B. Taste
    C. Smell
    D. Hardness
26. Which of the following pharmaceuticals is not correctly matched with its function?
    A. Antiseptics – pain killer
    B. Sedatives – sleep induction
    C. Disinfectants – stop infection
    D. Sulphonamides – antibiotics
27. Compared to inorganic compounds, organic compounds
    A. Have higher melting point
    B. Are more readily decomposed by heat
    C. Make very conductive water solution
    D. Are highly polar

28. Producer gas composed of

A. CO and H_2

B. CH_4 and H_2O

C. O_2 and H_2

D. N_2 and CO

29. Which of the pairs of compounds don't have the same general formula?

A. Cyclo alkane and non-cyclic alkene

B. Cyclo alkene and open chain alkyne

C. Dienes and open chain alkyne

D. Alcohols and aldehydes.

30. A compound has the molecular formula C_3H_6O. All of the following can be the structural formula for the compound except.

A. $CH_3 - COCH_3$

B. CH_3CH_2CHO

C. $CH_3CH_2CH_2OH$

D. $CH_2 = CH - OCH_3$

31. Which of the following is not an alkene or cyclo alkane with the molecular formula C_6H_{12}?

A. Cyclohexane

B. 1,3 – hexadiene

C. Methyl cylcopentane

D. 2,3 – dimethyl – 2 - butene

32. Which of the following does not represent a fertilizer?

A. Urea

B. DDT

C. DAP

D. Ammonium nitrate

33. Tincture of iodine is iodine dissolved in

A. Ethanol

B. Diethyl ether

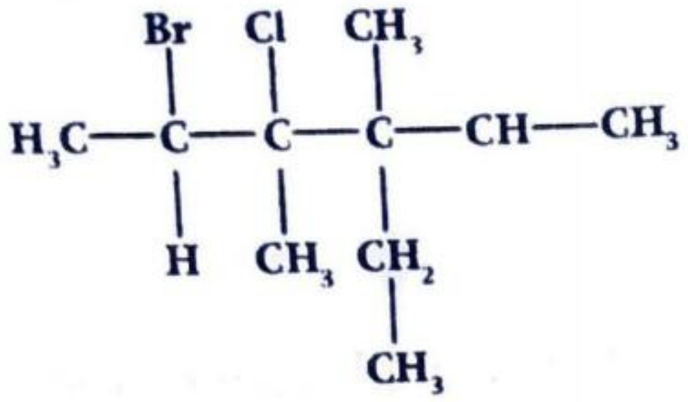
C. Chloroform

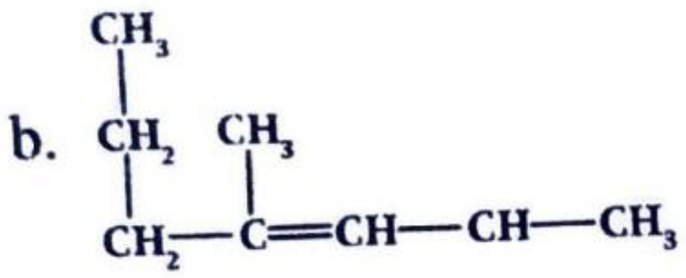
D. Acetone

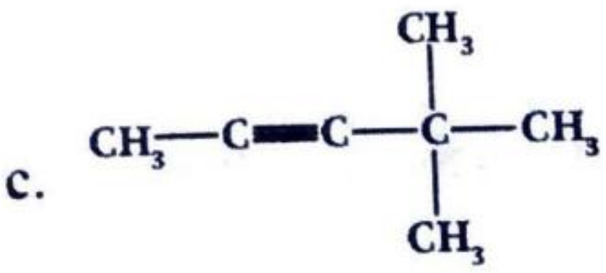
34. Benzene belongs to a class of hydrocarbons called
    A. Paraffins
    B. Olefins
    C. Aromatics
    D. Acetylene series
35. Which of the following is miscible with water in all proportions?
    A. CH_3CH_3
    B. $CH_2 = CH_2$
    C. CH_3CH_2OH
    D. CH_3OCH_3
36. Which of the following hydrocarbons does not belong to homologous series of alkenes?
    A. C_8H_{16}
    B. $C_{15}H_{10}$
    C. $C_{10}H_{20}$
    D. C_7H_{16}
37. Which of the following hydrocarbon is the most reactive?
    A. C_2H_2
    B. C_2H_4
    C. C_2H_6
    D. CH_4
38. The first synthetic organic compound was synthesized in the laboratory by:
    A. Wholer
    B. Dalton
    C. A. Kekule
    D. Wurtz
39. Ethyne decolorizes Br_2 in CCl_4. This is due to the formation of
    A. 1,1 – dibromoethane
    B. 1,2 – dibromo ethane
    C. 1,1,2,2 – tetrabromoethene
    D. 1,1,2,2, - tetra bromo ethane
40. Which of the following olefins exists in solid phase at room temperature?
    A. C_4H_8
    B. $C_{20}H_{40}$
    C. $C_{14}H_{28}$
    D. C_5H_{10}

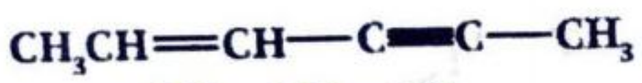
II. Give short answer

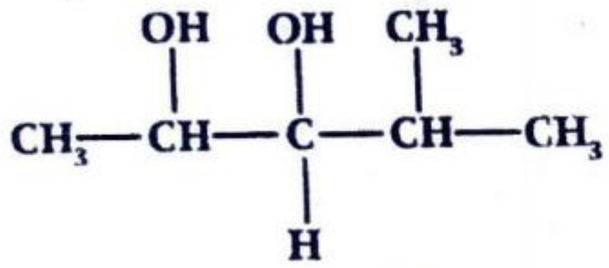
41. Write the IUPAC name for the following compounds

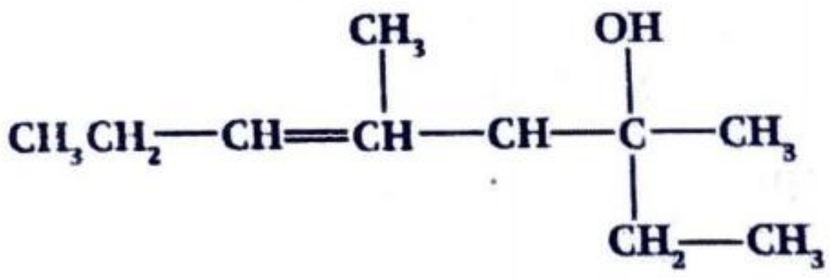
a. 

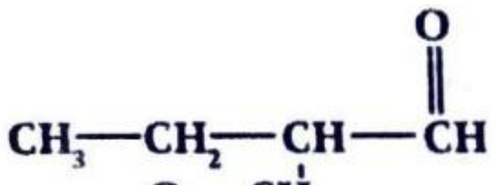
b. 

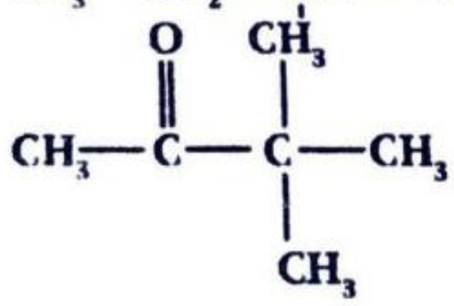
c. 

d. 

e. 

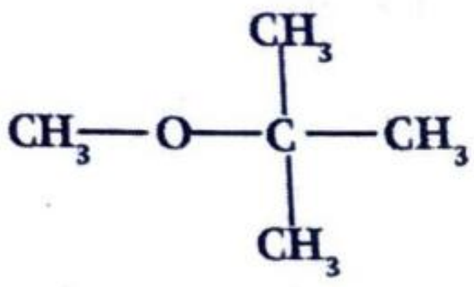
f. 

g. 

h. 

i. $CH_3(CH_2)_6COOH$

j. $CH_3COOCH_2CH_2CH_2CH_3$

k. 

42. Write the structural formula for

a. 3 – methyl – 2 – bromopentan – 1 – 01

b. Cyclohexanol

c. Cyclopentane carboxylic acid

d. 3 – heptynal

e. 1, 2, 3 – pentanetriol

f. Butanedioic acid

43. Complete the following chemical equation

a. $CH_3 - CH = CH_2 + H_2 \rightarrow$

b. $CH_3CH_2CH_2COONa + NaOH \rightarrow$

c. $CH_3CHBrCH_2Br + 2KOH \rightarrow$

d. $CH_3CHOHCH_3 \xrightarrow{H_2SO_4}$

e. $CH_3CH_2Cl + KOH \rightarrow$

f. $CH_3CH_2COOCH_3 + KOH \rightarrow$

g. $CH_3-C(CH_3)=CH_2 + H_2O \rightarrow$

h. $CH_3 - CHOH\ CH_2CH_3 \xrightarrow{[O]}$

i. $CH_3 - CH_2 - CH = CH_3 - CH_3 + HBr \rightarrow$

j. $CH_3CH_2CH_2OH + CH_3CH_2COOH \rightarrow$

44. State the advantages and disadvantages of pesticides.

45. Self – condensation of alcohols in the presence of acid catalyst produces _____.

UNIT 7

Important Inorganic compounds

Unit Outline:

7.1 Oxides

7.2 Acids and Bases

7.3 Salts

7.4 Chemical tests of some ions in salts

7.5 Plant Nutrients

Introduction

- Inorganic compounds are also called mineral compounds
- They melt or vapourize unlike organic compounds
- They are generally found in non-living things. Minerals and rocks are made up of inorganic compounds.

Classification of inorganic compounds

- Inorganic compounds are basically classified into four. These are:

1. Oxides
2. Acids
3. Bases
4. Salts

7.1 Oxides

- **Oxides**: are binary compounds containing oxygen.
- Oxygen reacts almost with all elements to form oxide except noble gases and noble metals such as gold, platinum, palladium etc.
  Binary compounds are formed from two types of elements only.
- When metals react with oxygen, **metallic oxides** are formed.

Metal + oxygen → metallic oxide

Example

Magnesium + oxygen ⟶ magnesiumoxide

$2Mg + O_2 \longrightarrow 2MgO$

- When non-metals react with oxygen, **non-metallic oxides** are formed.

Non-Metal + oxygen → Non-metallic oxide

Example

Sulphur+ oxygen ⟶ sulphurdioxide

$S + O_2 \longrightarrow SO_2$

Illustrative examples

1. Which of the following is not an oxide?

A. CO_2 B. CaO C. HNO_3 D. SiO_2

Explanation

➦ HNO_3 is not binary compound, it is ternary compound. As a result, it is not an oxide.

Answer C

2. All of the following are non-metallic oxides. Except

A. CO_2
B. CaO
C. SO_2
D. NO_2

Answer B

3. Classify the following as oxides or not oxides

a. BaO c. CuO e. H_2SO_4 g. NaOH
b. H_2O_2 d. HCl f. H_2O h. Fe_2O_3

Answer: a,b, c, f and h are oxides.

4. List some metals that do not react with oxygen to form oxides

Answer: Gold, platinium, palladium, osmium etc

5. Classify the following as metallic or non-metallic oxides

a. SO_3 c. V_2O_5 e. N_2O_5
b. K_2O d. CO_2 f. NiO_2

Answer:

➦ a, d, and e are non – metallic oxides.

➦ b, c and f are metallic oxides

Types of Oxides

➦ There are five types of oxides. These are:

1. Acidic oxides
2. Basic oxides
3. Amphoteric oxides
4. Neutral oxides
5. Peroxides

1. Acidic oxides /Acid anhydrides/

- Oxides that have acidic behavour are called **acidic oxides.**
- Most acidic oxides are non-metalic oxides.

Example

- SO_2- sulphurdioxide
- SO_3 – sulphurtrioxide
- CO_2 – carbondioxide
- NO_2- nitrogendioxide
- N_2O_3- Nitrogen trioxide
- N_2O_5- nitrogen pentoxide
- P_2O_3- phosphorous trioxide
- P_2O_5 – phosphorous pentoxide

Metalloid oxides are acidic oxides.

- B_2O_3 – boronoxide
- SiO_2 – silicondioxide
- GeO_2- germariumoxide

Few metallic oxides are acidic oxides

- CrO_3- chromhsm (VI) oxide
- Mn_2O_7 – manganese (VII) oxide

Chemical properties of acidic oxides

1. They react with water to form acids.

Acidic oxide + water $\longrightarrow$ acid

Example:

$SO_3 + H_2O \longrightarrow H_2SO_4$

To find acid anhydride of a given acid, use the following simple method.

Acid anhydride = Acid – water

First, balance the number of hydrogen atoms in water and acid.

Example:

determine the acid anhydride of HNO_3

- Acid anhydride = Acid – water

  $= HNO_3 - H_2O$, balance hydrogen

  $= 2HNO_3 - H_2O$

- $2H - 2H = Zero$

  $2N - Zero = N_2$

  $6O - O = O_5$

Thus, the acid anhydride of HNO_3 is N_2O_5

2. Acidic oxides react with basic oxides.

Acidic oxide + basic oxide $\longrightarrow$ salt

Example:

$SO_2 + CaO \longrightarrow CaSO_3$

3. Acidic oxides react with bases

Acidic oxide + base $\longrightarrow$ salt + water

Example:

$SO_3 + Ca(OH)_2 \longrightarrow CaSO_4 + H_2O$

2. Basic oxides

- Basic oxides dissolve in water to form bases
- They have basic behavior

All basic oxides are metallic oxides but all metallic oxides are not necessarily basic oxides

Chemical properties of basic oxides.

1. They react with water to form bases.

Basic oxide + water ⟶ base

Example

$MgO + H_2O \longrightarrow Mg(OH)_2$

2. They react with acidic oxide to form salts

Basic oxide + acidic oxide ⟶ salt

Example

$BaO + SO_3 \longrightarrow BaSO_4$

3. They react with acids to form salt and water

Basic oxide + Acid ⟶ salt + water

Example

$K_2O + 2HCl \longrightarrow 2KCl + H_2O$

To find the basic anhydride of a base:
Basic anhydride = Base − water

Example:

identify the basic anhydride of NaOH.

➥ Basic anhydride = base – water

$= NaOH - H_2O$, balance hydrogen atom

$= 2NaOH - H_2O$

$2H - 2H = zero$

➥ $2Na - zero = 2Na, 2O - O = O$

Thus, the basic anhydride of NaOH is Na_2O

Identification of acidic and basic oxides

Acidic and basic oxides in the laboratory can be identified by using indicators.

Indicators are substances that show colour change in basic and acidic solutions.

Indictor	Color in acidic solution	Color in basic solution
Litmus paper	Red	Blue
Phenol phthaleiun	Colorless	Pink/red
Methyl orange	Red	Yellow

Illustrative example

6. Which of the following is an acidic oxide?

A. CuO
C. SiO_2
B. Fe_2O_3
D. Na_2O

Explanation

➥ CuO, Fe_2O_3 and Na_2O are metallic oxides i.e. basic oxides.

Answer C

7. What compound doas "x" represent in the chemical equation shown below? (EGSECE . . . . 2001E.C)

$$X + H_2O \longrightarrow HNO_2 + HNO_3$$

A. NO_2
C. N_2O_5
B. NO
D. N_2O_3

Explanation

➥ NO_2 is mixed acid anhydride. It reacts with water and forms two types of acids, HNO_2 and HNO_3

Answer A

8. Which of the following is not basic oxide?

A. K_2O
C. Al_2O_3
B. CaO
D. BaO

Answer C

9. Identify the acid anhydride of the following acids

a. H_2SO_4 c. H_2CO_3 e. H_3PO_3

b. H_2SO_4 d. H_3PO_4 f. HNO_2

Explanation

a. Acid anhydride = Acid – water

$= H_2SO_4 - H_2O$

$= SO_3$

b. $= H_2SO_3 - H_2O = SO_2$

c. $= H_2CO_3 - H_2O = CO_2$

d. $= 2H_3PO_4 - 3H_2O = P_2O_5$

e. $= 2H_3PO_3 - 3H_2O = P_2O_3$

f. $= 2\,HNO_2 - H_2O = N_2O_3$

10. Identify the basic anhydride of the following bases

a. LiOH c. $Mg(OH)_2$ e. $Sr(OH)_2$

b. KOH d. $Ca(OH)_2$ f. $Fe(OH)_3$

Explanation

a. Basic anhydride = base – water

$= 2LiOH - H_2O$

$= Li_2O$

b. $= 2KOH - H_2O = K_2O$

c. $= Mg(OH)_2 - H_2O = MgO$

d. $= Ca(OH)_2 - H_2O = CaO$

e. $= Sr(OH)_2 - H_2O = SrO$

f. $= 2Fe(OH)_3 - 3H_2O = Fe_2O_3$

11. Complete and balance the following

a. $BaO + P_2O_5 \longrightarrow$

b. $SiO_2 + MgO \longrightarrow$

c. $CaO + H_2S \longrightarrow$

d. $SO_3 + LiOH \longrightarrow$

Solution

a. $3BaO + P_2O_5 \longrightarrow Ba_3(PO_4)_2$

b. $SiO_2 + MgO \longrightarrow MgSiO_3$

c. $SO_3 + 2LiOH \longrightarrow Li_2SO_4 + H_2O$

12. When magnesium is burnt with air, white ash is formed.when this ash is tasted with litmus paper, it turns to blue. What type of oxide is the product?

Answer : Basic oxide

3. Amphoteric oxides

- They display both acidic and basic properties .
- They react with acids and bases.

Examples:

Al_2O_3 – aluminum oxide

ZnO – Zinc oxide

PbO – lead (II) oxide

PbO_2 – lead (IV) oxide

SnO – tin (II) oxide

SnO_2 – tin (IV) oxide

BeO – beryllium oxide

- Al_2O_3 shows basic properties by reacting with acids

$Al_2O_3(s) + 6HCl(aq) \longrightarrow 2AlCl_3(aq) + 3H_2O(\ell)$

- It also exhibits acidic properties by reacting with bases:

$Al_2O_3(s) + 2NaOH(aq) + 3H_2O(\ell) \longrightarrow 2NaAl(OH)_4(aq)$

4. Neutral oxides

- These types of oxides have no basic and acidic behaviours.

Example

H_2O – water

CO – carbon monoxide

NO – Nitric oxide

N_2O – Nitrous oxide (loughing gas)

5. Peroxides

- These oxides are oxides in which the two oxygen atoms are linked to each other by "-O-O-" link and with atoms of other elements.
- The oxidation number of oxygen in peroxides is -1 but it is -2 in other oxides.

Example:

H_2O_2 – hydrogen peroide

Na_2O_2 – sodium peroxide

K_2O_2 – potassium peroxide

CaO_2 – calcium peroxide

SrO_2 – strontium period

BaO_2- barium peroxide

Chemical properties of peroxides

- Peroxides are oxidizing agents and react with other substances by losing oxygen

Illustrative Example

13. An oxide in which the two oxygen atoms are linked to each other by "-O-O" link and with atoms of the other elements is known as (EGSECE. . . . . 2008E.C)

A. An acidic
B. A basic
C. an amphoteric
D. a peroxide

Answer: D

14. Which of the following pair of oxides are amphoteric?

A. PbO and CaO_2
B. SnO and H_2O_2
C. Al_2O_3 and Na_2O_2
D. ZnO and PbO

Explanation

- PbO, Al_2O_3 and SnO are amphoturic oxides
- CaO_2, Na_2O_2 and H_2O_2 are peroxides

Answer – D

15. Which of the following is an amphoteric oxide?

A. P_4O_{10}
B. Na_2O_2
C. CaO
D. SnO_2

Answer - D

16. Which of the following is a neutral oxide?

A. NO_2

B. N_2O_3

C. NO

D. N_2O_5

Explanation:

From the five oxides of nitrogen, N_2O and NO are neutral oxides while NO_2, N_2O_3 and N_2O_5 are acidic oxides.

Answer C

17. Which of the following is true about oxides?

A. Oxides are ternary compounds of oxygen

B. All basic oxides are metallic oxides

C. Amphoteric oxides are neutral oxides

D. All non-metallic oxides are acidic oxides

Explanation

- Oxides are binary compounds of oxygen
- Amphoteric oxides have acidic and basic properties
- All non-metallic oxides are not necessarly acidic oxides. For example, H_2O, CO, H_2O_2 are non-metallic oxides but not acidic oxides.

Answer B

18. Write chemical equations to show the amphoteric properties of Al_2O_3 with HNO_3 and KOH.

Answer

- $Al_2O_3 + 6HNO_3 \longrightarrow 2Al(NO_3)_3 + 3H_2O$
  Acts as base salt
- $Al_2O_3 + KOH \longrightarrow 2KAlO_2 + H_2O$
  Acts as acid

19. List some chemical properties of peroxides.

Answer

- They react with dilute acid

➦ They react with other substances by losing oxygen

$2Na_2O_2 + 2CO_2 \longrightarrow 2Na_2CO_3 + O_2$

➦ They are powerful oxidizing agent

20. What type of oxide forms a salt with acid and base?

Answer: Amphoteric oxide

21. An oxide with a common name of laughing gas is _____

Answer : N_2O

22. The poisonous neutral oxide is _____

Answer: carbon monoxide /CO/

Preparation of oxides

➦ There are different methods used to prepare oxides

1. Direct combination of elements

Element + oxygen ⟶ oxide

Example

$S + O_2 \longrightarrow SO_2$

$2Mg + O_2 \longrightarrow 2MgO$

2. Decomposition of compounds by using heat

i **Decomposition of carbonates and hydrogen carbonates.**

➦ Decomposition of carbonates produce metallic oxides and carbondioxide.

Example: $CaCO_3 \xrightarrow{heat} CaO + CO_2$

➦ Metallic hydrogen earbonates decompose to give salt, carbondioxide and water.

Example: $CaCO_3 \longrightarrow Na_2CO_3 + CO_2 + H_2O$

ii **Decomposition of nitrates**

➦ Decomposition of metallic nitrates yield metallic oxide, nitrogen dioxide and oxygen.

Example: $2Zn(NO_3)_2 \longrightarrow 2ZnO + 4NO_2 + O_2$

iii decomposition of metallic hydroxides

➦ Decomposition of metallic hydroxides produce metallic oxide and water.

Example: $Pb(OH)_2 \longrightarrow PbO + H_2O$

iv Decomposition of oxyacids

$H_2CO_3 \longrightarrow H_2O + CO_2$

Illustrative examples

23. All of the following produce oxides. Except
    A. Decomposition of potassium chlorate
    B. Decomposition of magnesium hydroxide
    C. Decomposition of calcium carbonate
    D. Decomposition of carbonic acid

Explanation

$2KClO_3 \longrightarrow 2KCl + 3O_2$, no oxide formed

$Mg(OH)_2 \longrightarrow MgO + H_2O$

$CaCO_3 \longrightarrow CaO + CO_2$

Answer: A

24. Which of the following yields an oxide of carbon?
    A. $Zn(NO_3)_2$
    B. $Cu(OH)_2$
    C. $2NaHCO_3$
    D. H_2SO_3

Answer C

25. Write two carbonates that cannot be decomposed by heat

Answer Na_2CO_3 and K_2CO_3

26. Complete and balance the following equations
    a. $Al + O_2 \longrightarrow$
    b. $P_4 + O_2 \longrightarrow$
    c. $HNO_3 \longrightarrow$
    d. $Hg(NO_3)_2 \longrightarrow$
    e. $Cu(OH)_2 \longrightarrow$
    f. $MgCO_3 \longrightarrow$

Answer

a. $4Al + 3O_2 \longrightarrow 2Al_2O_3$

b. $P_4 + 5O_2 \longrightarrow P_4O_{10}$

c. $2Hg(NO_3)_2 \longrightarrow 2HgO + 4NO_2 + O_2$

d. $4HNO_3 \longrightarrow 4NO_2 + 2H_2O + O_2$

e. $Cu(OH)_2 \longrightarrow CuO + H_2O$

f. $MgCO_3 \longrightarrow MgO + CO_2$

7.2 Acids and Bases

➦ An acid and a base have different definition according to different scientists.

A. Arrhenius definition

➦ **Acid** is a substance that yields hydrogen ion (H^+) when dissolved in water.

Example:

HI, HBr, HCl, HNO_3, H_2SO_4, $HClO_4$, HF etc are Arrhenius acids

$HI \xrightarrow{H_2O} H^+ + I^-$

➦ Acids increase concentration of hydrogen ion in water

➦ **Base** is a substance that yields hydroxide ions (OH^-) when dissolved in water.

➦ base increases the concentration of hydroxide ion when dissolved in water.

Example:

LiOH, NaOH, KOH, $Ca(OH)_2$, $Ba(OH)_2$,etc

$LiOH \xrightarrow{H_2O} Li^+ + OH^-$

The acid – base reaction for Arrhenius definition is a
Acid + Base ⟶ salt + water

Limitations of Arrhenius definition

➦ The definition was restricted to water as a solvent. He did not explain the reaction of acids and bases in gaseous phase.

Example:

$HCl(s) + NH_3(g) \longrightarrow NH_4Cl(s)$

➦ He rejected insoluble acids without hydrogen.

B. Bronsted – Lowry definition

➦ Acid is proton (H^+) donor, so it must contain ionizable hydrogen.

➦ Base is proton acceptor, thus it should contain lone pair of electrons.

Example

$H_2S + H_2O \longrightarrow HS^- + H_3O^+$

- All Arrhenius acids are Bronsted – Lowry acids but all Bronsted – Lowry acids are not necessarlyArrhenius acids.
- All metal hydroxides are not Bronsted – Lowry bases but they contain hydroxide ion (OH^-) which is Bronsted – Lowry base.

➦ Every Bronsted acid has **a conjugate base** and every Bronsted base has a **conjugate acid**

➦ **Conjugate acid is** formed when a base accepts a proton.

Conjugate acid = base + H^+

➦ Conjugate acid has one more hydrogen and positive charge than its corresponding base.

Example:

the conjugate acid of H_2O is $H_2O + H^+ = H_3O^+$

➦ **Conjugate base** is formed when an acid donates a proton.

Conjugate base = Acid – H^+

Example:

the conjugate base of H_2O is $H_2O - H^+ = OH^-$

- Conjugate base has one fewer hydrogen and one more negative charge than its acid.
- The acid – base reaction according to Bronsted concept is :

$$Acid_1 + Base_1 \rightarrow Acid_2 + Base_2$$

$Acid_1 - Base_2$ } conjugate pairs
$Base_1 - Acid_2$ }

Example:

$$NH_3 + H_2O \rightleftharpoons NH^+_4 + OH^-$$

- NH^+_4 is conjugate acid of NH_3:OH^- is the conjugate base of H_2O
- NH_3 is conuugate base of $NH^+_4 \cdot H_2O$ is conjugate acid of OH^-

Illustrative Examples

27. Consider the following reversible reaction at equilibrium:

$$HCO^-_3 + H_2O \rightleftharpoons H_2CO_3 + OH^-$$

Which of the following are Bronsted – Lowering acids? (EGSECE . . . 2008E.C)

A. H_2O and H_2CO_3
B. HC_3 and H_2CO_3
C. H_2O and O
D. HC_3 and O

Explanation

For the forward reaction: H_2CO_3 is conjugate acid of HCO^-_3

OH^- is conjugate base of H_2O

For the reverse reaction: HCO^-_3 is conjugate base of H_2CO_3

H_2O is conjugate acid of OH^-

Thus, H_2O and H_2CO_3 are Bronsted – Lowery acids

Answer A

28. According to the Bronsted – Lowery definition, an acid is a substance that

A. Donatas proton(s) to a base

B. Accepts electron pair /s/ from a base

C. Donates electro n pair/s/ to a base

D. Accepts proton/s/ from a base

Answer A

29. Arrhenius acid is a substance that:

A. release hydronium ion in solution

B. release hydroxide ion in solution

C. accept proton from a base

D. donates electron pair to a base

Answer A

30. Identify the acid – base conjugate pairs in

a. $N_2H_4 + H_2O \rightleftharpoons N_2H_5 + OH^-$

b. $S^{2-} + HF \rightleftharpoons HS^- + F^-$

c. $H_2SO_4 + NH_3 \rightleftharpoons HSO_3^- + NH_4^+$

d. $HS^- + SO_3^{2-} \rightleftharpoons S^{2-} + HSO_3^-$

Answer

a. $N_2H_4, N_2H_5^+ ; H_2O, OH^-$

b. $S^2, HS^- ; HF, F$

c. $H_2SO_4, HSO_4^- ; NH_3, NH_4^+$

d. $HS^-, S^{2-} ; SO_3^{2-} ; HSO_3^-$

31. Determine the conjugate acid of:

a. P

b. H_2O

c. HS^-

d. HPO_4^{2-}

e. CN^-

f. NH_3

Explanation: Conjugate acid = base + H^+ , add hydrogen and charges

a. $P + H^+ = HP^+$

b. $H_2O + H^+ = H_3O^+$

c. $HS^- + H^+ = H_2S$

d. $HPO^{2-}_4 + H^+ = H_2PO_4^-$

e. $CN^- + H^+ = HCN$

f. $NH_3 + H^+ = NH_4^+$

32. Identify the conjugate bases of:

a. H_3PO_4

b. $HClO_4$

c. HCO_3^-

d. $N_2H^+_5$

e. NH_3

f. HS^-

Explanation

Conjugate base = Acid – H^+, subtract charges and hydrogen

a. $H_3PO_4 - H^+ = H_2PO^-_4$, here charge= $0 - (+1) = -1$

b. $HClO_4 - H^+ = ClO_4^-$

c. $HCO_3^- - H^+ = CO_3^{2-}$

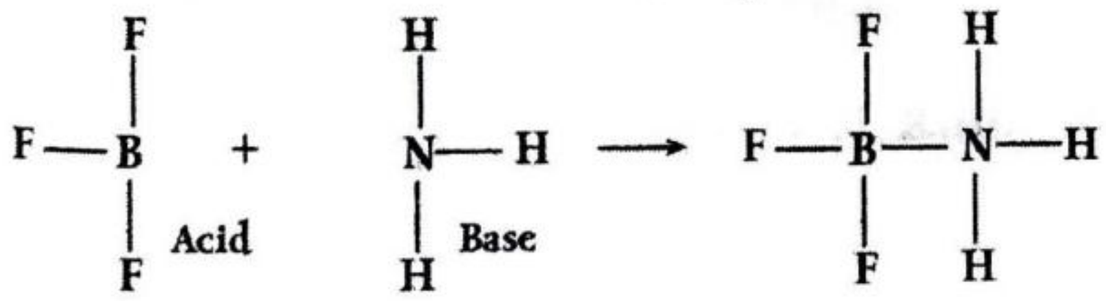
d. $N_2H_5^+ - H^+ = N_2H_4$

e. $NH_3 - H^+ = NH_2^-$

f. $HS^- - H^+ = S^{2-}$

C. Lewis definition

- **Acid** is a substance that can accept a pair of electrons.
- Lewis acids have vacant or unfulfilled subshells to accept a pair of electrons. They are electrophilic substances.
- **Base** is a substance that can donate a pair of electrons.
- Lewis bases have lone pair of electrons on the central atom. They are nucleophile.
- The significance of the Lewis concept is that it is much more general than other definitions, it includes many reactions that do not involve Bronsted acids. Consider, for example, the reaction between borontrifluoride (BF_3) and ammonia (NH_3).



The bond formed when a Lewis acid reacts with a Lewis base is called coordinate covalent bond.
Lewis bases are electron rich species while acids are electron deficient species.

Lewis acids include the following substances

1. Molecules whose centeral atoms contain below eight electrons.

Example: BF_3, $AlCl_3$, $BeCl_2$ etc

2. Molecules that expand their octet

Example: $SiCl_4$, PCl_5, SF_6, $FeCl_3$etc

3. Positive ions /cations/

Example: , Cu^{2+}, Ag^+ , H^+, etc

4. Some molecules with multiple bonds

Example: CO_2, SO_2, SO_3, etc

$CO_2 + H_2O \longrightarrow H_2CO_3$

5. Some metals

Example: Nickel

$Ni + 4CO \longrightarrow Ni(CO)_4$

Illustrative examples

33. Among the following, identify a substance that is capable of donating a pair of electrons

A. an Arrhenius acid
B. a Lewis acid
C. an Arrhenius base
D. a Lewis base

Answer D

34. Which of the following properties describes a Lewis acid?

A. Accepting proton/s/ from a base
B. Accepting electron pair/s/ from a base
C. Donating electron pair/s/ to a base
D. Donating proton/s/ to a base

Explanation: Lewis acid is electron pair acceptor

Answer B

35. Given the reaction: $FeCl_3 + Cl^- \longrightarrow FeCl^-_4$ which one is Lewis acid?

A. $FeCl_3$
B. Cl^-
C. $FeCl_4^-$
D. None

Explanation:Lewis acids and bases are found on the reactant side but not on the product side. Negative ions are Lewis bases.

Answer A

36. Classify each of the following species as a Lewis acid or a Lewis base.

a. H_2O d. I^- g. H^+

b. CO_2 e. NH_3 h. BH_3

c. SO_2 f. OH^-

Answer:

- b, c, g and h are Lewis acids
- a, d, e and f are Lewis bases.

37. All Bronsed – Lowry acids are Lewis acids, but the reverse is not true. Give two examples of Lewis acids that are not Bronsted – Lowry acids

Explanation

- To be a Bronsted – Lowry acid it must contain ionizable hydrogen to be donated. However, according to Lewis there are many acids without hydrogen. Example BCl_3, PCl_5, CO_2, SO_2etc

38. Describe the following reaction according to the Lewis concept of acids and bases.

$AlCl_3 + Cl^- \rightarrow AlCl_4^-$

Explanation:

$AlCl_3$ is a Lewis acid. Cl^- is a Lewis base and $AlCl_4^-$ is Lewis – acid – base complex.

39. What type of bond is formed when a Lewis acid reacts with a Lewis base?

Answer coordinate covalent bond or dative bond

40. Identify the Lewis Acid and base in the following reactions.

a. $Cu^{2+} + 4NH_3 \longrightarrow [Cu(NH_3)_4]^{2+}$

b. $BF_3 + NCl_3 \longrightarrow F_3BNCl_3$

c. $PCl_5 + Cl^- \longrightarrow PCl_6^-$

d. $Ni + 4CO \longrightarrow Ni(CO)_4$

Answer

Lewis acid	Lewis base
a. Cu^{2+}	NH_3
b. BF_3	NCl_3
c. PCl_5	Cl^-
d. Ni	CO

Properties of Acids

➦ Acids have the following general properties

1. They have sour taste
   ➦ This can be identified by eating fruits specificially lemon because it contains citric acid.
2. Most acids release hydrogen ion when dissolved in water
3. Acids change color of indicators

Example

➦ Blue litmus to red

➦ Phenolphthalein - colorless

➦ Methyl orange – red

➦ Universal indicator – yellow, orange, red

4. Dilute acids react with active metal and form salt and hydrogen gas.
   Dilute acid + active metal ⟶ salt + hydrogen gas

Example

$2HCl + Mg \longrightarrow MgCl_2 + H_2$

5. Acids react with bases and basic oxides to form salt and water
   Acid + base /basic oxide/ ⟶ salt + water

Example

$HBr + NaOH \longrightarrow NaBr + H_2O$

$2HCl + K_2O \longrightarrow 2KCl + H_2O$

6. Acids react with carbonates and hydrogen carbonates

Acid + carbonate/hydrogen carbonate⟶ salt + water + carbondioxide

Example

$CaCO_3 + 2HI \longrightarrow CaI_2 + CO_2 + H_2O$

$NaHCO_3 + HNO_3 \longrightarrow NaNO_3 + CO_2 + H_2O$

7. Acids have P^H value below seven

8. Acids transmit electricity in aqueous state

Illustrative examples

41. Which of the following is not true about acids?

A. They change the color of litmus to red

B. All acids release hydrogen ion in water

C. They have PH value below seven

D. They trarumit electricity in aqueous state

Answer: B

42. Which of the following gas is formed when dilute acids react with active metals?

A. H_2 B. CO_2 C. O_2 D. N_2

Explanation: When acids react with active metals, salt and hydrogen gas are formed.

Answer A

43. List some metals that do not react with acids to form salt and hydrogen gas

Answer

Gold, silver, platinum, palladium, osmium, etc

44. Complet and balance the following

a. $Al + HCl \longrightarrow$

b. $H_2SO_4 + NaOH \longrightarrow$

c. $HNO_3 + K_2O \longrightarrow$

d. $HBr + MgCO_3 \longrightarrow$

e. $HClO_4 + KHCO_3 \longrightarrow$

Explanation:

a. $2Al + 6HCl \longrightarrow 2AlCl_3 + 3H_2$

b. $H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$

c. $2HNO_3 + K_2O \longrightarrow 2KNO_3 + H_2O$

d. $2HBr + MgCO_3 \longrightarrow MgBr_2 + CO_2 + H_2O$

e. $HClO_3 + KHCO_3 \longrightarrow KClO_3 + CO_2 + H_2O$

45. Why acids transmit electricity in aqueous solution?

Explanation: When acids dissolve in water, ions of acids move freely to carry charge. That is the reason for acids in aqueous state to transmit electricity.

Classification of Acids

➥ There are different ways used to classify acids

A. **Based on the types of elements they contain:** acids are classified as binary acids and ternary acids.

Binary acids are acids formed from two types of elements only.

Example:

HCl, HF, HI, HBr, H_2S, H_2Se etc

Ternary acids are formed from three types of elements.

Example:

HNO_3, H_2SO_4, $HClO_4$ HCN, H_3PO_4etc

B. **Based on the number of ionizable hydrogen they contain:** acids are classified as monoprotic, diprotic or triprotic acids.

Monoprotic acids are acids that have only one ionizalbe hydrogen.

Example:

HNO_3, HCl, HI etc $HCl \longrightarrow H^+ + Cl^-$

Diprotic acids – have two ionizable hydrogens

Example:

H_2SO_4, H_2S, H_2CO_3, H_2SO_3, $H_2C_2O_4$etc

$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$

Triprotic acids contain three ionizable hydrogens in their formula.

Example

H_3PO_4, H_3AsO_4 ,etc$H_3PO_4 \longrightarrow 3H^+ + PO_4^{3-}$

Generally, acids containing more than one ionizable hydrogen are called polyprotic acids.

C. **Based on degree of ionization or dissociation:** acids are calssified as strong or weak acids.

Strong acids are strong electrolytes which, for most practical purposes, are assumed to ionize completely in water.

➦ Most strong acids are inorganic acids

For strong monoprotic acid, concentration of acid and ions are equal at equilibrium

Example:

HCl, HBr, HI, HNO_3, H_2SO_4 , $HClO_4$

➦ $[HCl] = [H_3O^+] = [Cl^-]$

Weakacids: ionize only to a limited extent in water.

Example:

HF, H_2S, H_2CO_3 , HCN, H_2SO_3, HNO_2, CH_3COOH, HCOOH. etc

$HF(aq) + H_2O(1) \rightleftharpoons H_3O^-(aq) + Cl^-(aq)$

At equilibrium: $[HF] >> (H_3O^+) = (F^-)$

D. **Based on the amount of acids in water:** acids are classified as dilute or concentrated acids.

Dilute acid has less solute and more solvent. Dilute acid can be strong acid or weak

Concentrated acid is a solution of acid with more acid and less solvent.

- Concentrated acid can be strong acid or weak acid

Concentration of solutions of acids

- It is the amount of solute (acid) present in a given quantitiy of solution. The most important concentration unit is **molarity.**

Molarity is the number of moles of solute in one liter of solution.

$$\text{Molarity} = \frac{\text{Moles of solute}}{\text{liters of solution}} \qquad \boxed{M = \frac{n}{v}}$$

Illustrative Examples

46. Which of the following statement is not true?
    A. A dilute solution of HCl is dissolved in a smaller volume of water
    B. A concentrated solution of HCl is dissolved in smaller volume of water
    C. A concentrated acid has relatively large amount of solute dissolved in the solvent
    D. A dilute acid solution has a relatively small amount of solute dissolved in the solvent

    Answer A

47. Which of the following statements is true about weak and dilute acids?
    A. Weak acids ionize only slightly in water whereas dilute acids contain a relatively small amount of acid dissolved in water.
    B. Weak acids contain a relatively large amount of acid dissolved in water where as dilute acids ionize completely or nearly completely in water

C. Weak acids ionize completely or nearly completely in water whereas dilute acids contain a relatively large amount of acid dissolved in water

D. Weak acids contain a relatively small amount of acid dissolved in water whereas dilute acids ionize only slightly in water

Explanation

- Weak acids ionize slightly in water
- Dilute acids can be strong or weak acids that contain relatively small amount of acid in water

Answer: A

48. Which of the following represents a ternary acid?

A. $HClO_4$ B. HF C. HCN D. HCl

***Explanation*:**

Ternary acids are also called oxyacids formed from three different types of elements.

Answer A

49. The strength of an acid can be determind by

A. Its degree of ionization in water

B. The number of oxygen atoms

C. The number of hydrogen atoms

D. Its molar concentration

Answer A

50. A certain base that was purchased from a super market ionizes almost completely in water. Then, it is a

A. Dilute base C. strong base

B. Concentrated based D. weak base

Answer C

51. The concentration of 10g of NaOH in 500ml of a solution is (at. Mass H=1, O=16, Na=23)

$$n = \frac{m}{M} = \frac{10g}{40g/mol} = 0.25mol$$

$$M = \frac{n}{v} = \frac{0.25mol}{0.5L} = 0.50M$$

52. Calculate the molarity of 24.5g of sulphuric acid in 400mL of sulphuric acid of solution.

Solution

m = 24.5g, molecular weight of H_2SO_4 = 98g/mol

= 0.625M

$$M = \frac{n}{v} = \frac{24.5g}{\frac{98g}{mol}.0.4L} = 0.625M$$

53. What is the mass of 1.5MKOH in 800cm^3 of solution?

Solution

$$M = \frac{m}{M_{wt.}V}, M = 1.5\frac{mol}{L}, V = 0.8L, M_{wt} = 56g/mol$$

$m = M \times M_{wt} \times V$

$$m = 1.5\frac{mol}{L} \times 56\frac{g}{mol} \times 0.5L = 67.2g \text{ of KOH}$$

thus, to prepare 1.5M KOH in 800cm^3 solution, you have to weigh 67.2g of KOH.

54. How many milliters of solution are required to prepare 0.02mol of 2.5MNaCl solution?

Solution

$$M = \frac{n}{v}, V = \frac{n}{M} = \frac{0.02mol}{2.5_{mol/L}} = 0.008L = 8mL$$

55. Classify the acids below as binary, ternary , strong or weak acids.

a. HNO_3
b. HI
c. HF
d. H_2CO_3
e. HCN
f. CH_3COOH
g. H_2S
h. $HClO_4$

Answer

- b, c and g are binary acids
- a, d, e, f and h are ternary acids
- a, b and h are strong acids
- c, d, e, f and g are weak acid,

56. Classify the following acids as monoprotic, diprotic or triprotic acids.

a. H_2S
b. H_2SO_4
c. H_3PO_4
d. H_2PO_2
e. H_3PO_2
f. HClO
g. HCOOH
h. HOOCCOOH

Explanation

- e, f and g are monoprotic acids
- a, b, d and h are diprotic acids
- c is triprotic acid

Note:

H_3PO_4, H_3PO_3 and H_3PO_2 are triprotic, diprotic and monoprotic acids respectively because in H_3PO_4 three hydrogens, in H_3PO_3 two hydrogen and in H_3PO_2 only one hydrogen are attached to oxygen atoms.

57. Explain the following

a. 99% nitric acid solution
b. 1% nitric acid solution

Explanation

a. 99% HNO_3 nitric acid and 1% water
b. 1% acid and 99% water

58. What is the mass of a 2M H_2SO_4 in a 250mL of solution?

Solution:

Solve by your own

Answer 49g

P^H and P^H scale

➥ The P^H of a solution is defined as the negative logarithm of the concentration of the hydrogen ion concentration (in mole per liter)

$$PH = -10g^{[H+]}$$
$$[H^+] = 10^{-PH}$$

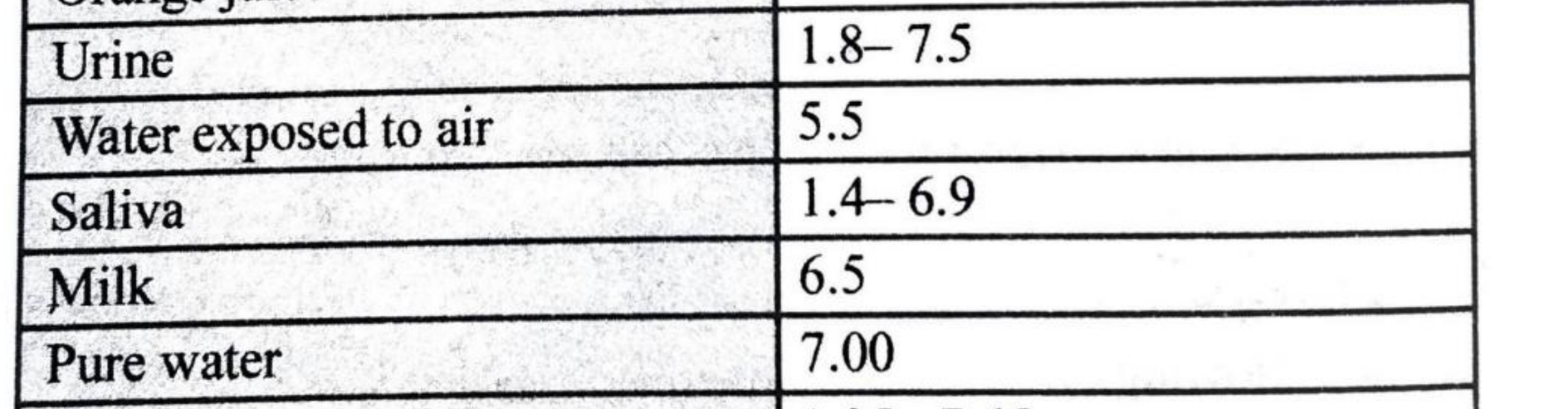
Notic that PH increases as $[H^+]$ decreases.

$P^H < 7$, acidic solution
$P^H > 7$, basic solution
$P^H = 7$, neutral solution

Table 7.1. The PHs of some common fluids

Sample	PH
Gastric juice in stomach	1.0– 2.0
Lemon juice	2. 4
Vinegar	3.0
Orange juice	3.5
Urine	1.8– 7.5
Water exposed to air	5.5
Saliva	1.4– 6.9
Milk	6.5
Pure water	7.00
Blood	1.35– 7.45
Tears	7.4
Milk of magnesia	10.6
House hold ammonia	11.5

The PH scale



➦ **A PH mater** is commonly used in the laboratory to determine the PH of a solution. Although many PH meters have scales marked from 1 to 14, PH values can in fact, be less than 1 and greater than 14

Illustrative Example

59. When the PH value of a solution changed from 2 to 1, the concentration of hydrogium ions

A. Increases by a factor of 10
B. Decreases by a factor of 10
C. Increases by a factor of 2
D. Decreases by a factor of 2

Solution

$PH = -10g\,[H_3O^+] \Rightarrow [H_3O^+] = 10^{-PH}$

$\Rightarrow PH_1 = 2 \quad , PH_2 = 1$

$$\Rightarrow \frac{[H_3O^+]_1}{[H_3O^+]_2} = \frac{10^{-PH_1}}{10^{-PH_2}} = \frac{10^{-2}}{10^{1}} \Rightarrow [H_3O^+]_2 = 10[H_3O^+]_1$$

Answer: A

60. Given the following table that contains information about three solutions

Solution	PH Value
1	10
2	7
3	5

Which list has the solutions placed in order of increasing hydrogen ion concentration? (EGSECE. . . . 2006E/C)

A. 3, 1, 2
B. 3, 2, 1
C. 1, 2, 3
D. 2, 1, 3

Explanation

➦ As P^H increases concentration of hydrogen ion decreases because they are inversely related.

Answer C

61. What is the P^H of a solution that has a hydrogen ion concentration 100 times greater than a solution with a P^H of 4?

Solution

Let the unknown P^H , and the given P^H be P^{H2}

$[H^+]_1 = 100\,[H^+]_2$ but $[H^+]_2 = 10^{-PH2} = 10^{-4}M$

$\Rightarrow [H^+]_1 = 100 \times 10^{-4} M = 10^{-2} M$

$\Rightarrow PH_1 = 10g[H^+]_1 = -10g10^{-2} = 2$

62. What is the PH of 0.02M HCl solution?

Solution

For strong monoprotic acid, $[acid]=[H^+]$

$\Rightarrow [H^+] = [HCl] = 0.02M = 2 \times 10^{-2}M$

$PH = -10g\,^{[H+]} = -10g^2 \times 10^{-2} = 2\text{-}10g^2 = 1.7$

63. Determine the PH of 0.05M H_2SO_4 solution assume complete ionization.

Solution

For strong polyprotic acids. Use stoichiometry to find the concentration of hydrogen ion.

$\Rightarrow H_2SO_4 \longrightarrow 2H^+ + SO_4^{2}$

1M 2M

0.05M ?

$\Rightarrow [H^+] = 2 \times 0.05H = 0.10M$

$\therefore P^H = -10g[H^+] = -10g0.1 = 1$

64. A 1M solution of HA, HB, HC and HD acids have PH values 1, 2, 3 and 4 respectively. Arrange these acids in increasing order of strength.

Explanation

With same concentration of acids, as PH increases, strength of acids decreases.

$\Rightarrow HD < HC < HB < HA$

General properties of Bases

- Bases have bitter taste
- They are slippery to touch
- They release hydroxide ion in water
- They change the color of indicators.

Example

Red litmus to blue

Phenphthalien to red

Methyl Orange to yellow

Universal indicator to blue or purple

- They are electrolyte i.e conduct electricity in aqueous or molten state.
- They have PH value above seven
- They react with acids and acidic oxides to form salt and water.

Example

$KOH + HCl \longrightarrow KCl + H_2O$

$2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$

Classification of Bases

- **Based on degree of ionization** bases are classified as strong or weak base.
- **Strong bases** ionize almost completely in water.

Example

$LiOH, NaOH, KOH, RbOH, Ca(OH)_2, Sr(OH)_2, Ba(OH)_2$

$LiOH \longrightarrow Li^+ + OH^-$

Weak bases ionize slightly when dissolved in water.

Example

NH_3, $Mg(OH)_2$, $Al(OH)_3$, $Fe(OH)_2$, etc

$NH_3 + H_2O \rightarrow NH_4^+ + OH^-$

➦ **Based on the amount of base in water,** bases are classified as concentrated or dilute base.

Concentrated base – contains relatively large amount or percentage of base.

Example:

98% NaOH solution is concentrated solution of NaOH.

Dilute base – contains relatively small amount or percentage of base.

Example:

2% NaOH solution is dilute solution of NaOH

➦ **Based on the number of hydroxide group they contain** bases are classified as monobasic bases, dibasic bases or tribasic bases.
Monobasic bases contain one hydroxyl group.

Example: NaOH, KOH, LiOH etc

Dibasic bases posses two hydroxyl group.

Example:

$Ca(OH)_2$, $Sr(OH)_2$, $Bn(OH)_2$, etc

Tribasic bases contain three hydroxyl group

Example:

$Fe(OH)_3$, $Al(OH)_3$etc

Illustrative Example

66. All of the following are strong bases. Except

A. RbOH
B. $Ca(OH)_2$
C. $Mg(OH)_2$
D. LiOH

Answer C

67. Which of the following is not true about the properties of bases?

A. They have bitter taste
B. They change blue litmus to red
C. They release hydroxide ion in water
D. They have PH value above seven

Answer B

POH

➦ POH is defined as the negative logarithm of the concentration of hydroxide ion

$$POH = -10g[O\bar{H}]$$

$$[O\bar{H}] = 10^{-POH}$$

Ionization of water

Pure water contains small number of hydronium and hydroxide ions due to self-ionization or autoionization of water.

$$H_2O + H_2O(1) \Rightarrow H_3O^+(aq) + OH^-(aq)$$

➦ we can apply the low of mass action and write:

$$k_c = \frac{[H_3O^+][O\bar{H}]}{[H_2O]^2}$$

For dilute solutions, the concentration for water is very large and remains constant

$$\Rightarrow Kc[H_2O]^2 = K_w = [H_3O^+][OH^-]$$

➦ K_{w-} is ion product of water

➦ At $25^0c \cdot K_w$ has the value of $1.0\times10^{-14}(mol/L)^2$

$\Rightarrow [H_3O^+][OH^-] = 10^{-14}(mol/L)^2$

Since equal amounts of H_3O^+ and OH^- ions form when water dissociates, we can readily calculate $[H_3O^+]$ and $[OH^-]$ in pure water at 25^0c as:

$[H_3O^+] = [OH^-]$

$$[H_3O^+] = \left[H_3O^+\right]^2 = 1.0\times10^{-14}\left(\frac{mol}{l}\right)^2$$

$$\Rightarrow [H_3O^+] = \left[H_3O^+\right] = 1.0\times10^{-7}\left(\frac{mol}{l}\right) = [O\bar{H}]$$

Relationship between PH and POH

From ion product of water we know :

$[H_3O^+] \times [OH^-] = 10^{-14}$ at 25^0c

Put logarithm on both sides of the equation

$\Rightarrow \log[H_3O^+][OH^-] = \log 10^{-14}$

$\log[H_3O^+] + \log\ [OH^-] = \log 10^{-14}$

multiply both sides by -1

$\Rightarrow -\log [H_3O^+] - \log [OH^-] = -\log 10^{-14}$

$\Rightarrow PH + POH = 14$

$PH = 14 - pOH$

$$PH = 14 - (-log[OH^-]) = 14 + log[OH^-]$$

Similarly

$POH = 14 + \log [H_3O^+]$

- PH and $[OH^-]$ are directly related.
- POH and $[H_3O^+]$ are directly related.

PH < 7 or POH > 7 . . ..
$[H_3O^+] > 10^{-7}$ or $[OH^-] < 10^{-7}$] acidic solution
PH>7 or POH < 7
$[OH^-] > 10^{-7}$ or $[H_3O^{+1}] < 10^{-7}$ basic solution
$[OH^-] = [H_3O^+] = 10^{-7}$
PH = POH = 7 neutral solution

Illustrative Examples

68. Which of the following is acidic solution?

A. $[H_3O^+] = 10^{-7}$ M

B. PH = 10.25

C. $[OH^-] = 10^{-8}$M

D. POH = 5.50

Explanation

For acidic solution, $[H_3O^+] > 10^{-7}$, PH <7, POH >7 and $[OH^-] < 10^{-7}$

Answer: C

69. A human blood has a PH of 7.4. Its POH will be (EGSECE….2001E.C)

A. 6.6

B. 3.98×10^{-8}

C. 6.51×10^{-7}

D. 7.4

Solution

PH + POH = 14

POH = 14 – PH = 14 - .4 = 6.6

Answer A

70. Assuming complete ionization, what is the PH of a 0.001M solution of KOH?

Solution

For strong monobasic basic: $[OH^-]$=[base]

$\Rightarrow [OH^-] = 0.001M = 10^{-3}M$

$PH = 14 + 10g\,[OH^-] = 14 + 10g\,[OH^-] = 14 + \log 10^{-3} = 11$